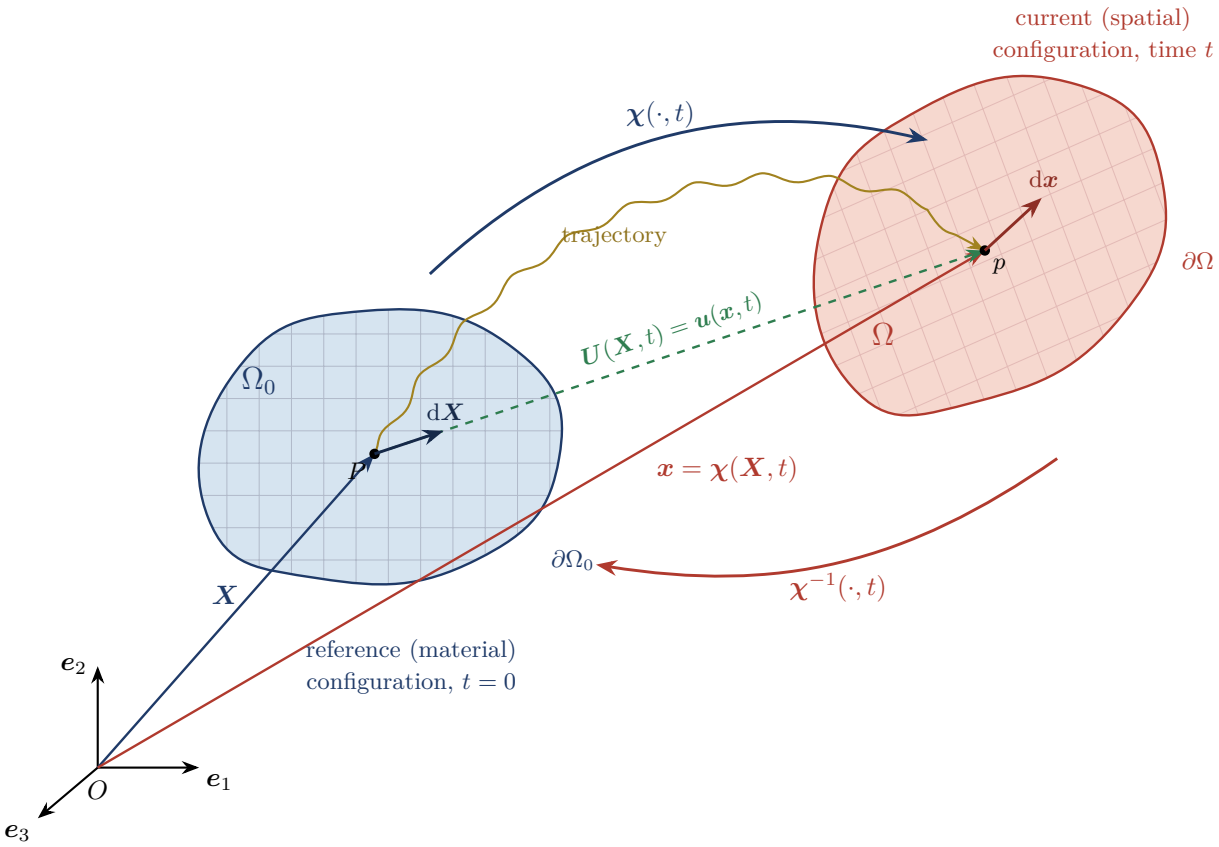
PHD COURSE · 2026

Politecnico di Milano · Università di Trento

Nonlinear Solid Mechanics

A continuum approach for engineers.
Lecture notes of the doctoral course



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Based on the classroom lectures, on the course slides and on G. A. Holzapfel, *Nonlinear Solid Mechanics. A Continuum Approach for Engineering*, Wiley, 2000.

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Course identification

Course	*An Introduction to Nonlinear Solid Mechanics* (NLSM)
Level	Doctoral (PhD) course, 5 credits (Italian system)
Institutions	Politecnico di Milano and Università degli Studi di Trento; open to PhD students in Italy and abroad
Lecturer	Prof. Anna Maria Pandolfi, Politecnico di Milano
Period	2–11 November 2026, Leonardo Campus (Milano), also streamed
Format	8 lectures of 3 hours + 4 hours of backup and verification; blackboard and slides
Main textbook	G. A. Holzapfel, *Nonlinear Solid Mechanics*, Wiley, 2000
Notes by	Lorenzo Vergata

How this book was written. Each chapter is a faithful, reorganised and corrected rewriting of the lectures, reconstructed from the recordings of the classes, integrated with the course slides (NLSM_01–NLSM_10) and with the reference textbooks listed in the Preface. Material that was only mentioned in class, left as an exercise, or added from the textbooks for completeness is marked explicitly ("In class" and "Complement" boxes), so that the reader always knows what was said by the lecturer and what was added.

Status. Draft version 0.1. This document grows lecture by lecture; chapters not yet written are announced in the table of contents of the Preface.

Preface

Aim of the course

Continuum mechanics explains successfully a wide variety of phenomena without describing the complexity of the internal microstructure of matter: the solid is regarded as a *macroscopic system*. This course is a rigorous introduction to *nonlinear* solid mechanics, with special attention to finite kinematics, material frame indifference (objectivity), and constitutive models formulated within a thermodynamic framework, followed by the analysis of several nonlinear material behaviours.

The course is addressed to PhD students, mostly of the first and second year. Many of the concepts reviewed at the beginning (vectors, tensors, stress, strain) are, in principle, known to every student of physics or engineering. They are nevertheless repeated carefully, because in nonlinear mechanics they are used continuously and in a more general setting (non-symmetric tensors, two different configurations, non-orthogonal bases), where intuition built on linear theory may fail.

Course outline

The lectures follow the sequence below; the corresponding chapters of this book are written lecture by lecture.

1. **Mathematical preliminaries.** Notation, vector and tensor algebra, tensor calculus (Chapter 1). *Dual basis*: general, non-orthogonal bases, covariant and contravariant components (Chapter 2).

2. **Kinematics of deformation.** Motion, deformation and strain; local kinematics; polar decomposition (Chapter 3). The aim is to understand precisely the difference between motion, deformation and strain.

3. **Statics.** The concept of stress, recalled because it is sometimes not completely clear even to advanced students (Chapter 4).

4. **Balance principles.** Conservation of mass, balance of linear and angular momentum, balance of energy; principle of virtual work, first and second law of thermodynamics (Chapter 5).

5. **Objectivity.** Material frame indifference (Chapter 6).

6. **Constitutive theory.** The additional equations needed to pose a continuum mechanics problem properly; thermodynamics of materials, Coleman–Noll procedure, thermodynamic potentials, kinetic relations, material classification (Chapter 7).

7. **Hyperelasticity.** Elastic symmetries, internal constraints; isotropic, transversely isotropic and anisotropic materials.

8. **Plasticity.** Finite plasticity, multiplicative decomposition of the deformation gradient, exponential and logarithmic mappings, J_2 and pressure-dependent plasticity.
9. **Viscosity.**
10. **Other behaviours** (if time allows): materials with an internal microstructure described through the phenomenological models introduced in the course, e.g. fibre-reinforced tissues and liquid crystals.

Textbooks

G. A. Holzapfel, *Nonlinear Solid Mechanics. A Continuum Approach for Engineering*, Wiley, Chichester, 2000.
The main textbook of the course. The lecturer studied on other books but, when preparing the course, found this one particularly well structured for teaching and adopted it. Most of the equations and *all of the notation* used in the course (and in this book) follow Holzapfel.

R. W. Ogden, *Non-Linear Elastic Deformations*, Ellis Horwood, 1984 (Dover, 1997).
An excellent book on continuum mechanics, useful to go deep into a topic. Its notation, however, is very mathematical and old-fashioned, and it is not easy to translate it into the standard ("Berkeley") notation used here and by most of the people who write papers on continuum mechanics today. Not easy to read.

J. E. Marsden and T. J. R. Hughes, *Mathematical Foundations of Elasticity*, Prentice-Hall, 1983 (Dover, 1994).
The lecturer had the honour of working with Jerry Marsden, a fantastic teacher, able to explain everything; his books, however, are very demanding. Not recommended as a *first* book on elasticity.

A. J. M. Spencer, *Continuum Mechanics*, Longman, 1980 (Dover, 2004); P. Chadwick, *Continuum Mechanics. Concise Theory and Problems*, Allen & Unwin, 1976 (Dover, 1999).
Two simple and compact books, good for a quick check of a concept.

L. E. Malvern, *Introduction to the Mechanics of a Continuous Medium*, Prentice-Hall, 1969; C. Truesdell and W. Noll, *The Non-Linear Field Theories of Mechanics*, Springer, 1965 (2004).
Classical treatises, incredibly good, on which a generation studied. They are less up to date on some developments that are central to the modern computational treatment of kinematics, for instance the systematic use of the polar decomposition of the deformation gradient that will be used extensively in this course. To be used to go deeply into specific concepts.

Full references are collected in the Bibliography.

Notation at a glance

The notation is that of Holzapfel's book and of the course slides. On the blackboard bold symbols cannot be reproduced, so the order of a tensor is indicated by the number of underlines.

Quantity	Book / slides	Blackboard	Components
Scalar	$\alpha, \beta, \gamma, \Phi$	α	–
Vector (1st order)	$\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{u}$	$\underline{a}$	a_i
2nd-order tensor	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$	$\underline{\underline{A}}$	A_{ij}
3rd-order tensor	$\boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{B}}$	$\underline{\underline{\underline{A}}}$	$\mathcal{A}_{ijk}$
4th-order tensor	$\mathbb{A}, \mathbb{B}, \mathbb{C}$	$\underline{\underline{\underline{\underline{A}}}}$	$\mathbb{A}_{ijkl}$

Indices of components range over $1, 2, 3$, and repeated indices are summed (Einstein convention); see Section 1.1. Lowercase indices $i, j, k, \dots$ (or $a, b, \dots$) refer to the current configuration, uppercase indices $A, B, \dots$ to the reference configuration (Chapter 3).

Conventions used in this book

- **Definition**, **Theorem**, **Example** and **Exercise** boxes follow the slides and the lectures.
- **In class** boxes collect comments, warnings and anecdotes given by the lecturer, which do not appear on the slides.
- **Complement** boxes contain material added from the textbooks (mostly Holzapfel) to complete a proof or a statement that in class was only sketched or left as an exercise.

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Part I

Mathematical Foundations

1

Mathematical Preliminaries

> **In class**
>
> It may look trivial to repeat these things at the first year of a PhD school, but it is better to repeat them: when they are used continuously, one easily forgets what they mean. Most of the content of this chapter is "known", but in nonlinear mechanics it is used in a more general setting than in linear elasticity, where everything is symmetric and one never has to wonder in which basis, or in which configuration, a quantity lives.

This chapter collects the algebraic and differential tools that are used throughout the course: notation, vectors and second-order tensors in an orthonormal basis, the main products between them, special classes of tensors (orthogonal, symmetric, skew, spherical, deviatoric), higher-order tensors, eigenvalue problems, change of basis, and the calculus of tensor functions and tensor fields. The extension to general (non-orthogonal) bases, i.e. the *dual basis*, is the subject of Chapter 2 and is only mentioned here where needed.

1.1 Notation

In the notes, in the slides and on the blackboard, different symbols are used for scalars, vectors and tensors of different order. The symbol itself tells the reader how many pieces of information the quantity carries.

Scalars
are denoted in general by Greek letters, $\alpha, \beta, \gamma, \ldots$ A scalar carries a single piece of information and has no index.

Vectors
are denoted by lowercase bold symbols, $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}, \ldots$ Bold means that the symbol carries more than one piece of information: a vector is an ordered sequence of scalars. On the blackboard bold cannot be reproduced, and a vector is *underlined once*, $\underline{a}$. The components of a vector are scalars, denoted by the same (non-bold) letter with one index, a_i, where the index i runs over the space dimension, 1 to 3. Not all vectors of the course live in the physical space, but most of them do, and they are characterised by three components with respect to a basis (introduced in Section 1.2).

Second-order tensors
are formally matrices, and are denoted by uppercase bold symbols $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \ldots$; on the blackboard they are *underlined twice*, $\underline{\underline{A}}$, to emphasise that they contain many components. Their components carry *two* indices, A_{ij}, each running from 1 to 3 for the tensors of this course.

Third-order tensors
appear when we take spatial derivatives of second-order tensors (e.g., the gradient of a tensor field). They are denoted by bold calligraphic letters $\boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{B}}, \ldots$, underlined three times on the blackboard, $\underline{\underline{\underline{A}}}$. Their components have three indices, $\mathcal{A}_{ijk}$: a third-order tensor does not have only rows and columns, but also *layers*.

Fourth-order tensors
appear when we differentiate a tensor with respect to a tensor (e.g., the elasticity tensor relating stress to strain). They are denoted by blackboard-bold letters $\mathbb{A}, \mathbb{B}, \mathbb{C}, \ldots$, underlined four times on the blackboard, and have four indices, $\mathbb{A}_{ijkl}$, each running from 1 to 3.

Remark 1.1. On the blackboard the components of third- and fourth-order tensors are often written with the same (roman) letter used for the components of second-order tensors, e.g., A_{ijkl}; the number of indices removes any ambiguity.

1.1.1 Index notation and summation convention

Definition 1.1 (Einstein summation convention)**.** Whenever an index is *repeated twice* in the same monomial (product term), summation over the range $1, 2, 3$ of that index is implied, and the summation symbol is omitted. A repeated index is called a *dummy* (or summation) index; an index appearing once is a *free* index.

For example,

$$a_i b_i \;=\; \sum_{i=1}^{3} a_i b_i \;=\; a_1 b_1 + a_2 b_2 + a_3 b_3. \tag{1.1}$$

The convention is essential to keep the calculations with tensors under control: without it one quickly loses track of the terms. A few rules follow:

- a dummy index can be renamed freely, $a_i b_i = a_k b_k$;
- an index may not appear more than twice in a monomial; when needed, use different letters, e.g., $(a_i b_i)(c_i d_i)$ must be written $a_i b_i\, c_k d_k$;
- each free index must appear in every term of an equation, and an equation with one free index stands for three scalar equations, one with two free indices for nine, and so on;
- the number of free indices of an expression equals the order of the resulting tensor (none: scalar; one: vector; two: second-order tensor…).

1.1.2 Symbols of the main operations

Table 1.1 collects the symbols of the products and of the differential operators used in the course; each one is discussed in detail in the following sections.

In class

Do not forget the dot in the scalar product. Some mathematicians argue that from the context one understands that two vectors produce a scalar, so the symbol is unnecessary. In mechanics the symbol is essential: it says explicitly that we want to *saturate* (contract) the index of the two vectors. The same holds for every product symbol: it is the symbol that tells which indices are contracted and which are expanded. The symbol $\times$ for the vector product is the one of the

Operation	Symbol	Index form	Result
Dot (scalar) product	$\boldsymbol{a}\cdot\boldsymbol{b}$	$a_i b_i$	scalar
Vector (cross) product	$\boldsymbol{a}\times\boldsymbol{b}$	$\varepsilon_{ijk}a_i b_j$	vector (k)
Dyadic (tensor) product	$\boldsymbol{a}\otimes\boldsymbol{b}$	$a_i b_j$	2nd-order tensor (ij)
Norm	$\lvert\boldsymbol{a}\rvert$	$\sqrt{a_i a_i}$	scalar
Single contraction	$\boldsymbol{AB}$, $\boldsymbol{Au}$	$A_{ik}B_{kj}$, $A_{ik}u_k$	tensor, vector
Double contraction	$\boldsymbol{A}:\boldsymbol{B}$	$A_{ij}B_{ij}$	scalar
Nabla (gradient) of a scalar	$\nabla\alpha$	$\alpha_{,i}$	vector
Divergence	$\nabla\cdot\boldsymbol{a}$	$a_{i,i}$	scalar
Curl	$\nabla\times\boldsymbol{a}$	$\varepsilon_{ijk}a_{j,i}$	vector (k)
Laplace operator	$\nabla^2=\nabla\cdot\nabla$	$(\bullet)_{,ii}$	same order

Table 1.1: Symbols and index forms of the main operations (slide 5). A comma followed by an index denotes partial differentiation with respect to the corresponding spatial coordinate, $(\bullet)_{,i}=\partial(\bullet)/\partial x_i$.

> "Berkeley" notation adopted here; other books use $\wedge$.

1.2 Orthonormal basis and representation of vectors

Every time we describe a physical quantity, the first thing to do is to specify a *reference system* (frame of reference) in the space in which we are working. We work in the three-dimensional Euclidean space $\mathbb{E}^3$ (identified with $\mathbb{R}^3$ once an origin and a basis are fixed), with a *right-handed orthonormal* system.

Definition 1.2 (Orthonormal basis)**.** A fixed set of three unit vectors $\{\boldsymbol{e}_1,\boldsymbol{e}_2,\boldsymbol{e}_3\}$ is an *orthonormal basis* if

$$\boldsymbol{e}_i\cdot\boldsymbol{e}_j=\delta_{ij},\qquad i,j=1,2,3, \tag{1.2}$$

where δ_{ij} is the *Kronecker delta*

$$\delta_{ij}=\begin{cases}1 & \text{if } i=j,\\ 0 & \text{if } i\neq j,\end{cases}\qquad \delta_{ii}=\delta_{11}+\delta_{22}+\delta_{33}=3. \tag{1.3}$$

Equation (1.2) expresses two facts: the basis vectors are mutually orthogonal ($\boldsymbol{e}_1\perp\boldsymbol{e}_2$, $\boldsymbol{e}_1\perp\boldsymbol{e}_3$, $\boldsymbol{e}_2\perp\boldsymbol{e}_3$) and each one has unit magnitude. This is the "normal" orthogonal Cartesian basis used in the vast majority of the course; the exception is the dual basis of Chapter 2.

The Kronecker delta has two indices: it can be seen as a matrix, whose diagonal terms are 1 and off-diagonal terms are 0, i.e. the *unit (identity) matrix*:

$$[\delta_{ij}]=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}\ \widehat{=}\ \boldsymbol{I}. \tag{1.4}$$

Strictly speaking the symbol "=" between a matrix of components and a tensor is not correct (one is the matrix notation, the other the tensor notation); here $\widehat{=}$ only means that the two objects have the same meaning. The identity $\boldsymbol{I}$ is very important in mechanics: when it is applied to a vector, nothing changes. Note also the *substitution* (or replacement) property of δ_{ij}:

$$\delta_{ij} u_j = u_i, \qquad \delta_{ij} A_{jk} = A_{ik}, \tag{1.5}$$

i.e. δ_{ij} "replaces" the dummy index j by i.

1.2.1 Components of a vector

In the chosen basis, any vector $\boldsymbol{u}$ is defined by its components:

$$\boldsymbol{u} = u_1 \boldsymbol{e}_1 + u_2 \boldsymbol{e}_2 + u_3 \boldsymbol{e}_3 = u_i \boldsymbol{e}_i, \tag{1.6}$$

where u_1 is the component of $\boldsymbol{u}$ in the direction of the axis x_1, etc. Geometrically (Fig. 1.1), $\boldsymbol{u}$ is the sum of the three vectors $u_1\boldsymbol{e}_1$, $u_2\boldsymbol{e}_2$ and $u_3\boldsymbol{e}_3$.

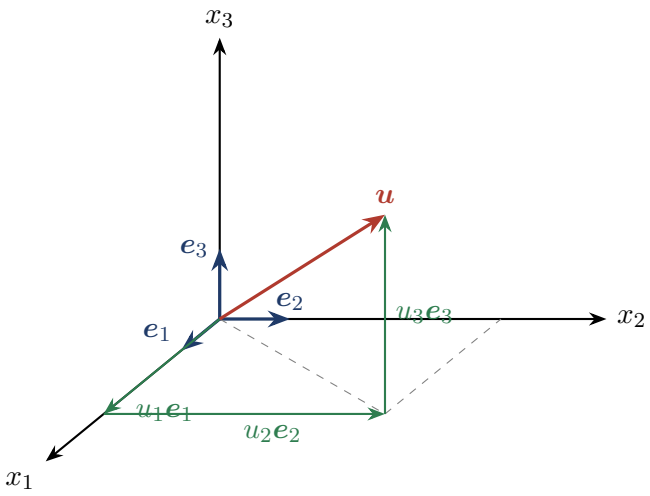


Figure 1.1: Representation of a vector in an orthonormal basis, $\boldsymbol{u} = u_i \boldsymbol{e}_i$.

What is commonly done in practice is to fix the reference system once and for all, and then to list only the components in an array (column matrix), $\{\boldsymbol{u}\} = \{u_1, u_2, u_3\}^{\mathrm{T}}$. The two representations carry the same information only because the basis is implied in the second one: in (1.6) the basis is explicitly present, in the array it is assumed.

> **In class**
>
> This looks trivial, and it is, for an orthonormal basis. Think, however, of a *non-orthogonal* reference system: there the decomposition of a vector into components is not trivial at all (it requires the dual basis). This is why the "correct" representation of a vector is always $\boldsymbol{u} = u_i \boldsymbol{e}_i$, with the basis written explicitly. When we move to matrices and tensors, the distinction between an object and the table of its components becomes even more important.

1.2.2 Permutation tensor

Definition 1.3 (Permutation (Levi-Civita, alternating) symbol)**.** The *permutation tensor* ε_{ijk} is the third-order quantity

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{if } (i,j,k) \text{ is an even (cyclic) permutation of } (1,2,3), \\ & \text{i.e. } (1,2,3),\ (2,3,1),\ (3,1,2); \\ -1 & \text{if } (i,j,k) \text{ is an odd permutation of } (1,2,3), \\ & \text{i.e. } (1,3,2),\ (3,2,1),\ (2,1,3); \\ 0 & \text{if two (or three) indices are equal.} \end{cases} \tag{1.7}$$

Being a third-order tensor, ε_{ijk} can be visualised as a cube of 27 numbers: a "matrix" with rows (i), columns (j) and layers (k). Writing the three layers $k = 1, 2, 3$ side by side:

$$\underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}}_{k=1} \qquad \underbrace{\begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}}_{k=2} \qquad \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{k=3} \tag{1.8}$$

The construction is easy to remember. In layer k, row k and column k are null (two indices would coincide), and so is the diagonal ($i = j$). Only two entries survive in each layer. For instance in the first layer, row 2 and column 3 give $(i,j,k) = (2,3,1)$, a cyclic permutation, hence $+1$; row 3 and column 2 give $(3,2,1)$, an inverted order, hence -1.

ε–δ identities

$$\varepsilon_{ijk}\varepsilon_{pqk} = \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}, \qquad \varepsilon_{ijk}\varepsilon_{ljk} = 2\delta_{il}, \qquad \varepsilon_{ijk}\varepsilon_{ijk} = 6. \tag{1.9}$$

The second identity follows from the first by setting $q = j$ and summing: $\delta_{ip}\delta_{jj} - \delta_{ij}\delta_{jp} = 3\delta_{ip} - \delta_{ip} = 2\delta_{ip}$; the third by setting $l = i$: $2\delta_{ii} = 6$. In words: saturating all three indices of the product of two permutation tensors gives 6 (the number of non-zero entries, each equal to $(\pm 1)^2$); saturating one index gives a combination of Kronecker deltas; saturating two indices gives twice the identity.

1.3 Basic vector algebra

Vectors are mathematical objects for which *more than one* kind of multiplication can be defined: one reduces two vectors to a scalar (dot product), one transforms two vectors into another vector (vector product), and one expands two vectors into a second-order tensor (dyadic product, Section 1.4).

1.3.1 Scalar (dot) product

Definition 1.4 (Scalar product)**.** The *scalar* (dot, inner) product of $\boldsymbol{u}$ and $\boldsymbol{v}$ is the scalar

$$\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{u} = |\boldsymbol{u}|\,|\boldsymbol{v}| \cos\theta(\boldsymbol{u}, \boldsymbol{v}), \tag{1.10}$$

where $\theta(\boldsymbol{u},\boldsymbol{v}) \in [0,\pi]$ is the angle between the two vectors.

Geometrically, $\boldsymbol{u}\cdot\boldsymbol{v}$ is the projection of one vector onto the direction of the other, multiplied by the length of the latter. A familiar example: if $\boldsymbol{v}$ is a force and $\boldsymbol{u}$ a displacement, the work done by the force is the scalar product of the two. Taking $|\boldsymbol{u}| = |\boldsymbol{v}| = 1$ one sees that $\boldsymbol{u}\cdot\boldsymbol{v}$ is exactly the cosine of the angle between them.

In components, using (1.2),

$$\boldsymbol{u}\cdot\boldsymbol{v} = (u_i\boldsymbol{e}_i)\cdot(v_j\boldsymbol{e}_j) = u_i v_j\,\boldsymbol{e}_i\cdot\boldsymbol{e}_j = u_i v_j \delta_{ij} = u_i v_i, \tag{1.11}$$

i.e. we take each component of $\boldsymbol{u}$, multiply it by the *same* component of $\boldsymbol{v}$, and sum: the dot product is a *contraction of indices*.

Components of a vector. To compute the component u_j of $\boldsymbol{u}$ along the axis j, take the scalar product of $\boldsymbol{u}$ with the unit vector of that axis:

$$\boldsymbol{u}\cdot\boldsymbol{e}_j = (u_i\boldsymbol{e}_i)\cdot\boldsymbol{e}_j = u_i\delta_{ij} = u_j. \tag{1.12}$$

Norm. The scalar product of a vector with itself gives the square of its length (Pythagoras' theorem):

$$|\boldsymbol{u}|^2 = \boldsymbol{u}\cdot\boldsymbol{u} = u_i\boldsymbol{e}_i\cdot u_j\boldsymbol{e}_j = u_i u_j\delta_{ij} = u_i u_i = u_1^2+u_2^2+u_3^2, \qquad |\boldsymbol{u}| = \sqrt{u_i u_i}. \tag{1.13}$$

1.3.2 Vector (cross) product

Definition 1.5 (Vector product)**.** The *vector* (cross) product $\boldsymbol{w} = \boldsymbol{u}\times\boldsymbol{v}$ is the vector

- orthogonal to the plane spanned by $\boldsymbol{u}$ and $\boldsymbol{v}$: $\boldsymbol{w}\cdot\boldsymbol{u} = \boldsymbol{w}\cdot\boldsymbol{v} = 0$;
- with magnitude equal to the area of the parallelogram spanned by $\boldsymbol{u}$ and $\boldsymbol{v}$: $|\boldsymbol{w}| = |\boldsymbol{u}|\,|\boldsymbol{v}|\sin\theta(\boldsymbol{u},\boldsymbol{v})$;
- oriented so that $(\boldsymbol{u},\boldsymbol{v},\boldsymbol{w})$ is right-handed.

It is anti-commutative: $\boldsymbol{u}\times\boldsymbol{v} = -\boldsymbol{v}\times\boldsymbol{u}$.

The most common way to compute it is the symbolic determinant

$$\boldsymbol{u}\times\boldsymbol{v} = \det\begin{bmatrix}\boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3\\ u_1 & u_2 & u_3\\ v_1 & v_2 & v_3\end{bmatrix} = (u_2v_3-u_3v_2)\,\boldsymbol{e}_1 + (u_3v_1-u_1v_3)\,\boldsymbol{e}_2 + (u_1v_2-u_2v_1)\,\boldsymbol{e}_3, \tag{1.14}$$

obtained by expanding along the first row (for the $\boldsymbol{e}_2$ term the columns are exchanged, to take the correct sign of the cofactor). The three brackets are the components along the directions 1, 2 and 3. With the permutation tensor the same result is written compactly as

Vector product in index notation

$$\boldsymbol{u}\times\boldsymbol{v} = \varepsilon_{ijk}u_i v_j\,\boldsymbol{e}_k, \qquad (\boldsymbol{u}\times\boldsymbol{v})_k = \varepsilon_{ijk}u_i v_j. \tag{1.15}$$

In $\varepsilon_{ijk}u_i v_j$ the index i appears twice (sum over i), j appears twice (sum over j), and k remains free: the result is a vector. One may verify, by expanding the sums, that (1.15) reproduces

(1.14) term by term; for instance, for $k = 1$ the only non-zero terms are $\varepsilon_{231}u_2v_3 + \varepsilon_{321}u_3v_2 = u_2v_3 - u_3v_2$.

Vector products of the basis vectors. Since the basis is orthonormal and right-handed, $\boldsymbol{e}_1 \times \boldsymbol{e}_2$ is orthogonal to the plane of $\boldsymbol{e}_1, \boldsymbol{e}_2$ and has the area of the unit square, 1: it is $\boldsymbol{e}_3$. Exchanging the order reverses the direction. In general

$$\begin{gathered} \boldsymbol{e}_1 \times \boldsymbol{e}_2 = \boldsymbol{e}_3, \quad \boldsymbol{e}_2 \times \boldsymbol{e}_3 = \boldsymbol{e}_1, \quad \boldsymbol{e}_3 \times \boldsymbol{e}_1 = \boldsymbol{e}_2, \\ \boldsymbol{e}_2 \times \boldsymbol{e}_1 = -\boldsymbol{e}_3, \quad \boldsymbol{e}_3 \times \boldsymbol{e}_2 = -\boldsymbol{e}_1, \quad \boldsymbol{e}_1 \times \boldsymbol{e}_3 = -\boldsymbol{e}_2, \\ \boldsymbol{e}_1 \times \boldsymbol{e}_1 = \boldsymbol{e}_2 \times \boldsymbol{e}_2 = \boldsymbol{e}_3 \times \boldsymbol{e}_3 = \boldsymbol{0}, \end{gathered} \tag{1.16}$$

which are summarised by

$$\boldsymbol{e}_i \times \boldsymbol{e}_j = \varepsilon_{ijk}\, \boldsymbol{e}_k, \qquad \varepsilon_{ijk} = (\boldsymbol{e}_i \times \boldsymbol{e}_j) \cdot \boldsymbol{e}_k. \tag{1.17}$$

Exercise 1.1. Prove (1.17) from the list above, and then use it to derive (1.15): $\boldsymbol{u} \times \boldsymbol{v} = u_i v_j\, \boldsymbol{e}_i \times \boldsymbol{e}_j = \varepsilon_{ijk} u_i v_j \boldsymbol{e}_k$. Check also that, with a *left-handed* basis, all the signs would be reversed.

▷ Solution on p. 133

1.3.3 Triple scalar (box) product

First take the vector product of $\boldsymbol{u}$ and $\boldsymbol{v}$, which is a vector, and then dot it with a third vector $\boldsymbol{w}$; the result is a scalar:

$$[\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}] \coloneqq (\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w} = |\boldsymbol{u}|\, |\boldsymbol{v}| \sin\theta_{uv}\ |\boldsymbol{w}| \cos\theta_{(u\times v)w} = \varepsilon_{ijk} u_i v_j w_k. \tag{1.18}$$

Geometrically, $|[\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}]|$ is the volume of the parallelepiped spanned by the three vectors (base area times height); the sign is positive when $(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w})$ is right-handed. The index form follows immediately from (1.15): $(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w} = (\boldsymbol{u} \times \boldsymbol{v})_k w_k = \varepsilon_{ijk} u_i v_j w_k$. This shows how convenient the permutation tensor is: something long to write becomes compact and easy to memorise. Since $\varepsilon_{ijk} = \varepsilon_{jki} = \varepsilon_{kij}$, the box product is invariant under cyclic permutations, $[\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}] = [\boldsymbol{v}, \boldsymbol{w}, \boldsymbol{u}] = [\boldsymbol{w}, \boldsymbol{u}, \boldsymbol{v}]$, and it is equal to the determinant of the matrix having $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ as rows.

1.4 Second-order tensors

Vectors are familiar objects: we have used them since high school, and many physical quantities (displacements, forces, velocities) are vectors. Matrices are much less intuitive: they are tables of numbers, but what do they *represent*?

Definition 1.6 (Second-order tensor)**.** A *second-order tensor* $\boldsymbol{A}$ is a *linear transformation* (linear mapping) that acts on a vector $\boldsymbol{u}$ to generate another vector $\boldsymbol{v}$:

$$\boldsymbol{v} = \boldsymbol{A}\boldsymbol{u}, \qquad v_i = A_{ij} u_j, \tag{1.19}$$

with $\boldsymbol{A}(\alpha\boldsymbol{u} + \beta\boldsymbol{w}) = \alpha\boldsymbol{A}\boldsymbol{u} + \beta\boldsymbol{A}\boldsymbol{w}$ for all scalars α, β and vectors $\boldsymbol{u}, \boldsymbol{w}$.

This is exactly what one does when solving a linear system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$: the matrix of coefficients $\boldsymbol{A}$ is applied to the unknown vector $\boldsymbol{x}$ and gives the known vector $\boldsymbol{b}$. In index form, $A_{ij}u_j$ is the row-by-column product: the index j is contracted and disappears, the index i remains free.

1.4.1 Dyadic product

How can a tensor be generated? Through the combination of *dyads*.

Definition 1.7 (Dyad)**.** The *dyadic* (tensor, outer) product of two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ is the second-order tensor $\boldsymbol{u} \otimes \boldsymbol{v}$ with components

$$(\boldsymbol{u} \otimes \boldsymbol{v})_{ij} = u_i v_j, \qquad [\boldsymbol{u} \otimes \boldsymbol{v}] = \begin{bmatrix} u_1v_1 & u_1v_2 & u_1v_3 \\ u_2v_1 & u_2v_2 & u_2v_3 \\ u_3v_1 & u_3v_2 & u_3v_3 \end{bmatrix}. \tag{1.20}$$

The dyadic product is an *expansion of indices*, as opposed to the dot product, which is a contraction: from two first-order objects we obtain a second-order one. In matrix algebra, it corresponds to the column-by-row product $\{\boldsymbol{u}\}\{\boldsymbol{v}\}^{\mathrm{T}}$, whereas the row-by-column product $\{\boldsymbol{u}\}^{\mathrm{T}}\{\boldsymbol{v}\}$ gives the scalar $\boldsymbol{u} \cdot \boldsymbol{v}$. The dyadic product is not commutative:

$$\boldsymbol{W} = \boldsymbol{u} \otimes \boldsymbol{v} \neq \boldsymbol{v} \otimes \boldsymbol{u}, \qquad W_{ij} = u_i v_j \neq v_i u_j. \tag{1.21}$$

In this sense, vectors are often called *first-order tensors*.

Definition 1.8 (Dyadic)**.** A *dyadic* is a linear combination of dyads with scalar coefficients:

$$\boldsymbol{W} = \sum_{k=1}^{N} \alpha_k \, \boldsymbol{u}_k \otimes \boldsymbol{v}_k, \qquad \text{e.g.,} \qquad \boldsymbol{W} = \alpha \, \boldsymbol{u} \otimes \boldsymbol{v} + \beta \, \boldsymbol{w} \otimes \boldsymbol{x}. \tag{1.22}$$

In general a tensor cannot be expressed as a single dyad (a single dyad has rank one), but it can always be expressed as a dyadic.

In particular, any second-order tensor can be expressed as a dyadic formed by the basis vectors:

Representation of a second-order tensor

$$\boldsymbol{A} = A_{ij} \, \boldsymbol{e}_i \otimes \boldsymbol{e}_j = \sum_{i=1}^{3} \sum_{j=1}^{3} A_{ij} \, \boldsymbol{e}_i \otimes \boldsymbol{e}_j. \tag{1.23}$$

This is the analogue of (1.6): the nine components A_{ij} multiply the nine dyads $\boldsymbol{e}_i \otimes \boldsymbol{e}_j$, which carry the two directions i and j. Of course we may also write simply $\boldsymbol{A} = [A_{ij}]$, $i, j = 1, 2, 3$, but the correct representation is (1.23), where the basis in which the components are defined is explicit.

1.4.2 Interpretation of the dyad

Apply the dyad $\boldsymbol{u} \otimes \boldsymbol{v}$ to a third vector $\boldsymbol{w}$, as one multiplies a matrix by a vector. In indices:

$$n_i = (\boldsymbol{u} \otimes \boldsymbol{v})_{ij} w_j = (u_i v_j) w_j = u_i (v_j w_j), \tag{1.24}$$

where the order of the factors can be changed freely because they are scalars. The term $v_j w_j$ is the scalar product $\boldsymbol{v} \cdot \boldsymbol{w}$, hence

Action of a dyad

$$\boldsymbol{n} = (\boldsymbol{u} \otimes \boldsymbol{v})\boldsymbol{w} = \boldsymbol{u}\,(\boldsymbol{v} \cdot \boldsymbol{w}) = (\boldsymbol{v} \cdot \boldsymbol{w})\,\boldsymbol{u}. \tag{1.25}$$

The dyad transforms the vector $\boldsymbol{w}$ into a vector oriented as the *first* vector of the dyad, $\boldsymbol{u}$, with a length proportional to the projection of $\boldsymbol{w}$ on the *second* vector, $\boldsymbol{v}$. The rule to remember is: *take the second vector of the dyad and dot it with the external vector*. This trick will be used many times.

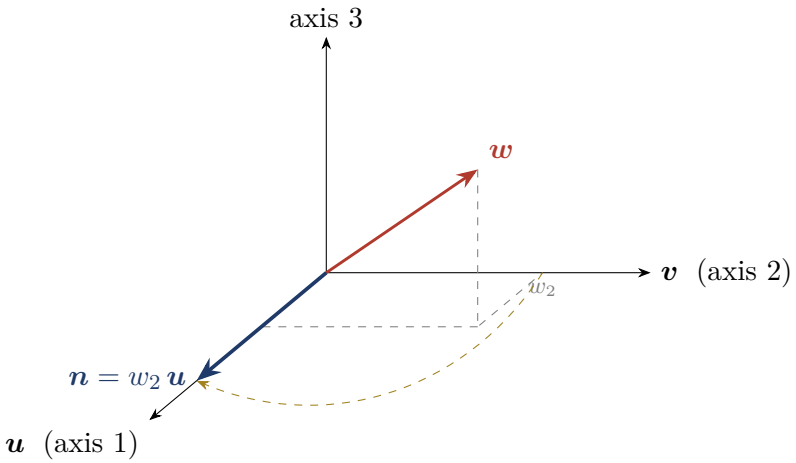


Figure 1.2: Interpretation of a dyad (slide 9) for unit, orthogonal vectors $\boldsymbol{u}$ and $\boldsymbol{v}$: the dyad $\boldsymbol{u} \otimes \boldsymbol{v}$ maps $\boldsymbol{w}$ into $\boldsymbol{n} = (\boldsymbol{v} \cdot \boldsymbol{w})\boldsymbol{u}$, a vector in the direction of $\boldsymbol{u}$ whose length is the component of $\boldsymbol{w}$ along $\boldsymbol{v}$.

Example 1.1 (Dyad in Cartesian coordinates)**.** Let $\boldsymbol{u} = \{u_1, 0, 0\}^{\mathrm{T}}$, $\boldsymbol{v} = \{0, v_2, 0\}^{\mathrm{T}}$, $\boldsymbol{w} = \{w_1, w_2, w_3\}^{\mathrm{T}}$. Then

$$[\boldsymbol{u} \otimes \boldsymbol{v}] = \begin{bmatrix} 0 & u_1 v_2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad (\boldsymbol{u} \otimes \boldsymbol{v})\boldsymbol{w} = \begin{bmatrix} 0 & u_1 v_2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} w_1 \\ w_2 \\ w_3 \end{Bmatrix} = \begin{Bmatrix} u_1 v_2 w_2 \\ 0 \\ 0 \end{Bmatrix} = (\boldsymbol{v} \cdot \boldsymbol{w})\,\boldsymbol{u}.$$

The vector $\boldsymbol{w}$ is transformed into a vector in the direction of $\boldsymbol{u}$ (Fig. 1.2). If $\boldsymbol{u}$ and $\boldsymbol{v}$ are unit vectors, the length of $\boldsymbol{n}$ equals the component of $\boldsymbol{w}$ in the direction of $\boldsymbol{v}$.

In class

The figure on slide 9 has been corrected recently: the version downloaded early from the course website may still show the old (wrong) picture.

1.4.3 Cartesian tensors and the identity tensor

When a tensor is resolved along orthonormal basis vectors, as in (1.23), it is called a *Cartesian tensor*. In particular, the *unit* (identity) tensor is

Identity tensor

$$\boldsymbol{I} = \delta_{ij}\,\boldsymbol{e}_i \otimes \boldsymbol{e}_j = \boldsymbol{e}_j \otimes \boldsymbol{e}_j = \boldsymbol{e}_1 \otimes \boldsymbol{e}_1 + \boldsymbol{e}_2 \otimes \boldsymbol{e}_2 + \boldsymbol{e}_3 \otimes \boldsymbol{e}_3. \tag{1.26}$$

The second equality follows from the substitution property: δ_{ij} means that everywhere we have the index i we may replace it with j. Wherever the identity tensor appears, it can be replaced by $\boldsymbol{e}_j \otimes \boldsymbol{e}_j$; one verifies immediately that $\boldsymbol{I}\boldsymbol{u} = (\boldsymbol{e}_j \otimes \boldsymbol{e}_j)\boldsymbol{u} = (\boldsymbol{e}_j \cdot \boldsymbol{u})\boldsymbol{e}_j = u_j \boldsymbol{e}_j = \boldsymbol{u}$.

A dyad and a tensor can be expressed in *matrix notation* (square brackets) by entering their nine Cartesian components in a table:

$$[\boldsymbol{A}] = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}. \tag{1.27}$$

Such a table must always be read as *the collection of the components of the tensor in a given basis*; the tensor itself is $A_{ij}\boldsymbol{e}_i \otimes \boldsymbol{e}_j$. Disregarding the basis is legitimate only as long as everybody agrees on which basis is being used.

Components of a tensor. We have seen in (1.12) how to compute the components of a vector. For a tensor, apply $\boldsymbol{A}$ to $\boldsymbol{e}_j$ and use (1.25):

$$\boldsymbol{A}\boldsymbol{e}_j = A_{il}(\boldsymbol{e}_i \otimes \boldsymbol{e}_l)\boldsymbol{e}_j = A_{il}(\boldsymbol{e}_l \cdot \boldsymbol{e}_j)\boldsymbol{e}_i = A_{il}\delta_{lj}\boldsymbol{e}_i = A_{ij}\boldsymbol{e}_i. \tag{1.28}$$

The vector $\boldsymbol{A}\boldsymbol{e}_j$ has three components (one for each i): it is the j-th column of $[\boldsymbol{A}]$. Now pre-multiply by $\boldsymbol{e}_i$ (scalar product); since $\boldsymbol{e}_i \cdot \boldsymbol{e}_i = 1$ (no sum),

Components of a tensor

$$A_{ij} = \boldsymbol{e}_i \cdot \boldsymbol{A}\boldsymbol{e}_j. \tag{1.29}$$

To find the components of a tensor, then, one needs *two projections*, along two (in general different) orientations. This is what makes tensors less intuitive than vectors, and it is the origin of the two kinds of components (normal and shear) of the stress tensor.

1.5 Definiteness, trace and transpose

1.5.1 Positive and negative definiteness

In the constitutive theory the behaviour of materials is often characterised by an energy, which is a scalar expressed in terms of a tensor. We therefore need the notions of definiteness (here for second-order tensors; for higher-order tensors the idea is similar but more involved).

Definition 1.9 (Definiteness)**.** A second-order tensor $\boldsymbol{A}$ is said to be

- *positive definite* if $\boldsymbol{v} \cdot \boldsymbol{A}\boldsymbol{v} > 0$ for every $\boldsymbol{v} \neq \boldsymbol{0}$;
- *positive semi-definite* if $\boldsymbol{v} \cdot \boldsymbol{A}\boldsymbol{v} \geq 0$ for every $\boldsymbol{v} \neq \boldsymbol{0}$;
- *negative definite* (*semi-definite*) if $\boldsymbol{v} \cdot \boldsymbol{A}\boldsymbol{v} < 0$ (≤ 0) for every $\boldsymbol{v} \neq \boldsymbol{0}$.

The condition $\boldsymbol{v} \neq \boldsymbol{0}$ is essential: for $\boldsymbol{v} = \boldsymbol{0}$ the quadratic form vanishes trivially. Positive or negative definiteness is related to the eigenvalues of the tensor (Section 1.14): a *symmetric*

tensor is positive definite if and only if all its eigenvalues are strictly positive.

Remark 1.2. Only the symmetric part of $\boldsymbol{A}$ (Section 1.10) contributes to $\boldsymbol{v}\cdot\boldsymbol{A}\boldsymbol{v}$, because $\boldsymbol{v}\cdot\boldsymbol{W}\boldsymbol{v}=0$ for every skew $\boldsymbol{W}$. Definiteness is therefore a property of $\frac{1}{2}(\boldsymbol{A}+\boldsymbol{A}^{\mathrm{T}})$.

1.5.2 Trace

Definition 1.10 (Trace)**.** The *trace* of a dyad is the scalar product of its two vectors; by linearity, the trace of a (square) tensor is the sum of its diagonal terms:

$$\operatorname{tr}(\boldsymbol{u}\otimes\boldsymbol{v})=\boldsymbol{u}\cdot\boldsymbol{v}=u_iv_i,\qquad \operatorname{tr}\boldsymbol{A}=A_{ij}\operatorname{tr}(\boldsymbol{e}_i\otimes\boldsymbol{e}_j)=A_{ij}\,\boldsymbol{e}_i\cdot\boldsymbol{e}_j=A_{ij}\delta_{ij}=A_{ii}. \tag{1.30}$$

Since A_{ii} already contains a repeated index, the sum $A_{11}+A_{22}+A_{33}$ is implied. Equivalently, $\operatorname{tr}\boldsymbol{A}=\boldsymbol{A}:\boldsymbol{I}$ (Section 1.6): the double contraction takes every A_{ij} and multiplies it by δ_{ij}, which replaces j with i, giving A_{ii}.

1.5.3 Transpose

For a matrix, the transpose is simply the matrix with exchanged indices. For a tensor the definition is more subtle.

Definition 1.11 (Transpose)**.** The *transpose* $\boldsymbol{A}^{\mathrm{T}}$ of a tensor $\boldsymbol{A}$ is the unique tensor such that, for all vectors $\boldsymbol{u},\boldsymbol{v}$,

$$\boldsymbol{v}\cdot\boldsymbol{A}^{\mathrm{T}}\boldsymbol{u}=\boldsymbol{u}\cdot\boldsymbol{A}\boldsymbol{v}\ (=\boldsymbol{A}\boldsymbol{v}\cdot\boldsymbol{u}). \tag{1.31}$$

In words: exchanging $\boldsymbol{u}$ and $\boldsymbol{v}$ in the bilinear form, we obtain the same result if we use the transpose. Taking $\boldsymbol{u}=\boldsymbol{e}_i$, $\boldsymbol{v}=\boldsymbol{e}_j$ and (1.29),

$$(\boldsymbol{A}^{\mathrm{T}})_{ji}=\boldsymbol{e}_j\cdot\boldsymbol{A}^{\mathrm{T}}\boldsymbol{e}_i=\boldsymbol{e}_i\cdot\boldsymbol{A}\boldsymbol{e}_j=A_{ij},\qquad\text{i.e.}\qquad(\boldsymbol{A}^{\mathrm{T}})_{ij}=A_{ji}, \tag{1.32}$$

which is what we already knew: the transpose has exchanged indices. The point is that the *definition* is (1.31); the index rule is a consequence in an orthonormal basis. Moreover,

$$(\boldsymbol{A}^{\mathrm{T}})^{\mathrm{T}}=\boldsymbol{A},\qquad(\boldsymbol{u}\otimes\boldsymbol{v})^{\mathrm{T}}=\boldsymbol{v}\otimes\boldsymbol{u},\qquad(\boldsymbol{A}\boldsymbol{B})^{\mathrm{T}}=\boldsymbol{B}^{\mathrm{T}}\boldsymbol{A}^{\mathrm{T}},\qquad(\alpha\boldsymbol{A}+\beta\boldsymbol{B})^{\mathrm{T}}=\alpha\boldsymbol{A}^{\mathrm{T}}+\beta\boldsymbol{B}^{\mathrm{T}}. \tag{1.33}$$

1.6 Contraction products

The scalar product is the most familiar example of a *contraction of an index*. Contractions can be defined also between tensors.

1.6.1 Single contraction

Definition 1.12 (Single contraction)**.** The *single index contraction* (dot product, or simply product) between two tensors, or between a tensor and a vector, is written without any symbol and yields a tensor or a vector, respectively:

$$\boldsymbol{C}=\boldsymbol{A}\boldsymbol{B}\neq\boldsymbol{B}\boldsymbol{A},\quad C_{ik}=A_{ij}B_{jk};\qquad\qquad\boldsymbol{v}=\boldsymbol{A}\boldsymbol{u}\neq\boldsymbol{u}\boldsymbol{A},\quad v_i=A_{ik}u_k. \tag{1.34}$$

The two notations are consistent: just as $\boldsymbol{A}$ applied to $\boldsymbol{u}$ gives a vector, $\boldsymbol{A}$ applied to $\boldsymbol{B}$ gives a tensor, defined by $(\boldsymbol{AB})\boldsymbol{u} = \boldsymbol{A}(\boldsymbol{Bu})$ for every $\boldsymbol{u}$ (composition of linear maps). In indices, one index is eliminated (contracted), the index j in $A_{ij}B_{jk}$, and the result keeps the two outer indices i and k. This is the usual row-by-column product of matrices learned in the first year of university. The product of tensors is associative but, in general, *not commutative*.

1.6.2 Double contraction

Definition 1.13 (Double contraction)**.** The *double contraction* of two second-order tensors contracts both indices and yields a scalar:

$$\boldsymbol{A} : \boldsymbol{B} = A_{ij}B_{ij} = \operatorname{tr}(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{B}) = \operatorname{tr}(\boldsymbol{A}\boldsymbol{B}^{\mathrm{T}}) = \operatorname{tr}(\boldsymbol{B}\boldsymbol{A}^{\mathrm{T}}) = \operatorname{tr}(\boldsymbol{B}^{\mathrm{T}}\boldsymbol{A}) = \boldsymbol{B} : \boldsymbol{A}. \tag{1.35}$$

Every component A_{ij} is multiplied by the same component B_{ij} and, since both i and j are repeated, everything is summed: nine terms. A familiar example is the *internal work* (or the stress power) in the principle of virtual work: stress and strain are two tensors, and to obtain a scalar we contract both indices, $\boldsymbol{\sigma} : \boldsymbol{\varepsilon} = \sigma_{ij}\varepsilon_{ij}$.

Exercise 1.2. Prove the chain of equalities (1.35). Hint: $\operatorname{tr}(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{B}) = (\boldsymbol{A}^{\mathrm{T}}\boldsymbol{B})_{ii} = (\boldsymbol{A}^{\mathrm{T}})_{ij}B_{ji} = A_{ji}B_{ji}$. Since the trace only collects diagonal terms, one may transpose and reorder the product as shown and always obtain the same result. Prove also that $\boldsymbol{u} \cdot \boldsymbol{A}\boldsymbol{v} = \boldsymbol{A} : (\boldsymbol{u} \otimes \boldsymbol{v})$ and $(\boldsymbol{u} \otimes \boldsymbol{v}) : (\boldsymbol{w} \otimes \boldsymbol{x}) = (\boldsymbol{u} \cdot \boldsymbol{w})(\boldsymbol{v} \cdot \boldsymbol{x})$.

▷ Solution on p. 133

Norm of a tensor. As for vectors, the double contraction defines a norm:

$$|\boldsymbol{A}| = (\boldsymbol{A} : \boldsymbol{A})^{1/2} = (A_{ij}A_{ij})^{1/2} \geq 0, \tag{1.36}$$

the square root of the sum of the squares of all nine components.

Remark 1.3. Do not confuse the single contraction $\boldsymbol{AB}$ (a tensor) with the double contraction $\boldsymbol{A} : \boldsymbol{B}$ (a scalar). Tensors can be combined in different ways depending on the physics to be represented, and the symbol between them is what tells which.

1.7 A small digression: groups of linear transformations

Some concepts used later (in the kinematics and in the objectivity) require the notion of group. Only the basic definitions are needed.

Definition 1.14 (Group)**.** A *group* is a set X together with a binary operation $(x, y) \mapsto xy$ that associates to two elements of X an element of X, such that

1. the operation is *associative*: $(xy)z = x(yz)$ for all $x, y, z \in X$;
2. an *identity element* $e \in X$ exists: $xe = ex = x$ for all $x \in X$;
3. every element has an *inverse*: for all $x \in X$ there is $x^{-1} \in X$ such that $xx^{-1} = x^{-1}x = e$.

By fixing a basis in $\mathbb{R}^n$, we identify the set $\mathrm{Lin}(\mathbb{R}^n, \mathbb{R}^n)$ of the linear mappings of $\mathbb{R}^n$ into itself with the set of all square $n \times n$ matrices (for us, $n = 3$: the 3×3 tensors used so far, which transform a vector into a vector). Matrix multiplication can be taken as the binary operation, and it represents the composition of linear mappings. It is associative and the identity matrix is the identity element; nevertheless $\mathrm{Lin}(\mathbb{R}^n, \mathbb{R}^n)$ *fails to be a group*, because not every linear mapping has an inverse. A restriction is required.

Definition 1.15 (General linear group)**.** The *general linear group* $\mathrm{GL}(n)$ is the group of the invertible linear mappings over $\mathbb{R}^n$:

$$\mathrm{GL}(n) = \{\boldsymbol{A} \in \mathrm{Lin}(\mathbb{R}^n, \mathbb{R}^n) \ : \ \det \boldsymbol{A} \neq 0\}. \tag{1.37}$$

From linear algebra, the existence of the inverse $\boldsymbol{A}^{-1}$ ($\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I}$) is equivalent to a non-zero determinant. Invertible tensors are the only ones that we need to describe deformations (Chapter 3).

1.8 Determinant and inverse of a tensor

The determinant of a tensor is the determinant of the matrix of its components (in any orthonormal basis; it is an invariant, Section 1.14). For a 3×3 matrix, the rule of Sarrus is the easiest to memorise: repeat the first two columns to the right of the matrix, sum the products along the three "descending" diagonals and subtract the products along the three "ascending" ones:

$$\det \boldsymbol{A} = A_{11}A_{22}A_{33} + A_{12}A_{23}A_{31} + A_{13}A_{21}A_{32} - A_{13}A_{22}A_{31} - A_{11}A_{23}A_{32} - A_{12}A_{21}A_{33}. \tag{1.38}$$

With the permutation tensor, the same expression can be written in a compact form. Fixing the column indices $1, 2, 3$ (expansion along the columns, the same structure as the vector product (1.14)):

Determinant

$$\det \boldsymbol{A} = \det [\boldsymbol{A}] = \varepsilon_{ijk} A_{i1} A_{j2} A_{k3} = \frac{1}{6} \varepsilon_{ijk} \varepsilon_{lrs} A_{il} A_{jr} A_{ks}. \tag{1.39}$$

The first form is obtained by focusing on the three columns; the second form, which treats all columns on the same footing, uses a second permutation tensor ε_{lrs} in place of the fixed column indices $1, 2, 3$. Indeed, the general identity $\varepsilon_{ijk} A_{il} A_{jr} A_{ks} = \varepsilon_{lrs} \det \boldsymbol{A}$ holds (exchanging two columns changes the sign); multiplying both sides by ε_{lrs} and using $\varepsilon_{lrs}\varepsilon_{lrs} = 6$ from (1.9) gives the factor $1/6$. This form is convenient, for instance, when coding the calculation of a determinant.

Properties.

$$\det(\boldsymbol{A}\boldsymbol{B}) = \det \boldsymbol{A} \, \det \boldsymbol{B}, \qquad \det(\boldsymbol{A}^{\mathrm{T}}) = \det \boldsymbol{A}, \qquad \det(\alpha \boldsymbol{A}) = \alpha^3 \det \boldsymbol{A}. \tag{1.40}$$

Definition 1.16 (Inverse)**.** If $\det \boldsymbol{A} = 0$ the tensor is *singular*. For a non-singular tensor

a unique *inverse* $\boldsymbol{A}^{-1}$ exists such that

$$\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I} = \boldsymbol{A}^{-1}\boldsymbol{A}, \tag{1.41}$$

and $\boldsymbol{A}$ is said to be *invertible*. The general linear group $\mathrm{GL}(n)$ is the group of the invertible linear mappings over $\mathbb{R}^n$.

We shortly denote the transpose of the inverse (equal to the inverse of the transpose) as

$$(\boldsymbol{A}^{-1})^{\mathrm{T}} = (\boldsymbol{A}^{\mathrm{T}})^{-1} \coloneqq \boldsymbol{A}^{-\mathrm{T}}. \tag{1.42}$$

Moreover $(\boldsymbol{A}\boldsymbol{B})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$ and $\det(\boldsymbol{A}^{-1}) = (\det \boldsymbol{A})^{-1}$, from (1.40).

1.9 Orthogonal tensors

Among the invertible linear transformations, a special class is extremely important in mechanics, because it describes *rotations*. We look for tensors that preserve lengths and angles: the change of length and the change of angle are precisely what a deformation produces (Chapter 3), so transformations that preserve both are those that do *not* deform.

Definition 1.17 (Orthogonal tensor)**.** A tensor $\boldsymbol{Q}$ (the symbol $\boldsymbol{Q}$ is generally reserved for orthogonal tensors) is *orthogonal* if it preserves the length of vectors and the angle between them:

$$|\boldsymbol{Q}\boldsymbol{u}| = |\boldsymbol{u}| \quad \forall \boldsymbol{u} \in \mathbb{R}^n, \qquad \boldsymbol{Q}\boldsymbol{u} \cdot \boldsymbol{Q}\boldsymbol{v} = \boldsymbol{u} \cdot \boldsymbol{v} \quad \forall \boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n. \tag{1.43}$$

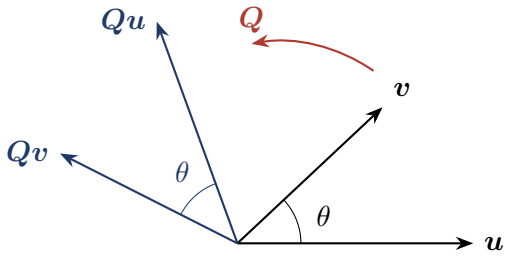


Figure 1.3: An orthogonal tensor preserves lengths and angles (slide 15).

Consequences. Preserving the length means $|\boldsymbol{Q}\boldsymbol{u}|^2 = |\boldsymbol{u}|^2$. In indices, $\boldsymbol{Q}\boldsymbol{u}$ has components $Q_{ij}u_j$ and, to compute its square length, we dot it with itself, keeping the free index i as the one to be summed:

$$|\boldsymbol{Q}\boldsymbol{u}|^2 = (Q_{ij}u_j)(Q_{ik}u_k) = u_j\, Q_{ij}Q_{ik}\, u_k = u_j\, (\boldsymbol{Q}^{\mathrm{T}})_{ji}Q_{ik}\, u_k = \boldsymbol{u} \cdot \boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{u}. \tag{1.44}$$

The index i is the one used for the multiplication of Q_{ij} and Q_{ik}; since it is the *first* index of both, the first factor must be transposed to obtain a row-by-column product. Thus

$$|\boldsymbol{Q}\boldsymbol{u}|^2 = \boldsymbol{Q}\boldsymbol{u} \cdot \boldsymbol{Q}\boldsymbol{u} = \boldsymbol{u} \cdot \boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{u} = \boldsymbol{u} \cdot \boldsymbol{u} = |\boldsymbol{u}|^2 \quad \forall \boldsymbol{u} \qquad \Longrightarrow \qquad \boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q} = \boldsymbol{I}. \tag{1.45}$$

(The implication follows because $\boldsymbol{u}\cdot(\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}-\boldsymbol{I})\boldsymbol{u}=0$ for all $\boldsymbol{u}$ and $\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}-\boldsymbol{I}$ is symmetric.) Since by definition $\boldsymbol{Q}^{-1}\boldsymbol{Q}=\boldsymbol{I}$, the condition of length preservation requires

Orthogonal tensor

$$\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}=\boldsymbol{Q}\boldsymbol{Q}^{\mathrm{T}}=\boldsymbol{I},\qquad \boldsymbol{Q}^{\mathrm{T}}=\boldsymbol{Q}^{-1}, \tag{1.46}$$

i.e. the inverse of an orthogonal tensor is simply its transpose, which is very easy to compute. Conversely, (1.46) implies both conditions (1.43).

Determinant. Computing the determinant of $\boldsymbol{I}=\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}$ and using (1.40) (the determinant does not change under transposition, although $\boldsymbol{Q}^{\mathrm{T}}\neq\boldsymbol{Q}$):

$$1=\det\boldsymbol{I}=\det(\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q})=\det\boldsymbol{Q}^{\mathrm{T}}\det\boldsymbol{Q}=(\det\boldsymbol{Q})^2\qquad\Longrightarrow\qquad\det\boldsymbol{Q}=\pm1,\quad\boldsymbol{Q}\in\mathrm{GL}(n). \tag{1.47}$$

- If $\det\boldsymbol{Q}=+1$, $\boldsymbol{Q}$ is a *proper orthogonal* tensor, or *rotation* tensor: it rotates vectors preserving lengths, angles and orientation.
- If $\det\boldsymbol{Q}=-1$, $\boldsymbol{Q}$ is an *improper* orthogonal tensor, and contains a *reflection*: like an object seen in a mirror, rotated "in the wrong way" (a right hand becomes a left hand). Reflections cannot represent the physical rotation of a body.

Orthogonal groups. If $\boldsymbol{Q}$ and $\boldsymbol{R}$ are orthogonal, so is their product: $(\boldsymbol{Q}\boldsymbol{R})^{\mathrm{T}}(\boldsymbol{Q}\boldsymbol{R})=\boldsymbol{R}^{\mathrm{T}}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{R}=\boldsymbol{R}^{\mathrm{T}}\boldsymbol{R}=\boldsymbol{I}$. The set of orthogonal mappings is therefore a subgroup of $\mathrm{GL}(n)$, closed under multiplication, called the *orthogonal group* $\mathrm{O}(n)$. The subgroup of $\mathrm{O}(n)$ that preserves the orientation, i.e. of the tensors with determinant $+1$, is the *special orthogonal group* $\mathrm{SO}(n)$ (the rotations). The remaining elements of $\mathrm{O}(n)$, with determinant -1, correspond to reflections; they do not form a group (the product of two reflections has determinant $+1$).

1.10 Symmetric and skew tensors

Proposition 1.1 (Additive decomposition)**.** *Any second-order tensor $\boldsymbol{A}$ can always be uniquely decomposed into a symmetric tensor $\boldsymbol{S}$ and a skew (antisymmetric) tensor $\boldsymbol{W}$:*

$$\boldsymbol{A}=\boldsymbol{S}+\boldsymbol{W},$$
$$\boldsymbol{S}=\operatorname{sym}\boldsymbol{A}=\tfrac{1}{2}(\boldsymbol{A}+\boldsymbol{A}^{\mathrm{T}}),\quad S_{ij}=S_{ji};\qquad\boldsymbol{W}=\operatorname{skew}\boldsymbol{A}=\tfrac{1}{2}(\boldsymbol{A}-\boldsymbol{A}^{\mathrm{T}}),\quad W_{ij}=-W_{ji}. \tag{1.48}$$

Proof. Summing, $\frac{1}{2}\boldsymbol{A}+\frac{1}{2}\boldsymbol{A}=\boldsymbol{A}$ and $\frac{1}{2}\boldsymbol{A}^{\mathrm{T}}-\frac{1}{2}\boldsymbol{A}^{\mathrm{T}}=\boldsymbol{O}$, so the identity always holds. Uniqueness: if $\boldsymbol{A}=\boldsymbol{S}'+\boldsymbol{W}'$ is another such decomposition, then $\boldsymbol{A}^{\mathrm{T}}=\boldsymbol{S}'-\boldsymbol{W}'$, whence $\boldsymbol{S}'=\frac{1}{2}(\boldsymbol{A}+\boldsymbol{A}^{\mathrm{T}})=\boldsymbol{S}$ and $\boldsymbol{W}'=\boldsymbol{W}$. ∎

In a symmetric tensor the indices can be exchanged freely; in a skew tensor exchanging the indices changes the sign. Hence a skew tensor has *zero diagonal* ($W_{ii}=-W_{ii}$ with no sum implies $W_{ii}=0$) and only three independent components. The meaning and importance of symmetric tensors (stress, strain) will be discussed later.

Axial vector. Any skew tensor $\boldsymbol{W}$ behaves like a three-component vector. Introducing the *axial* (dual) vector $\boldsymbol{\omega}$ of $\boldsymbol{W}$, for every vector $\boldsymbol{u}$:

Skew tensor and axial vector

$$\boldsymbol{W}\boldsymbol{u} = \boldsymbol{\omega} \times \boldsymbol{u}, \qquad W_{ij} = \varepsilon_{kij}\,\omega_k = -\varepsilon_{ijk}\,\omega_k, \qquad \omega_k = -\tfrac{1}{2}\varepsilon_{ijk}W_{ij}, \qquad |\boldsymbol{\omega}| = \tfrac{1}{\sqrt{2}}|\boldsymbol{W}|. \tag{1.49}$$

Check: $(\boldsymbol{W}\boldsymbol{u})_i = W_{ij}u_j = -\varepsilon_{ijk}\omega_k u_j = \varepsilon_{ikj}\omega_k u_j = (\boldsymbol{\omega}\times\boldsymbol{u})_i$; and, from (1.9), $-\tfrac{1}{2}\varepsilon_{ijk}W_{ij} = \tfrac{1}{2}\varepsilon_{ijk}\varepsilon_{ijl}\omega_l = \tfrac{1}{2}\,2\delta_{kl}\omega_l = \omega_k$.

Example 1.2. For

$$[\boldsymbol{W}] = \begin{bmatrix} 0 & \alpha & -\beta \\ -\alpha & 0 & \gamma \\ \beta & -\gamma & 0 \end{bmatrix}, \quad \boldsymbol{u} = \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix}, \quad \boldsymbol{\omega} = -\begin{Bmatrix} \gamma \\ \beta \\ \alpha \end{Bmatrix}, \quad \boldsymbol{W}\boldsymbol{u} = \begin{Bmatrix} \alpha u_2 - \beta u_3 \\ \gamma u_3 - \alpha u_1 \\ \beta u_1 - \gamma u_2 \end{Bmatrix} = \boldsymbol{\omega}\times\boldsymbol{u},$$

and $|\boldsymbol{\omega}|^2 = \alpha^2 + \beta^2 + \gamma^2 = \tfrac{1}{2}|\boldsymbol{W}|^2$.

In class

The remaining properties of skew tensors on the slides are skipped, because they are not used in the course. Skew tensors will reappear in the kinematics (spin tensor) and in the linearisation of rotations.

1.11 Projection tensors

Whenever we have a vector and an orientation, we may want the components of the vector along that orientation and orthogonal to it. This is fundamental for the definition of stress, so it is worth repeating.

Consider a smooth surface and a point P on it; if the surface is sufficiently regular we can define the unit normal $\boldsymbol{n}$ at P (the slides call it $\boldsymbol{e}$). Any vector $\boldsymbol{u}$ at P can be decomposed into a part parallel to $\boldsymbol{n}$ and a part lying in the plane normal to $\boldsymbol{n}$ (Fig. 1.4):

$$\boldsymbol{u} = \boldsymbol{u}_{\parallel} + \boldsymbol{u}_{\perp}. \tag{1.50}$$

Parallel part. The *scalar* parallel component is the projection $\boldsymbol{u}\cdot\boldsymbol{n}$; the parallel *vector* is this scalar times $\boldsymbol{n}$. Using the action of a dyad (1.25) backwards (with the two vectors of the dyad equal to $\boldsymbol{n}$):

$$\boldsymbol{u}_{\parallel} = (\boldsymbol{u}\cdot\boldsymbol{n})\,\boldsymbol{n} = (\boldsymbol{n}\otimes\boldsymbol{n})\,\boldsymbol{u} \coloneqq \boldsymbol{P}_{\boldsymbol{n}}^{\parallel}\boldsymbol{u}, \qquad (u_{\parallel})_i = n_i\,n_j\,u_j. \tag{1.51}$$

(If you do not trust the manipulation, check it in indices: $(\boldsymbol{u}\cdot\boldsymbol{n})n_i = u_j n_j n_i = (n_i n_j)u_j$.)

Perpendicular part. By difference, introducing the identity tensor to collect $\boldsymbol{u}$ as a common factor:

$$\boldsymbol{u}_{\perp} = \boldsymbol{u} - \boldsymbol{u}_{\parallel} = \boldsymbol{I}\boldsymbol{u} - (\boldsymbol{n}\otimes\boldsymbol{n})\boldsymbol{u} = (\boldsymbol{I} - \boldsymbol{n}\otimes\boldsymbol{n})\,\boldsymbol{u} \coloneqq \boldsymbol{P}_{\boldsymbol{n}}^{\perp}\boldsymbol{u}, \qquad (u_{\perp})_i = (\delta_{ij} - n_i n_j)\,u_j. \tag{1.52}$$

Projection tensors

$$\boldsymbol{P}_{\boldsymbol{n}}^{\parallel} = \boldsymbol{n}\otimes\boldsymbol{n}, \qquad \boldsymbol{P}_{\boldsymbol{n}}^{\perp} = \boldsymbol{I} - \boldsymbol{n}\otimes\boldsymbol{n}, \qquad \boldsymbol{P}_{\boldsymbol{n}}^{\parallel} + \boldsymbol{P}_{\boldsymbol{n}}^{\perp} = \boldsymbol{I}. \tag{1.53}$$

$\boldsymbol{P}_{\boldsymbol{n}}^{\parallel}$ is the *parallel projection* tensor and $\boldsymbol{P}_{\boldsymbol{n}}^{\perp}$ the *orthogonal* (perpendicular) projection tensor. Both are symmetric and idempotent ($\boldsymbol{P}\boldsymbol{P} = \boldsymbol{P}$: projecting twice is the same as projecting once), and $\boldsymbol{P}_{\boldsymbol{n}}^{\parallel}\boldsymbol{P}_{\boldsymbol{n}}^{\perp} = \boldsymbol{O}$.

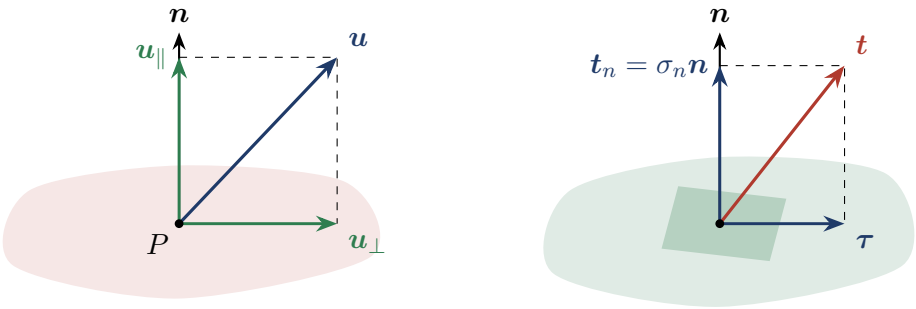


Figure 1.4: Left: decomposition of a vector $\boldsymbol{u}$ into its components parallel and orthogonal to a unit vector $\boldsymbol{n}$. Right: the same decomposition applied to the traction vector $\boldsymbol{t} = \boldsymbol{\sigma}\boldsymbol{n}$ on a cut with normal $\boldsymbol{n}$ gives the normal and shear stress (slide 17).

Application: normal and shear stress. These expressions are exactly those giving the normal and shear components of the stress vector. By Cauchy's theorem (Section 4.2) the stress (traction) vector $\boldsymbol{t}$ at a point $\boldsymbol{x}$ of a surface with normal $\boldsymbol{n}$ depends on the position *and* on the normal, $\boldsymbol{t} = \boldsymbol{t}(\boldsymbol{x}, \boldsymbol{n})$; Cauchy proved that the dependence on the normal is linear, through the application of a tensor, the Cauchy stress tensor $\boldsymbol{\sigma}(\boldsymbol{x})$, to the normal: $\boldsymbol{t}(\boldsymbol{x}, \boldsymbol{n}) = \boldsymbol{\sigma}(\boldsymbol{x})\boldsymbol{n}$. Decomposing $\boldsymbol{t}$ with the projectors,

$$\boldsymbol{t}_n = (\boldsymbol{n}\otimes\boldsymbol{n})\boldsymbol{t} = \sigma_n\,\boldsymbol{n}, \quad \sigma_n = \boldsymbol{n}\cdot\boldsymbol{\sigma}\boldsymbol{n}, \qquad \boldsymbol{\tau} = (\boldsymbol{I} - \boldsymbol{n}\otimes\boldsymbol{n})\boldsymbol{t}, \tag{1.54}$$

the normal stress vector and the tangential (shear) stress vector. The same decomposition is used for any vector, for example for a displacement $\boldsymbol{\delta}$ across a cohesive surface, decomposed into an opening $\delta_n = |\boldsymbol{\delta}_{\parallel}|$ and a sliding $\boldsymbol{\delta}_S = \boldsymbol{\delta}_{\perp}$ (slide 17).

> **In class**
>
> If the Cauchy stress is not familiar, it is worth reviewing it before the chapter on stress: it is supposed to be a very basic part of the background for this course.

1.12 Spherical and deviatoric tensors

Any second-order tensor $\boldsymbol{A}$, and in particular the stress tensor, can be decomposed into a *spherical* (volumetric) part and a *deviatoric* part:

Spherical–deviatoric decomposition

$$\boldsymbol{A} = \alpha\boldsymbol{I} + \operatorname{dev}\boldsymbol{A}, \qquad \alpha = \tfrac{1}{3}\operatorname{tr}\boldsymbol{A} = \tfrac{1}{3}(\boldsymbol{I}:\boldsymbol{A}) = \tfrac{1}{3}A_{ii}, \tag{1.55}$$

$$\operatorname{dev}(\bullet) = (\bullet) - \tfrac{1}{3}\operatorname{tr}(\bullet)\,\boldsymbol{I}, \qquad \operatorname{dev}(\bullet)_{ij} = (\bullet)_{ij} - \tfrac{1}{3}(\bullet)_{kk}\,\delta_{ij}. \tag{1.56}$$

The trace $\operatorname{tr} \boldsymbol{A} = 3\alpha$ is a scalar, called the *first invariant* of $\boldsymbol{A}$ (Section 1.14). Once α is known, the spherical part $\alpha \boldsymbol{I}$ is immediately defined, and the deviator is obtained by difference.

Example 1.3 (Stress tensor)**.** For the stress tensor, α is the mean (isotropic) stress $p = \frac{1}{3}\operatorname{tr}\boldsymbol{\sigma} = \frac{1}{3}(\sigma_{11} + \sigma_{22} + \sigma_{33})$, and $\boldsymbol{\sigma} = p\boldsymbol{I} + \boldsymbol{s}$ with deviatoric stress

$$\boldsymbol{s} = \boldsymbol{\sigma} - p\boldsymbol{I} = \begin{bmatrix} \sigma_{11} - p & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} - p & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} - p \end{bmatrix} = \begin{bmatrix} \frac{2\sigma_{11} - \sigma_{22} - \sigma_{33}}{3} & \tau_{12} & \tau_{13} \\ \tau_{21} & \frac{2\sigma_{22} - \sigma_{33} - \sigma_{11}}{3} & \tau_{23} \\ \tau_{31} & \tau_{32} & \frac{2\sigma_{33} - \sigma_{11} - \sigma_{22}}{3} \end{bmatrix},$$

where $\tau_{ij} = \sigma_{ij}$, $i \neq j$, are the shear stresses. The deviatoric stress is used, for example, in the von Mises theory of plasticity, and many constitutive relations are written in terms of deviatoric quantities. (Sign convention: here p is the mean stress, positive in tension; in fluid mechanics and in incompressible elasticity the pressure is often defined with the opposite sign.)

Properties.

- The spherical part of a second-order tensor is also called *isotropic*: it is the same in every direction.
- The off-diagonal elements of a spherical tensor are null (in every orthonormal basis).
- The deviatoric part of a second-order tensor is also called *isochoric* (in the kinematics, it does not produce volume changes to first order).
- The trace of a deviatoric tensor is null: $\operatorname{tr}(\operatorname{dev}\boldsymbol{A}) = \operatorname{tr}\boldsymbol{A} - \frac{1}{3}\operatorname{tr}\boldsymbol{A}\ \operatorname{tr}\boldsymbol{I} = 0$.
- The two parts are orthogonal: $\alpha\boldsymbol{I} : \operatorname{dev}\boldsymbol{A} = \alpha \operatorname{tr}(\operatorname{dev}\boldsymbol{A}) = 0$.

1.13 Higher-order tensors

1.13.1 Third-order tensors

A third-order tensor has $3^3 = 27$ components and is represented as

$$\boldsymbol{\mathcal{A}} = \mathcal{A}_{ijk}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k. \tag{1.57}$$

The double contraction between a third-order and a second-order tensor gives a vector:

$$\boldsymbol{\mathcal{A}} : \boldsymbol{B} = \mathcal{A}_{ijk} B_{jk}\, \boldsymbol{e}_i. \tag{1.58}$$

The permutation tensor ε_{ijk} and the gradient of a tensor field (Section 1.17) are third-order tensors. Apart from these, third-order tensors are seldom used in this course.

1.13.2 Fourth-order tensors

A fourth-order tensor has $3^4 = 81$ components and is represented as

$$\mathbb{A} = \mathbb{A}_{ijkl}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k \otimes \boldsymbol{e}_l, \tag{1.59}$$

a direct generalisation of (1.23), in which all the basis vectors that define the tensor appear explicitly. The double contraction between a fourth-order and a second-order tensor gives a second-order tensor:

$$\mathbb{A} : \boldsymbol{B} = \mathbb{A}_{ijkl} B_{kl}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j. \tag{1.60}$$

This is the typical tensor of the constitutive relations, which link a second-order tensor (the stress) to another (the strain). The *generalised Hooke's law* reads

$$\boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}, \qquad \sigma_{ij} = \mathbb{C}_{ijkl}\, \varepsilon_{kl}, \tag{1.61}$$

where the double contraction saturates the two indices of the strain.

Remark 1.4 (Voigt notation). In engineering practice the stress and strain tensors are often stored in two 6×1 arrays (Voigt notation), and the fourth-order tensor $\mathbb{C}$ becomes a 6×6 matrix. This is very common and convenient when describing material symmetries (anisotropy), see Appendix A. In general, however, we keep the tensor form (1.61).

Dyadic product of two second-order tensors. A fourth-order tensor can be obtained by expanding both indices of two second-order tensors:

$$\mathbb{D} = \boldsymbol{A} \otimes \boldsymbol{B}, \qquad \mathbb{D}_{ijkl} = A_{ij} B_{kl}, \qquad (\boldsymbol{A} \otimes \boldsymbol{B}) : \boldsymbol{C} = (\boldsymbol{B} : \boldsymbol{C})\, \boldsymbol{A}. \tag{1.62}$$

1.13.3 Transpose and identity fourth-order tensors

Definition 1.18 (Transpose of a fourth-order tensor)**.** The (unique) transpose $\mathbb{A}^{\mathrm{T}}$ of $\mathbb{A}$ is defined by

$$\boldsymbol{B} : \mathbb{A}^{\mathrm{T}} : \boldsymbol{C} = \boldsymbol{C} : \mathbb{A} : \boldsymbol{B} \qquad \forall\, \boldsymbol{B}, \boldsymbol{C}, \tag{1.63}$$

so that $(\mathbb{A}^{\mathrm{T}})_{ijkl} = \mathbb{A}_{klij}$, $(\mathbb{A}^{\mathrm{T}})^{\mathrm{T}} = \mathbb{A}$ and $(\boldsymbol{A} \otimes \boldsymbol{B})^{\mathrm{T}} = \boldsymbol{B} \otimes \boldsymbol{A}$.

This definition is a direct extension of (1.31); it is not used frequently in the course. Two *distinct* fourth-order identity tensors are important.

Fourth-order identity tensors

$$\boldsymbol{A} = \mathbb{I} : \boldsymbol{A}, \quad \mathbb{I} = \delta_{ik}\delta_{jl}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k \otimes \boldsymbol{e}_l = \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_i \otimes \boldsymbol{e}_j, \quad (\mathbb{I})_{ijkl} = \delta_{ik}\delta_{jl}, \tag{1.64}$$

$$\boldsymbol{A}^{\mathrm{T}} = \bar{\mathbb{I}} : \boldsymbol{A}, \quad \bar{\mathbb{I}} = \delta_{il}\delta_{jk}\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k \otimes \boldsymbol{e}_l = \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_i, \quad (\bar{\mathbb{I}})_{ijkl} = \delta_{il}\delta_{jk}. \tag{1.65}$$

The first one leaves a tensor as it is; the second one gives its transpose: it preserves the information, but "scrambles" rows and columns. Check: $(\mathbb{I} : \boldsymbol{A})_{ij} = \delta_{ik}\delta_{jl}A_{kl} = A_{ij}$ and $(\bar{\mathbb{I}} : \boldsymbol{A})_{ij} = \delta_{il}\delta_{jk}A_{kl} = A_{ji}$. In both the components are products of two Kronecker deltas with all four indices different; what changes is the position of the unit terms in the tensor. Note that $\mathbb{I}^{\mathrm{T}} = \mathbb{I}$, so that $\bar{\mathbb{I}} \neq \mathbb{I}^{\mathrm{T}}$: the transposer is *not* the transpose of the identity.

Complement (Holzapfel)

The *symmetric* fourth-order identity is $\mathbb{I}_{\mathrm{s}} = \frac{1}{2}(\mathbb{I}+\bar{\mathbb{I}})$, with $(\mathbb{I}_{\mathrm{s}})_{ijkl} = \frac{1}{2}(\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk})$, so that $\mathbb{I}_{\mathrm{s}} : \boldsymbol{A} = \operatorname{sym} \boldsymbol{A}$; it acts as the identity on the space of symmetric tensors (stress, strain). It is convenient to introduce also the product $\odot$ of two second-order tensors, $(\boldsymbol{A} \odot \boldsymbol{B})_{ijkl} = \frac{1}{2}(A_{ik}B_{jl} + A_{il}B_{jk})$, so that $\mathbb{I}_{\mathrm{s}} = \boldsymbol{I} \odot \boldsymbol{I}$.

Deviatoric projector. The deviatoric part of a second-order tensor can be written by introducing a fourth-order projection tensor $\mathbb{P}$:

Fourth-order deviatoric projector

$$\operatorname{dev} \boldsymbol{A} = \boldsymbol{A} - \tfrac{1}{3}(\boldsymbol{I} : \boldsymbol{A})\boldsymbol{I} = \left(\mathbb{I} - \tfrac{1}{3}\boldsymbol{I} \otimes \boldsymbol{I}\right) : \boldsymbol{A} = \mathbb{P} : \boldsymbol{A}, \qquad \mathbb{P}_{ijkl} = \delta_{ik}\delta_{jl} - \tfrac{1}{3}\delta_{ij}\delta_{kl}. \quad (1.66)$$

Exercise 1.3. Starting from (1.55), derive (1.66). Verify also that $\mathbb{P} : \mathbb{P} = \mathbb{P}$ (a projector) and $\mathbb{P} : \boldsymbol{I} = \boldsymbol{O}$.

▷ Solution on p. 134

1.14 Eigenvalues and eigenvectors

To simplify the constitutive relations it is very often convenient to use principal stresses and principal strains, i.e. to solve the eigenvalue problem of the tensor at hand. We are used to symmetric matrices (stress and strain in linear elasticity are symmetric), whose eigenvalues are always real. The tensors of this course, however, are *not always symmetric* (the deformation gradient is not), so we recall the general definitions.

Definition 1.19 (Eigenvalue, spectrum)**.** An *eigenvalue* of a tensor $\boldsymbol{A}$ is a complex number $\lambda \in \mathbb{C}$ such that $\boldsymbol{A} - \lambda\boldsymbol{I}$ is *not* invertible, i.e.

$$\det(\boldsymbol{A} - \lambda\boldsymbol{I}) = 0. \quad (1.67)$$

The *spectrum* $\sigma(\boldsymbol{A})$ is the collection of all eigenvalues of $\boldsymbol{A}$ (for us, $n = 3$ eigenvalues counted with their multiplicity).

The link with the usual formulation is the following. The eigenvalue problem asks for vectors $\boldsymbol{u}$ such that $(\boldsymbol{A} - \lambda\boldsymbol{I})\boldsymbol{u} = \boldsymbol{0}$. This homogeneous system always admits the trivial solution $\boldsymbol{u} = \boldsymbol{0}$; a non-trivial solution exists if and only if the matrix of the system is singular, $\det(\boldsymbol{A} - \lambda\boldsymbol{I}) = 0$. The non-zero vectors $\boldsymbol{u}$ so obtained are the eigenvectors, which depend on the solution of the eigenvalue problem.

1.14.1 Characteristic polynomial and invariants

Expanding the determinant one obtains the *characteristic polynomial*. With the sign convention of the slides,

Characteristic equation

$$p_{\boldsymbol{A}}(\lambda) = \det(\lambda\boldsymbol{I} - \boldsymbol{A}) = \lambda^3 - I_1\lambda^2 + I_2\lambda - I_3 = 0 \qquad (\det(\boldsymbol{A} - \lambda\boldsymbol{I}) = -p_{\boldsymbol{A}}(\lambda)), \quad (1.68)$$

and $\lambda \in \sigma(\boldsymbol{A}) \iff p_{\boldsymbol{A}}(\lambda) = 0$. The coefficients are the three *principal invariants* of $\boldsymbol{A}$:

$$I_1(\boldsymbol{A}) = \operatorname{tr} \boldsymbol{A} = A_{ii} = \lambda_1 + \lambda_2 + \lambda_3, \quad (1.69)$$

$$I_2(\boldsymbol{A}) = \tfrac{1}{2}\left[(\operatorname{tr} \boldsymbol{A})^2 - \operatorname{tr}(\boldsymbol{A}^2)\right] = \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1, \quad (1.70)$$

$$I_3(\boldsymbol{A}) = \det \boldsymbol{A} = \varepsilon_{ijk}A_{1i}A_{2j}A_{3k} = \lambda_1\lambda_2\lambda_3. \quad (1.71)$$

A practical rule to compute them from the matrix $[\boldsymbol{A}]$:

- I_1 is the sum of the determinants of the 1×1 submatrices taken on the main diagonal: $A_{11} + A_{22} + A_{33}$, the trace;
- I_2 is the sum of the determinants of the three *principal* 2×2 submatrices, obtained by deleting a row and the column with the same index (their diagonal lies on the diagonal of $\boldsymbol{A}$):

$$I_2 = (A_{11}A_{22} - A_{12}A_{21}) + (A_{22}A_{33} - A_{23}A_{32}) + (A_{33}A_{11} - A_{31}A_{13});$$

- I_3 is the determinant of the whole matrix.

When the matrix is diagonal (i.e. in the principal basis, for a diagonalisable tensor) the invariants take the very simple forms in terms of the eigenvalues written on the right of (1.69)–(1.71).

Invariance. The characteristic polynomial is invariant under similarity transformations: for every $\boldsymbol{Q} \in \mathrm{GL}(3)$, $p_{\boldsymbol{Q}\boldsymbol{A}\boldsymbol{Q}^{-1}}(\lambda) = \det(\boldsymbol{Q}(\lambda\boldsymbol{I} - \boldsymbol{A})\boldsymbol{Q}^{-1}) = p_{\boldsymbol{A}}(\lambda)$. Hence its coefficients I_1, I_2, I_3 are invariants, and so are the eigenvalues: *eigenvalues characterise the physical nature of a tensor and do not depend on the coordinates*. In particular they do not change under a rotation of the basis (Section 1.15).

Nature of the roots. The characteristic equation is a cubic with real coefficients (since $\boldsymbol{A} \in \mathrm{Lin}(\mathbb{R}^3, \mathbb{R}^3)$): its three roots are either all real, or one real and two *complex conjugate*, $\lambda = a \pm \mathrm{i}\, b$, with $\mathrm{i} = \sqrt{-1}$ the imaginary unit. Moreover:

- $\boldsymbol{A}$ and $\boldsymbol{A}^{\mathrm{T}}$ have the same eigenvalues, since $\det(\boldsymbol{A}^{\mathrm{T}} - \lambda\boldsymbol{I}) = \det\big((\boldsymbol{A} - \lambda\boldsymbol{I})^{\mathrm{T}}\big) = \det(\boldsymbol{A} - \lambda\boldsymbol{I})$;
- if $\boldsymbol{A}$ is invertible, the eigenvalues of $\boldsymbol{A}^{-1}$ are the reciprocals of those of $\boldsymbol{A}$: from $\boldsymbol{A}\boldsymbol{u} = \lambda\boldsymbol{u}$, $\boldsymbol{A}^{-1}\boldsymbol{u} = \lambda^{-1}\boldsymbol{u}$ ($\lambda \neq 0$ since $\det \boldsymbol{A} = \lambda_1\lambda_2\lambda_3 \neq 0$).

Complement (Cayley–Hamilton theorem)

Every tensor satisfies its own characteristic equation: $\boldsymbol{A}^3 - I_1\boldsymbol{A}^2 + I_2\boldsymbol{A} - I_3\boldsymbol{I} = \boldsymbol{O}$. It allows, for instance, to express $\boldsymbol{A}^{-1} = I_3^{-1}(\boldsymbol{A}^2 - I_1\boldsymbol{A} + I_2\boldsymbol{I})$ and is used repeatedly in isotropic hyperelasticity.

1.14.2 Eigenvectors of general tensors

Definition 1.20 (Right and left eigenvectors)**.** A *right* eigenvector $\boldsymbol{u} \in \mathbb{C}^n$ and a *left* eigenvector $\boldsymbol{v} \in \mathbb{C}^n$ associated with an eigenvalue λ of $\boldsymbol{A}$ satisfy, respectively,

$$(\boldsymbol{A} - \lambda\boldsymbol{I})\,\boldsymbol{u} = \boldsymbol{0}, \qquad (\boldsymbol{A}^{\mathrm{T}} - \lambda\boldsymbol{I})\,\boldsymbol{v} = \boldsymbol{0}. \tag{1.72}$$

When $\boldsymbol{A}$ is not symmetric, the problem for $\boldsymbol{A}^{\mathrm{T}}$ is different from that for $\boldsymbol{A}$: the eigenvalues are the same, but the eigenvectors are not. We obtain two sets of eigenvectors, the right ones $\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3$ and the left ones $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$ (Greek indices $\alpha, \beta = 1, 2, 3$ are used to label them, to distinguish them from components). In general the eigenvectors of one set are *not* mutually orthogonal. However:

Proposition 1.2 (Bi-orthogonality)**.** *A right eigenvector* $\boldsymbol{u}_1$ *and a left eigenvector* $\boldsymbol{v}_2$ *associated with* distinct *eigenvalues are orthogonal:*

$$(\boldsymbol{A} - \lambda_1 \boldsymbol{I})\boldsymbol{u}_1 = \boldsymbol{0}, \quad (\boldsymbol{A}^{\mathrm{T}} - \lambda_2 \boldsymbol{I})\boldsymbol{v}_2 = \boldsymbol{0}, \quad \lambda_1 \neq \lambda_2 \quad \Longrightarrow \quad \boldsymbol{u}_1 \cdot \boldsymbol{v}_2 = 0. \tag{1.73}$$

Proof. $\boldsymbol{v}_2 \cdot \boldsymbol{A}\boldsymbol{u}_1 = \lambda_1\, \boldsymbol{v}_2 \cdot \boldsymbol{u}_1$; but also, by the definition of transpose, $\boldsymbol{v}_2 \cdot \boldsymbol{A}\boldsymbol{u}_1 = \boldsymbol{u}_1 \cdot \boldsymbol{A}^{\mathrm{T}}\boldsymbol{v}_2 = \lambda_2\, \boldsymbol{u}_1 \cdot \boldsymbol{v}_2$. Subtracting, $(\lambda_1 - \lambda_2)\, \boldsymbol{u}_1 \cdot \boldsymbol{v}_2 = 0$. ■

Definition 1.21 (Diagonalisable tensor)**.** $\boldsymbol{A}$ is *diagonalisable* if it has n linearly independent eigenvectors. In such a case one can define a right and a left eigenbasis with normalised eigenvectors such that

$$\boldsymbol{u}_\alpha \cdot \boldsymbol{v}_\beta = \delta_{\alpha\beta}, \qquad \alpha, \beta = 1, \ldots, n. \tag{1.74}$$

In words, the orthogonality condition is not respected *within* a set of eigenvectors, but each eigenvector of one set is orthogonal to the other two eigenvectors of the other set. The two sets form a pair of *dual bases*: this is precisely the construction needed to describe geometry in reference systems whose basis vectors are not orthogonal, the subject of Chapter 2.

Spectral decomposition (general diagonalisable tensor)

$$\boldsymbol{A} = \sum_{\alpha=1}^{n} \lambda_\alpha\, \boldsymbol{u}_\alpha \otimes \boldsymbol{v}_\alpha, \qquad \boldsymbol{A}^{\mathrm{T}} = \sum_{\alpha=1}^{n} \lambda_\alpha\, \boldsymbol{v}_\alpha \otimes \boldsymbol{u}_\alpha. \tag{1.75}$$

The set of eigenvalues is the *spectrum*, hence the name. The two expressions have the same structure, with the two vectors of each dyad exchanged. Check with (1.25) and (1.74): $\boldsymbol{A}\boldsymbol{u}_\beta = \sum_\alpha \lambda_\alpha (\boldsymbol{v}_\alpha \cdot \boldsymbol{u}_\beta)\boldsymbol{u}_\alpha = \lambda_\beta \boldsymbol{u}_\beta$.

1.14.3 Symmetric tensors and spectral decomposition

For a symmetric tensor everything becomes easier: $\boldsymbol{A} = \boldsymbol{A}^{\mathrm{T}}$, so the right and left eigenvectors coincide, $\boldsymbol{u}_\alpha = \boldsymbol{v}_\alpha$. Moreover:

- the eigenvalues are *real* numbers;
- the eigenvectors $\boldsymbol{n}_1, \boldsymbol{n}_2, \boldsymbol{n}_3$ form an *orthonormal* basis (the principal basis), and the tensor is always diagonalisable.

Writing the identity tensor (1.26) in the principal basis and using $\boldsymbol{A}\boldsymbol{n}_i = \lambda_i \boldsymbol{n}_i$:

Spectral decomposition of a symmetric tensor

$$\boldsymbol{I} = \sum_{i=1}^{3} \boldsymbol{n}_i \otimes \boldsymbol{n}_i, \qquad \boldsymbol{A} = \boldsymbol{A}\boldsymbol{I} = \sum_{i=1}^{3} (\boldsymbol{A}\boldsymbol{n}_i) \otimes \boldsymbol{n}_i = \sum_{i=1}^{3} \lambda_i\, \boldsymbol{n}_i \otimes \boldsymbol{n}_i. \tag{1.76}$$

(The sums are written explicitly because the index i appears three times.) This is something we do routinely without thinking about it: when we compute the principal stresses and put ourselves in the principal reference system, we write the stress tensor as a diagonal matrix with the principal stresses on the diagonal, $\boldsymbol{\sigma} = \sum_i \sigma_i\, \boldsymbol{n}_i \otimes \boldsymbol{n}_i$.

Repeated eigenvalues.

- *Two equal eigenvalues*, $\lambda_1 = \lambda_2 = \lambda \neq \lambda_3$. Only the eigenvector $\boldsymbol{n}_3$ associated with λ_3 is uniquely determined: for the double eigenvalue, the procedure of solution leads to a degenerate system (all equations proportional to one) whose solutions span the whole plane normal to $\boldsymbol{n}_3$. Any pair of orthonormal vectors $\boldsymbol{n}_1, \boldsymbol{n}_2$ in that plane can be taken: there are infinitely many principal reference systems. Since $\boldsymbol{n}_1 \otimes \boldsymbol{n}_1 + \boldsymbol{n}_2 \otimes \boldsymbol{n}_2 = \boldsymbol{I} - \boldsymbol{n}_3 \otimes \boldsymbol{n}_3$,

$$\boldsymbol{A} = \lambda\,(\boldsymbol{I} - \boldsymbol{n}_3 \otimes \boldsymbol{n}_3) + \lambda_3\,\boldsymbol{n}_3 \otimes \boldsymbol{n}_3 = \lambda\,\boldsymbol{P}^{\perp}_{\boldsymbol{n}_3} + \lambda_3\,\boldsymbol{P}^{\parallel}_{\boldsymbol{n}_3}, \tag{1.77}$$

  a combination of the projection tensors of Section 1.11.

- *Three equal eigenvalues*, $\lambda_1 = \lambda_2 = \lambda_3 = \lambda$:

$$\boldsymbol{A} = \lambda\,\boldsymbol{I}. \tag{1.78}$$

  Such a tensor is the same in all directions; a stress of this kind is called *isotropic* (hydrostatic). Any orthonormal basis is a principal basis.

> **In class**
>
> Slide 23 is left to the reader. A good exercise: assume a (fictitious) "spectral decomposition" of the identity tensor in the principal basis of $\boldsymbol{A}$, as in (1.76), start from $\boldsymbol{A} = \boldsymbol{A}\boldsymbol{I}$ and use the definition of eigenvalue to obtain the spectral form; then repeat the reasoning for the case of two coincident eigenvalues, where the parallel and perpendicular projectors appear.

1.15 Transformation laws

Vectors and tensors are *invariant* upon a change of basis; their *components*, however, depend on the coordinate system introduced. In a different basis, the same vector (tensor) is described by different components.

1.15.1 Transformation of basis vectors

Consider two orthonormal bases $\{\boldsymbol{e}_i\}$ and $\{\tilde{\boldsymbol{e}}_i\}$, the second obtained from the first by a rotation $\boldsymbol{Q}$ (Fig. 1.5). Define (pay attention to the meaning of the indices)

$$Q_{ij} = \boldsymbol{e}_i \cdot \tilde{\boldsymbol{e}}_j, \tag{1.79}$$

the cosine of the angle between the old axis i and the new axis j. Using the dyadic representation $\boldsymbol{Q} = Q_{jk}\,\boldsymbol{e}_j \otimes \boldsymbol{e}_k$:

$$\tilde{\boldsymbol{e}}_i = \boldsymbol{Q}\boldsymbol{e}_i = Q_{jk}(\boldsymbol{e}_j \otimes \boldsymbol{e}_k)\boldsymbol{e}_i = Q_{jk}(\boldsymbol{e}_k \cdot \boldsymbol{e}_i)\boldsymbol{e}_j = Q_{jk}\delta_{ki}\boldsymbol{e}_j = Q_{ji}\,\boldsymbol{e}_j, \tag{1.80}$$

$$\boldsymbol{e}_i = \boldsymbol{Q}^{\mathrm{T}}\tilde{\boldsymbol{e}}_i = Q_{kj}(\tilde{\boldsymbol{e}}_j \otimes \tilde{\boldsymbol{e}}_k)\tilde{\boldsymbol{e}}_i = Q_{kj}(\tilde{\boldsymbol{e}}_k \cdot \tilde{\boldsymbol{e}}_i)\tilde{\boldsymbol{e}}_j = Q_{kj}\delta_{ki}\tilde{\boldsymbol{e}}_j = Q_{ij}\,\tilde{\boldsymbol{e}}_j, \tag{1.81}$$

where in the second line $\boldsymbol{Q}$ has been written in the rotated basis (its components are the same in the two bases, $\tilde{\boldsymbol{e}}_j \cdot \boldsymbol{Q}\tilde{\boldsymbol{e}}_k = \boldsymbol{Q}\boldsymbol{e}_j \cdot \boldsymbol{Q}(\boldsymbol{Q}\boldsymbol{e}_k) = \boldsymbol{e}_j \cdot \boldsymbol{Q}\boldsymbol{e}_k$, see below). $[\boldsymbol{Q}]$ is an orthogonal matrix containing the components Q_{ij} of the proper orthogonal tensor $\boldsymbol{Q}$; it is called the *transformation matrix*. In the following, $[\boldsymbol{u}]$ (or $\{\boldsymbol{u}\}$) denotes the column matrix of the components of a vector and $[\boldsymbol{A}]$ the matrix of the components of a tensor.

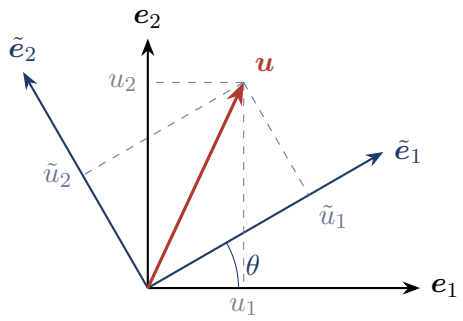


Figure 1.5: The same vector $\boldsymbol{u}$ described in two orthonormal bases related by a rotation $\tilde{\boldsymbol{e}}_i = \boldsymbol{Q}\boldsymbol{e}_i$. The vector does not change; its components do.

1.15.2 Vector transformation law

Describe the same vector in the two systems, $\boldsymbol{u} = u_j \boldsymbol{e}_j = \tilde{u}_j \tilde{\boldsymbol{e}}_j$. Using (1.12):

Vector transformation law

$$\begin{gathered} \tilde{u}_i = \boldsymbol{u} \cdot \tilde{\boldsymbol{e}}_i = Q_{ji}\, \boldsymbol{u} \cdot \boldsymbol{e}_j = Q_{ji} u_j, \qquad u_i = \boldsymbol{u} \cdot \boldsymbol{e}_i = Q_{ij}\, \boldsymbol{u} \cdot \tilde{\boldsymbol{e}}_j = Q_{ij} \tilde{u}_j; \\ [\tilde{\boldsymbol{u}}] = [\boldsymbol{Q}]^{\mathrm{T}} [\boldsymbol{u}], \quad [\boldsymbol{u}] = [\boldsymbol{Q}] [\tilde{\boldsymbol{u}}]. \end{gathered} \tag{1.82}$$

Note that $[\tilde{\boldsymbol{u}}]$ is *not* a new vector: it is the same vector $\boldsymbol{u}$, seen in the new reference system (in the slides the square brackets stress that only the components are involved). Note also that the basis vectors transform with $\boldsymbol{Q}$ ($\tilde{\boldsymbol{e}}_i = \boldsymbol{Q}\boldsymbol{e}_i$), whereas the components transform with $\boldsymbol{Q}^{\mathrm{T}}$: the relation between the basis vectors is "reversed" with respect to the relation between the components. This is not strange, it follows from the definition $\boldsymbol{u} = u_j \boldsymbol{e}_j = \tilde{u}_j \tilde{\boldsymbol{e}}_j$, and it will be met again with the dual basis.

1.15.3 Tensor transformation law

Describe the same tensor in the two systems, $\boldsymbol{A} = A_{km}\, \boldsymbol{e}_k \otimes \boldsymbol{e}_m = \tilde{A}_{ij}\, \tilde{\boldsymbol{e}}_i \otimes \tilde{\boldsymbol{e}}_j$. Using (1.29):

Tensor transformation law

$$\begin{gathered} \tilde{A}_{ij} = \tilde{\boldsymbol{e}}_i \cdot \boldsymbol{A}\tilde{\boldsymbol{e}}_j = (Q_{ki}\boldsymbol{e}_k) \cdot \boldsymbol{A}(Q_{mj}\boldsymbol{e}_m) = Q_{ki}Q_{mj}\, (\boldsymbol{e}_k \cdot \boldsymbol{A}\boldsymbol{e}_m) = Q_{ki}Q_{mj}A_{km}, \\ \left[\tilde{\boldsymbol{A}}\right] = [\boldsymbol{Q}]^{\mathrm{T}} [\boldsymbol{A}] [\boldsymbol{Q}]; \end{gathered} \tag{1.83}$$

$$A_{ij} = Q_{ik}Q_{jm}\tilde{A}_{km}, \qquad [\boldsymbol{A}] = [\boldsymbol{Q}] \left[\tilde{\boldsymbol{A}}\right] [\boldsymbol{Q}]^{\mathrm{T}}. \tag{1.84}$$

Each index of the tensor transforms with one factor Q; a fourth-order tensor transforms with four factors, $\tilde{\mathbb{A}}_{ijkl} = Q_{ai}Q_{bj}Q_{ck}Q_{dl}\mathbb{A}_{abcd}$.

Definition 1.22 (Isotropic tensor)**.** A tensor is said to be *isotropic* if its components are the same under arbitrary rotations of the basis vectors. Examples: every scalar; the second-order tensors $\alpha\boldsymbol{I}$; the fourth-order tensors $\mathbb{I}$, $\bar{\mathbb{I}}$, $\boldsymbol{I} \otimes \boldsymbol{I}$ and their linear combinations. There are no non-zero isotropic vectors.

In class

This part is long and, to do it carefully, requires some time. The students are invited to work through the derivations above (it is not difficult, but it takes some time) and to ask the lecturer in case of doubts. The key concept: changing reference system does not change the vector or the tensor, it only changes how we *see* their components.

1.16 Scalar, vector and tensor functions

In this course we need derivatives with respect to time (related to velocities and rates) and derivatives with respect to coordinates or to other tensors (gradients). The functions we deal with return scalars, vectors or tensors (we limit ourselves to second order), and their arguments may in turn be scalars, vectors or tensors:

$$\begin{aligned} &\Phi(t), \quad \boldsymbol{v}(t) = v_i(t)\,\boldsymbol{e}_i, \quad \boldsymbol{A}(t) = A_{ij}(t)\,\boldsymbol{e}_i \otimes \boldsymbol{e}_j, \\ &\Phi(\boldsymbol{u}), \quad \boldsymbol{v}(\boldsymbol{u}) = v_i(\boldsymbol{u})\,\boldsymbol{e}_i, \quad \boldsymbol{A}(\boldsymbol{u}) = A_{ij}(\boldsymbol{u})\,\boldsymbol{e}_i \otimes \boldsymbol{e}_j, \\ &\Phi(\boldsymbol{B}), \quad \boldsymbol{v}(\boldsymbol{B}) = v_i(\boldsymbol{B})\,\boldsymbol{e}_i, \quad \boldsymbol{A}(\boldsymbol{B}) = A_{ij}(\boldsymbol{B})\,\boldsymbol{e}_i \otimes \boldsymbol{e}_j. \end{aligned} \tag{1.85}$$

For vector- and tensor-valued functions we assume that the *components*, in a fixed basis $\boldsymbol{e}_i$, are functions of the argument; the basis itself does not depend on the argument.

1.16.1 Derivatives with respect to a scalar

The first (n-th) derivatives with respect to a scalar argument, for instance the time t, are computed as derivatives of the components, the reference system being independent of time. The dot denotes the time derivative:

$$\dot{\Phi} = \frac{\mathrm{d}\Phi}{\mathrm{d}t}, \qquad \dot{\boldsymbol{u}}(t) = \frac{\mathrm{d}\boldsymbol{u}}{\mathrm{d}t} = \dot{u}_i(t)\,\boldsymbol{e}_i, \qquad \dot{\boldsymbol{A}}(t) = \frac{\mathrm{d}\boldsymbol{A}}{\mathrm{d}t} = \dot{A}_{ij}(t)\,\boldsymbol{e}_i \otimes \boldsymbol{e}_j, \tag{1.86}$$

$$\frac{\mathrm{d}^n\boldsymbol{u}}{\mathrm{d}t^n} = \frac{\mathrm{d}^n u_i(t)}{\mathrm{d}t^n}\,\boldsymbol{e}_i, \qquad \frac{\mathrm{d}^n\boldsymbol{A}}{\mathrm{d}t^n} = \frac{\mathrm{d}^n A_{ij}(t)}{\mathrm{d}t^n}\,\boldsymbol{e}_i \otimes \boldsymbol{e}_j. \tag{1.87}$$

The usual product rules hold, with attention to the order of the factors: $\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{A}\boldsymbol{B}) = \dot{\boldsymbol{A}}\boldsymbol{B} + \boldsymbol{A}\dot{\boldsymbol{B}}$, $\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{u}\cdot\boldsymbol{v}) = \dot{\boldsymbol{u}}\cdot\boldsymbol{v} + \boldsymbol{u}\cdot\dot{\boldsymbol{v}}$, $\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{A}^{\mathrm{T}}) = \dot{\boldsymbol{A}}^{\mathrm{T}}$.

Derivative of the inverse. Once we know $\dot{\boldsymbol{A}}$, what is the time derivative of $\boldsymbol{A}^{-1}$? Start from the definition $\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I}$ and differentiate; the derivative of $\boldsymbol{I}$ is the zero tensor:

$$\dot{\boldsymbol{A}}\,\boldsymbol{A}^{-1} + \boldsymbol{A}\,\frac{\mathrm{d}(\boldsymbol{A}^{-1})}{\mathrm{d}t} = \boldsymbol{O} \quad\Longrightarrow\quad \boldsymbol{A}\,\frac{\mathrm{d}(\boldsymbol{A}^{-1})}{\mathrm{d}t} = -\dot{\boldsymbol{A}}\,\boldsymbol{A}^{-1}. \tag{1.88}$$

To make $\boldsymbol{A}$ disappear on the left, pre-multiply both sides by $\boldsymbol{A}^{-1}$ ($\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}$):

Derivative of the inverse

$$\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{A}^{-1}(t)) = \dot{\overline{\boldsymbol{A}^{-1}}} = -\boldsymbol{A}^{-1}\dot{\boldsymbol{A}}\,\boldsymbol{A}^{-1}. \tag{1.89}$$

Even derivatives with respect to a scalar, when tensors are involved, are “a big business”. This rule will be applied in the kinematics.

1.16.2 Gradient of a scalar-valued tensor function

Consider a nonlinear scalar-valued function $\Phi(\boldsymbol{A})$, which returns a scalar for each tensor $\boldsymbol{A}$. A typical example is the strain energy, from which the stress tensor is derived. We want to approximate Φ at $\boldsymbol{A}$ with a linear function: use a first-order Taylor expansion

$$\Phi(\boldsymbol{A} + \mathrm{d}\boldsymbol{A}) = \Phi(\boldsymbol{A}) + \frac{\partial \Phi(\boldsymbol{A})}{\partial \boldsymbol{A}} : \mathrm{d}\boldsymbol{A} + o(\mathrm{d}\boldsymbol{A}), \qquad \mathrm{d}\Phi = \frac{\partial \Phi(\boldsymbol{A})}{\partial \boldsymbol{A}} : \mathrm{d}\boldsymbol{A}, \tag{1.90}$$

where $\mathrm{d}\Phi$ is the *total differential* and $o(\mathrm{d}\boldsymbol{A})$ the remainder. The Landau order symbol $o(\bullet)$ denotes a small error that tends to zero faster than its argument:

$$\lim_{\mathrm{d}\boldsymbol{A}\to\boldsymbol{O}} \frac{o(\mathrm{d}\boldsymbol{A})}{|\mathrm{d}\boldsymbol{A}|} = 0. \tag{1.91}$$

The second-order tensor

$$\frac{\partial \Phi(\boldsymbol{A})}{\partial \boldsymbol{A}} = \operatorname{grad} \Phi(\boldsymbol{A}), \qquad \Big(\frac{\partial \Phi}{\partial \boldsymbol{A}}\Big)_{kl} = \frac{\partial \Phi}{\partial A_{kl}} = \Phi_{,kl}(\boldsymbol{A}), \tag{1.92}$$

is called the *gradient* (or derivative) of Φ at $\boldsymbol{A}$; the double contraction with $\mathrm{d}\boldsymbol{A}$ is needed to obtain a scalar. The gradient is derived as the first variation of Φ (see the directional derivative below). Simple examples: $\partial(\operatorname{tr}\boldsymbol{A})/\partial\boldsymbol{A} = \boldsymbol{I}$, $\partial(\boldsymbol{A}:\boldsymbol{A})/\partial\boldsymbol{A} = 2\boldsymbol{A}$, $\partial(\boldsymbol{u}\cdot\boldsymbol{A}\boldsymbol{v})/\partial\boldsymbol{A} = \boldsymbol{u}\otimes\boldsymbol{v}$.

1.16.3 Gradient of a tensor-valued tensor function

Consider now a nonlinear tensor-valued tensor function $\boldsymbol{A}(\boldsymbol{B})$, for instance the stress as a function of the strain. To evaluate how much $\boldsymbol{A}$ varies when the argument changes a little, assume sufficient regularity and use the Taylor expansion:

$$\boldsymbol{A}(\boldsymbol{B} + \mathrm{d}\boldsymbol{B}) = \boldsymbol{A}(\boldsymbol{B}) + \frac{\partial \boldsymbol{A}(\boldsymbol{B})}{\partial \boldsymbol{B}} : \mathrm{d}\boldsymbol{B} + o(\mathrm{d}\boldsymbol{B}). \tag{1.93}$$

The derivative of a (second-order) tensor with respect to a (second-order) tensor is a *fourth-order* tensor; to go back to a second-order tensor a double contraction with $\mathrm{d}\boldsymbol{B}$ is needed. This fourth-order tensor is the *gradient* of $\boldsymbol{A}$ with respect to its argument, and we must specify with respect to which variable it is taken:

$$\frac{\partial \boldsymbol{A}(\boldsymbol{B})}{\partial \boldsymbol{B}} = \operatorname{grad}_{\boldsymbol{B}} \boldsymbol{A}, \qquad \Big(\frac{\partial \boldsymbol{A}}{\partial \boldsymbol{B}}\Big)_{ijkl} = \frac{\partial A_{ij}(\boldsymbol{B})}{\partial B_{kl}} = A_{ij,kl}(\boldsymbol{B}). \tag{1.94}$$

> **In class**
>
> Be careful with the indices: the derivative of A_{ij} cannot be taken with respect to B_{ij}, which would consider only the components with the same indices. Each component of $\boldsymbol{A}$ is a function of *all* the components of $\boldsymbol{B}$, so all components of $\boldsymbol{A}$ must be differentiated with respect to all components of $\boldsymbol{B}$, and different indices must be used: the result has four indices. The typical example is the material tangent stiffness, the fourth-order constitutive tensor $\mathbb{C} = \partial\boldsymbol{\sigma}/\partial\boldsymbol{\varepsilon}$.

Notable derivatives.

Derivatives to be remembered

$$\frac{\partial \det \boldsymbol{A}}{\partial \boldsymbol{A}} = \det \boldsymbol{A}\ \boldsymbol{A}^{-\mathrm{T}}, \qquad \frac{\partial \det \boldsymbol{A}}{\partial A_{ij}} = \det \boldsymbol{A}\ A^{-1}_{ji} \qquad \text{(Jacobi's formula)}, \tag{1.95}$$

$$\frac{\partial \boldsymbol{A}}{\partial \boldsymbol{A}} = \mathbb{I}, \qquad \frac{\partial A_{ij}}{\partial A_{kl}} = \delta_{ik}\delta_{jl} = (\mathbb{I})_{ijkl}, \qquad \frac{\partial \boldsymbol{A}^{\mathrm{T}}}{\partial \boldsymbol{A}} = \bar{\mathbb{I}}, \tag{1.96}$$

$$\frac{\partial A^{-1}_{ij}}{\partial A_{kl}} = -A^{-1}_{ik}A^{-1}_{lj}, \tag{1.97}$$

$$\text{if } \boldsymbol{A} = \boldsymbol{A}^{\mathrm{T}}: \qquad \Big(\frac{\partial \boldsymbol{A}^{-1}}{\partial \boldsymbol{A}}\Big)_{ijkl} = -\tfrac{1}{2}(A^{-1}_{ik}A^{-1}_{lj} + A^{-1}_{il}A^{-1}_{kj}).$$

Jacobi's formula will be used many times for large strains (the determinant of the deformation gradient measures the volume change). In class it was not proved ("it is not easy to prove, but it must be learned by heart"); Holzapfel proves it with a trick. A short proof is given below.

Complement (proof of Jacobi's formula and of (1.97))

For invertible $\boldsymbol{A}$ and any $\boldsymbol{B}$, using $\det(\boldsymbol{A}+\varepsilon\boldsymbol{B}) = \det\boldsymbol{A}\,\det(\boldsymbol{I}+\varepsilon\boldsymbol{A}^{-1}\boldsymbol{B})$ and the characteristic polynomial (1.68) with $\lambda = -1/\varepsilon$,

$$\det(\boldsymbol{I}+\varepsilon\boldsymbol{C}) = 1 + \varepsilon\, I_1(\boldsymbol{C}) + \varepsilon^2 I_2(\boldsymbol{C}) + \varepsilon^3 I_3(\boldsymbol{C}),$$

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\det(\boldsymbol{A}+\varepsilon\boldsymbol{B})\Big|_{\varepsilon=0} = \det\boldsymbol{A}\ \mathrm{tr}(\boldsymbol{A}^{-1}\boldsymbol{B}) = \det\boldsymbol{A}\ \boldsymbol{A}^{-\mathrm{T}} : \boldsymbol{B}.$$

Since the left-hand side is $\frac{\partial \det \boldsymbol{A}}{\partial \boldsymbol{A}} : \boldsymbol{B}$ for every $\boldsymbol{B}$ (directional derivative, Section 1.17.2), (1.95) follows. For the inverse, differentiating $\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I}$ as in (1.89), $\mathrm{d}(\boldsymbol{A}^{-1}) = -\boldsymbol{A}^{-1}\,\mathrm{d}\boldsymbol{A}\,\boldsymbol{A}^{-1}$, i.e. $\mathrm{d}A^{-1}_{ij} = -A^{-1}_{ik}\ \mathrm{d}A_{kl}\ A^{-1}_{lj}$. If $\boldsymbol{A}$ is symmetric, only symmetric increments are admissible and the result is symmetrised with respect to (k,l); in Holzapfel's notation, $\partial\boldsymbol{A}^{-1}/\partial\boldsymbol{A} = -\boldsymbol{A}^{-1}\odot\boldsymbol{A}^{-1}$.

1.17 Scalar, vector and tensor fields

A scalar field $\Phi(\boldsymbol{x})$, a vector field $\boldsymbol{u}(\boldsymbol{x})$ or a tensor field $\boldsymbol{A}(\boldsymbol{x})$ assign a scalar, a vector or a tensor, respectively, to each point $\boldsymbol{x}$ of a domain Ω (for instance, the temperature, the displacement and the stress at the points of a body).

1.17.1 Gradient of a scalar field

The Taylor expansion of a smooth scalar field introduces the gradient:

$$\Phi(\boldsymbol{x} + \mathrm{d}\boldsymbol{x}) = \Phi(\boldsymbol{x}) + \frac{\partial\Phi(\boldsymbol{x})}{\partial\boldsymbol{x}}\cdot\mathrm{d}\boldsymbol{x} + o(\mathrm{d}\boldsymbol{x}). \tag{1.98}$$

The total differential of Φ is

$$\mathrm{d}\Phi = \frac{\partial\Phi(\boldsymbol{x})}{\partial\boldsymbol{x}}\cdot\mathrm{d}\boldsymbol{x} = \frac{\partial\Phi(\boldsymbol{x})}{\partial x_i}\,\mathrm{d}x_i = \frac{\partial\Phi}{\partial x_1}\,\mathrm{d}x_1 + \frac{\partial\Phi}{\partial x_2}\,\mathrm{d}x_2 + \frac{\partial\Phi}{\partial x_3}\,\mathrm{d}x_3. \tag{1.99}$$

Introduce the *vector operator*, or *nabla operator*

Nabla operator

$$\nabla(\bullet) = \frac{\partial(\bullet)}{\partial x_i}\,\boldsymbol{e}_i = \frac{\partial(\bullet)}{\partial x_1}\boldsymbol{e}_1 + \frac{\partial(\bullet)}{\partial x_2}\boldsymbol{e}_2 + \frac{\partial(\bullet)}{\partial x_3}\boldsymbol{e}_3, \qquad \operatorname{grad}\Phi = \nabla\Phi = \Phi_{,i}\,\boldsymbol{e}_i. \tag{1.100}$$

The nabla is a kind of vector, because it contains the derivatives with respect to the three space variables. In Einstein notation, the derivative with respect to the spatial coordinate x_i is denoted by a comma followed by the index, $\Phi_{,i} = \partial\Phi/\partial x_i$. The gradient of a scalar field (e.g., the temperature) is a vector field.

1.17.2 Directional derivative

Consider a level surface of the scalar field through $\boldsymbol{x}$,

$$\Phi(\boldsymbol{x}) = \text{const} \quad \Longleftrightarrow \quad \mathrm{d}\Phi = 0, \tag{1.101}$$

assumed smooth, so that its normal $\boldsymbol{n}$ is defined at $\boldsymbol{x}$ (Fig. 1.6). For every $\mathrm{d}\boldsymbol{x}$ tangent to the surface, $\mathrm{d}\Phi = \operatorname{grad}\Phi \cdot \mathrm{d}\boldsymbol{x} = 0$: the gradient is a vector *normal* to the level surface, and the unit normal is

$$\boldsymbol{n} = \frac{\operatorname{grad}\Phi}{|\operatorname{grad}\Phi|}. \tag{1.102}$$

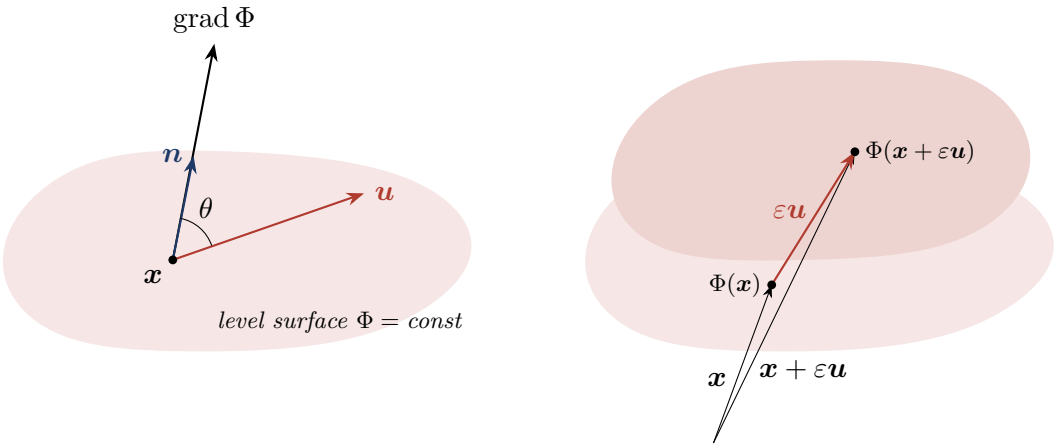


Figure 1.6: Directional derivative of a scalar field (slides 30–31). Left: projection of the gradient along $\boldsymbol{u}$. Right: rate of change of Φ along the straight line through $\boldsymbol{x}$ in the direction of $\boldsymbol{u}$.

Definition 1.23 (Directional (Gâteaux) derivative)**.** For any vector $\boldsymbol{u}$ forming an angle θ with $\boldsymbol{n}$, the *directional derivative* (Gâteaux derivative) of Φ at $\boldsymbol{x}$ in the direction of $\boldsymbol{u}$ is the scalar product

$$\mathrm{GD} = \operatorname{grad}\Phi \cdot \boldsymbol{u} = |\operatorname{grad}\Phi|\,|\boldsymbol{u}|\cos\theta. \tag{1.103}$$

In practice we take the gradient and project it along the given direction: we measure how the scalar field changes along that direction. If $\boldsymbol{u}$ is tangent to the level surface, the directional derivative is zero (the gradient has no component in the tangent plane); for unit $\boldsymbol{u}$ it is maximum in the direction of $\boldsymbol{n}$ and minimum in the direction of $-\boldsymbol{n}$. The *normal derivative* is the maximum of all the directional derivatives at $\boldsymbol{x}$ for unit vectors $\boldsymbol{u}$:

$$\mathrm{ND} = \operatorname{grad}\Phi \cdot \boldsymbol{n} = |\operatorname{grad}\Phi|. \tag{1.104}$$

Alternative definition. The directional derivative can be defined from another point of view, which does *not* require the normal to the surface. Starting from $\boldsymbol{x}$, move a little along $\boldsymbol{u}$, to the point $\boldsymbol{x}+\varepsilon\boldsymbol{u}$, with ε a small scalar parameter. The directional derivative is the rate of change of Φ along the straight line through $\boldsymbol{x}$ in the direction of $\boldsymbol{u}$:

Directional derivative

$$D_{\boldsymbol{u}}\Phi(\boldsymbol{x}) = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Phi(\boldsymbol{x}+\varepsilon\boldsymbol{u})\Big|_{\varepsilon=0}. \tag{1.105}$$

Differentiate first with respect to ε, applying the chain rule (derivative of Φ with respect to its argument $\boldsymbol{x}+\varepsilon\boldsymbol{u}$, times the derivative of the argument with respect to ε, which is $\boldsymbol{u}$), and only then set $\varepsilon = 0$:

$$D_{\boldsymbol{u}}\Phi(\boldsymbol{x}) = \frac{\partial\Phi}{\partial\boldsymbol{x}}\Big|_{\boldsymbol{x}+\varepsilon\boldsymbol{u}} \cdot \boldsymbol{u}\,\Big|_{\varepsilon=0} = \operatorname{grad}\Phi(\boldsymbol{x})\cdot\boldsymbol{u}, \tag{1.106}$$

exactly the same result as before, obtained in a different way. Since the derivative of Φ produces a vector, a dot is needed to contract it with $\boldsymbol{u}$.

Complement (Gâteaux derivative of general functions)

Definition (1.105) applies to any function, not only to fields. For a scalar-valued tensor function, $D_{\boldsymbol{B}}\Phi(\boldsymbol{A}) = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Phi(\boldsymbol{A}+\varepsilon\boldsymbol{B})\big|_{\varepsilon=0} = \frac{\partial\Phi}{\partial\boldsymbol{A}} : \boldsymbol{B}$; for a tensor-valued one, $D_{\boldsymbol{B}}\boldsymbol{A}(\boldsymbol{B}_0) = \frac{\partial\boldsymbol{A}}{\partial\boldsymbol{B}}\big|_{\boldsymbol{B}_0} : \boldsymbol{B}$. This is the tool used for the *linearisation* of the nonlinear equations of mechanics (Holzapfel, Sect. 1.8 and Chap. 8).

1.17.3 Divergence and curl of a vector field

Definition 1.24 (Divergence)**.** The *divergence* of a vector field is the scalar field produced by the vector operator *dotted* into the vector field:

$$\operatorname{div}\boldsymbol{u} = \nabla\cdot\boldsymbol{u} = \frac{\partial u_j}{\partial x_i}\,\boldsymbol{e}_j\cdot\boldsymbol{e}_i = \frac{\partial u_j}{\partial x_i}\delta_{ji} = \frac{\partial u_i}{\partial x_i} = u_{i,i} = \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} + \frac{\partial u_3}{\partial x_3}. \tag{1.107}$$

The divergence cannot be applied to a scalar (it would make no sense): it applies to a vector (or a tensor) and reduces its order by one. Each component u_i is differentiated with respect to the coordinate with the same index, and the repeated index implies the sum. When $\operatorname{div}\boldsymbol{u} = 0$ the field is said to be *solenoidal* (divergence-free).

Definition 1.25 (Curl)**.** The *curl* (rotor) of a vector field is the vector field produced by the vector operator *crossed* into the vector field:

$$\operatorname{curl}\boldsymbol{u} = \nabla\times\boldsymbol{u} = \det\begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 \\ \partial/\partial x_1 & \partial/\partial x_2 & \partial/\partial x_3 \\ u_1 & u_2 & u_3 \end{bmatrix} = \frac{\partial u_j}{\partial x_i}\,\boldsymbol{e}_i\times\boldsymbol{e}_j = \varepsilon_{ijk}\frac{\partial u_j}{\partial x_i}\,\boldsymbol{e}_k, \tag{1.108}$$

$$\operatorname{curl}\boldsymbol{u} = \Big(\frac{\partial u_3}{\partial x_2} - \frac{\partial u_2}{\partial x_3}\Big)\boldsymbol{e}_1 + \Big(\frac{\partial u_1}{\partial x_3} - \frac{\partial u_3}{\partial x_1}\Big)\boldsymbol{e}_2 + \Big(\frac{\partial u_2}{\partial x_1} - \frac{\partial u_1}{\partial x_2}\Big)\boldsymbol{e}_3. \tag{1.109}$$

When $\operatorname{curl}\boldsymbol{u} = \boldsymbol{0}$ the field is said to be *irrotational* (conservative, curl-free). The following

identities hold:

$$\operatorname{curl}\operatorname{grad}\Phi = \mathbf{0}, \qquad \operatorname{div}\operatorname{curl}\boldsymbol{u} = 0. \tag{1.110}$$

A vector field such that $\boldsymbol{u} = \operatorname{grad}\Phi$ is therefore automatically irrotational, and Φ is called the *potential* of $\boldsymbol{u}$.

1.17.4 Gradient of a vector field

Definition 1.26 (Gradient of a vector field)**.** The gradient of a smooth vector field $\boldsymbol{u}(\boldsymbol{x})$ is the second-order tensor field

$$\operatorname{grad}\boldsymbol{u} = \frac{\partial \boldsymbol{u}}{\partial \boldsymbol{x}} = \frac{\partial u_i}{\partial x_j}\,\boldsymbol{e}_i \otimes \boldsymbol{e}_j, \qquad [\operatorname{grad}\boldsymbol{u}] = \begin{bmatrix} \frac{\partial u_1}{\partial x_1} & \frac{\partial u_1}{\partial x_2} & \frac{\partial u_1}{\partial x_3} \\ \frac{\partial u_2}{\partial x_1} & \frac{\partial u_2}{\partial x_2} & \frac{\partial u_2}{\partial x_3} \\ \frac{\partial u_3}{\partial x_1} & \frac{\partial u_3}{\partial x_2} & \frac{\partial u_3}{\partial x_3} \end{bmatrix}. \tag{1.111}$$

Note that

$$\operatorname{tr}(\operatorname{grad}\boldsymbol{u}) = \operatorname{grad}\boldsymbol{u} : \boldsymbol{I} = \operatorname{div}\boldsymbol{u}, \qquad \operatorname{grad}^{\mathrm{T}}\boldsymbol{u} = (\operatorname{grad}\boldsymbol{u})^{\mathrm{T}} = \frac{\partial u_j}{\partial x_i}\,\boldsymbol{e}_i \otimes \boldsymbol{e}_j. \tag{1.112}$$

By the Taylor expansion, $\boldsymbol{u}(\boldsymbol{x} + \mathrm{d}\boldsymbol{x}) = \boldsymbol{u}(\boldsymbol{x}) + \operatorname{grad}\boldsymbol{u}\ \mathrm{d}\boldsymbol{x} + o(\mathrm{d}\boldsymbol{x})$: the gradient maps the increment of position into the increment of the field. With $\boldsymbol{u}$ the displacement, its symmetric part is the infinitesimal strain tensor of linear elasticity, $\boldsymbol{\varepsilon} = \operatorname{sym}\operatorname{grad}\boldsymbol{u}$.

Remark 1.5 (Nabla and the order of the indices). With the nabla operator, (1.111) is written by some authors (and on slide 33) as $\nabla \otimes \boldsymbol{u}$, by others as $\boldsymbol{u} \otimes \nabla$; the two writings differ by a transposition, depending on whether the derivative index is attached on the left or on the right. In this book the gradient is *always* defined by the index expression in (1.111), as in Holzapfel: the first index is the component, the second the derivative ($[\operatorname{grad}\boldsymbol{u}]_{ij} = u_{i,j}$).

1.17.5 Divergence and gradient of a tensor field

Definition 1.27 (Divergence of a tensor field)**.** The divergence of a tensor field is the vector field produced by the vector operator dotted into the tensor field:

$$\operatorname{div}\boldsymbol{A} = \frac{\partial A_{ik}}{\partial x_j}(\boldsymbol{e}_i \otimes \boldsymbol{e}_k)\boldsymbol{e}_j = \frac{\partial A_{ik}}{\partial x_j}\delta_{kj}\,\boldsymbol{e}_i = \frac{\partial A_{ij}}{\partial x_j}\,\boldsymbol{e}_i. \tag{1.113}$$

The typical example is the linear momentum balance (equilibrium equation), $\operatorname{div}\boldsymbol{\sigma} + \boldsymbol{b} = \mathbf{0}$. An alternative definition,

$$\operatorname{div}\boldsymbol{A} = \frac{\partial A_{ji}}{\partial x_j}\,\boldsymbol{e}_i, \tag{1.114}$$

contracting the derivative with the *first* index, is used in some books for the linear momentum balance differential equation; it is the divergence of $\boldsymbol{A}^{\mathrm{T}}$ according to (1.113). For a symmetric tensor such as the Cauchy stress the two coincide; for non-symmetric stress tensors (the first Piola–Kirchhoff stress) attention is needed.

The gradient of a smooth tensor field $\boldsymbol{A}(\boldsymbol{x})$ is a third-order tensor field:

$$\operatorname{grad}\boldsymbol{A} = \frac{\partial A_{ij}}{\partial x_k}\,\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k, \qquad \operatorname{tr}(\operatorname{grad}\boldsymbol{A}) = \operatorname{grad}\boldsymbol{A} : \boldsymbol{I} = \operatorname{div}\boldsymbol{A}, \tag{1.115}$$

where the double contraction with $\boldsymbol{I}$ contracts the last two indices.

Laplace operator. The vector operator dotted into itself:

$$\nabla^2(\bullet) = \nabla \cdot \nabla(\bullet) = \nabla \cdot \frac{\partial(\bullet)}{\partial x_i}\boldsymbol{e}_i = \frac{\partial^2(\bullet)}{\partial x_i \partial x_j}\,\boldsymbol{e}_i \cdot \boldsymbol{e}_j = \frac{\partial^2(\bullet)}{\partial x_i \partial x_j}\,\delta_{ij} = \frac{\partial^2(\bullet)}{\partial x_i^2} = (\bullet)_{,ii}. \tag{1.116}$$

Some books denote the Laplace operator by Δ; here ∇^2 is used. It does not change the order of the field to which it is applied.

1.17.6 Curl of a tensor field

The curl of a tensor field is defined using a relation based on the curl of a vector. For an arbitrary constant vector $\boldsymbol{c}$ and a vector field $\boldsymbol{u}$,

$$(\nabla \times \boldsymbol{u}) \cdot \boldsymbol{c} = \nabla \cdot (\boldsymbol{u} \times \boldsymbol{c}), \qquad (\nabla \times \boldsymbol{u}) \cdot \boldsymbol{c} = \varepsilon_{ijk} \frac{\partial u_j}{\partial x_i} \boldsymbol{e}_k \cdot c_m \boldsymbol{e}_m = \varepsilon_{ijk} \frac{\partial u_j}{\partial x_i} c_k = \varepsilon_{ijk} \frac{\partial (u_j c_k)}{\partial x_i}. \tag{1.117}$$

The relation is extended to a second-order tensor $\boldsymbol{F}$ by introducing the vector $\boldsymbol{cF}$ (i.e. $\boldsymbol{F}^{\mathrm{T}}\boldsymbol{c}$, with components $c_m F_{mj}$):

$$(\nabla \times \boldsymbol{F})\,\boldsymbol{c} = \nabla \times (\boldsymbol{cF}), \qquad \begin{aligned} \nabla \times (\boldsymbol{cF}) &= \varepsilon_{ijk} \frac{\partial (c_m F_{mj})}{\partial x_i} \boldsymbol{e}_k = \varepsilon_{ijk} \frac{\partial F_{mj}}{\partial x_i} c_m \boldsymbol{e}_k \\ &= \varepsilon_{ijk} \frac{\partial F_{mj}}{\partial x_i} (\boldsymbol{e}_k \otimes \boldsymbol{e}_m)\,\boldsymbol{c}. \end{aligned} \tag{1.118}$$

Thus the curl of a second-order tensor $\boldsymbol{F}$ is defined as

$$\nabla \times \boldsymbol{F} = \operatorname{curl} \boldsymbol{F} = \varepsilon_{ijk} \frac{\partial F_{mj}}{\partial x_i}\, \boldsymbol{e}_k \otimes \boldsymbol{e}_m. \tag{1.119}$$

Example: the *compatibility equations* for the deformation gradient. If $\boldsymbol{F}$ is the gradient of a vector field, $F_{mj} = \partial \chi_m / \partial x_j$, then $\operatorname{curl} \boldsymbol{F} = \boldsymbol{O}$ because ε_{ijk} is antisymmetric in (i, j) while $\partial^2 \chi_m / \partial x_i \partial x_j$ is symmetric.

1.17.7 Special differential equations and Hessian

The Laplace operator applied to a scalar field Φ yields another scalar field, as in Laplace's and Poisson's equations,

$$\nabla^2 \Phi = 0, \qquad \nabla^2 \Phi = \Psi. \tag{1.120}$$

The *Hessian* operator (gradient of a gradient) leads to a symmetric second-order tensor:

$$\nabla\nabla(\bullet) = \nabla \otimes \nabla(\bullet) = \nabla \otimes \frac{\partial(\bullet)}{\partial x_i}\boldsymbol{e}_i = \frac{\partial^2(\bullet)}{\partial x_i \partial x_j}\,\boldsymbol{e}_i \otimes \boldsymbol{e}_j. \tag{1.121}$$

Useful properties (to be remembered in the calculation of the tangent stiffness):

$$\nabla^2(\boldsymbol{u} \cdot \boldsymbol{v}) = \nabla^2 \boldsymbol{u} \cdot \boldsymbol{v} + 2\,\operatorname{grad} \boldsymbol{u} : \operatorname{grad} \boldsymbol{v} + \boldsymbol{u} \cdot \nabla^2 \boldsymbol{v}, \qquad \operatorname{curl} \operatorname{curl} \boldsymbol{u} = \operatorname{grad}(\operatorname{div} \boldsymbol{u}) - \nabla^2 \boldsymbol{u}. \tag{1.122}$$

(The middle term is a double contraction, $2u_{i,k}v_{i,k}$.) If a vector field is both solenoidal and irrotational it is said to be *harmonic*:

$$\operatorname{div}\boldsymbol{u} = 0 \text{ and } \operatorname{curl}\boldsymbol{u} = \boldsymbol{0} \quad \Longrightarrow \quad \nabla^2\boldsymbol{u} = \boldsymbol{0}. \tag{1.123}$$

1.18 Integral theorems

The divergence theorem is used very often, in particular to derive the local form of the equilibrium (balance) equations from their integral form.

Theorem 1.3 (Divergence (Gauss) theorem)**.** *Let $\boldsymbol{u}$ and $\boldsymbol{A}$ be smooth vector and tensor fields defined over a 3D region of volume V bounded by the closed surface S with outward unit normal $\boldsymbol{n}$. Then*

$$\int_S \boldsymbol{u}\cdot\boldsymbol{n}\,\mathrm{d}S = \int_V \operatorname{div}\boldsymbol{u}\,\mathrm{d}V, \qquad \int_S u_i n_i\,\mathrm{d}S = \int_V \frac{\partial u_i}{\partial x_i}\,\mathrm{d}V, \tag{1.124}$$

$$\int_S \boldsymbol{A}\boldsymbol{n}\,\mathrm{d}S = \int_V \operatorname{div}\boldsymbol{A}\,\mathrm{d}V, \qquad \int_S A_{ij}n_j\,\mathrm{d}S = \int_V \frac{\partial A_{ij}}{\partial x_j}\,\mathrm{d}V. \tag{1.125}$$

The first integral is the total (outward normal) *flux* of $\boldsymbol{u}$ out of the boundary surface S enclosing V. Written in components, it is clear what happens: on the surface we have the scalar product $u_i n_i$; in the volume, each component u_i is differentiated with respect to the coordinate with the same index. For a tensor the theorem is the same, only slightly more complicated in the indices.

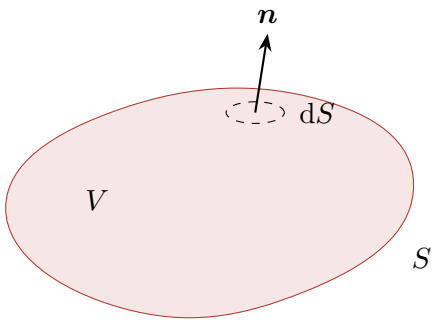


Figure 1.7: Region of volume V bounded by the closed surface S with outward unit normal $\boldsymbol{n}$ (slide 37).

1.19 Exercises

The following exercises were suggested during the lecture ("it is a very good exercise to learn how to use the indices").

Exercise 1.4. Prove the ε–δ identity $\varepsilon_{ijk}\varepsilon_{pqk} = \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}$ by checking the possible cases, and use it to prove $\boldsymbol{u}\times(\boldsymbol{v}\times\boldsymbol{w}) = (\boldsymbol{u}\cdot\boldsymbol{w})\boldsymbol{v} - (\boldsymbol{u}\cdot\boldsymbol{v})\boldsymbol{w}$.

▷ Solution on p. 134

Exercise 1.5. Prove, using components, that $(\boldsymbol{u}\times\boldsymbol{v})\cdot\boldsymbol{w}=\varepsilon_{ijk}u_iv_jw_k$ and that the box product equals the determinant of the matrix whose rows are $\boldsymbol{u}$, $\boldsymbol{v}$, $\boldsymbol{w}$.

▷ Solution on p. 134

Exercise 1.6. Prove that $\operatorname{tr}\boldsymbol{A}=\boldsymbol{A}:\boldsymbol{I}$, that $\operatorname{tr}(\boldsymbol{u}\otimes\boldsymbol{v})=\boldsymbol{u}\cdot\boldsymbol{v}$, and that $\operatorname{tr}(\boldsymbol{AB})=\operatorname{tr}(\boldsymbol{BA})$.

▷ Solution on p. 134

Exercise 1.7. Verify the components of ε_{ijk} in (1.8) and prove the expression (1.39) of the determinant, starting from (1.38).

▷ Solution on p. 135

Exercise 1.8. Prove that for a skew tensor $\boldsymbol{W}$, $\boldsymbol{v}\cdot\boldsymbol{W}\boldsymbol{v}=0$ for all $\boldsymbol{v}$, and that $\boldsymbol{S}:\boldsymbol{W}=0$ for every symmetric $\boldsymbol{S}$ and skew $\boldsymbol{W}$.

▷ Solution on p. 135

Exercise 1.9. Derive the vector transformation law (1.82) starting from $\boldsymbol{u}=u_1\boldsymbol{e}_1+u_2\boldsymbol{e}_2+u_3\boldsymbol{e}_3=\tilde{u}_1\tilde{\boldsymbol{e}}_1+\tilde{u}_2\tilde{\boldsymbol{e}}_2+\tilde{u}_3\tilde{\boldsymbol{e}}_3$, and specialise it to a rotation by an angle θ about $\boldsymbol{e}_3$ (Fig. 1.5), for which $\tilde{u}_1=u_1\cos\theta+u_2\sin\theta$, $\tilde{u}_2=-u_1\sin\theta+u_2\cos\theta$, $\tilde{u}_3=u_3$.

▷ Solution on p. 135

Exercise 1.10. Show that the eigenvalues of a symmetric tensor are real and that eigenvectors associated with distinct eigenvalues are orthogonal. *Hint:* take the complex conjugate of $\boldsymbol{An}=\lambda\boldsymbol{n}$ and compute $\bar{\boldsymbol{n}}\cdot\boldsymbol{An}$ in two ways.

▷ Solution on p. 135

2

Dual Bases

In class

The dual basis had been postponed on the first day. The lecturer introduced it at the start of the second day, warning that it is an abstract topic: "it is not difficult, but it requires some patience". Its purpose in the course is twofold. First, it explains *why* vectors and tensors can have different kinds of components (covariant and contravariant), something that is invisible in an orthonormal basis. Second, it gives the language to treat the deformation of a body, which turns an orthonormal frame attached to the material into a non-orthogonal one. For the reader who wants to go deeper, the lecturer recommended the small book by Gel'fand [2], which is very clear and almost free of heavy formalism.

In Chapter 1 all vectors and tensors were represented in an orthonormal basis $\{\boldsymbol{e}_i\}$, and the position of an index (up or down) was irrelevant. Many problems of mechanics are, however, more naturally described in a basis whose vectors are neither orthogonal nor of unit length: think of crystal lattices, of fibres embedded in a matrix, of shells described by curvilinear coordinates, or of a material grid that has been deformed (Fig. 3.1). This chapter introduces the tools needed to work in such bases.

2.1 Vector spaces and Euclidean spaces

A **vector space** (or linear space) $\mathcal{V}$ over $\mathbb{R}$ is a set of elements, called vectors, closed with respect to two operations, the sum $\boldsymbol{u}+\boldsymbol{v}\in\mathcal{V}$ and the multiplication by a scalar $\alpha\boldsymbol{u}\in\mathcal{V}$, which satisfy the usual axioms (commutativity and associativity of the sum, existence of the zero vector and of the opposite, distributivity, $1\,\boldsymbol{u}=\boldsymbol{u}$). A vector space has no notion of length or angle: it only tells us how to combine vectors linearly. The maximum number of linearly independent vectors is the *dimension* n of the space; any set of n independent vectors is a *basis*.

Definition 2.1 (Euclidean space)**.** A vector space $\mathcal{V}$ is a **Euclidean space** if it is equipped with an *inner product*, i.e., a real function $(\boldsymbol{u},\boldsymbol{v})\mapsto \boldsymbol{u}\cdot\boldsymbol{v}$ such that, for all $\boldsymbol{u},\boldsymbol{v},\boldsymbol{w}\in\mathcal{V}$ and $\alpha,\beta\in\mathbb{R}$,

$$\boldsymbol{u}\cdot\boldsymbol{v}=\boldsymbol{v}\cdot\boldsymbol{u}, \qquad \text{(symmetry)} \tag{2.1}$$

$$\boldsymbol{u}\cdot(\alpha\boldsymbol{v}+\beta\boldsymbol{w})=\alpha\,\boldsymbol{u}\cdot\boldsymbol{v}+\beta\,\boldsymbol{u}\cdot\boldsymbol{w}, \qquad \text{(linearity)} \tag{2.2}$$

$$\boldsymbol{u}\cdot\boldsymbol{u}\geq 0,\quad \boldsymbol{u}\cdot\boldsymbol{u}=0\iff\boldsymbol{u}=\boldsymbol{0}. \qquad \text{(positive definiteness)} \tag{2.3}$$

The inner product induces the norm $|\boldsymbol{u}|=(\boldsymbol{u}\cdot\boldsymbol{u})^{1/2}$ and the angle θ between two vectors, $\cos\theta=\boldsymbol{u}\cdot\boldsymbol{v}/(|\boldsymbol{u}|\,|\boldsymbol{v}|)$. These are the metric notions (length, angle, orthogonality) that we use every day without thinking; in a vector space without an inner product they simply do not

exist. The three-dimensional space of classical mechanics is a Euclidean space, denoted $\mathbb{E}$ or $\mathbb{R}^3$.

2.2 Linear forms and the dual space

Definition 2.2 (Linear form)**.** A **linear form** (or linear function, or covector) on $\mathcal{V}$ is a scalar-valued function $f : \mathcal{V} \to \mathbb{R}$ such that

$$f(\alpha \boldsymbol{u} + \beta \boldsymbol{v}) = \alpha f(\boldsymbol{u}) + \beta f(\boldsymbol{v}) \qquad \forall\, \boldsymbol{u}, \boldsymbol{v} \in \mathcal{V},\ \alpha, \beta \in \mathbb{R}. \tag{2.4}$$

Let $\{\boldsymbol{g}_1, \dots, \boldsymbol{g}_n\}$ be a basis of $\mathcal{V}$. Any vector can be written as $\boldsymbol{u} = \xi^1 \boldsymbol{g}_1 + \cdots + \xi^n \boldsymbol{g}_n = \xi^i \boldsymbol{g}_i$, and by linearity

$$f(\boldsymbol{u}) = \xi^i f(\boldsymbol{g}_i) \coloneqq \xi^i a_i, \qquad a_i = f(\boldsymbol{g}_i). \tag{2.5}$$

A linear form is thus completely determined by the n numbers a_i, its values on the basis vectors. These numbers are *coefficients* attached to a given basis, not intrinsic quantities: if we change the basis, they change.

2.2.1 Change of basis: coefficients are not components

Let $\{\boldsymbol{g}'_i\}$ be a second basis, related to the first by

$$\boldsymbol{g}'_i = \alpha_i^{\ k}\, \boldsymbol{g}_k, \tag{2.6}$$

where $[\alpha_i^{\ k}]$ is an invertible $n \times n$ matrix. The coefficients of the linear form in the new basis are

$$a'_i = f(\boldsymbol{g}'_i) = \alpha_i^{\ k} f(\boldsymbol{g}_k) = \alpha_i^{\ k} a_k : \tag{2.7}$$

they transform *like the basis vectors*. The coordinates of a vector behave differently. From $\boldsymbol{u} = \xi^k \boldsymbol{g}_k = \xi'^i \boldsymbol{g}'_i = \xi'^i \alpha_i^{\ k} \boldsymbol{g}_k$ and the uniqueness of the representation,

$$\xi^k = \alpha_i^{\ k}\, \xi'^i \quad \Longleftrightarrow \quad \xi'^i = (\alpha^{-1})_k^{\ i}\, \xi^k : \tag{2.8}$$

the coordinates transform with the *inverse* (transpose) matrix, i.e., "against" the basis.

Definition 2.3 (Covariant and contravariant quantities)**.** Quantities that, under a change of basis, transform like the basis vectors (Eq. (2.7)) are called **covariant**; they carry *lower* indices. Quantities that transform with the inverse law (Eq. (2.8)) are called **contravariant**; they carry *upper* indices.

In class

The lecturer insisted on the word *coefficient*: the a_i of the linear form and the ξ^i of the vector are both lists of numbers attached to a basis, but they are of a different nature, and the nature is revealed only by how they change when the basis changes. In an orthonormal basis, and for orthogonal changes of basis, $\boldsymbol{\alpha}^{-1} = \boldsymbol{\alpha}^{\mathrm{T}}$ and the two laws coincide: this is why the distinction is invisible in Chapter 1.

2.2.2 The dual space

Linear forms can be summed and multiplied by scalars, $(f+g)(\boldsymbol{u}) = f(\boldsymbol{u}) + g(\boldsymbol{u})$, $(\alpha f)(\boldsymbol{u}) = \alpha f(\boldsymbol{u})$: they form themselves a vector space of dimension n, the **dual space** $\mathcal{V}'$ (also written $\mathcal{V}^*$). Its elements are called *covariant vectors* or covectors, while the elements of $\mathcal{V}$ are called *contravariant vectors*.

Given a vector $\boldsymbol{u} \in \mathcal{V}$ and a form $f \in \mathcal{V}'$, the number $f(\boldsymbol{u}) = a_i \xi^i$ is bilinear in the pair $(f, \boldsymbol{u})$ and is *independent of the basis*, because the two transformation laws compensate: $a'_i \xi'^i = \alpha_i{}^k a_k (\alpha^{-1})^i_m \xi^m = \delta^k_m a_k \xi^m = a_k \xi^k$. It is called the *scalar product* between the dual spaces, and it does not need any metric.

2.2.3 Euclidean spaces: identification of $\mathcal{V}$ and $\mathcal{V}'$

In a Euclidean space every vector $\boldsymbol{v}$ defines a linear form through the inner product, $f_{\boldsymbol{v}}(\boldsymbol{u}) = \boldsymbol{v} \cdot \boldsymbol{u}$. Conversely, every linear form can be written in this way for a unique $\boldsymbol{v}$ (this is the finite dimensional version of the Riesz representation theorem). The map $\boldsymbol{v} \mapsto f_{\boldsymbol{v}}$ is a linear bijection, an *isomorphism*, between $\mathcal{V}$ and $\mathcal{V}'$. Thanks to it, in a Euclidean space we may think of covariant and contravariant vectors as two descriptions of the *same* object, and the scalar product between the dual spaces becomes identical to the inner product. This is the starting point of the construction of the dual basis.

2.3 Covariant and contravariant bases

From now on we work in $\mathbb{E}$ ($n = 3$).

Definition 2.4 (Covariant basis)**.** A set of three vectors $\{\boldsymbol{g}_1, \boldsymbol{g}_2, \boldsymbol{g}_3\}$ such that

$$(\boldsymbol{g}_1 \times \boldsymbol{g}_2) \cdot \boldsymbol{g}_3 \neq 0 \tag{2.9}$$

is a basis of $\mathbb{E}$, called the **covariant basis**. The vectors are in general neither orthogonal to each other nor of unit length.

The condition on the box (triple) product states that the three vectors are not coplanar, i.e., they are linearly independent. We denote $g \coloneqq (\boldsymbol{g}_1 \times \boldsymbol{g}_2) \cdot \boldsymbol{g}_3$; if $g > 0$ the basis is right-handed.

Definition 2.5 (Contravariant (dual) basis)**.** The **contravariant basis** $\{\boldsymbol{g}^1, \boldsymbol{g}^2, \boldsymbol{g}^3\}$ associated with $\{\boldsymbol{g}_i\}$ is defined by

$$\boxed{\boldsymbol{g}^i \cdot \boldsymbol{g}_j = \delta^i_j} \qquad i, j = 1, 2, 3, \tag{2.10}$$

where δ^i_j is the *mixed Kronecker delta*, equal to 1 if $i = j$ and to 0 otherwise.

Condition Eq. (2.10) says that $\boldsymbol{g}^1$ is orthogonal to $\boldsymbol{g}_2$ and $\boldsymbol{g}_3$ and has projection one on $\boldsymbol{g}_1$, and similarly for the others. The solution is explicit:

$$\boldsymbol{g}^1 = \frac{\boldsymbol{g}_2 \times \boldsymbol{g}_3}{g}, \qquad \boldsymbol{g}^2 = \frac{\boldsymbol{g}_3 \times \boldsymbol{g}_1}{g}, \qquad \boldsymbol{g}^3 = \frac{\boldsymbol{g}_1 \times \boldsymbol{g}_2}{g}, \qquad g = (\boldsymbol{g}_1 \times \boldsymbol{g}_2) \cdot \boldsymbol{g}_3. \tag{2.11}$$

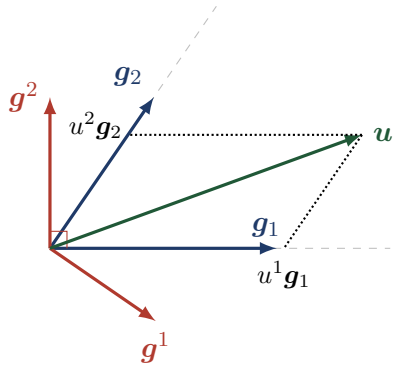


Figure 2.1: A covariant basis $\{\boldsymbol{g}_1, \boldsymbol{g}_2\}$ (blue) in the plane and its dual (contravariant) basis $\{\boldsymbol{g}^1, \boldsymbol{g}^2\}$ (red): $\boldsymbol{g}^1 \perp \boldsymbol{g}_2$, $\boldsymbol{g}^2 \perp \boldsymbol{g}_1$, $\boldsymbol{g}^i \cdot \boldsymbol{g}_i = 1$. The contravariant components u^i of a vector $\boldsymbol{u}$ are obtained with the parallelogram rule along the covariant basis vectors.

Indeed, $\boldsymbol{g}^1 \cdot \boldsymbol{g}_1 = (\boldsymbol{g}_2 \times \boldsymbol{g}_3) \cdot \boldsymbol{g}_1 / g = 1$ by the cyclic property of the box product, while $\boldsymbol{g}^1 \cdot \boldsymbol{g}_2 = \boldsymbol{g}^1 \cdot \boldsymbol{g}_3 = 0$ because $\boldsymbol{g}_2 \times \boldsymbol{g}_3$ is orthogonal to both. The two bases are called *reciprocal* or *dual* to each other: the relation is symmetric, and $\boldsymbol{g}_i$ can be recovered from $\boldsymbol{g}^i$ with the same formulas. One also finds $(\boldsymbol{g}^1 \times \boldsymbol{g}^2) \cdot \boldsymbol{g}^3 = 1/g$.

Remark 2.1. If the covariant basis is orthonormal, $\boldsymbol{g}_i = \boldsymbol{e}_i$, then $g = 1$ and Eq. (2.11) gives $\boldsymbol{g}^i = \boldsymbol{e}_i$: the two bases coincide, and so do covariant and contravariant components. Everything in Chapter 1 is a special case of what follows.

2.4 Metric coefficients

Definition 2.6 (Metric coefficients)**.** The **metric coefficients** are the inner products of the basis vectors among themselves:

$$g_{ij} \coloneqq \boldsymbol{g}_i \cdot \boldsymbol{g}_j, \qquad g^{ij} \coloneqq \boldsymbol{g}^i \cdot \boldsymbol{g}^j. \tag{2.12}$$

Both matrices are symmetric, $g_{ij} = g_{ji}$, $g^{ij} = g^{ji}$, by the symmetry of the inner product, and positive definite, since $\xi^i g_{ij} \xi^j = \left|\xi^i \boldsymbol{g}_i\right|^2 > 0$ for $\xi \neq 0$. They contain all the metric information of the basis:

$$g_{11} = |\boldsymbol{g}_1|^2, \quad g_{22} = |\boldsymbol{g}_2|^2, \quad g_{33} = |\boldsymbol{g}_3|^2, \qquad \cos\theta_{ij} = \frac{g_{ij}}{\sqrt{g_{ii}}\sqrt{g_{jj}}} \quad \text{(no sum)}, \tag{2.13}$$

where θ_{ij} is the angle between $\boldsymbol{g}_i$ and $\boldsymbol{g}_j$. Diagonal terms measure *lengths*, off-diagonal terms measure *angles*: this interpretation will reappear identically for the Cauchy–Green tensor in Section 3.9.

2.4.1 Raising and lowering indices

Each vector of one basis can be expressed in the other basis. Write the trial representation $\boldsymbol{g}^i = a^{ik} \boldsymbol{g}_k$ and take the inner product with $\boldsymbol{g}^j$:

$$g^{ij} = \boldsymbol{g}^i \cdot \boldsymbol{g}^j = a^{ik}\, \boldsymbol{g}_k \cdot \boldsymbol{g}^j = a^{ik} \delta^j_k = a^{ij}.$$

Repeating with $\boldsymbol{g}_i = b_{ik} \boldsymbol{g}^k$ gives $b_{ij} = g_{ij}$. Hence

Change between covariant and contravariant bases

$$\boldsymbol{g}^i = g^{ij}\,\boldsymbol{g}_j, \qquad \boldsymbol{g}_i = g_{ij}\,\boldsymbol{g}^j, \qquad g^{ij}g_{jk} = \delta^i_k. \tag{2.14}$$

The last relation follows by substituting one expression into the other: $\boldsymbol{g}^i = g^{ij}g_{jk}\boldsymbol{g}^k$, so $g^{ij}g_{jk} = \delta^i_k$. In matrix form,

$$\left[g^{ij}\right] = [g_{ij}]^{-1}\,: \tag{2.15}$$

the contravariant metric is the inverse of the covariant one. The metric coefficients act as "index-raising" (g^{ij}) and "index-lowering" (g_{ij}) operators. One also has $\det[g_{ij}] = g^2$, since $[g_{ij}] = [G]^{\mathrm{T}}[G]$ where $[G]$ collects the Cartesian components of the $\boldsymbol{g}_i$ by columns.

2.5 Components of a vector

A vector $\boldsymbol{u}$ can be decomposed along either basis:

$$\boldsymbol{u} = u^i\,\boldsymbol{g}_i = u_i\,\boldsymbol{g}^i. \tag{2.16}$$

The u^i are the **contravariant components** and the u_i the **covariant components** of $\boldsymbol{u}$. Taking inner products with the basis vectors and using Eq. (2.10),

$$u^i = \boldsymbol{u}\cdot\boldsymbol{g}^i, \qquad u_i = \boldsymbol{u}\cdot\boldsymbol{g}_i. \tag{2.17}$$

Note the geometric meaning, visible in Fig. 2.1: the contravariant components are the parallelogram (affine) coordinates along the $\boldsymbol{g}_i$, while the covariant components are the orthogonal projections on the $\boldsymbol{g}_i$ (multiplied by $|\boldsymbol{g}_i|$).

The two sets are related by the metric:

$$u^i = \boldsymbol{u}\cdot\boldsymbol{g}^i = u_j\,\boldsymbol{g}^j\cdot\boldsymbol{g}^i = g^{ij}u_j, \qquad u_i = g_{ij}u^j. \tag{2.18}$$

A vector has therefore six components, but only three of them are independent.

Index rules

- A *free* index must appear on both sides of an equation, and at the same level (up or down).
- A *summation* (dummy) index always appears once up and once down: contraction is only possible between a covariant and a contravariant index. Expressions such as u_iv_i are meaningless in a general basis.
- The metric coefficients raise or lower an index: $u^i = g^{ij}u_j$, $u_i = g_{ij}u^j$.

2.5.1 Inner product and norm

With the rules above,

$$\boldsymbol{u}\cdot\boldsymbol{v} = u^i\boldsymbol{g}_i\cdot v_j\boldsymbol{g}^j = u^iv_j\delta^j_i = u^iv_i = u_iv^i = g_{ij}u^iv^j = g^{ij}u_iv_j, \tag{2.19}$$

and

$$|\boldsymbol{u}| = (u^iu_i)^{1/2} = (g_{ij}u^iu^j)^{1/2}. \tag{2.20}$$

The mixed form $u^i v_i$ is the cheapest one: it does not involve the metric. This is the scalar product between dual spaces of Section 2.2, which in a Euclidean space coincides with the inner product.

Example 2.1 (An oblique basis in the plane)**.** Let $\boldsymbol{g}_1 = 2\boldsymbol{e}_1$, $\boldsymbol{g}_2 = \boldsymbol{e}_1 + \boldsymbol{e}_2$ (in the plane, with $\boldsymbol{g}_3 = \boldsymbol{e}_3$). Then $g_{11} = 4$, $g_{12} = 2$, $g_{22} = 2$, so that $[g_{ij}] = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$, $[g^{ij}] = \frac{1}{4}\begin{bmatrix} 2 & -2 \\ -2 & 4 \end{bmatrix}$, and $\cos\theta_{12} = 2/(2\sqrt{2}) = 1/\sqrt{2}$ ($\theta_{12} = 45°$). The dual basis is $\boldsymbol{g}^1 = g^{11}\boldsymbol{g}_1 + g^{12}\boldsymbol{g}_2 = \frac{1}{2}(\boldsymbol{e}_1 - \boldsymbol{e}_2)$, $\boldsymbol{g}^2 = g^{21}\boldsymbol{g}_1 + g^{22}\boldsymbol{g}_2 = \boldsymbol{e}_2$; one checks $\boldsymbol{g}^1 \cdot \boldsymbol{g}_1 = 1$, $\boldsymbol{g}^1 \cdot \boldsymbol{g}_2 = 0$, $\boldsymbol{g}^2 \cdot \boldsymbol{g}_1 = 0$, $\boldsymbol{g}^2 \cdot \boldsymbol{g}_2 = 1$. For $\boldsymbol{u} = 3\boldsymbol{e}_1 + \boldsymbol{e}_2$: $u_1 = \boldsymbol{u} \cdot \boldsymbol{g}_1 = 6$, $u_2 = \boldsymbol{u} \cdot \boldsymbol{g}_2 = 4$, $u^1 = \boldsymbol{u} \cdot \boldsymbol{g}^1 = 1$, $u^2 = \boldsymbol{u} \cdot \boldsymbol{g}^2 = 1$. Indeed $\boldsymbol{u} = \boldsymbol{g}_1 + \boldsymbol{g}_2$, and $u^i u_i = 6 + 4 = 10 = |\boldsymbol{u}|^2$.

2.6 Second-order tensors in a general basis

Using the dyadic product of Chapter 1, the nine dyads built with either basis provide four equivalent representations of a second-order tensor:

Components of a second-order tensor

$$\boldsymbol{A} = A^{ij}\,\boldsymbol{g}_i \otimes \boldsymbol{g}_j = A_{ij}\,\boldsymbol{g}^i \otimes \boldsymbol{g}^j = A^{i}_{\cdot j}\,\boldsymbol{g}_i \otimes \boldsymbol{g}^j = A_{i}^{\cdot j}\,\boldsymbol{g}^i \otimes \boldsymbol{g}_j, \tag{2.21}$$

with

$$A^{ij} = \boldsymbol{g}^i \cdot \boldsymbol{A}\boldsymbol{g}^j, \quad A_{ij} = \boldsymbol{g}_i \cdot \boldsymbol{A}\boldsymbol{g}_j, \quad A^{i}_{\cdot j} = \boldsymbol{g}^i \cdot \boldsymbol{A}\boldsymbol{g}_j, \quad A_{i}^{\cdot j} = \boldsymbol{g}_i \cdot \boldsymbol{A}\boldsymbol{g}^j. \tag{2.22}$$

A^{ij} are the *contravariant*, A_{ij} the *covariant* and $A^{i}_{\cdot j}$, $A_{i}^{\cdot j}$ the *mixed* components. The dot in the mixed components keeps track of the order of the indices, which matters when $\boldsymbol{A}$ is not symmetric. A tensor has $4 \times 9 = 36$ components, of which only nine are independent: they are related by the metric, e.g.,

$$A^{ij} = g^{ik} A_k^{\cdot j} = g^{ik} g^{jl} A_{kl}, \qquad A_{ij} = g_{ik} g_{jl} A^{kl}, \qquad A^{i}_{\cdot j} = g^{ik} A_{kj} = g_{jk} A^{ik}. \tag{2.23}$$

The identity tensor and the metric. For $\boldsymbol{A} = \boldsymbol{I}$ the components Eq. (2.22) give $I_{ij} = \boldsymbol{g}_i \cdot \boldsymbol{g}_j = g_{ij}$, $I^{ij} = g^{ij}$, $I^{i}_{\cdot j} = \delta^i_j$:

$$\boldsymbol{I} = g_{ij}\,\boldsymbol{g}^i \otimes \boldsymbol{g}^j = g^{ij}\,\boldsymbol{g}_i \otimes \boldsymbol{g}_j = \boldsymbol{g}_i \otimes \boldsymbol{g}^i = \boldsymbol{g}^i \otimes \boldsymbol{g}_i. \tag{2.24}$$

The metric coefficients are the covariant and contravariant components of the identity: this is why they are called *metric tensor* in differential geometry.

Replacement rule. Contracting with the mixed Kronecker delta replaces an index: $A_{ij}\delta^j_k = A_{ik}$, $\delta^i_j v^j = v^i$.

Operations. Using Eq. (2.10) each time two basis vectors meet in a contraction:

$$\boldsymbol{A}\boldsymbol{u} = A^{ij}u_j\,\boldsymbol{g}_i = A^i_{\cdot j}u^j\,\boldsymbol{g}_i, \tag{2.25}$$

$$(\boldsymbol{A}\boldsymbol{B})_{ij} = A_{ik}B^k_{\cdot j} = A_i^{\cdot k}B_{kj}, \tag{2.26}$$

$$\boldsymbol{A}:\boldsymbol{B} = A_{ij}B^{ij} = A^{ij}B_{ij} = A^i_{\cdot j}B_i^{\cdot j}, \tag{2.27}$$

$$\operatorname{tr}\boldsymbol{A} = \boldsymbol{A}:\boldsymbol{I} = A^i_{\cdot i} = g_{ij}A^{ij} = g^{ij}A_{ij}, \tag{2.28}$$

$$(\boldsymbol{A}^{\mathrm{T}})^{ij} = A^{ji}, \qquad (\boldsymbol{A}^{\mathrm{T}})^i_{\cdot j} = A^{\cdot i}_j. \tag{2.29}$$

Note that the trace is the contraction of a *mixed* component: in a general basis A_{ii} (and A^{ii}) are not invariants.

Remark 2.2 (Determinant). The determinant is also computed from the mixed components, $\det\boldsymbol{A} = \det\left[A^i_{\cdot j}\right]$. For the covariant components one has $\det[A_{ij}] = g^2\det\boldsymbol{A}$, so the matrix of covariant (or of contravariant) components does not directly give the determinant.

2.7 Spectral decomposition revisited

The dual basis gives the natural setting for the spectral decomposition of a non-symmetric tensor anticipated in Section 1.14, Eq. (1.75).

Theorem 2.1 (Spectral decomposition of a diagonalisable tensor)**.** *Let $\boldsymbol{A}$ have three independent (right) eigenvectors $\boldsymbol{u}_\alpha$, $\boldsymbol{A}\boldsymbol{u}_\alpha = \lambda_\alpha\boldsymbol{u}_\alpha$, $\alpha = 1,2,3$, and let $\{\boldsymbol{u}^\alpha\}$ be their dual basis, $\boldsymbol{u}_\alpha\cdot\boldsymbol{u}^\beta = \delta^\beta_\alpha$. Then the $\boldsymbol{u}^\alpha$ are the eigenvectors of $\boldsymbol{A}^{\mathrm{T}}$ with the same eigenvalues, $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{u}^\alpha = \lambda_\alpha\boldsymbol{u}^\alpha$, and*

$$\boldsymbol{A} = \sum_{\alpha=1}^{3}\lambda_\alpha\,\boldsymbol{u}_\alpha\otimes\boldsymbol{u}^\alpha. \tag{2.30}$$

Proof. Any $\boldsymbol{x}$ can be expanded along the eigenvectors, $\boldsymbol{x} = \sum_\alpha x^\alpha\boldsymbol{u}_\alpha$, with $x^\alpha = \boldsymbol{x}\cdot\boldsymbol{u}^\alpha$. Then

$$\boldsymbol{A}\boldsymbol{x} = \sum_\alpha x^\alpha\boldsymbol{A}\boldsymbol{u}_\alpha = \sum_\alpha\lambda_\alpha(\boldsymbol{x}\cdot\boldsymbol{u}^\alpha)\boldsymbol{u}_\alpha = \Big(\sum_\alpha\lambda_\alpha\boldsymbol{u}_\alpha\otimes\boldsymbol{u}^\alpha\Big)\boldsymbol{x},$$

and since $\boldsymbol{x}$ is arbitrary, Eq. (2.30) follows. Transposing, $\boldsymbol{A}^{\mathrm{T}} = \sum_\alpha\lambda_\alpha\boldsymbol{u}^\alpha\otimes\boldsymbol{u}_\alpha$, so that $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{u}^\beta = \sum_\alpha\lambda_\alpha\boldsymbol{u}^\alpha(\boldsymbol{u}_\alpha\cdot\boldsymbol{u}^\beta) = \lambda_\beta\boldsymbol{u}^\beta$. ∎

In class

The proof on the slide (no. 56) states the ingredients in an order that is a bit confusing (the eigenvectors of $\boldsymbol{A}^{\mathrm{T}}$ are introduced before being needed). The version above starts from the right eigenvectors, builds their dual basis, and *derives* that the dual vectors are the left eigenvectors. When $\boldsymbol{A}$ is symmetric the eigenvectors can be chosen orthonormal, the two bases coincide, and one recovers $\boldsymbol{A} = \sum_\alpha\lambda_\alpha\boldsymbol{n}_\alpha\otimes\boldsymbol{n}_\alpha$.

Complement Why this matters for continuum mechanics

Take an orthonormal material grid $\{\boldsymbol{E}_A\}$ in the reference configuration and deform it with a homogeneous deformation $\boldsymbol{F}$ (Chapter 3). The deformed grid vectors $\boldsymbol{g}_A = \boldsymbol{F}\boldsymbol{E}_A$ form a covariant basis, generally oblique. Their metric coefficients are $g_{AB} = \boldsymbol{F}\boldsymbol{E}_A\cdot\boldsymbol{F}\boldsymbol{E}_B = \boldsymbol{E}_A\cdot\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}\boldsymbol{E}_B = C_{AB}$: the *right Cauchy–Green tensor* $\boldsymbol{C}$ introduced in Section 3.9 is nothing but the metric of the

deformed material grid. The dual basis $\boldsymbol{g}^A = \boldsymbol{F}^{-\mathrm{T}}\boldsymbol{E}_A$ transforms like the normals to material surfaces (Nanson's formula, Section 3.10.2). This is the point of view of convected coordinates adopted by Marsden and Hughes [5].

2.8 Exercises

Exercise 2.1. Given $\boldsymbol{g}_1 = \boldsymbol{e}_1 + \boldsymbol{e}_2$, $\boldsymbol{g}_2 = \boldsymbol{e}_2 + \boldsymbol{e}_3$, $\boldsymbol{g}_3 = \boldsymbol{e}_1 + \boldsymbol{e}_3$: (a) verify that they form a basis and compute g; (b) compute $[g_{ij}]$ and $[g^{ij}]$; (c) compute the dual basis with Eq. (2.11) and check it against Eq. (2.14); (d) find the covariant and contravariant components of $\boldsymbol{u} = \boldsymbol{e}_1 + 2\boldsymbol{e}_2 + 3\boldsymbol{e}_3$ and verify $u^i u_i = |\boldsymbol{u}|^2$.

▷ Solution on p. 136

Exercise 2.2. Prove that $(\boldsymbol{g}^1 \times \boldsymbol{g}^2) \cdot \boldsymbol{g}^3 = 1/g$ and that $\det[g_{ij}] = g^2$.

▷ Solution on p. 136

Exercise 2.3. Show that $\operatorname{tr}\boldsymbol{A} = A^i_{\cdot i}$ is invariant under a change of basis Eq. (2.6), while A_{ii} is not. Give a counterexample using the basis of the example in Section 2.5.

▷ Solution on p. 136

Exercise 2.4. Let $\boldsymbol{A} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ (Cartesian components, in the plane). Compute its right eigenvectors, their dual basis and verify Eq. (2.30) and that the dual vectors are eigenvectors of $\boldsymbol{A}^{\mathrm{T}}$.

▷ Solution on p. 137

Part II

Kinematics, Stress and Balance Principles

3

Kinematics

> **In class**
>
> The lecturer jumped directly from the mathematical preliminaries to the kinematics, postponing the dual basis to the next lecture, because the kinematics of deformation is, in her opinion, the most instructive section of the course. This chapter collects what was presented from the end of the first day to the middle of the third day.

The kinematics studies the motion and the deformation of a body without asking for their causes. Its main goal, in this course, is to understand precisely the difference between *motion*, *deformation* and *strain*: concepts that everyone who studied a little physics should know, but which in the nonlinear setting must be handled with care.

3.1 The continuum body

We consider a *body* made of matter, but at this stage we do not care about the nature of the matter. The body is a *macroscopic system*, characterised by (piecewise) continuous mass and volume, which are functions of *continuum particles*: it occupies a particular volume and has a particular boundary shape.

We disregard everything that happens at the microscopic level: for instance the presence of atoms and molecules, and the particular microstructure or sub-components of the body. We start with the simplest assumption; otherwise things become much more complicated.

Definition 3.1 (Continuum particle)**.** A *continuum particle* is a small portion of the continuum surrounding a material point P. It must not be confused with

- the Newtonian point mass;
- the discrete atoms and molecules.

Our focus will be on the behaviour of the small amount of material that surrounds a point (the *neighbourhood* of the point), rather than on the motion of the entire body.

Reference frame. We use a rectangular right-handed system of coordinate axes with fixed origin O and orthonormal basis $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$, the same introduced in Section 1.2; all tensors are defined with respect to it according to Chapter 1.

3.1.1 Compatible motion

We cannot describe every kinematics that can happen in reality. To set up a mathematical framework in which the equations are valid, we make an important assumption: the motion

is *compatible*. As in the fundamental courses of mechanics, where the definition of strain is based on this assumption, compatibility means that the body

- *does not break* (no fracture): a point P before the motion ends up, after the motion, in a unique point; one point remains one point;
- *does not interpenetrate*: two distinct points P and Q evolve during the motion into two distinct points p and q.

3.2 Configurations

When we consider a motion, we refer to the sequence of spatial regions occupied by the body $\mathcal{B}$ in time (without breaking and without interpenetration). At each time t the body occupies a different region $\Omega_0, \ldots, \Omega$, and each particle of the body is located at a geometrical point having a position in $\Omega_0, \ldots, \Omega$.

Definition 3.2 (Configurations)**.**

- The *configuration at time* t is the region Ω occupied by the body at that time.
- The *initial configuration* corresponds to the time $t = 0$; the region occupied is denoted Ω_0.
- The *current configuration* is the generic configuration at time $t > 0$, occupying the region Ω.
- The *reference* (or *natural*) configuration corresponds to a fixed reference time, and is the configuration to which we refer the description of the motion.

The reference configuration does not necessarily coincide with the initial one. Think of a balloon: originally it is flat; then it is inflated and reaches the state in which it works well (for playing soccer, or for whatever use), and this is a convenient reference configuration. Then, for some reason, the balloon is inflated more, or a hole is opened and it loses some air, and the configuration changes. The reference configuration is just something useful for us. In most of the course, for the sake of simplicity, we assume that the reference and the initial configurations coincide (this is not always the case, especially in dynamics).

The reference configuration is often called the *material* configuration, the current one the *spatial* configuration: "material" and "spatial" are used as synonyms of "reference" and "current". The boundary of the reference configuration is denoted $\partial\Omega_0$, that of the current one $\partial\Omega$.

3.2.1 Material and spatial position vectors

Reference configuration. The point that identifies the position of a particle P in the reference configuration is denoted by the *referential position vector* $\boldsymbol{X}$. If the basis $\{\boldsymbol{E}_A\}$ is fixed, the particle P can be labelled univocally by its position $\boldsymbol{X}$. The *material* (or reference) coordinates X_A are the components of $\boldsymbol{X}$ along that basis:

$$\boldsymbol{X} = X_A \, \boldsymbol{E}_A. \tag{3.1}$$

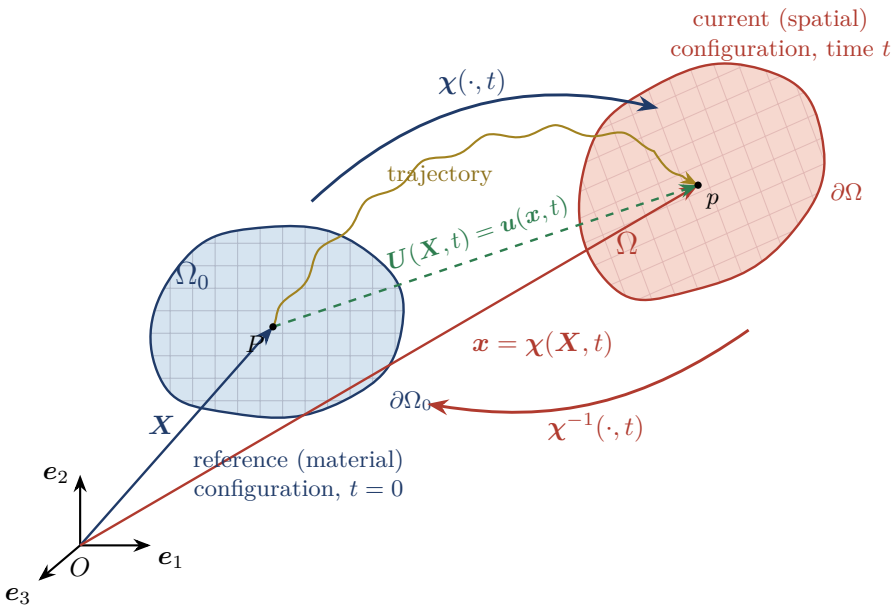


Figure 3.1: Reference and current configurations of a continuum body, motion $\boldsymbol{\chi}$ and its inverse, position vectors, displacement and trajectory of a particle (slides 59–64). A single fixed frame $\{\boldsymbol{e}_i\}$ is used for both configurations.

Upper case symbols are used for reference quantities: the position $\boldsymbol{X}$, the coordinates X_1, X_2, X_3, the region Ω_0 (in some cases the index 0 labels the reference configuration). From now on the notation x_1, x_2, x_3 for the coordinates of a generic point is abandoned, because it would become confusing.

Current configuration. The point that identifies the position of the same particle in the current configuration is denoted by the *current position vector* $\boldsymbol{x}$, and the particle can be labelled by $\boldsymbol{x}$ as well. The *spatial* (or current) coordinates x_a are its components along a fixed basis $\{\boldsymbol{e}_a\}$:

$$\boldsymbol{x} = x_a \, \boldsymbol{e}_a. \tag{3.2}$$

Lower case symbols are used for current quantities (the material point is the same, but its position is different, and it is denoted by p).

> **In class**
>
> Throughout the course the same reference system is used for the reference and for the current configuration, $\boldsymbol{E}_A = \boldsymbol{e}_a$. This is not always the case: in many situations it may be useful to introduce two different reference systems (as shown on some slides), but this is complicated enough, and it is better to keep a unique reference system. The distinction of the *indices*, however, is kept: uppercase indices $(A, B, \dots)$ for components in the reference configuration, lowercase indices $(a, b, \dots)$ for components in the current one.

3.3 Motion

3.3.1 Mappings

Every configuration, even the reference one, can be thought of as obtained through a mapping of an ideal "parent" shape of the body $\mathcal{B}$, a transformation of coordinates (slide 62). Assume that

- the map $\boldsymbol{\kappa}_0$ defines a one-to-one correspondence between a particle P in $\mathcal{B}$ and the point $\boldsymbol{X}$ in Ω_0 that it occupies at time $t = 0$:

$$\boldsymbol{X} = \boldsymbol{\kappa}_0(P, t = 0) = \boldsymbol{\kappa}_0(P), \qquad P = \boldsymbol{\kappa}_0^{-1}(\boldsymbol{X}, t = 0) = \boldsymbol{\kappa}_0^{-1}(\boldsymbol{X}); \tag{3.3}$$

- the map $\boldsymbol{\kappa}$ defines a one-to-one correspondence between a particle P in $\mathcal{B}$ and the point $\boldsymbol{x}$ in Ω that it occupies at time t:

$$\boldsymbol{x} = \boldsymbol{\kappa}(P, t), \qquad P = \boldsymbol{\kappa}^{-1}(\boldsymbol{x}, t). \tag{3.4}$$

Since the motion is regular, both maps can be inverted.

3.3.2 Motion and inverse motion

Substituting $P = \boldsymbol{\kappa}_0^{-1}(\boldsymbol{X})$ in $\boldsymbol{x} = \boldsymbol{\kappa}(P, t)$, i.e. going back to the parent configuration through the inverse mapping and then forward to the current one, the position occupied by the particle at time t is

Motion

$$\boldsymbol{x} = \boldsymbol{\kappa}\big[\boldsymbol{\kappa}_0^{-1}(\boldsymbol{X}), t\big] = \boldsymbol{\chi}(\boldsymbol{X}, t), \qquad x_a = \chi_a(X_1, X_2, X_3, t). \tag{3.5}$$

The vector field $\boldsymbol{\chi} = \boldsymbol{\kappa} \circ \boldsymbol{\kappa}_0^{-1}$ is called the *motion* and specifies the place $\boldsymbol{x}$ of $\boldsymbol{X}$ for every fixed t. It is assumed regular, possessing all the necessary derivatives with respect to time and position. One is therefore not obliged to connect directly the reference configuration to the current one: every configuration can be defined independently, and the regularity of the motion allows to describe the connection between them in this way.

Trajectory. Since there is no fracture and no interpenetration, in going from P to p the particle follows a *trajectory*, a continuous sequence of points in space. The parametric equation of motion (3.5), with $\boldsymbol{X}$ fixed and t varying, gives all the sequential positions of P in time; the sequence of positions is the trajectory (*path line*).

Inverse motion. If the motion is regular we are also able to go back. Since the body neither breaks nor interpenetrates, if we can describe how P moves to p through $\boldsymbol{\chi}$, we can go back from p to P through the inverse motion. We assume $\boldsymbol{\chi}$ uniquely invertible:

Inverse motion

$$\boldsymbol{X} = \boldsymbol{\chi}^{-1}(\boldsymbol{x}, t), \qquad X_A = \chi_A^{-1}(x_1, x_2, x_3, t). \tag{3.6}$$

Note the change of independent variables: in the inverse motion the independent variables are the spatial coordinates x_a (and t) and the dependent ones are the material coordinates X_A. Regular means here that $\boldsymbol{\chi}$ is smooth enough (continuous, differentiable as many times as needed, invertible) for all the mathematical operations required by the geometrical description:

compatibility is precisely the assumption that allows us to define a motion and an inverse motion.

Deformable bodies. A motion changes the position, the orientation and the shape of a body. A continuum body able to change its shape is called *deformable*.

3.3.3 Motion versus deformation

Deformation

A *deformation* (inverse deformation) is a body motion $\boldsymbol{\chi}$ (inverse motion $\boldsymbol{\chi}^{-1}$) at a *fixed* time t, i.e. independent of time.

The motion depends strictly on time. If, instead, we fix the time at which we look at the body, the time is no longer a variable but a parameter: $\boldsymbol{x}$ is a function of $\boldsymbol{X}$ only, and $\boldsymbol{X}$ a function of $\boldsymbol{x}$ only. A deformation compares a fixed configuration (the current one, fixed because the time is fixed) with the reference configuration. Time matters in that it brings the body from one configuration to the other; but when we speak about deformation we do not look at time, only at the spatial positions.

Definition 3.3 (Deformation)**.** A deformation of a body $\mathcal{B}$ is an *injective* (one-to-one) mapping between the reference configuration and the configuration at time t:

$$\boldsymbol{\chi} : \mathcal{B} \to \mathbb{R}^3, \qquad \boldsymbol{x} = \boldsymbol{\chi}(\boldsymbol{X}, t), \ t \text{ fixed}. \tag{3.7}$$

Injectivity rules out deformations involving tearing or interpenetration of matter: it is the mathematical form of the compatibility assumption of Section 3.1.1.

Composition of deformations. Deformations compose by composition of mappings: a motion $\boldsymbol{\chi}$ can always be thought of as the combination of two motions $\boldsymbol{\chi}'$ and $\boldsymbol{\chi}''$, applied in sequence:

$$\boldsymbol{\chi}(\boldsymbol{X}, t) = \boldsymbol{\chi}''(\boldsymbol{\chi}'(\boldsymbol{X}, t), t) = (\boldsymbol{\chi}'' \circ \boldsymbol{\chi}')(\boldsymbol{X}, t). \tag{3.8}$$

To combine two motions, the second motion is applied to the result of the first. This composition will be used when discussing objectivity (superposed rigid body motions).

3.3.4 Examples of deformations

Example 3.1 (Stretching of a cube)**.**

$$x_1 = \lambda_1 X_1, \qquad x_2 = \lambda_2 X_2, \qquad x_3 = \lambda_3 X_3, \qquad \lambda_i > 0, \ i = 1, 2, 3. \tag{3.9}$$

Every point is moved, along each coordinate direction, proportionally to the parameter λ_i, called the *stretch ratio* (or stretch). A cube with equal edges is transformed into a parallelepiped with edges $\lambda_1, \lambda_2, \lambda_3$ times the original ones: if the three stretches are different, the cube may become very elongated in one direction.

Example 3.2 (Pure shear)**.**

$$x_1 = X_1 + \gamma X_2, \qquad x_2 = X_2, \qquad x_3 = X_3, \tag{3.10}$$

where γ is the shear parameter: planes $X_2 = \text{const}$ slide parallel to $\boldsymbol{e}_1$ by an amount proportional to their height. (This deformation is often called *simple shear* in the literature.)

Example 3.3 (Affine deformation)**.** For any two points $\boldsymbol{X}, \boldsymbol{Y}$ of the reference configuration, with images $\boldsymbol{x}, \boldsymbol{y}$,

$$\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{F}(\boldsymbol{Y} - \boldsymbol{X}), \tag{3.11}$$

where $\boldsymbol{F}$ is a constant tensor (better defined later as the deformation gradient). *Rigid body* kinematics is necessarily an affine deformation that preserves distances:

$$\begin{gathered} |\boldsymbol{y} - \boldsymbol{x}| = |\boldsymbol{Y} - \boldsymbol{X}|, \\ |\boldsymbol{y} - \boldsymbol{x}|^2 = |\boldsymbol{F}(\boldsymbol{Y} - \boldsymbol{X})|^2 = (\boldsymbol{Y} - \boldsymbol{X}) \cdot \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}(\boldsymbol{Y} - \boldsymbol{X}) = |\boldsymbol{Y} - \boldsymbol{X}|^2 \iff \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} = \boldsymbol{I}, \end{gathered} \tag{3.12}$$

i.e. $\boldsymbol{F}$ must be an orthogonal tensor (Section 1.9), a rotation if orientation is preserved.

Example 3.4 (Motions; slide 68)**.**

1. Describe the motion of the point P of a block sheared by an angle $\theta(t) = \omega t$: $x_1 = X_1 + \tan(\omega t)\, X_2$, $x_2 = X_2$, $x_3 = X_3$.
2. Identify the path lines of the motion $\boldsymbol{x} = \alpha t^3 \boldsymbol{a} + \boldsymbol{X}$ (α a scalar, $\boldsymbol{a}$ a unit vector), i.e. $x_i = \alpha t^3 a_i + X_i$: straight lines through $\boldsymbol{X}$ parallel to $\boldsymbol{a}$.

In class

The examples of slides 67–68 are left to the students, who are invited to read them carefully; only the first one (stretching of a cube) was discussed in class.

3.4 Displacement

Consider only the particle P, at $\boldsymbol{X}$ in the reference configuration and at $\boldsymbol{x}$ in the current one. The vector that joins the original position with the new one is the *displacement*.

Definition 3.4 (Displacement field)**.** The displacement field relates the positions of a particle in the reference and in the current configuration. In the material description (Lagrangian form),

$$\boldsymbol{U}(\boldsymbol{X}, t) = \boldsymbol{x}(\boldsymbol{X}, t) - \boldsymbol{X} = \boldsymbol{\chi}(\boldsymbol{X}, t) - \boldsymbol{X}; \tag{3.13}$$

in the spatial description (Eulerian form),

$$\boldsymbol{u}(\boldsymbol{x}, t) = \boldsymbol{x} - \boldsymbol{X}(\boldsymbol{x}, t) = \boldsymbol{x} - \boldsymbol{\chi}^{-1}(\boldsymbol{x}, t). \tag{3.14}$$

They are related by means of the motion:

$$\boldsymbol{U}(\boldsymbol{X}, t) = \boldsymbol{U}\left[\boldsymbol{\chi}^{-1}(\boldsymbol{x}, t), t\right] = \boldsymbol{u}(\boldsymbol{x}, t). \tag{3.15}$$

$\boldsymbol{U}$ and $\boldsymbol{u}$ have the same values: they are two different ways of expressing the same quantity, convenient respectively from the reference (material) and from the spatial point of view. The capital letter is used when everything depends on $\boldsymbol{X}$, the small letter when the independent variables are $\boldsymbol{x}$. Their components are referred to material and spatial coordinates, respectively, U_A and u_a. The displacement vector bridges the two configurations: it can be seen both as a reference vector and as a spatial vector.

Trajectory versus displacement. The displacement is very different from the trajectory (Fig. 3.1).

- The *trajectory* is the sequence of positions occupied by the particle during the motion. It contains all the information about the motion: whatever we want to know about it is contained in the trajectories of all the (infinitely many) particles of the body.
- The *displacement* is a vector, a straight arrow: it only compares the current position with the original one, and loses track of everything that happened in between. Using only the displacement, we miss something about the motion.

For the deformation, however, this is exactly what we want: we do not want to keep track of the history, only to compare two configurations. The displacement is therefore sufficient.

3.5 Descriptions of motion

The classical literature distinguishes four descriptions of motion (this classification is taken from Malvern's book; it is less common in modern texts).

Material description
characterised by the *particle* X (not its position) and the current time t.

Referential description
characterised by the material coordinates $\boldsymbol{X}$ (the position of the particle at the origin of the motion) and the current time t. If the reference configuration coincides with the initial one it is called *Lagrangian*, or (extending the previous term) *material* description. One considers a particle and observes what happens to it during the motion. This is the natural point of view for solids.

Spatial description
characterised by the spatial coordinates $\boldsymbol{x}$ and the current time t: one considers a location in space and observes what passes there as time changes (*Eulerian* description). It is very often used for fluids, which do not have a proper shape.

Relative description
characterised by the current position $\boldsymbol{x}$ occupied by the particle and by a reference time τ, when the particle was occupying the position $\boldsymbol{\xi}$; the motion is described using $\boldsymbol{\xi}$ as the dependent variable. One observes the motion of other systems from a moving one, and the true motion must be reconstructed through the relative motion. It is common in robotics and mechatronics, not in this course.

Definition 3.5 (Material and spatial fields)**.** A *material field* $\Phi(\boldsymbol{X}, t)$ has as independent variables the material positions $\boldsymbol{X}$, with coordinates X_A, and the time t. A *spatial field*

$\phi(\boldsymbol{x}, t)$ has as independent variables the spatial positions $\boldsymbol{x}$, with coordinates x_a, and the time t.

From now on, material (reference) quantities are denoted by capital letters and spatial (current) quantities by small letters, both for the symbols and for the indices.

3.6 Velocity and acceleration

To speak about velocity and acceleration we need again the time, which we had set aside when discussing deformation.

Definition 3.6 (Material velocity and acceleration)**.** In the material description, the velocity and the acceleration are the first and second derivatives of the motion $\boldsymbol{\chi}$ with respect to time, holding $\boldsymbol{X}$ fixed:

$$\boldsymbol{V}(\boldsymbol{X}, t) = \frac{\partial \boldsymbol{\chi}(\boldsymbol{X}, t)}{\partial t}, \qquad \boldsymbol{A}(\boldsymbol{X}, t) = \frac{\partial \boldsymbol{V}(\boldsymbol{X}, t)}{\partial t} = \frac{\partial^2 \boldsymbol{\chi}(\boldsymbol{X}, t)}{\partial t^2}. \tag{3.16}$$

$\boldsymbol{V}$ and $\boldsymbol{A}$ are functions of the material coordinates X_A and represent the time rate of change of position and of velocity of the particle $\boldsymbol{X}$ at time t. Since $\boldsymbol{\chi}(\boldsymbol{X}, t) = \boldsymbol{X} + \boldsymbol{U}(\boldsymbol{X}, t)$ and $\boldsymbol{X}$ does not change with time, the velocity is equivalently the time derivative of the displacement:

$$\boldsymbol{V}(\boldsymbol{X}, t) = \frac{\partial \boldsymbol{U}(\boldsymbol{X}, t)}{\partial t}, \qquad V_A = \frac{\partial U_A}{\partial t}. \tag{3.17}$$

In class

In class the velocity was first (mistakenly) introduced as the time derivative of the displacement; the correct *definition* is the time derivative of the motion $\boldsymbol{\chi}$, which, as shown above, gives the same result because the reference coordinates do not change.

3.6.1 Spatial velocity and acceleration

In the spatial (Eulerian) description, velocity and acceleration are expressed as functions of the spatial coordinates x_a. The two descriptions are related through the motion $\boldsymbol{\chi}$ and its inverse:

Material and spatial velocity and acceleration

$$\begin{aligned} \boldsymbol{v}(\boldsymbol{x}, t) &= \boldsymbol{V}(\boldsymbol{\chi}^{-1}(\boldsymbol{x}, t), t), \qquad & \boldsymbol{a}(\boldsymbol{x}, t) &= \boldsymbol{A}(\boldsymbol{\chi}^{-1}(\boldsymbol{x}, t), t), \\ \boldsymbol{V}(\boldsymbol{X}, t) &= \boldsymbol{v}(\boldsymbol{\chi}(\boldsymbol{X}, t), t), \qquad & \boldsymbol{A}(\boldsymbol{X}, t) &= \boldsymbol{a}(\boldsymbol{\chi}(\boldsymbol{X}, t), t). \end{aligned} \tag{3.18}$$

$\boldsymbol{V}$ and $\boldsymbol{v}$, $\boldsymbol{A}$ and $\boldsymbol{a}$ have the same *values*, but they are referred to different arguments: material coordinates (V_A, A_A) and spatial coordinates (v_a, a_a), respectively. The spatial velocity $\boldsymbol{v}(\boldsymbol{x}, t)$ is the velocity of the particle that, at time t, happens to be at the place $\boldsymbol{x}$.

Example 3.5 (Uniaxial motion)**.** Consider the motion, with α a constant,

$$x_1 = X_1(1 + \alpha t^3), \qquad x_2 = X_2, \qquad x_3 = X_3.$$

The material velocity and acceleration are $V_1 = 3\alpha t^2 X_1$, $A_1 = 6\alpha t X_1$, $V_2 = V_3 = A_2 = A_3 = 0$. The inverse motion is $X_1 = x_1/(1+\alpha t^3)$, so that the spatial fields are

$$v_1 = \frac{3\alpha t^2}{1+\alpha t^3}\, x_1, \qquad a_1 = \frac{6\alpha t}{1+\alpha t^3}\, x_1, \qquad v_2 = v_3 = a_2 = a_3 = 0.$$

Note that $a_1 \neq \partial v_1/\partial t$: in the spatial description the acceleration is *not* the partial time derivative of the velocity (see Eq. (3.22) below).

Example 3.6 (Expansion of an incompressible hollow sphere)**.** A hollow sphere of incompressible material has, in the reference configuration, inner radius a_0; its inner radius changes in time as $a(t)$, and the motion is radial. A material sphere of radius R is carried into a sphere of radius $r(t)$; since the volume of material between the cavity and that sphere does not change,

$$\tfrac{4}{3}\pi(R^3 - a_0^3) = \tfrac{4}{3}\pi(r^3(t) - a^3(t)) \quad \Longrightarrow$$
$$r(t) = R\left(\frac{a^3(t) - a_0^3}{R^3} + 1\right)^{1/3}, \qquad R = r(t)\left(1 - \frac{a^3(t) - a_0^3}{r^3(t)}\right)^{1/3}.$$

Since the motion is radial, $\boldsymbol{x} = (r/R)\boldsymbol{X}$:

$$\boldsymbol{x}(\boldsymbol{X}, t) = \left(\frac{a^3(t) - a_0^3}{R^3} + 1\right)^{1/3} \boldsymbol{X}, \qquad R = |\boldsymbol{X}|\,.$$

Differentiating with respect to time at fixed $\boldsymbol{X}$,

$$\boldsymbol{V}(\boldsymbol{X}, t) = \left(\frac{a^3(t) - a_0^3}{R^3} + 1\right)^{-2/3} \frac{a^2(t)\dot{a}(t)}{R^3}\, \boldsymbol{X}.$$

To obtain the spatial velocity, note that the bracket equals r^3/R^3 and that $\boldsymbol{X} = (R/r)\boldsymbol{x}$; hence

$$\boldsymbol{v}(\boldsymbol{x}, t) = \frac{R^2}{r^2}\,\frac{a^2\dot{a}}{R^3}\,\frac{R}{r}\,\boldsymbol{x} = \frac{a^2(t)\dot{a}(t)}{r^3}\,\boldsymbol{x}, \qquad r = |\boldsymbol{x}|\,.$$

The radial speed is $v_r = a^2\dot{a}/r^2$: the flux $4\pi r^2 v_r = 4\pi a^2 \dot{a}$ through any sphere is the same, as expected for an incompressible material.

In class

The spatial velocity printed on the slide (no. 71) contains an extra factor $\left(1 - (a^3 - a_0^3)/r^3\right)^{-1/3} = r/R$; the correct result, derived above, is $\boldsymbol{v} = a^2\dot{a}\,\boldsymbol{x}/r^3$. The lecturer left as an exercise the two-dimensional version of this example (expansion of a hollow disc preserving its area), see Exercise 3.1.

3.7 Material and spatial derivatives

In class

The slides on time derivatives and gradients (nos. 72–77) were covered very quickly before the lunch break; the lecturer recommended reading them, and this section develops them following Holzapfel [3, Sect. 2.4].

Any physical quantity (density, temperature, velocity, stress) can be described either as a material field $\mathcal{F}(\boldsymbol{X}, t)$ or as a spatial field $f(\boldsymbol{x}, t)$; they take the same value on the same particle, $\mathcal{F}(\boldsymbol{X}, t) = f(\boldsymbol{\chi}(\boldsymbol{X}, t), t)$. Correspondingly, there are two kinds of time derivatives and two kinds of gradients.

3.7.1 Material fields

Definition 3.7 (Material time derivative of a material field)**.** The **material time derivative** (also total, or substantial derivative) of a material field $\mathcal{F}$ is its derivative with respect to time holding $\boldsymbol{X}$ fixed:

$$\dot{\mathcal{F}}(\boldsymbol{X}, t) = \frac{\mathrm{D}\mathcal{F}(\boldsymbol{X}, t)}{\mathrm{D}t} = \left(\frac{\partial \mathcal{F}(\boldsymbol{X}, t)}{\partial t}\right)_{\boldsymbol{X}}. \tag{3.19}$$

It represents the rate at which $\mathcal{F}$ changes in time as seen by an observer who follows the particle along its path line. The subscript $\boldsymbol{X}$ is usually dropped.

Definition 3.8 (Material gradient)**.** The *material gradient* of $\mathcal{F}$ is its derivative with respect to $\boldsymbol{X}$ holding t fixed,

$$\operatorname{Grad} \mathcal{F}(\boldsymbol{X}, t) = \frac{\partial \mathcal{F}(\boldsymbol{X}, t)}{\partial \boldsymbol{X}}, \tag{3.20}$$

and similarly one defines the material divergence $\operatorname{Div} \mathcal{F}$ and the material curl $\operatorname{Curl} \mathcal{F}$, with the definitions of Section 1.17 applied with respect to the coordinates X_A. They are differential operators with respect to the reference configuration and are denoted with capital initials.

Remark 3.1. Slide 72 also states that $D\mathcal{F}/Dt$ "equals the directional derivative of $\mathcal{F}$ in the direction of $\boldsymbol{V}$", $\operatorname{Grad} \mathcal{F} \cdot \boldsymbol{V}$. This should be read with care: for a material field, the material time derivative is simply the partial derivative at fixed $\boldsymbol{X}$, and $\operatorname{Grad} \mathcal{F} \cdot \boldsymbol{V}$ is a different quantity (it compares values at different particles). The interpretation of the material derivative as a directional derivative along the particle path is correct for *spatial* fields, where it gives the convective term of Eq. (3.21).

3.7.2 Spatial fields

Definition 3.9 (Spatial time derivative and spatial gradient)**.** The *spatial* (or local) time derivative of a spatial field f is $\partial f(\boldsymbol{x}, t)/\partial t$, holding $\boldsymbol{x}$ fixed: the rate of change of f seen by an observer standing at the place $\boldsymbol{x}$. The *spatial gradient* is $\operatorname{grad} f(\boldsymbol{x}, t) = \partial f(\boldsymbol{x}, t)/\partial \boldsymbol{x}$, holding t fixed; similarly $\operatorname{div} f$ and $\operatorname{curl} f$ (small initials).

3.7.3 Material time derivative of a spatial field

The material time derivative can also be applied to a spatial field: it is the derivative of f with respect to time holding $\boldsymbol{X}$ fixed, i.e., following the particle that is at $\boldsymbol{x}$ at time t:

$$\dot{f}(\boldsymbol{x},t) = \frac{\mathrm{D}f(\boldsymbol{x},t)}{\mathrm{D}t} = \left(\frac{\partial f[\boldsymbol{\chi}(\boldsymbol{X},t),t]}{\partial t}\right)_{\boldsymbol{X}=\boldsymbol{\chi}^{-1}(\boldsymbol{x},t)}.$$

Time appears twice, explicitly and through the position of the particle. By the chain rule,

$$\dot{f}(\boldsymbol{x},t) = \left(\frac{\partial f(\boldsymbol{x},t)}{\partial t}\right)_{\boldsymbol{x}} + \left(\frac{\partial f(\boldsymbol{x},t)}{\partial \boldsymbol{x}}\right)_{t} \cdot \left(\frac{\partial \boldsymbol{\chi}(\boldsymbol{X},t)}{\partial t}\right)_{\boldsymbol{X}=\boldsymbol{\chi}^{-1}(\boldsymbol{x},t)},$$

and, recognising the spatial gradient and the spatial velocity,

Material time derivative of a spatial field

$$\dot{f} = \frac{\mathrm{D}f}{\mathrm{D}t} = \frac{\partial f(\boldsymbol{x},t)}{\partial t} + \operatorname{grad} f(\boldsymbol{x},t)\cdot\boldsymbol{v}(\boldsymbol{x},t), \qquad \dot{f} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x_b}v_b. \tag{3.21}$$

The material time derivative of a spatial field is the sum of the *local* (spatial) time derivative and of the *convective* rate of change, which accounts for the fact that the particle changes its position.

For a spatial vector field $\boldsymbol{u}(\boldsymbol{x},t)$ the same argument gives $\dot{\boldsymbol{u}} = \partial\boldsymbol{u}/\partial t + (\operatorname{grad}\boldsymbol{u})\boldsymbol{v}$. Applied to the velocity field itself, it gives the spatial acceleration:

Spatial acceleration

$$\boldsymbol{a}(\boldsymbol{x},t) = \dot{\boldsymbol{v}}(\boldsymbol{x},t) = \frac{\partial\boldsymbol{v}(\boldsymbol{x},t)}{\partial t} + \operatorname{grad}\boldsymbol{v}(\boldsymbol{x},t)\,\boldsymbol{v}(\boldsymbol{x},t), \qquad a_a = \frac{\partial v_a}{\partial t} + \frac{\partial v_a}{\partial x_b}v_b. \tag{3.22}$$

The acceleration is the sum of the *local acceleration* $\partial\boldsymbol{v}/\partial t$ and of the *convective acceleration* $(\operatorname{grad}\boldsymbol{v})\boldsymbol{v}$. This relation allows one to compute the acceleration field from the velocity field alone, without knowing the motion: this is what is done in fluid mechanics, where the motion is not tracked. In compact notation,

$$\boldsymbol{v} = \dot{\boldsymbol{x}} = \frac{\partial\boldsymbol{x}}{\partial t}, \quad v_a = \dot{x}_a; \qquad \boldsymbol{a} = \dot{\boldsymbol{v}} = \ddot{\boldsymbol{x}}, \quad a_a = \dot{v}_a = \ddot{x}_a; \qquad \frac{\mathrm{D}(\bullet)}{\mathrm{D}t} = \frac{\partial(\bullet)}{\partial t} + \operatorname{grad}(\bullet)\,\boldsymbol{v}. \tag{3.23}$$

Example 3.7 (Uniaxial motion, continued)**.** For the motion of Example 3.5, $\partial v_1/\partial t = x_1(6\alpha t - 3\alpha^2 t^4)/(1+\alpha t^3)^2$ and $(\partial v_1/\partial x_1)v_1 = 9\alpha^2 t^4 x_1/(1+\alpha t^3)^2$; their sum is $6\alpha t\, x_1/(1+\alpha t^3) = a_1$, as found before.

Definition 3.10 (Special motions)**.** A motion is called

- *steady* if the spatial velocity field does not depend on time: $\partial\boldsymbol{v}/\partial t = \boldsymbol{0}$, $\boldsymbol{v} = \boldsymbol{v}(\boldsymbol{x})$, hence $\boldsymbol{a} = (\operatorname{grad}\boldsymbol{v})\boldsymbol{v}$;
- *uniform* if the velocity field does not depend on position: $\operatorname{grad}\boldsymbol{v} = \boldsymbol{O}$, $\boldsymbol{v} = \boldsymbol{v}(t)$, hence $\boldsymbol{a} = \partial\boldsymbol{v}/\partial t$;
- *potential* if there exists a scalar field $\Phi(\boldsymbol{x},t)$, the potential, such that $\boldsymbol{v} = \operatorname{grad}\Phi$.

A steady motion can have a non-zero acceleration (think of a fluid accelerating in a converging pipe with constant flow), and a uniform motion has the same acceleration at all points.

Example 3.8 (Slide 77)**.** Consider the spatial scalar field $\Phi(\boldsymbol{x}, t) = -t^{-1}/\left|\boldsymbol{x}\right|$, $\boldsymbol{x} \neq \boldsymbol{0}$, $t > 0$, and an observer moving with velocity

$$\boldsymbol{v} = \frac{x_1 x_3}{t\, e^t}\boldsymbol{e}_1 + \frac{x_2 x_3}{t\, e^t}\boldsymbol{e}_2 + \frac{x_3^2}{t\, e^t}\boldsymbol{e}_3.$$

We have $\partial\Phi/\partial t = t^{-2}/\left|\boldsymbol{x}\right|$ and $\operatorname{grad}\Phi = t^{-1}\boldsymbol{x}/\left|\boldsymbol{x}\right|^3$. Since $\boldsymbol{x}\cdot\boldsymbol{v} = (x_1^2+x_2^2+x_3^2)x_3/(t\, e^t) = \left|\boldsymbol{x}\right|^2 x_3/(t\, e^t)$,

$$\dot{\Phi} = \frac{t^{-2}}{\left|\boldsymbol{x}\right|} + t^{-1}\frac{\boldsymbol{x}\cdot\boldsymbol{v}}{\left|\boldsymbol{x}\right|^3} = \frac{t^{-2}}{\left|\boldsymbol{x}\right|} + \frac{t^{-2}x_3}{\left|\boldsymbol{x}\right| e^t} = \frac{t^{-2}}{\left|\boldsymbol{x}\right|}\left(1 + \frac{x_3}{e^t}\right).$$

3.8 Deformation gradient

In class

The second part of the lecture entered the core of the kinematics: the *local* description of the deformation. The lecturer introduced it by recalling what we need to solve a boundary value problem in solid mechanics: equilibrium (balance) equations, compatibility, constitutive equations and boundary conditions. Kinematics provides the compatibility part and the strain measures that enter the constitutive equations.

3.8.1 Principle of local action

Continuum mechanics assumes the **principle of local action**: the state of a material point (for instance its stress) depends only on the motion of an arbitrarily small neighbourhood of that point, not on distant points. Hence we need to describe what happens to the neighbourhood of a point, i.e., to infinitesimal material line, area and volume elements. Non-local theories, in which distant points interact, are needed only for materials with a microstructure; they will be mentioned in the last lecture.

3.8.2 Deformation of material curves

Consider a smooth material curve in the reference configuration, parametrised by $\xi \in \mathbb{R}$,

$$\boldsymbol{X} = \boldsymbol{\Gamma}(\xi),$$

and its image in the current configuration,

$$\boldsymbol{x} = \boldsymbol{\gamma}(\xi, t) = \boldsymbol{\chi}(\boldsymbol{\Gamma}(\xi), t).$$

The tangent vectors to the two curves are $\mathrm{d}\boldsymbol{X} = \boldsymbol{\Gamma}'(\xi)\,\mathrm{d}\xi$ and $\mathrm{d}\boldsymbol{x} = \boldsymbol{\gamma}'(\xi, t)\,\mathrm{d}\xi$, where a prime denotes the derivative with respect to ξ. By the chain rule,

$$\boldsymbol{\gamma}'(\xi, t) = \frac{\partial\boldsymbol{\chi}(\boldsymbol{X}, t)}{\partial\boldsymbol{X}}\,\boldsymbol{\Gamma}'(\xi) = \boldsymbol{F}(\boldsymbol{X}, t)\,\boldsymbol{\Gamma}'(\xi),$$

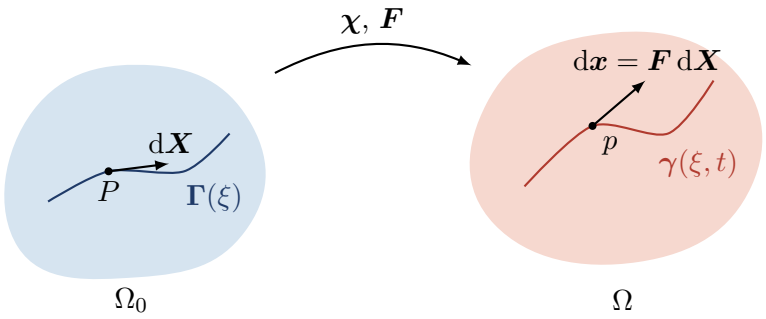


Figure 3.2: A material curve $\boldsymbol{\Gamma}$ and its image $\boldsymbol{\gamma}$: the deformation gradient maps the tangent $\mathrm{d}\boldsymbol{X}$ at P into the tangent $\mathrm{d}\boldsymbol{x}$ at $p = \boldsymbol{\chi}(P)$.

so that the tangent vectors are related by a linear map.

Definition 3.11 (Deformation gradient)**.** The **deformation gradient** is the second-order tensor

$$\boldsymbol{F}(\boldsymbol{X}, t) = \frac{\partial \boldsymbol{\chi}(\boldsymbol{X}, t)}{\partial \boldsymbol{X}} = \operatorname{Grad} \boldsymbol{x}(\boldsymbol{X}, t), \qquad F_{aA} = \frac{\partial \chi_a}{\partial X_A} = \frac{\partial x_a}{\partial X_A}, \tag{3.24}$$

which transforms a material line element $\mathrm{d}\boldsymbol{X}$ into the spatial line element $\mathrm{d}\boldsymbol{x}$:

$$\boxed{\mathrm{d}\boldsymbol{x} = \boldsymbol{F}(\boldsymbol{X}, t)\ \mathrm{d}\boldsymbol{X}}, \qquad \mathrm{d}x_a = F_{aA}\,\mathrm{d}X_A. \tag{3.25}$$

$\boldsymbol{F}$ is the primary measure of deformation. It is a *two-point tensor*: one leg in the current configuration (index a) and one in the reference configuration (index A),

$$\boldsymbol{F} = F_{aA}\, \boldsymbol{e}_a \otimes \boldsymbol{E}_A. \tag{3.26}$$

It maps vectors tangent to material curves in Ω_0 into vectors tangent to the deformed curves in Ω. In general $\boldsymbol{F}$ is *not* symmetric (the two indices live in different spaces) and its value depends on the point $\boldsymbol{X}$ and on time.

3.8.3 An alternative derivation

The lecturer also presented the derivation she uses in her undergraduate course. Take a point P (position $\boldsymbol{X}$) and a point Q in a small neighbourhood (position $\boldsymbol{X} + \mathrm{d}\boldsymbol{X}$), where "small" means that the distance $d = |\,\mathrm{d}\boldsymbol{X}|$ is one to three orders of magnitude smaller than the characteristic size D of the body. After the deformation, P goes to $\boldsymbol{x} = \boldsymbol{\chi}(\boldsymbol{X})$ and Q to $\boldsymbol{y} = \boldsymbol{\chi}(\boldsymbol{X} + \mathrm{d}\boldsymbol{X})$. A Taylor expansion gives

$$\boldsymbol{y} = \boldsymbol{\chi}(\boldsymbol{X}) + \frac{\partial \boldsymbol{\chi}}{\partial \boldsymbol{X}}\,\mathrm{d}\boldsymbol{X} + o(|\,\mathrm{d}\boldsymbol{X}|) = \boldsymbol{x} + \boldsymbol{F}\,\mathrm{d}\boldsymbol{X} + o(|\,\mathrm{d}\boldsymbol{X}|),$$

so that, to first order, $\mathrm{d}\boldsymbol{x} = \boldsymbol{y} - \boldsymbol{x} = \boldsymbol{F}\,\mathrm{d}\boldsymbol{X}$.

3.8.4 Inverse deformation gradient and Jacobian

Using the inverse motion $\boldsymbol{X} = \boldsymbol{\chi}^{-1}(\boldsymbol{x}, t)$ the same argument gives

$$\mathrm{d}\boldsymbol{X} = \boldsymbol{F}^{-1}(\boldsymbol{x}, t)\ \mathrm{d}\boldsymbol{x}, \qquad \boldsymbol{F}^{-1} = \operatorname{grad} \boldsymbol{X} = \frac{\partial \boldsymbol{\chi}^{-1}(\boldsymbol{x}, t)}{\partial \boldsymbol{x}}, \qquad F^{-1}_{Aa} = \frac{\partial X_A}{\partial x_a}. \tag{3.27}$$

Pay attention to the order of the indices: the first index of $\boldsymbol{F}^{-1}$ is the material one. Indeed $F_{aA}F^{-1}_{Ab} = \partial x_a/\partial x_b = \delta_{ab}$ and $F^{-1}_{Aa}F_{aB} = \partial X_A/\partial X_B = \delta_{AB}$.

The invertibility of $\boldsymbol{F}$ requires $\det \boldsymbol{F} \neq 0$. The matrix $[F_{aA}]$ is the *Jacobian matrix* of the transformation $\boldsymbol{\chi}$ and its determinant

$$J(\boldsymbol{X}, t) = \det \boldsymbol{F}(\boldsymbol{X}, t) \tag{3.28}$$

is the **Jacobian determinant**, or simply the *Jacobian*. As shown in Section 3.10, J is the ratio between current and reference volumes, $\mathrm{d}v = J\,\mathrm{d}V$; since volumes must remain positive (the material cannot interpenetrate, nor can a volume be annihilated), we require

$$J = \det \boldsymbol{F} > 0. \tag{3.29}$$

Definition 3.12 (Homogeneous and isochoric deformations)**.** A deformation is *homogeneous* if $\boldsymbol{F}$ does not depend on $\boldsymbol{X}$ (it may depend on time): every part of the body deforms in the same way, and the motion has the affine form $\boldsymbol{x} = \boldsymbol{F}(t)\boldsymbol{X} + \boldsymbol{c}(t)$. A deformation is *isochoric* (volume preserving) if $J = 1$.

Example 3.9 (A homogeneous deformation in the plane, slide 83)**.** Consider

$$x_1 = 4 - 2X_1 - X_2, \qquad x_2 = 2 + \tfrac{3}{2}X_1 - \tfrac{1}{2}X_2.$$

Then

$$[\boldsymbol{F}] = \begin{bmatrix} -2 & -1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -4 & -2 \\ 3 & -1 \end{bmatrix}, \qquad J = 1 + \tfrac{3}{2} = \tfrac{5}{2}, \qquad \left[\boldsymbol{F}^{-1}\right] = \frac{1}{5}\begin{bmatrix} -1 & 2 \\ -3 & -4 \end{bmatrix}.$$

The deformation is homogeneous, with area increase $5/2$. The material vector $\boldsymbol{A} = \frac{1}{\sqrt{2}}(\boldsymbol{E}_1 + \boldsymbol{E}_2)$ becomes $\boldsymbol{a} = \boldsymbol{F}\boldsymbol{A} = \frac{1}{\sqrt{2}}(-3\boldsymbol{e}_1 + \boldsymbol{e}_2)$, of length $\sqrt{5}$; the spatial vector $\boldsymbol{b} = \boldsymbol{e}_1$ is the image of $\boldsymbol{B} = \boldsymbol{F}^{-1}\boldsymbol{b} = -\frac{1}{5}(\boldsymbol{E}_1 + 3\boldsymbol{E}_2)$, of length $\sqrt{2/5}$.

3.8.5 Displacement gradient

From $\boldsymbol{x} = \boldsymbol{X} + \boldsymbol{U}(\boldsymbol{X}, t)$ (Section 3.4), the material displacement gradient is

$$\operatorname{Grad} \boldsymbol{U} = \boldsymbol{F} - \boldsymbol{I}, \qquad \frac{\partial U_a}{\partial X_A} = F_{aA} - \delta_{aA}, \tag{3.30}$$

where we used $\partial X_B/\partial X_A = \delta_{AB}$ (always use different indices when differentiating a coordinate with respect to another!). Similarly, from $\boldsymbol{X} = \boldsymbol{x} - \boldsymbol{u}(\boldsymbol{x}, t)$, the spatial displacement gradient is

$$\operatorname{grad} \boldsymbol{u} = \boldsymbol{I} - \boldsymbol{F}^{-1}. \tag{3.31}$$

If there is no motion, or the motion is a rigid translation, $\boldsymbol{F} = \boldsymbol{I}$ and both gradients vanish: the deformation gradient cannot distinguish a body at rest from a translated one, which is correct, because neither is deformed.

3.9 Material strain tensors

3.9.1 Stretch vector and right Cauchy–Green tensor

Let $\mathrm{d}\boldsymbol{X} = \boldsymbol{a}_0\,\mathrm{d}\varepsilon$, where $\boldsymbol{a}_0$ is a unit material vector (the direction of a fibre) and $\mathrm{d}\varepsilon$ its infinitesimal length. The deformed fibre is

$$\mathrm{d}\boldsymbol{x} = \boldsymbol{F}\boldsymbol{a}_0\,\mathrm{d}\varepsilon \coloneqq \boldsymbol{\lambda}_{\boldsymbol{a}_0}\,\mathrm{d}\varepsilon, \qquad \boldsymbol{\lambda}_{\boldsymbol{a}_0} = \boldsymbol{F}\boldsymbol{a}_0.$$

$\boldsymbol{\lambda}_{\boldsymbol{a}_0}$ is the *stretch vector* in the direction $\boldsymbol{a}_0$, and its length $\lambda = |\boldsymbol{\lambda}_{\boldsymbol{a}_0}| = |\mathrm{d}\boldsymbol{x}|\,/\,|\mathrm{d}\boldsymbol{X}|$ is the **stretch ratio**, or simply *stretch*. Its square is

$$\lambda^2 = \boldsymbol{F}\boldsymbol{a}_0 \cdot \boldsymbol{F}\boldsymbol{a}_0 = \boldsymbol{a}_0 \cdot \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}\boldsymbol{a}_0 = \boldsymbol{a}_0 \cdot \boldsymbol{C}\boldsymbol{a}_0, \tag{3.32}$$

where we introduced the **right Cauchy–Green tensor**

Right Cauchy–Green tensor

$$\boldsymbol{C} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}, \qquad C_{AB} = F_{aA}F_{aB}, \qquad \det\boldsymbol{C} = J^2 > 0. \tag{3.33}$$

In index form $\lambda^2 = a_{0A}C_{AB}a_{0B}$: the index a of $\boldsymbol{F}$ is contracted, so $\boldsymbol{C}$ has two material indices, it is a *material* tensor. It is the reason why the product is $\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}$ and not $\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}$: one has to multiply "a column of $\boldsymbol{F}$ by a column of $\boldsymbol{F}$".

The fibre is called

$$\text{extended if } \lambda > 1, \qquad \text{unstretched if } \lambda = 1, \qquad \text{compressed if } \lambda < 1.$$

Properties of $\boldsymbol{C}$. $\boldsymbol{C}$ is symmetric, $\boldsymbol{C}^{\mathrm{T}} = (\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F})^{\mathrm{T}} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} = \boldsymbol{C}$, and positive definite, since for every $\boldsymbol{u} \neq \boldsymbol{0}$ $\boldsymbol{u}\cdot\boldsymbol{C}\boldsymbol{u} = |\boldsymbol{F}\boldsymbol{u}|^2 > 0$ ($\boldsymbol{F}$ is invertible). Its inverse,

$$\boldsymbol{B} \coloneqq \boldsymbol{C}^{-1} = \boldsymbol{F}^{-1}\boldsymbol{F}^{-\mathrm{T}}, \tag{3.34}$$

is called the **Piola deformation tensor**. For a rigid motion $\boldsymbol{F} = \boldsymbol{Q}$ is orthogonal and $\boldsymbol{C} = \boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q} = \boldsymbol{I}$: $\lambda = 1$ in all directions.

Complement Stretch and engineering strain

In a uniaxial test with initial length l_0 and final length l, the engineering (small) strain is $\varepsilon = (l - l_0)/l_0$ while the stretch is $\lambda = l/l_0$. Hence $\lambda = 1 + \varepsilon$: no deformation corresponds to $\lambda = 1$, not to $\lambda = 0$.

3.9.2 Deformation and strain

In class

In English two different words are used: *deformation* and *strain*. Italian has only one word (*deformazione*), and the distinction is easily lost. Deformation measures, like $\boldsymbol{F}$ and $\boldsymbol{C}$, are equal to the identity when there is no motion; strain measures, like $\boldsymbol{E}$ below, are equal to zero when there is no motion (or a rigid motion). Strain is the quantity that enters constitutive equations, since a body that is not strained should not be stressed.

The change of the squared length of a fibre is

$$(\lambda\,\mathrm{d}\varepsilon)^2 - \mathrm{d}\varepsilon^2 = \mathrm{d}\boldsymbol{x}\cdot\mathrm{d}\boldsymbol{x} - \mathrm{d}\boldsymbol{X}\cdot\mathrm{d}\boldsymbol{X} = \mathrm{d}\boldsymbol{X}\cdot(\boldsymbol{C}-\boldsymbol{I})\,\mathrm{d}\boldsymbol{X} = 2\,\mathrm{d}\boldsymbol{X}\cdot\boldsymbol{E}\,\mathrm{d}\boldsymbol{X},$$

which defines the **Green–Lagrange strain tensor**

Green–Lagrange strain tensor

$$\boldsymbol{E} = \tfrac{1}{2}(\boldsymbol{C}-\boldsymbol{I}) = \tfrac{1}{2}(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}-\boldsymbol{I}), \qquad E_{AB} = \tfrac{1}{2}(F_{aA}F_{aB}-\delta_{AB}). \tag{3.35}$$

$\boldsymbol{E}$ is symmetric and material, and vanishes for a rigid motion. With $\boldsymbol{F} = \boldsymbol{I} + \operatorname{Grad}\boldsymbol{U}$, it can be written in terms of the displacement as

$$\boldsymbol{E} = \tfrac{1}{2}(\operatorname{Grad}\boldsymbol{U} + \operatorname{Grad}^{\mathrm{T}}\boldsymbol{U} + \operatorname{Grad}^{\mathrm{T}}\boldsymbol{U}\ \operatorname{Grad}\boldsymbol{U}), \tag{3.36}$$

whose linear part is the small strain tensor of Appendix A. In computations (for instance in finite elements) the lecturer prefers to work directly with $\boldsymbol{F}$, which is computed from the nodal positions, rather than with the displacements.

3.9.3 Principal stretches

Being symmetric and positive definite, $\boldsymbol{C}$ has three real positive eigenvalues, which are the squares of the *principal stretches* λ_α, and three orthonormal eigenvectors $\hat{\boldsymbol{N}}_\alpha$, the principal referential directions:

$$\boldsymbol{C}\hat{\boldsymbol{N}}_\alpha = \lambda_\alpha^2\hat{\boldsymbol{N}}_\alpha, \qquad \boldsymbol{C} = \sum_{\alpha=1}^{3}\lambda_\alpha^2\,\hat{\boldsymbol{N}}_\alpha\otimes\hat{\boldsymbol{N}}_\alpha. \tag{3.37}$$

In the principal frame $[\boldsymbol{C}] = \operatorname{diag}(\lambda_1^2,\lambda_2^2,\lambda_3^2)$: along the principal directions the fibres are only stretched, and right angles between them are preserved. In a general frame the diagonal components of $\boldsymbol{C}$ are related to the change of length of the coordinate fibres ($C_{11} = \lambda^2$ for $\boldsymbol{a}_0 = \boldsymbol{E}_1$), and the off-diagonal components to the change of angle between them (see Eq. (3.39) below), exactly as the metric coefficients of Section 2.4.

> **In class**
>
> In class the λ_α were called "eigenvalues of $\boldsymbol{F}$". This is not correct in general: $\boldsymbol{F}$ is not symmetric and its eigenvalues need not be real. The λ_α are the eigenvalues of the right stretch tensor $\boldsymbol{U} = \sqrt{\boldsymbol{C}}$, which will be introduced with the polar decomposition $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U}$.

3.9.4 Changes of length and angle

From Eq. (3.32), the stretch in the direction $\boldsymbol{a}_0$ is $\lambda^2 = C_{IJ}a_{0I}a_{0J}$. The length of a deformed material curve $\boldsymbol{X} = \boldsymbol{\Gamma}(\xi)$, $\xi\in[\xi_1,\xi_2]$, is therefore

$$l = \int_{\xi_1}^{\xi_2}\left|\boldsymbol{F}\boldsymbol{\Gamma}'(\xi)\right|\,\mathrm{d}\xi = \int_{\xi_1}^{\xi_2}\sqrt{C_{IJ}\,\Gamma'_I\Gamma'_J}\,\mathrm{d}\xi\,: \tag{3.38}$$

$\boldsymbol{C}$ plays the role of the metric of the deformed body expressed in material coordinates.

Consider two material fibres $\mathrm{d}\boldsymbol{X}$ and $\mathrm{d}\boldsymbol{Y}$ forming the angle Θ, with $\cos\Theta = \mathrm{d}\boldsymbol{X}\cdot$

$\mathrm{d}\boldsymbol{Y}/(|\,\mathrm{d}\boldsymbol{X}|\,|\,\mathrm{d}\boldsymbol{Y}|)$. After the deformation they form the angle θ, with

$$\cos\theta = \frac{\boldsymbol{F}\,\mathrm{d}\boldsymbol{X}\cdot\boldsymbol{F}\,\mathrm{d}\boldsymbol{Y}}{|\boldsymbol{F}\,\mathrm{d}\boldsymbol{X}|\,|\boldsymbol{F}\,\mathrm{d}\boldsymbol{Y}|} = \frac{\mathrm{d}\boldsymbol{X}\cdot\boldsymbol{C}\,\mathrm{d}\boldsymbol{Y}}{\sqrt{\mathrm{d}\boldsymbol{X}\cdot\boldsymbol{C}\,\mathrm{d}\boldsymbol{X}}\sqrt{\mathrm{d}\boldsymbol{Y}\cdot\boldsymbol{C}\,\mathrm{d}\boldsymbol{Y}}}. \tag{3.39}$$

For $\mathrm{d}\boldsymbol{X} \parallel \boldsymbol{E}_1$ and $\mathrm{d}\boldsymbol{Y} \parallel \boldsymbol{E}_2$, $\cos\theta = C_{12}/\sqrt{C_{11}C_{22}}$: compare with Eq. (2.13).

Example 3.10 (Fibre-reinforced rubber box, slide 94)**.** A rubber box is reinforced by two families of inextensible fibres lying in the plane X_1X_2, at angles $\pm A$ with the X_1 axis. The box is deformed as

$$x_1 = \mu X_1, \qquad x_2 = \lambda X_2, \qquad x_3 = X_3,$$

so that $[\boldsymbol{F}] = \mathrm{diag}(\mu, \lambda, 1)$, $J = \lambda\mu$ and $[\boldsymbol{C}] = \mathrm{diag}(\mu^2, \lambda^2, 1)$. The fibre direction is $\boldsymbol{a}_0 = \cos A\,\boldsymbol{E}_1 \pm \sin A\,\boldsymbol{E}_2$; inextensibility requires

$$\boldsymbol{a}_0\cdot\boldsymbol{C}\boldsymbol{a}_0 = \mu^2\cos^2 A + \lambda^2\sin^2 A = 1 \quad\Longrightarrow\quad \mu^2 = \frac{1-\lambda^2\sin^2 A}{\cos^2 A}.$$

The deformed fibres form angles $\pm\alpha$ with the x_1 axis, and the angle 2α between them follows from Eq. (3.39) (the denominators are 1 because the fibres are inextensible):

$$\cos 2\alpha = \mu^2\cos^2 A - \lambda^2\sin^2 A = 1 - 2\lambda^2\sin^2 A.$$

Stretching the box in the X_2 direction ($\lambda > 1$) forces it to contract in the X_1 direction and opens the angle between the fibres; λ cannot exceed $1/\sin A$.

3.10 Volume and area changes

3.10.1 Volume change

Consider an infinitesimal parallelepiped in Ω_0 with edges $\mathrm{d}\boldsymbol{X}^{(1)}$, $\mathrm{d}\boldsymbol{X}^{(2)}$, $\mathrm{d}\boldsymbol{X}^{(3)}$ (a right-handed triad). Its volume is the box product

$$\mathrm{d}V = (\,\mathrm{d}\boldsymbol{X}^{(1)} \times \mathrm{d}\boldsymbol{X}^{(2)})\cdot\mathrm{d}\boldsymbol{X}^{(3)} = \varepsilon_{IJK}\,\mathrm{d}X_I^{(1)}\,\mathrm{d}X_J^{(2)}\,\mathrm{d}X_K^{(3)}.$$

The deformed edges are $\mathrm{d}\boldsymbol{x}^{(i)} = \boldsymbol{F}\,\mathrm{d}\boldsymbol{X}^{(i)}$, and the deformed volume is

$$\mathrm{d}v = \varepsilon_{ijk}\,F_{iI}\,\mathrm{d}X_I^{(1)}\,F_{jJ}\,\mathrm{d}X_J^{(2)}\,F_{kK}\,\mathrm{d}X_K^{(3)}.$$

Using the index definition of the determinant (Chapter 1), $\varepsilon_{ijk}F_{iI}F_{jJ}F_{kK} = \det\boldsymbol{F}\,\varepsilon_{IJK}$, hence

Volume change

$$\mathrm{d}v = J\,\mathrm{d}V, \qquad J = \det\boldsymbol{F} > 0. \tag{3.40}$$

A deformation is isochoric if and only if $J = 1$ everywhere. Note also $J^{-1} = \det\boldsymbol{F}^{-1}$.

3.10.2 Area change: Nanson's formula

Consider an infinitesimal material surface element of area $\mathrm{d}S$ and unit normal $\boldsymbol{N}$, and set $\mathrm{d}\boldsymbol{S} = \boldsymbol{N}\,\mathrm{d}S$; in the current configuration it becomes $\mathrm{d}\boldsymbol{s} = \boldsymbol{n}\,\mathrm{d}s$. Build a cylinder with base $\mathrm{d}\boldsymbol{S}$ and generator $\mathrm{d}\boldsymbol{X}$ (any material vector not lying in the surface). The reference and current volumes of the cylinder are

$$\mathrm{d}V = \mathrm{d}\boldsymbol{S}\cdot\mathrm{d}\boldsymbol{X}, \qquad \mathrm{d}v = \mathrm{d}\boldsymbol{s}\cdot\mathrm{d}\boldsymbol{x}.$$

Using Eq. (3.40) and $\mathrm{d}\boldsymbol{x} = \boldsymbol{F}\,\mathrm{d}\boldsymbol{X}$,

$$\mathrm{d}\boldsymbol{s}\cdot\boldsymbol{F}\,\mathrm{d}\boldsymbol{X} = J\,\mathrm{d}\boldsymbol{S}\cdot\mathrm{d}\boldsymbol{X} \quad\Longrightarrow\quad (\boldsymbol{F}^{\mathrm{T}}\,\mathrm{d}\boldsymbol{s} - J\,\mathrm{d}\boldsymbol{S})\cdot\mathrm{d}\boldsymbol{X} = 0$$

for every $\mathrm{d}\boldsymbol{X}$, which gives

Nanson's formula

$$\mathrm{d}\boldsymbol{s} = J\boldsymbol{F}^{-\mathrm{T}}\,\mathrm{d}\boldsymbol{S}, \qquad \boldsymbol{n}\,\mathrm{d}s = J\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{N}\,\mathrm{d}S, \qquad n_a\,\mathrm{d}s = JF^{-1}_{Aa}N_A\,\mathrm{d}S. \tag{3.41}$$

The index form is the *Piola transformation* of the oriented area element. Taking the norm, $\mathrm{d}s = J\left|\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{N}\right|\mathrm{d}S = J\sqrt{\boldsymbol{N}\cdot\boldsymbol{C}^{-1}\boldsymbol{N}}\;\mathrm{d}S$.

In class

Normals do *not* transform with $\boldsymbol{F}$: in general $\boldsymbol{n} \nparallel \boldsymbol{F}\boldsymbol{N}$. The lecturer mentioned having found this mistake in a review paper. The reason is visible in Nanson's formula: tangent vectors transform with $\boldsymbol{F}$, normals with $\boldsymbol{F}^{-\mathrm{T}}$, like the covariant and contravariant bases of Chapter 2.

Complement Volume change from Nanson's formula

Conversely, assuming Eq. (3.41), the volume change follows immediately: $\mathrm{d}v = \mathrm{d}\boldsymbol{s}\cdot\mathrm{d}\boldsymbol{x} = J\boldsymbol{F}^{-\mathrm{T}}\,\mathrm{d}\boldsymbol{S}\cdot\boldsymbol{F}\,\mathrm{d}\boldsymbol{X} = J\,\mathrm{d}\boldsymbol{S}\cdot\boldsymbol{F}^{-1}\boldsymbol{F}\,\mathrm{d}\boldsymbol{X} = J\,\mathrm{d}V$. This is the alternative proof of slide 91. An independent proof of Nanson's formula uses the identity $\boldsymbol{F}^{\mathrm{T}}\operatorname{cof}\boldsymbol{F} = J\boldsymbol{I}$ for the cofactor $\operatorname{cof}\boldsymbol{F} = J\boldsymbol{F}^{-\mathrm{T}}$ and the vector product $\mathrm{d}\boldsymbol{S} = \mathrm{d}\boldsymbol{X}^{(1)}\times\mathrm{d}\boldsymbol{X}^{(2)}$: $\boldsymbol{F}\boldsymbol{a}\times\boldsymbol{F}\boldsymbol{b} = \operatorname{cof}\boldsymbol{F}(\boldsymbol{a}\times\boldsymbol{b})$.

3.11 Spatial strain tensors

In class

The spatial counterpart of the previous section was left as an exercise; it is summarised here following Holzapfel [3, Sect. 2.5].

Let $\boldsymbol{a}$ be a unit spatial vector and $\mathrm{d}\boldsymbol{x} = \boldsymbol{a}\,\mathrm{d}\varepsilon$. The corresponding material fibre is $\mathrm{d}\boldsymbol{X} = \boldsymbol{F}^{-1}\boldsymbol{a}\,\mathrm{d}\varepsilon$, and the *inverse stretch vector* is $\boldsymbol{\lambda}_{\boldsymbol{a}}^{-1} = \boldsymbol{F}^{-1}\boldsymbol{a}$, with length $\lambda^{-1} = |\,\mathrm{d}\boldsymbol{X}|\,/\,|\,\mathrm{d}\boldsymbol{x}|$:

$$\lambda^{-2} = \boldsymbol{F}^{-1}\boldsymbol{a}\cdot\boldsymbol{F}^{-1}\boldsymbol{a} = \boldsymbol{a}\cdot\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{-1}\boldsymbol{a} = \boldsymbol{a}\cdot\boldsymbol{b}^{-1}\boldsymbol{a}, \tag{3.42}$$

where

Left Cauchy–Green tensor

$$\boldsymbol{b} = \boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}, \qquad b_{ab} = F_{aA}F_{bA}, \qquad \det\boldsymbol{b} = J^2 > 0. \tag{3.43}$$

$\boldsymbol{b}$ is the **left Cauchy–Green tensor** (or Finger tensor); it is symmetric, positive definite,

Table 3.1: Summary of deformation and strain measures introduced so far.

	Definition	Type	No motion
Deformation gradient	$\boldsymbol{F} = \operatorname{Grad} \boldsymbol{x}$	two-point	$\boldsymbol{I}$
Right Cauchy–Green	$\boldsymbol{C} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}$	material	$\boldsymbol{I}$
Piola	$\boldsymbol{B} = \boldsymbol{C}^{-1}$	material	$\boldsymbol{I}$
Green–Lagrange	$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I})$	material	$\boldsymbol{O}$
Left Cauchy–Green	$\boldsymbol{b} = \boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}$	spatial	$\boldsymbol{I}$
Euler–Almansi	$\boldsymbol{e} = \frac{1}{2}(\boldsymbol{I} - \boldsymbol{b}^{-1})$	spatial	$\boldsymbol{O}$

and *spatial* (both indices are spatial: the material index is contracted). The change of squared length, now measured on the current fibre, is

$$\mathrm{d}\boldsymbol{x} \cdot \mathrm{d}\boldsymbol{x} - \mathrm{d}\boldsymbol{X} \cdot \mathrm{d}\boldsymbol{X} = \mathrm{d}\boldsymbol{x} \cdot (\boldsymbol{I} - \boldsymbol{b}^{-1})\,\mathrm{d}\boldsymbol{x} = 2\,\mathrm{d}\boldsymbol{x} \cdot \boldsymbol{e}\,\mathrm{d}\boldsymbol{x},$$

which defines the **Euler–Almansi strain tensor**

Euler–Almansi strain tensor

$$\boldsymbol{e} = \tfrac{1}{2}(\boldsymbol{I} - \boldsymbol{b}^{-1}) = \tfrac{1}{2}(\boldsymbol{I} - \boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{-1}), \qquad e_{ab} = \tfrac{1}{2}(\delta_{ab} - F^{-1}_{Aa}F^{-1}_{Ab}). \tag{3.44}$$

Note the inverse: the formula $\frac{1}{2}(\boldsymbol{I} - \boldsymbol{b})$, which was written in class, is not correct, since it does not measure the change of squared length of the current fibre $\mathrm{d}\boldsymbol{x}$. The two strain tensors are related by

$$\boldsymbol{E} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{e}\,\boldsymbol{F}, \qquad \boldsymbol{e} = \boldsymbol{F}^{-\mathrm{T}}\boldsymbol{E}\boldsymbol{F}^{-1}, \tag{3.45}$$

as follows from $\mathrm{d}\boldsymbol{X} \cdot \boldsymbol{E}\,\mathrm{d}\boldsymbol{X} = \mathrm{d}\boldsymbol{x} \cdot \boldsymbol{e}\,\mathrm{d}\boldsymbol{x}$ with $\mathrm{d}\boldsymbol{x} = \boldsymbol{F}\,\mathrm{d}\boldsymbol{X}$. This is an example of *pull-back* ($\boldsymbol{e} \mapsto \boldsymbol{E}$) and *push-forward* ($\boldsymbol{E} \mapsto \boldsymbol{e}$) operations, the subject of the next lecture.

In class

The second day ended here (slide 97).

3.12 Push-forward and pull-back operations

In class

The third day opened by skipping the spatial derivation of the strain measures (identical in spirit to the material one, and summarised in Section 3.11) and by briefly mentioning push-forward and pull-back operations (slides 98–99). They are "simply operations done by pre- and post-multiplying by the deformation gradient or its inverse", to move a quantity from one configuration to the other.

Definition 3.13 (Push-forward and pull-back)**.** The **push-forward** $\boldsymbol{\chi}_*(\bullet)$ transforms a quantity (vector or tensor) based on the reference configuration into the corresponding quantity based on the current configuration. The **pull-back** $\boldsymbol{\chi}_*^{-1}(\bullet)$ is the inverse operation.

We already met an example at the end of Section 3.11. Since $\boldsymbol{e} = \frac{1}{2}(\boldsymbol{I} - \boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{-1})$,

$$\boldsymbol{e} = \boldsymbol{F}^{-\mathrm{T}}\Big[\boldsymbol{F}^{\mathrm{T}}\tfrac{1}{2}(\boldsymbol{I} - \boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{-1})\boldsymbol{F}\Big]\boldsymbol{F}^{-1} = \boldsymbol{F}^{-\mathrm{T}}\,\boldsymbol{E}\,\boldsymbol{F}^{-1} = \boldsymbol{\chi}_*(\boldsymbol{E}), \tag{3.46}$$

$$\boldsymbol{E} = \boldsymbol{F}^{\mathrm{T}}\Big[\boldsymbol{F}^{-\mathrm{T}}\tfrac{1}{2}(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} - \boldsymbol{I})\boldsymbol{F}^{-1}\Big]\boldsymbol{F} = \boldsymbol{F}^{\mathrm{T}}\,\boldsymbol{e}\,\boldsymbol{F} = \boldsymbol{\chi}_*^{-1}(\boldsymbol{e}). \tag{3.47}$$

A tensor is a combination of dyads, i.e., of *two* vectors, so $\boldsymbol{F}$ (or its inverse) must be applied twice, once per leg. The form of the operation depends on the nature of the tensor. Recalling Chapter 2: tangent vectors (line elements) transform with $\boldsymbol{F}$, like a covariant basis $\boldsymbol{g}_i$, while normals and gradients transform with $\boldsymbol{F}^{-\mathrm{T}}$, like the contravariant basis $\boldsymbol{g}^i$. The tensors $\boldsymbol{E}$, $\boldsymbol{e}$, $\boldsymbol{C}$ and $\boldsymbol{b}^{-1}$ are *covariant* (they are meant to be contracted twice with line elements, as in $\mathrm{d}\boldsymbol{X}\cdot\boldsymbol{E}\,\mathrm{d}\boldsymbol{X}$), while $\boldsymbol{C}^{-1}$, $\boldsymbol{b}$ and most of the stress tensors are *contravariant*.

Push-forward and pull-back

covariant tensors:	$\boldsymbol{\chi}_*(\bullet)^{\mathrm{cv}} = \boldsymbol{F}^{-\mathrm{T}}(\bullet)^{\mathrm{cv}}\boldsymbol{F}^{-1}$,	$\boldsymbol{\chi}_*^{-1}(\bullet)^{\mathrm{cv}} = \boldsymbol{F}^{\mathrm{T}}(\bullet)^{\mathrm{cv}}\boldsymbol{F}$,
contravariant tensors:	$\boldsymbol{\chi}_*(\bullet)^{\mathrm{ctv}} = \boldsymbol{F}(\bullet)^{\mathrm{ctv}}\boldsymbol{F}^{\mathrm{T}}$,	$\boldsymbol{\chi}_*^{-1}(\bullet)^{\mathrm{ctv}} = \boldsymbol{F}^{-1}(\bullet)^{\mathrm{ctv}}\boldsymbol{F}^{-\mathrm{T}}$,
covariant vectors:	$\boldsymbol{\chi}_*(\bullet)^{\mathrm{cv}} = \boldsymbol{F}^{-\mathrm{T}}(\bullet)^{\mathrm{cv}}$,	$\boldsymbol{\chi}_*^{-1}(\bullet)^{\mathrm{cv}} = \boldsymbol{F}^{\mathrm{T}}(\bullet)^{\mathrm{cv}}$,
contravariant vectors:	$\boldsymbol{\chi}_*(\bullet)^{\mathrm{ctv}} = \boldsymbol{F}(\bullet)^{\mathrm{ctv}}$,	$\boldsymbol{\chi}_*^{-1}(\bullet)^{\mathrm{ctv}} = \boldsymbol{F}^{-1}(\bullet)^{\mathrm{ctv}}$.

For instance, $\boldsymbol{b} = \boldsymbol{F}\boldsymbol{I}\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{\chi}_*(\boldsymbol{I})$ as a contravariant tensor, and $\boldsymbol{b}^{-1} = \boldsymbol{F}^{-\mathrm{T}}\boldsymbol{C}\boldsymbol{F}^{-1} = \boldsymbol{\chi}_*(\boldsymbol{C})$ as a covariant one. The double contraction of a covariant and a contravariant tensor is invariant under these operations, e.g., $\boldsymbol{\chi}_*^{-1}(\boldsymbol{A}) : \boldsymbol{\chi}_*^{-1}(\boldsymbol{B}) = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{F} : \boldsymbol{F}^{-1}\boldsymbol{B}\boldsymbol{F}^{-\mathrm{T}} = \boldsymbol{A} : \boldsymbol{B}$ for $\boldsymbol{A}$ covariant and $\boldsymbol{B}$ contravariant: this is the reason why stresses (contravariant) and strains (covariant) pair naturally in the expression of work.

Remark 3.2. Since in this course the same Cartesian basis is used for both configurations, the distinction between covariant and contravariant components disappears, and the only thing to remember is which formula to use. The distinction becomes essential when working with convected coordinates, shells and curved surfaces.

3.13 Polar decomposition

The deformation gradient is not symmetric, and we have seen that only part of the information it contains is related to the deformation: a rigid rotation $\boldsymbol{F} = \boldsymbol{Q}$ produces no strain. The polar decomposition separates the two parts.

Theorem 3.1 (Polar decomposition)**.** *For any tensor $\boldsymbol{F}$ with $\det\boldsymbol{F} > 0$ there exist a unique proper orthogonal tensor $\boldsymbol{R}$ and unique symmetric positive definite tensors $\boldsymbol{U}$ and $\boldsymbol{v}$ such that*

$$\boxed{\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U} = \boldsymbol{v}\boldsymbol{R}}. \tag{3.48}$$

Proof. $\boldsymbol{C} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}$ is symmetric and positive definite (Section 3.9); hence it has a unique symmetric positive definite square root $\boldsymbol{U} = \sqrt{\boldsymbol{C}}$ (computed with its spectral decomposition, Section 3.14). Set $\boldsymbol{R} = \boldsymbol{F}\boldsymbol{U}^{-1}$. Then $\boldsymbol{R}^{\mathrm{T}}\boldsymbol{R} = \boldsymbol{U}^{-1}\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}\boldsymbol{U}^{-1} = \boldsymbol{U}^{-1}\boldsymbol{U}^2\boldsymbol{U}^{-1} = \boldsymbol{I}$, and $\det\boldsymbol{R} = \det\boldsymbol{F}/\det\boldsymbol{U} = J/J = 1$: $\boldsymbol{R}$ is a rotation. If $\boldsymbol{F} = \boldsymbol{R}'\boldsymbol{U}'$ were another decomposition, $\boldsymbol{C} = \boldsymbol{U}'\boldsymbol{R}'^{\mathrm{T}}\boldsymbol{R}'\boldsymbol{U}' = \boldsymbol{U}'^2$, and uniqueness of the square root gives $\boldsymbol{U}' = \boldsymbol{U}$, $\boldsymbol{R}' = \boldsymbol{R}$. Finally

$\boldsymbol{v} = \boldsymbol{R}\boldsymbol{U}\boldsymbol{R}^{\mathrm{T}}$ is symmetric positive definite and satisfies $\boldsymbol{v}\boldsymbol{R} = \boldsymbol{R}\boldsymbol{U} = \boldsymbol{F}$; uniqueness of $\boldsymbol{v}$ follows from $\boldsymbol{b} = \boldsymbol{v}^2$ as before. ■

$\boldsymbol{U}$
: is the **right stretch tensor** (material): "right" because it stands on the right of $\boldsymbol{R}$.

$\boldsymbol{v}$ is the **left stretch tensor** (spatial).

$\boldsymbol{R}$
: is the **rotation tensor**, a two-point tensor like $\boldsymbol{F}$, with $\det \boldsymbol{R} = +1$.

The stretch tensors measure the local change of shape, i.e., the stretching along their mutually orthogonal eigenvectors; $\boldsymbol{U}$ acts on the reference configuration, $\boldsymbol{v}$ on the current one. $\boldsymbol{R}$ measures the local change of orientation. Of the nine components of $\boldsymbol{F}$, three go into $\boldsymbol{R}$ (three angles are enough to define the orientation of an orthonormal triad) and six into $\boldsymbol{U}$ (or $\boldsymbol{v}$).

Remark 3.3 (Notation). Following Holzapfel, the left stretch tensor is denoted $\boldsymbol{v}$, the same symbol as the spatial velocity; the context always makes the meaning clear. Some authors use $\boldsymbol{V}$, which here denotes the material velocity.

Relations. From $\boldsymbol{R}\boldsymbol{U} = \boldsymbol{v}\boldsymbol{R}$ and $\boldsymbol{R}^{\mathrm{T}}\boldsymbol{R} = \boldsymbol{I}$,

$$\boldsymbol{U} = \boldsymbol{R}^{\mathrm{T}}\boldsymbol{v}\boldsymbol{R}, \qquad \boldsymbol{v} = \boldsymbol{R}\boldsymbol{U}\boldsymbol{R}^{\mathrm{T}}. \tag{3.49}$$

Moreover

$$\boldsymbol{C} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} = \boldsymbol{U}\boldsymbol{R}^{\mathrm{T}}\boldsymbol{R}\boldsymbol{U} = \boldsymbol{U}^2, \qquad \boldsymbol{b} = \boldsymbol{F}\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{v}\boldsymbol{R}\boldsymbol{R}^{\mathrm{T}}\boldsymbol{v} = \boldsymbol{v}^2, \tag{3.50}$$

$$\det \boldsymbol{U} = \det \boldsymbol{v} = J, \qquad \det \boldsymbol{R} = 1. \tag{3.51}$$

The rotation disappears from $\boldsymbol{C}$ and $\boldsymbol{b}$: this is why $\boldsymbol{C}$ "contains the stretches", as observed in Section 3.9. Also $\boldsymbol{C}$ and $\boldsymbol{b}$ are related by $\boldsymbol{b} = \boldsymbol{R}\boldsymbol{C}\boldsymbol{R}^{\mathrm{T}}$.

> **In class**
>
> The lecturer stressed why this matters: rotations do not imply deformation and must not enter the constitutive relations. If one works directly with $\boldsymbol{F}$, the rotation $\boldsymbol{R}$ has to be removed with special care; working with $\boldsymbol{C}$ or $\boldsymbol{b}$ (and the strains $\boldsymbol{E}$, $\boldsymbol{e}$ built from them) the rotation is automatically cancelled. This is the starting point of the discussion on *objectivity*.

Interpretation. A material line element is mapped as

$$\mathrm{d}\boldsymbol{x} = \boldsymbol{F}\,\mathrm{d}\boldsymbol{X} = \boldsymbol{R}(\boldsymbol{U}\,\mathrm{d}\boldsymbol{X}) = \boldsymbol{v}(\boldsymbol{R}\,\mathrm{d}\boldsymbol{X}). \tag{3.52}$$

When a product of tensors is applied to a vector, the rightmost one acts first. In the right decomposition, the neighbourhood of the point is first stretched by $\boldsymbol{U}$ (change of length and shape) and then rigidly rotated by $\boldsymbol{R}$; in the left decomposition, it is first rotated and then stretched by $\boldsymbol{v}$ (Fig. 3.3). For a rigid rotation $\boldsymbol{F} = \boldsymbol{R}$, $\boldsymbol{U} = \boldsymbol{v} = \boldsymbol{I}$; when $\boldsymbol{R} = \boldsymbol{I}$ the deformation is a *pure stretch*, $\boldsymbol{F} = \boldsymbol{U} = \boldsymbol{v}$.

The stretch vector of Section 3.9 becomes $\boldsymbol{\lambda}_{\boldsymbol{a}_0} = \boldsymbol{F}\boldsymbol{a}_0 = \boldsymbol{R}\boldsymbol{U}\boldsymbol{a}_0$ and

$$\lambda^2 = \boldsymbol{R}\boldsymbol{U}\boldsymbol{a}_0 \cdot \boldsymbol{R}\boldsymbol{U}\boldsymbol{a}_0 = \boldsymbol{a}_0 \cdot \boldsymbol{U}^2\boldsymbol{a}_0 = \boldsymbol{a}_0 \cdot \boldsymbol{C}\boldsymbol{a}_0 : \tag{3.53}$$

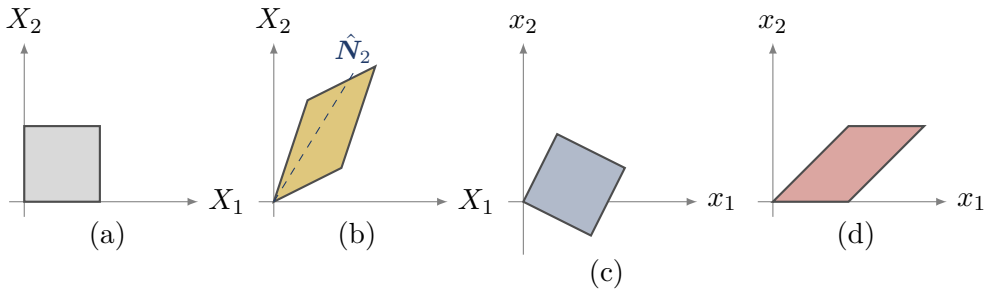


Figure 3.3: Plane polar decomposition of the simple shear $\boldsymbol{F} = \boldsymbol{I} + \gamma\, \boldsymbol{e}_1 \otimes \boldsymbol{E}_2$ with $\gamma = 1$. Right decomposition $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U}$: the unit square (a) is first stretched by $\boldsymbol{U}$ (b) and then rotated by $\boldsymbol{R}$ (d). Left decomposition $\boldsymbol{F} = \boldsymbol{v}\boldsymbol{R}$: the square is first rotated by $\boldsymbol{R}$ (c), here clockwise by 26.57°, and then stretched by $\boldsymbol{v}$ (d). See Example 3.11.

the stretch depends only on $\boldsymbol{U}$. The tensors $\boldsymbol{C}$ and $\boldsymbol{E}$, which depend only on $\boldsymbol{U}$, are true strain measures; $\boldsymbol{F}$ is more general and holds all the information (stretch and rotation).

Example 3.11 (Polar decomposition of simple shear)**.** For the simple shear $x_1 = X_1 + \gamma X_2$, $x_2 = X_2$, $x_3 = X_3$, in the plane,

$$[\boldsymbol{F}] = \begin{bmatrix} 1 & \gamma \\ 0 & 1 \end{bmatrix}, \quad [\boldsymbol{C}] = \begin{bmatrix} 1 & \gamma \\ \gamma & 1+\gamma^2 \end{bmatrix}, \quad [\boldsymbol{b}] = \begin{bmatrix} 1+\gamma^2 & \gamma \\ \gamma & 1 \end{bmatrix}.$$

Writing $\boldsymbol{R}$ as a rotation by the (counterclockwise) angle β and imposing that $\boldsymbol{U} = \boldsymbol{R}^{\mathrm{T}}\boldsymbol{F}$ is symmetric gives $\tan\beta = -\gamma/2$. With $s = \sqrt{4+\gamma^2}$,

$$[\boldsymbol{R}] = \frac{1}{s}\begin{bmatrix} 2 & \gamma \\ -\gamma & 2 \end{bmatrix}, \quad [\boldsymbol{U}] = \frac{1}{s}\begin{bmatrix} 2 & \gamma \\ \gamma & 2+\gamma^2 \end{bmatrix}, \quad [\boldsymbol{v}] = \frac{1}{s}\begin{bmatrix} 2+\gamma^2 & \gamma \\ \gamma & 2 \end{bmatrix}.$$

One checks $\boldsymbol{U}^2 = \boldsymbol{C}$, $\boldsymbol{v}^2 = \boldsymbol{b}$, $\det \boldsymbol{U} = \det \boldsymbol{v} = 1$ (simple shear is isochoric). The principal stretches are $\lambda_{1,2} = \frac{1}{2}(s \pm \gamma)$, with $\lambda_1\lambda_2 = 1$. For $\gamma = 1$: $\beta = -26.57°$, $\lambda_1 = 1.618$, $\lambda_2 = 0.618$ (Fig. 3.3). For small γ, $\beta \approx -\gamma/2$: the infinitesimal rotation of the linear theory.

Complement Compatibility conditions for $\boldsymbol{F}$

Given a field $\boldsymbol{F}(\boldsymbol{X})$, is there a motion $\boldsymbol{\chi}$ with $\operatorname{Grad}\boldsymbol{\chi} = \boldsymbol{F}$? If $\boldsymbol{\chi}$ is twice continuously differentiable, the second derivatives commute, $\partial F_{iA}/\partial X_B = \partial F_{iB}/\partial X_A$, i.e.,

$$\operatorname{Curl}\boldsymbol{F} = \boldsymbol{O} \qquad \Big(\varepsilon_{ABC}\,\frac{\partial F_{iC}}{\partial X_B} = 0\Big). \tag{3.54}$$

This necessary condition is also sufficient on a simply connected body (slide 104). It is the large-strain analogue of the Saint-Venant compatibility conditions of linear elasticity, which guarantee that a given small strain field $\boldsymbol{\varepsilon}$ derives from a displacement field. Compatibility can also be expressed for $\boldsymbol{C}$, in terms of the vanishing of the Riemann curvature tensor built with $\boldsymbol{C}$ as metric (via the Christoffel symbols).

Hadamard condition (slide 105). If two parts of a body, separated by a plane π with normal $\boldsymbol{N}$, undergo two different homogeneous deformations $\boldsymbol{F}^+$ and $\boldsymbol{F}^-$, continuity across π requires that every vector $\boldsymbol{T}$ tangent to π be mapped into the same spatial vector: $\boldsymbol{F}^+\boldsymbol{T} = \boldsymbol{F}^-\boldsymbol{T}$ for all $\boldsymbol{T} \perp \boldsymbol{N}$.

Hence $\boldsymbol{F}^+ - \boldsymbol{F}^-$ annihilates the plane, and

$$\boldsymbol{F}^+ - \boldsymbol{F}^- = \boldsymbol{a} \otimes \boldsymbol{N} \tag{3.55}$$

for some vector $\boldsymbol{a}$: the jump of the deformation gradient must be a rank-one tensor. The same condition holds point-wise on any smooth surface of discontinuity; it is the basis of the theory of martensitic microstructures and of shear bands.

3.14 Spectral representation and functions of tensors

For a diagonalisable tensor $\boldsymbol{A}$ with eigenvalues λ_α, right eigenvectors $\boldsymbol{u}_\alpha$ and dual (left) eigenvectors $\boldsymbol{u}^\alpha$ (Section 2.7), $\boldsymbol{A} = \sum_\alpha \lambda_\alpha \boldsymbol{u}_\alpha \otimes \boldsymbol{u}^\alpha$. Since $\boldsymbol{u}^\alpha \cdot \boldsymbol{u}_\beta = \delta^\alpha_\beta$, $\boldsymbol{A}^2 = \sum_\alpha \lambda_\alpha^2 \boldsymbol{u}_\alpha \otimes \boldsymbol{u}^\alpha$, and in general every power, and therefore every function expressible as a power series, acts on the eigenvalues only:

Functions of a diagonalisable tensor

$$f(\boldsymbol{A}) = \sum_\alpha f(\lambda_\alpha)\, \boldsymbol{u}_\alpha \otimes \boldsymbol{u}^\alpha, \tag{3.56}$$

$$\sqrt{\boldsymbol{A}} = \sum_\alpha \sqrt{\lambda_\alpha}\, \boldsymbol{u}_\alpha \otimes \boldsymbol{u}^\alpha, \quad \exp \boldsymbol{A} = \sum_\alpha e^{\lambda_\alpha}\, \boldsymbol{u}_\alpha \otimes \boldsymbol{u}^\alpha, \quad \ln \boldsymbol{A} = \sum_\alpha \ln \lambda_\alpha\, \boldsymbol{u}_\alpha \otimes \boldsymbol{u}^\alpha.$$

The square root and the logarithm require $\lambda_\alpha > 0$, which is the case for $\boldsymbol{C}$, $\boldsymbol{b}$, $\boldsymbol{U}$, $\boldsymbol{v}$. For a symmetric tensor the two bases coincide and are orthonormal. The exponential and the logarithm admit the power series

$$\exp \boldsymbol{A} = \sum_{k=0}^{\infty} \frac{1}{k!} \boldsymbol{A}^k, \qquad \ln(\boldsymbol{I} + \boldsymbol{A}) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \boldsymbol{A}^k \quad (\text{for } |\lambda_\alpha| < 1). \tag{3.57}$$

In class

Why the exponential and the logarithm? Some constitutive theories, in particular in finite plasticity, map the deformation into logarithmic form. Large stretches have small logarithms, so the machinery developed for plasticity in linearised kinematics can be reused on $\ln \lambda_\alpha$; the result is mapped back with the exponential. "It is not so simple, but the concept is easy."

3.14.1 Principal stretches and principal directions

For the symmetric positive definite tensor $\boldsymbol{U}$, the eigenvalue problem gives the principal stretches $\lambda_\alpha > 0$ and the orthonormal eigenvectors $\hat{\boldsymbol{N}}_\alpha$, the *principal referential directions*; $\boldsymbol{C} = \boldsymbol{U}^2$ has the same eigenvectors and eigenvalues λ_α^2:

$$\boldsymbol{U}\hat{\boldsymbol{N}}_\alpha = \lambda_\alpha \hat{\boldsymbol{N}}_\alpha, \qquad \boldsymbol{C}\hat{\boldsymbol{N}}_\alpha = \lambda_\alpha^2 \hat{\boldsymbol{N}}_\alpha. \tag{3.58}$$

Using $\boldsymbol{v} = \boldsymbol{R}\boldsymbol{U}\boldsymbol{R}^{\mathrm{T}}$, $\boldsymbol{v}(\boldsymbol{R}\hat{\boldsymbol{N}}_\alpha) = \boldsymbol{R}\boldsymbol{U}\hat{\boldsymbol{N}}_\alpha = \lambda_\alpha(\boldsymbol{R}\hat{\boldsymbol{N}}_\alpha)$: $\boldsymbol{v}$ and $\boldsymbol{b}$ have the *same* eigenvalues as $\boldsymbol{U}$ and $\boldsymbol{C}$, and eigenvectors, the *principal spatial directions*,

$$\hat{\boldsymbol{n}}_\alpha = \boldsymbol{R}\hat{\boldsymbol{N}}_\alpha. \tag{3.59}$$

The spectral decompositions read

Table 3.2: The most used members of the Seth–Hill family.

m	Name	Lagrangian	Eulerian
0	Hencky (logarithmic)	$\boldsymbol{E}^{(0)} = \ln \boldsymbol{U} = \frac{1}{2}\ln \boldsymbol{C}$	$\boldsymbol{e}^{(0)} = \ln \boldsymbol{v} = \frac{1}{2}\ln \boldsymbol{b}$
1	Biot	$\boldsymbol{E}^{(1)} = \boldsymbol{U} - \boldsymbol{I}$	$\boldsymbol{e}^{(1)} = \boldsymbol{v} - \boldsymbol{I}$
2	Green–Lagrange	$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{U}^2 - \boldsymbol{I}) = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I})$	$\frac{1}{2}(\boldsymbol{v}^2 - \boldsymbol{I}) = \frac{1}{2}(\boldsymbol{b} - \boldsymbol{I})$
-2	Euler–Almansi	$\frac{1}{2}(\boldsymbol{I} - \boldsymbol{U}^{-2}) = \frac{1}{2}(\boldsymbol{I} - \boldsymbol{C}^{-1})$	$\boldsymbol{e} = \frac{1}{2}(\boldsymbol{I} - \boldsymbol{v}^{-2}) = \frac{1}{2}(\boldsymbol{I} - \boldsymbol{b}^{-1})$

Spectral decompositions of the stretch tensors

$$\boldsymbol{U} = \sum_{\alpha=1}^{3} \lambda_\alpha \hat{\boldsymbol{N}}_\alpha \otimes \hat{\boldsymbol{N}}_\alpha, \qquad \boldsymbol{C} = \sum_{\alpha=1}^{3} \lambda_\alpha^2 \hat{\boldsymbol{N}}_\alpha \otimes \hat{\boldsymbol{N}}_\alpha, \tag{3.60}$$

$$\boldsymbol{v} = \sum_{\alpha=1}^{3} \lambda_\alpha \hat{\boldsymbol{n}}_\alpha \otimes \hat{\boldsymbol{n}}_\alpha, \qquad \boldsymbol{b} = \sum_{\alpha=1}^{3} \lambda_\alpha^2 \hat{\boldsymbol{n}}_\alpha \otimes \hat{\boldsymbol{n}}_\alpha, \tag{3.61}$$

$$\boldsymbol{F} = \sum_{\alpha=1}^{3} \lambda_\alpha \hat{\boldsymbol{n}}_\alpha \otimes \hat{\boldsymbol{N}}_\alpha, \qquad \boldsymbol{R} = \sum_{\alpha=1}^{3} \hat{\boldsymbol{n}}_\alpha \otimes \hat{\boldsymbol{N}}_\alpha. \tag{3.62}$$

The last two are not proper spectral decompositions (the two legs live in different configurations), but they show clearly that $\boldsymbol{F}$ maps each principal referential direction into the corresponding principal spatial direction, stretched by λ_α, and $\boldsymbol{R}$ rotates $\hat{\boldsymbol{N}}_\alpha$ into $\hat{\boldsymbol{n}}_\alpha$. All these quantities depend on the point and on time. This completes the remark made in Section 3.9: the λ_α are the eigenvalues of $\boldsymbol{U}$ and $\boldsymbol{v}$, not of $\boldsymbol{F}$.

3.15 General strain measures

Any strain measure must reduce to the small strain tensor $\boldsymbol{\varepsilon}$ when the displacement gradient is small: this is the minimal requirement for a good strain measure. Seth (1964) and Hill (1968–1978) proposed a family satisfying it.

Definition 3.14 (Seth–Hill strain measures)**.** For m a real number,

$$\boldsymbol{E}^{(m)} = \begin{cases} \dfrac{1}{m}(\boldsymbol{U}^m - \boldsymbol{I}), & m \neq 0, \\ \ln \boldsymbol{U}, & m = 0, \end{cases} \qquad \boldsymbol{e}^{(m)} = \begin{cases} \dfrac{1}{m}(\boldsymbol{v}^m - \boldsymbol{I}), & m \neq 0, \\ \ln \boldsymbol{v}, & m = 0. \end{cases} \tag{3.63}$$

The first family is Lagrangian (material), the second Eulerian (spatial). By the spectral decomposition, $\boldsymbol{E}^{(m)} = \sum_\alpha \frac{1}{m}(\lambda_\alpha^m - 1)\hat{\boldsymbol{N}}_\alpha \otimes \hat{\boldsymbol{N}}_\alpha$; with $\lambda_\alpha = 1 + \varepsilon_\alpha$ and ε_α small, $\frac{1}{m}(\lambda_\alpha^m - 1) \approx \varepsilon_\alpha$ for every m (and $\ln \lambda_\alpha \approx \varepsilon_\alpha$ for $m = 0$), which shows that all of them reduce to the small strain. The case $m = 0$ is defined as the limit of the others.

Only the Green–Lagrange (material) and the Euler–Almansi (spatial) strains were derived in Sections 3.9 and 3.11; Table 3.2 shows that each has a counterpart in the other configuration. Thanks to the logarithm, the Hencky strains decompose additively into volumetric and isochoric parts, $\ln \boldsymbol{U} = \frac{1}{3}\ln J\, \boldsymbol{I} + \ln(J^{-1/3}\boldsymbol{U})$, and play an important role in nonlinear constitutive theories.

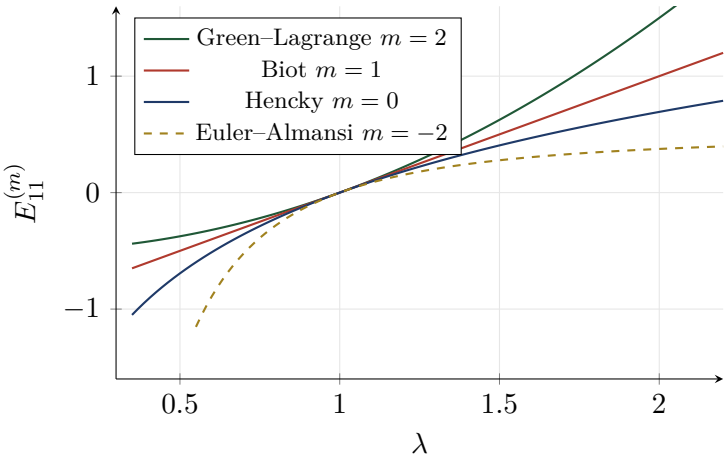


Figure 3.4: Strain measures of the Seth–Hill family in uniaxial stretch (Example 3.12).

Example 3.12 (Uniaxial stretch, slide 109)**.** For $[\boldsymbol{F}] = \mathrm{diag}(\lambda, 1, 1)$, with $\lambda = L/L_0$, the rotation is $\boldsymbol{R} = \boldsymbol{I}$, $\boldsymbol{U} = \boldsymbol{F}$, $\boldsymbol{C} = \mathrm{diag}(\lambda^2, 1, 1)$, and the only non-zero component of each strain is

$$E_{11}^{(0)} = \ln\lambda, \qquad E_{11}^{(1)} = \lambda - 1, \qquad E_{11} = \tfrac{1}{2}(\lambda^2 - 1), \qquad E_{11}^{(-2)} = \tfrac{1}{2}(1 - \lambda^{-2}).$$

All have unit slope at $\lambda = 1$ (Fig. 3.4). The Biot strain is linear in λ; the Hencky strain "softens" (its slope decreases) while the Green–Lagrange strain "stiffens". The measures are different but equally legitimate: what matters is to associate to each strain measure the suitable (work-conjugate) stress measure in the constitutive relation.

3.15.1 Special classes of deformations

The following homogeneous deformations (slide 113) are the building blocks of the experimental characterisation of materials:

- uniaxial extension ($\lambda_1 > 1$) or compression ($\lambda_1 < 1$): $x_1 = \lambda_1 X_1$, $x_2 = X_2$, $x_3 = X_3$;
- general triaxial (uniform) deformation: $x_1 = \lambda_1 X_1$, $x_2 = \lambda_2 X_2$, $x_3 = \lambda_3 X_3$;
- uniform expansion (contraction): $x_i = \lambda X_i$, $J = \lambda^3$;
- biaxial isochoric deformation: $x_1 = \lambda_1 X_1$, $x_2 = \lambda_2 X_2$, $x_3 = X_3/(\lambda_1\lambda_2)$, $J = 1$;
- equibiaxial isochoric deformation: $x_1 = \lambda X_1$, $x_2 = \lambda X_2$, $x_3 = \lambda^{-2} X_3$;
- pure shear (isochoric): $x_1 = \lambda X_1$, $x_2 = \lambda^{-1} X_2$, $x_3 = X_3$; the incompressible uniaxial extension is instead $x_1 = \lambda X_1$, $x_2 = \lambda^{-1/2} X_2$, $x_3 = \lambda^{-1/2} X_3$.

In all of them $\boldsymbol{F}$ is diagonal: $\boldsymbol{R} = \boldsymbol{I}$ and $\boldsymbol{F} = \boldsymbol{U} = \boldsymbol{v}$.

3.16 Rates of deformation

3.16.1 Velocity gradient

Definition 3.15 (Spatial velocity gradient)**.** The **spatial velocity gradient** is

$$\boldsymbol{l}(\boldsymbol{x},t) = \operatorname{grad}\boldsymbol{v}(\boldsymbol{x},t) = \frac{\partial \boldsymbol{v}(\boldsymbol{x},t)}{\partial \boldsymbol{x}}, \qquad l_{ab} = \frac{\partial v_a}{\partial x_b}. \tag{3.64}$$

It describes how the velocity changes from point to point: two neighbouring particles at $\boldsymbol{x}$ and $\boldsymbol{x} + \mathrm{d}\boldsymbol{x}$ have relative velocity $\mathrm{d}\boldsymbol{v} = \boldsymbol{l}\,\mathrm{d}\boldsymbol{x}$.

3.16.2 Rate of the deformation gradient

The material time derivative of $\boldsymbol{F}$ is obtained by exchanging the derivatives with respect to t and $\boldsymbol{X}$, which is allowed because $\boldsymbol{X}$ does not depend on time:

$$\dot{\boldsymbol{F}} = \frac{\mathrm{D}\boldsymbol{F}(\boldsymbol{X},t)}{\mathrm{D}t} = \frac{\partial}{\partial t}\frac{\partial \boldsymbol{\chi}(\boldsymbol{X},t)}{\partial \boldsymbol{X}} = \frac{\partial}{\partial \boldsymbol{X}}\frac{\partial \boldsymbol{\chi}(\boldsymbol{X},t)}{\partial t} = \operatorname{Grad}\boldsymbol{V}(\boldsymbol{X},t), \tag{3.65}$$

the material gradient of the velocity. The same result can be obtained as a directional derivative of $\boldsymbol{F}$ in the direction of the velocity:

$$D_{\boldsymbol{v}}\boldsymbol{F}(\boldsymbol{X},t) = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\frac{\partial(\boldsymbol{x}+\varepsilon\boldsymbol{v})}{\partial \boldsymbol{X}}\Big|_{\varepsilon=0} = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Big(\frac{\partial \boldsymbol{x}}{\partial \boldsymbol{X}} + \varepsilon\frac{\partial \boldsymbol{v}}{\partial \boldsymbol{X}}\Big)\Big|_{\varepsilon=0} = \frac{\partial \boldsymbol{v}}{\partial \boldsymbol{X}} = \dot{\boldsymbol{F}}. \tag{3.66}$$

In class

The second form is recalled because it is the one met in the principle of virtual work (or virtual power), where the deformation gradient is perturbed in the direction of a velocity (or virtual displacement) field. The velocity is one physical vector, called $\boldsymbol{V}$ when expressed in material coordinates and $\boldsymbol{v}$ in spatial coordinates.

By the chain rule, $\partial \boldsymbol{v}/\partial \boldsymbol{x} = (\partial \boldsymbol{v}/\partial \boldsymbol{X})(\partial \boldsymbol{X}/\partial \boldsymbol{x})$, hence

Velocity gradient and rate of F

$$\boldsymbol{l} = \dot{\boldsymbol{F}}\boldsymbol{F}^{-1}, \qquad \dot{\boldsymbol{F}} = \boldsymbol{l}\boldsymbol{F}, \qquad l_{ab} = \dot{F}_{aA}F^{-1}_{Ab}, \qquad \dot{F}_{aA} = l_{ab}F_{bA}. \tag{3.67}$$

The material index of $\dot{\boldsymbol{F}}$ is saturated by $\boldsymbol{F}^{-1}$: $\boldsymbol{l}$ is a purely spatial tensor. Differentiating $\boldsymbol{F}\boldsymbol{F}^{-1} = \boldsymbol{I}$,

$$\dot{\overline{\boldsymbol{F}^{-1}}} = -\boldsymbol{F}^{-1}\boldsymbol{l}, \qquad \dot{\overline{\boldsymbol{F}^{-\mathrm{T}}}} = -\boldsymbol{l}^{\mathrm{T}}\boldsymbol{F}^{-\mathrm{T}}. \tag{3.68}$$

3.16.3 Rate of deformation and spin

The velocity gradient decomposes additively into its symmetric and skew parts:

Rate of deformation and spin tensors

$$\boldsymbol{l} = \boldsymbol{d} + \boldsymbol{w}, \qquad \boldsymbol{d} = \tfrac{1}{2}(\boldsymbol{l} + \boldsymbol{l}^{\mathrm{T}}) = \boldsymbol{d}^{\mathrm{T}}, \qquad \boldsymbol{w} = \tfrac{1}{2}(\boldsymbol{l} - \boldsymbol{l}^{\mathrm{T}}) = -\boldsymbol{w}^{\mathrm{T}}. \tag{3.69}$$

$\boldsymbol{d}$ is the **rate of deformation tensor** and $\boldsymbol{w}$ the **spin** (or vorticity) tensor. Both are spatial; many authors write them $\boldsymbol{L}$, $\boldsymbol{D}$, $\boldsymbol{W}$. The axial vector $\boldsymbol{\omega}$ of $\boldsymbol{w}$, $\boldsymbol{w}\boldsymbol{u} = \boldsymbol{\omega}\times\boldsymbol{u}$ for all $\boldsymbol{u}$, is the *angular velocity* vector, and $2\boldsymbol{\omega} = \operatorname{curl}\boldsymbol{v}$ is the vorticity vector.

The physical meaning of $\boldsymbol{d}$ is the rate of stretching. For a line element, $\mathrm{d}\boldsymbol{x} = \boldsymbol{F}\,\mathrm{d}\boldsymbol{X}$ and

$$\overline{\mathrm{d}\boldsymbol{x}} = \dot{\boldsymbol{F}}\,\mathrm{d}\boldsymbol{X} = \boldsymbol{l}\boldsymbol{F}\,\mathrm{d}\boldsymbol{X} = \boldsymbol{l}\,\mathrm{d}\boldsymbol{x}, \tag{3.70}$$

so that $\overline{|\mathrm{d}\boldsymbol{x}|^2} = 2\,\mathrm{d}\boldsymbol{x}\cdot\boldsymbol{l}\,\mathrm{d}\boldsymbol{x} = 2\,\mathrm{d}\boldsymbol{x}\cdot\boldsymbol{d}\,\mathrm{d}\boldsymbol{x}$ (the skew part does not contribute to a quadratic form). With $\mathrm{d}\boldsymbol{x} = \lambda\,\boldsymbol{n}\,\mathrm{d}\varepsilon$ ($\boldsymbol{n}$ unit),

$$\frac{\dot{\lambda}}{\lambda} = \boldsymbol{n}\cdot\boldsymbol{d}\boldsymbol{n}\,: \tag{3.71}$$

the normal components of $\boldsymbol{d}$ are the stretching rates. A motion is rigid (at the current instant) if and only if $\boldsymbol{d} = \boldsymbol{O}$.

Complement Rates and the polar decomposition

From $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U}$, $\boldsymbol{l} = \dot{\boldsymbol{F}}\boldsymbol{F}^{-1} = \dot{\boldsymbol{R}}\boldsymbol{R}^{\mathrm{T}} + \boldsymbol{R}\dot{\boldsymbol{U}}\boldsymbol{U}^{-1}\boldsymbol{R}^{\mathrm{T}}$. The term $\boldsymbol{\Omega}_R = \dot{\boldsymbol{R}}\boldsymbol{R}^{\mathrm{T}}$ is skew (differentiate $\boldsymbol{R}\boldsymbol{R}^{\mathrm{T}} = \boldsymbol{I}$), so

$$\boldsymbol{d} = \boldsymbol{R}\,\mathrm{sym}\big(\dot{\boldsymbol{U}}\boldsymbol{U}^{-1}\big)\boldsymbol{R}^{\mathrm{T}}, \qquad \boldsymbol{w} = \boldsymbol{\Omega}_R + \boldsymbol{R}\,\mathrm{skew}\big(\dot{\boldsymbol{U}}\boldsymbol{U}^{-1}\big)\boldsymbol{R}^{\mathrm{T}}.$$

In general the spin $\boldsymbol{w}$ differs from the rate of rotation $\boldsymbol{\Omega}_R$ of the polar decomposition: they coincide only when $\boldsymbol{U}$ and $\dot{\boldsymbol{U}}$ are coaxial. Slides 117–119, not discussed in class, develop these relations in the principal frames.

3.16.4 Rates of the strain tensors and of volume

Differentiating $\boldsymbol{E} = \frac{1}{2}(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} - \boldsymbol{I})$ and using $\dot{\boldsymbol{F}} = \boldsymbol{l}\boldsymbol{F}$,

$$\dot{\boldsymbol{E}} = \tfrac{1}{2}(\dot{\boldsymbol{F}}^{\mathrm{T}}\boldsymbol{F} + \boldsymbol{F}^{\mathrm{T}}\dot{\boldsymbol{F}}) = \tfrac{1}{2}\boldsymbol{F}^{\mathrm{T}}(\boldsymbol{l}^{\mathrm{T}} + \boldsymbol{l})\boldsymbol{F} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{d}\boldsymbol{F}, \qquad \dot{\boldsymbol{C}} = 2\dot{\boldsymbol{E}} = 2\boldsymbol{F}^{\mathrm{T}}\boldsymbol{d}\boldsymbol{F}. \tag{3.72}$$

The material strain rate is the pull-back of the rate of deformation (as a covariant tensor). For the spatial tensors, $\dot{\boldsymbol{b}} = \boldsymbol{l}\boldsymbol{b} + \boldsymbol{b}\boldsymbol{l}^{\mathrm{T}}$ and $\dot{\boldsymbol{e}} = \boldsymbol{d} - \boldsymbol{l}^{\mathrm{T}}\boldsymbol{e} - \boldsymbol{e}\boldsymbol{l}$.

Rate of the Jacobian. Using the derivative of the determinant, $\partial(\det\boldsymbol{F})/\partial\boldsymbol{F} = J\boldsymbol{F}^{-\mathrm{T}}$ (Chapter 1), and $\dot{\boldsymbol{F}} = \boldsymbol{l}\boldsymbol{F}$,

$$\dot{J} = \frac{\partial\det\boldsymbol{F}}{\partial\boldsymbol{F}} : \dot{\boldsymbol{F}} = J\boldsymbol{F}^{-\mathrm{T}} : \boldsymbol{l}\boldsymbol{F} = J\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{\mathrm{T}} : \boldsymbol{l} = J\boldsymbol{I} : \boldsymbol{l} = J\,\mathrm{tr}\,\boldsymbol{l},$$

where we moved $\boldsymbol{F}$ across the double contraction with $\boldsymbol{A} : (\boldsymbol{B}\boldsymbol{C}) = (\boldsymbol{A}\boldsymbol{C}^{\mathrm{T}}) : \boldsymbol{B}$. Since $\mathrm{tr}\,\boldsymbol{l} = \mathrm{tr}\,\mathrm{grad}\,\boldsymbol{v} = \mathrm{div}\,\boldsymbol{v}$ and the trace of the skew part $\boldsymbol{w}$ vanishes,

Rate of the volume ratio

$$\dot{J} = J\,\mathrm{div}\,\boldsymbol{v} = J\,\mathrm{tr}\,\boldsymbol{d}, \qquad \dot{J} = J\,v_{a,a} = J\,d_{aa}. \tag{3.73}$$

A motion is isochoric if and only if $\mathrm{div}\,\boldsymbol{v} = 0$. For the oriented area element $\mathrm{d}\boldsymbol{s} = J\boldsymbol{F}^{-\mathrm{T}}\,\mathrm{d}\boldsymbol{S}$ one finds similarly $\overline{\mathrm{d}\boldsymbol{s}} = (\mathrm{div}\,\boldsymbol{v}\,\boldsymbol{I} - \boldsymbol{l}^{\mathrm{T}})\,\mathrm{d}\boldsymbol{s}$.

In class

In class, the derivation of $\dot{J}$ (slide 152) was done on the blackboard at the beginning of the balance principles, together with Eq. (3.70), because both were needed for the continuity equation. A

student noticed that the Jacobian had been forgotten in the intermediate lines and that the result is the *trace* of $\boldsymbol{l}$, not its divergence. Here they are collected in the kinematics.

Complement Linearisation versus rates

Taking rates and linearising are mathematically identical operations (both are directional derivatives). The correspondence is (slide 124):

Linearised kinematics	Rates
displacement u_a	spatial velocity v_a
displacement gradient $u_{a,b}$	velocity gradient $l_{ab} = v_{a,b}$
small strain $\varepsilon_{ab} = \frac{1}{2}(u_{a,b} + u_{b,a})$	rate of deformation $d_{ab} = \frac{1}{2}(v_{a,b} + v_{b,a})$
infinitesimal rotation $\frac{1}{2}(u_{a,b} - u_{b,a})$	spin $w_{ab} = \frac{1}{2}(v_{a,b} - v_{b,a})$

The *Lie time derivative* of a spatial field (slides 122–123) is obtained by pulling it back to the reference configuration, taking the material time derivative, and pushing the result forward: $\mathcal{L}_{\boldsymbol{v}}(\bullet) = \boldsymbol{\chi}_*\big(\frac{\mathrm{D}}{\mathrm{D}t}\boldsymbol{\chi}_*^{-1}(\bullet)\big)$. For instance, for the covariant tensor $\boldsymbol{e}$, $\mathcal{L}_{\boldsymbol{v}}\boldsymbol{e} = \boldsymbol{F}^{-\mathrm{T}}\dot{\boldsymbol{E}}\boldsymbol{F}^{-1} = \boldsymbol{d}$.

In class

The third day closed the kinematics at slide 116 (with slide 115 left as an exercise) and moved to the stress (Chapter 4).

3.17 Exercises

Exercise 3.1. Repeat Example 3.6 for a hollow disc of incompressible material in plane motion (the area is preserved). Show that $r^2 - a^2 = R^2 - a_0^2$, compute $\boldsymbol{x}(\boldsymbol{X}, t)$, $\boldsymbol{V}(\boldsymbol{X}, t)$ and show that $\boldsymbol{v}(\boldsymbol{x}, t) = a\dot{a}\,\boldsymbol{x}/\,|\boldsymbol{x}|^2$. Compute $\boldsymbol{F}$ and verify $J = 1$.

▷ Solution on p. 137

Exercise 3.2. For Example 3.9, compute $\boldsymbol{C}$, $\boldsymbol{E}$, $\boldsymbol{b}$ and $\boldsymbol{e}$; verify $\lambda^2 = \boldsymbol{A}\cdot\boldsymbol{C}\boldsymbol{A} = 5$ for the direction $\boldsymbol{A} = \frac{1}{\sqrt{2}}(\boldsymbol{E}_1 + \boldsymbol{E}_2)$, and compute the principal stretches.

▷ Solution on p. 138

Exercise 3.3. Compute the spatial acceleration of Example 3.6 with Eq. (3.22) and compare it with the material acceleration $\boldsymbol{A} = \partial\boldsymbol{V}/\partial t$ expressed in spatial coordinates.

▷ Solution on p. 138

Exercise 3.4. Prove that $\boldsymbol{E} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{e}\boldsymbol{F}$ and that, for small displacement gradients, $\boldsymbol{E}$ and $\boldsymbol{e}$ both reduce to the small strain tensor $\boldsymbol{\varepsilon} = \operatorname{sym}\operatorname{Grad}\boldsymbol{U}$.

▷ Solution on p. 138

Exercise 3.5. For the simple shear $x_1 = X_1 + \gamma X_2$, $x_2 = X_2$, $x_3 = X_3$, compute $\boldsymbol{F}$, J, $\boldsymbol{C}$, $\boldsymbol{b}$, the stretch of the fibre $\boldsymbol{E}_2$ and the change of the angle between $\boldsymbol{E}_1$ and $\boldsymbol{E}_2$.

▷ Solution on p. 139

Exercise 3.6. Verify the relations of slide 103: $\boldsymbol{v} = \boldsymbol{R}\boldsymbol{U}\boldsymbol{R}^{\mathrm{T}}$, $\boldsymbol{b} = \boldsymbol{R}\boldsymbol{C}\boldsymbol{R}^{\mathrm{T}}$, $\boldsymbol{v}^{-1} = \boldsymbol{R}\boldsymbol{U}^{-1}\boldsymbol{R}^{\mathrm{T}}$, and show that $\boldsymbol{E} = \boldsymbol{R}^{\mathrm{T}}(\frac{1}{2}(\boldsymbol{b} - \boldsymbol{I}))\boldsymbol{R}$.

▷ Solution on p. 139

Exercise 3.7. For Example 3.11, verify $\tan\beta = -\gamma/2$ by imposing the symmetry of $\boldsymbol{R}^{\mathrm{T}}\boldsymbol{F}$, compute the principal stretches and the angles of the principal referential and spatial directions, and check that they differ by β.

▷ Solution on p. 139

Exercise 3.8. For the uniaxial stretch of Example 3.12, compute also the Eulerian measures of Table 3.2 and plot them against λ.

▷ Solution on p. 139

Exercise 3.9. For the motion of Example 3.5, compute $\boldsymbol{l}$, $\boldsymbol{d}$, $\boldsymbol{w}$, $\dot{\boldsymbol{F}}$, and verify $\dot{\boldsymbol{F}} = \boldsymbol{l}\boldsymbol{F}$ and $\dot{J} = J \operatorname{div} \boldsymbol{v}$.

▷ Solution on p. 140

Exercise 3.10. Prove that $\overline{\dot{\boldsymbol{F}^{-1}}} = -\boldsymbol{F}^{-1}\boldsymbol{l}$ and $\overline{\dot{\mathrm{d}\boldsymbol{s}}} = (\operatorname{div}\boldsymbol{v}\,\boldsymbol{I} - \boldsymbol{l}^{\mathrm{T}})\,\mathrm{d}\boldsymbol{s}$ (slides 115 and 152).

▷ Solution on p. 140

4

The Concept of Stress

In class

The lecturer assumed that the Cauchy stress is known from the basic courses ("if someone does not know what the Cauchy stress is, please let me know") and focused on the new point: in finite deformations the Cauchy stress is not the only stress measure, and we need to know which stress produces the correct work, or power, with each of the strain and strain-rate measures of Chapter 3. In small strain the work is $\boldsymbol{\sigma} : \boldsymbol{\varepsilon}$; in large deformations the Cauchy stress $\boldsymbol{\sigma}$ is the partner of the rate of deformation $\boldsymbol{d}$ in the expression of power, $\boldsymbol{\sigma} : \boldsymbol{d}$, and other stresses pair with the other strain rates (Chapter 5).

4.1 Traction vectors

Consider a deformable body that at time t occupies the region Ω with boundary $\partial\Omega$ (at $t = 0$, Ω_0 and $\partial\Omega_0$). External forces act on (part of) the boundary and internal forces act across any internal surface of the body. Forces are *densities*: *surface tractions* are forces per unit area, *body forces* are forces per unit volume; the two have different physical dimensions. Their definition requires the material to be seen as a continuum: if the microstructure matters, the Cauchy stress cannot be defined in this way.

Take a point P with position $\boldsymbol{x}$ in the current configuration ($\boldsymbol{X}$ in the reference one) and cut the body with a plane through P with unit normal $\boldsymbol{n}$ ($\boldsymbol{N}$ in the reference configuration). The normal is that of an arbitrary internal cut, and must not be confused with the outward normal to the external surface of the body. The cut divides the body into two parts that exchange forces across the surface. Let $\mathrm{d}s$ be a small surface element around P and $\mathrm{d}\boldsymbol{f}$ the resultant force exerted across it by one part on the other; resultant couples are neglected (non-polar continuum).

Definition 4.1 (Cauchy traction vector)**.** The **Cauchy (true) traction vector** is the force per unit *current* area:

$$\mathrm{d}\boldsymbol{f} = \boldsymbol{t}(\boldsymbol{x}, t, \boldsymbol{n})\,\mathrm{d}s. \tag{4.1}$$

This is *Cauchy's postulate*: the traction depends on the position, on time and on the cut only through its normal $\boldsymbol{n}$ (not on the curvature of the surface). The vector $\boldsymbol{t}$ is also called stress vector, contact force or, on the boundary, load. It is an internal force, not an external load.

Now go back to the reference configuration. The same point P is at $\boldsymbol{X}$, the element $\mathrm{d}s$ was the element $\mathrm{d}S$ with normal $\boldsymbol{N}$, the image of $\mathrm{d}s$ before the motion. Piola's idea was to take the *same* force $\mathrm{d}\boldsymbol{f}$ and refer it to the reference area.

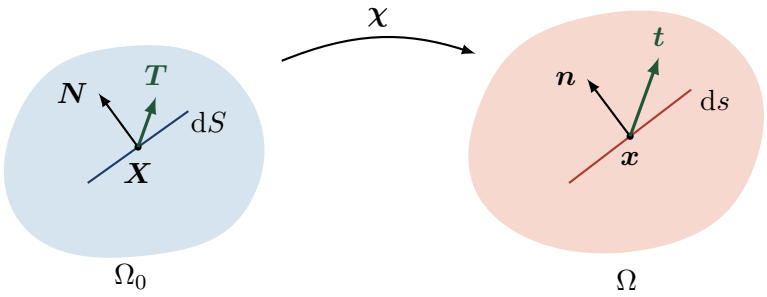


Figure 4.1: Cauchy traction $\boldsymbol{t}$ (force per unit current area $\mathrm{d}s$, normal $\boldsymbol{n}$) and first Piola–Kirchhoff traction $\boldsymbol{T}$ (the same force per unit reference area $\mathrm{d}S$, normal $\boldsymbol{N}$): $\boldsymbol{t}\,\mathrm{d}s = \boldsymbol{T}\,\mathrm{d}S$.

Definition 4.2 (First Piola–Kirchhoff traction vector)**.** The **first Piola–Kirchhoff (nominal) traction vector** $\boldsymbol{T}$ is defined by

$$\mathrm{d}\boldsymbol{f} = \boldsymbol{T}(\boldsymbol{X}, t, \boldsymbol{N})\,\mathrm{d}S. \tag{4.2}$$

$\boldsymbol{T}$ has the same direction as $\boldsymbol{t}$ but a different magnitude. It is a *pseudo* traction vector, which does not describe the actual force intensity, and is a function of $\boldsymbol{X}$ and $\boldsymbol{N}$.

In class

"It is absurd": the force $\mathrm{d}\boldsymbol{f}$ acts in the current configuration, where the body is in equilibrium; moving it to the reference configuration, where nothing is happening yet, makes no physical sense. And yet this is exactly what is convenient to do, because the reference configuration is known, while the current one is an unknown of the problem.

4.2 Cauchy's stress theorem

Theorem 4.1 (Cauchy's stress theorem)**.** *There exist unique second-order tensor fields* $\boldsymbol{\sigma}$ *and* $\boldsymbol{P}$ *such that*

$$\boldsymbol{t}(\boldsymbol{x}, t, \boldsymbol{n}) = \boldsymbol{\sigma}(\boldsymbol{x}, t)\,\boldsymbol{n}, \quad t_a = \sigma_{ab} n_b; \qquad \boldsymbol{T}(\boldsymbol{X}, t, \boldsymbol{N}) = \boldsymbol{P}(\boldsymbol{X}, t)\,\boldsymbol{N}, \quad T_a = P_{aA} N_A. \tag{4.3}$$

$\boldsymbol{\sigma}$ is the **Cauchy stress tensor** (true stress), a symmetric spatial tensor; $\boldsymbol{P}$ is the **first Piola–Kirchhoff stress tensor** (nominal stress, Piola stress), a non-symmetric two-point tensor. The content of the theorem is that if the traction depends on the normal, it must depend on it *linearly*. The proof, based on the equilibrium of a small tetrahedron, is classical and is not repeated here (see Holzapfel [3, Sect. 3.1]). As a consequence of Newton's third law (action and reaction),

$$\boldsymbol{t}(\boldsymbol{x}, t, \boldsymbol{n}) = -\boldsymbol{t}(\boldsymbol{x}, t, -\boldsymbol{n}), \qquad \boldsymbol{T}(\boldsymbol{X}, t, \boldsymbol{N}) = -\boldsymbol{T}(\boldsymbol{X}, t, -\boldsymbol{N}), \tag{4.4}$$

i.e., the two parts of the body exchange equal and opposite forces across the cut. In matrix notation $\{\boldsymbol{t}\} = [\boldsymbol{\sigma}]\,\{\boldsymbol{n}\}$.

4.2.1 Index convention

Projecting $\boldsymbol{\sigma}$ on an orthonormal basis, $\sigma_{ab} = \boldsymbol{e}_a \cdot \boldsymbol{\sigma}\boldsymbol{e}_b$, and $\boldsymbol{t}_{(\boldsymbol{e}_b)} = \boldsymbol{\sigma}\boldsymbol{e}_b = \sigma_{ab}\boldsymbol{e}_a$: the b-th column of $[\boldsymbol{\sigma}]$ contains the components of the traction acting on the plane with normal $\boldsymbol{e}_b$,

$$[\boldsymbol{\sigma}] = \left[\left\{\boldsymbol{t}_{(\boldsymbol{e}_1)}\right\} \quad \left\{\boldsymbol{t}_{(\boldsymbol{e}_2)}\right\} \quad \left\{\boldsymbol{t}_{(\boldsymbol{e}_3)}\right\} \right] = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}. \tag{4.5}$$

Because of the symmetry, only six components are independent.

> **In class**
>
> Unlike the convention used in most engineering textbooks (first index = normal to the plane, second index = direction of the component), in this course the *first index denotes the direction of the component and the second the plane* on which the traction acts. This is the convention consistent with $t_a = \sigma_{ab} n_b$, with the sum on the second index. For the symmetric Cauchy stress the two conventions give the same numbers, but for the non-symmetric $\boldsymbol{P}$ the distinction matters.

4.3 Relations between Cauchy and Piola stresses

The force is the same, $\mathrm{d}\boldsymbol{f} = \boldsymbol{t}\,\mathrm{d}s = \boldsymbol{T}\,\mathrm{d}S$, so $\boldsymbol{\sigma}\boldsymbol{n}\,\mathrm{d}s = \boldsymbol{P}\boldsymbol{N}\,\mathrm{d}S$. Using Nanson's formula Eq. (3.41), $\boldsymbol{n}\,\mathrm{d}s = J\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{N}\,\mathrm{d}S$,

$$J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{N}\,\mathrm{d}S = \boldsymbol{P}\boldsymbol{N}\,\mathrm{d}S \qquad \forall\,\boldsymbol{N},$$

which gives the *Piola transformation*

First Piola–Kirchhoff stress

$$\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}, \qquad P_{aA} = J\sigma_{ab}F^{-1}_{Ab}; \qquad \boldsymbol{\sigma} = J^{-1}\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}}, \qquad \sigma_{ab} = J^{-1}P_{aA}F_{bA}. \tag{4.6}$$

To obtain the two-point tensor $\boldsymbol{P}$ a Piola transformation is performed on the *second* index of $\boldsymbol{\sigma}$ (the one related to the normal), which becomes a material index; the first index (the direction of the force) stays spatial. The inverse relation is obtained by multiplying by $\boldsymbol{F}^{\mathrm{T}}$ on the right.

$\boldsymbol{P}$ is in general *not* symmetric. The symmetry of $\boldsymbol{\sigma}$ ($\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\mathrm{T}}$) translates into a condition on the product

$$\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{F}\boldsymbol{P}^{\mathrm{T}}, \tag{4.7}$$

not on $\boldsymbol{P}$ itself.

4.4 Normal and shear stresses; principal stresses

Let $\boldsymbol{t}_{\boldsymbol{n}} = \boldsymbol{\sigma}\boldsymbol{n}$ be the traction on a surface element with normal $\boldsymbol{n}$, and $\boldsymbol{m}$ a unit vector in the plane of the element in the direction of the tangential part of $\boldsymbol{t}_{\boldsymbol{n}}$. The traction decomposes into a component parallel and one orthogonal to the normal,

$$\boldsymbol{t}_{\boldsymbol{n}} = \boldsymbol{t}^{\parallel}_{\boldsymbol{n}} + \boldsymbol{t}^{\perp}_{\boldsymbol{n}}, \qquad \boldsymbol{t}^{\parallel}_{\boldsymbol{n}} = (\boldsymbol{n}\cdot\boldsymbol{t}_{\boldsymbol{n}})\boldsymbol{n} = \boldsymbol{P}^{\parallel}_{\boldsymbol{n}}\boldsymbol{t}_{\boldsymbol{n}}, \qquad \boldsymbol{t}^{\perp}_{\boldsymbol{n}} = (\boldsymbol{m}\cdot\boldsymbol{t}_{\boldsymbol{n}})\boldsymbol{m} = \boldsymbol{P}^{\perp}_{\boldsymbol{n}}\boldsymbol{t}_{\boldsymbol{n}}, \tag{4.8}$$

with the projection operators of Section 1.11

$$\boldsymbol{P}_{\boldsymbol{n}}^{\parallel} = \boldsymbol{n} \otimes \boldsymbol{n}, \qquad \boldsymbol{P}_{\boldsymbol{n}}^{\perp} = \boldsymbol{I} - \boldsymbol{n} \otimes \boldsymbol{n}. \tag{4.9}$$

(Do not confuse these projectors with the first Piola–Kirchhoff stress $\boldsymbol{P}$.) The magnitudes

$$\sigma = \boldsymbol{n} \cdot \boldsymbol{t}_{\boldsymbol{n}} = \boldsymbol{n} \cdot \boldsymbol{\sigma}\boldsymbol{n} = n_a \sigma_{ab} n_b, \qquad \tau = \boldsymbol{m} \cdot \boldsymbol{t}_{\boldsymbol{n}} = \boldsymbol{m} \cdot \boldsymbol{\sigma}\boldsymbol{n} = m_a \sigma_{ab} n_b \tag{4.10}$$

are the *normal* and *tangential* (shear) stress, and $\boldsymbol{t}_{\boldsymbol{n}}^{\parallel} = \sigma \boldsymbol{n}$, $\boldsymbol{t}_{\boldsymbol{n}}^{\perp} = \tau \boldsymbol{m}$, $\sigma^2 + \tau^2 = |\boldsymbol{t}_{\boldsymbol{n}}|^2$. These are the quantities plotted in Mohr's circles.

Extremal normal stresses. To find the normals that make σ stationary, use a Lagrange multiplier for the constraint $|\boldsymbol{n}|^2 = 1$:

$$\mathcal{L}(\boldsymbol{n}, \lambda) = \boldsymbol{n} \cdot \boldsymbol{\sigma}\boldsymbol{n} - \lambda(\boldsymbol{n} \cdot \boldsymbol{n} - 1), \qquad \frac{\partial \mathcal{L}}{\partial \boldsymbol{n}} = 2(\boldsymbol{\sigma}\boldsymbol{n} - \lambda \boldsymbol{n}) = \boldsymbol{0}.$$

This is the eigenvalue problem $(\boldsymbol{\sigma} - \lambda \boldsymbol{I})\boldsymbol{n} = \boldsymbol{0}$: since $\boldsymbol{\sigma}$ is symmetric it has three real eigenvalues $\sigma_1, \sigma_2, \sigma_3$, the *principal stresses*, and three mutually orthogonal unit eigenvectors $\hat{\boldsymbol{n}}_\alpha$, the *principal directions*. On the principal planes the traction is purely normal ($\tau = 0$). The spectral form is

$$\boldsymbol{\sigma} = \sum_{\alpha=1}^{3} \sigma_\alpha \, \hat{\boldsymbol{n}}_\alpha \otimes \hat{\boldsymbol{n}}_\alpha. \tag{4.11}$$

Examples of states of stress. The state of stress at a point is the set of all pairs $(\boldsymbol{t}, \boldsymbol{n})$. The stress is *homogeneous* if it does not depend on $\boldsymbol{x}$. Typical states are (slide 143):

- pure normal stress (tension or compression): $\boldsymbol{\sigma} = \sigma \, \boldsymbol{n} \otimes \boldsymbol{n}$;
- pure shear stress: $\boldsymbol{\sigma} = \tau(\boldsymbol{n} \otimes \boldsymbol{m} + \boldsymbol{m} \otimes \boldsymbol{n})$, with $\boldsymbol{n} \perp \boldsymbol{m}$;
- hydrostatic stress: $\boldsymbol{\sigma} = -p\boldsymbol{I}$, for which every direction is principal;
- plane stress: $\boldsymbol{\sigma}\boldsymbol{e}_3 = \boldsymbol{0}$, i.e., $\sigma_{13} = \sigma_{23} = \sigma_{33} = 0$.

4.5 Alternative stress tensors

Definition 4.3 (Kirchhoff stress)**.** The **Kirchhoff stress tensor** is the Cauchy stress weighted by the volume ratio:

$$\boldsymbol{\tau} = J\boldsymbol{\sigma}, \qquad \tau_{ab} = J\sigma_{ab}. \tag{4.12}$$

$\boldsymbol{\tau}$ is symmetric and spatial; the Jacobian inside accounts for the change of volume. Both $\boldsymbol{\sigma}$ and $\boldsymbol{\tau}$ are contravariant tensors parametrised by spatial coordinates.

Definition 4.4 (Second Piola–Kirchhoff stress)**.** The **second Piola–Kirchhoff stress tensor** $\boldsymbol{S}$ is the pull-back of the (contravariant) Kirchhoff stress:

$$\boldsymbol{S} = \boldsymbol{\chi}_*^{-1}(\boldsymbol{\tau}) = \boldsymbol{F}^{-1}\boldsymbol{\tau}\boldsymbol{F}^{-\mathrm{T}}, \qquad S_{AB} = F_{Aa}^{-1}\tau_{ab}F_{Bb}^{-1}. \tag{4.13}$$

Conversely, the Kirchhoff stress is the push-forward of $\boldsymbol{S}$, $\boldsymbol{\tau} = \boldsymbol{\chi}_*(\boldsymbol{S}) = \boldsymbol{F}\boldsymbol{S}\boldsymbol{F}^{\mathrm{T}}$. $\boldsymbol{S}$ is a symmetric contravariant *material* tensor, and

$$\boldsymbol{S} = J\boldsymbol{F}^{-1}\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}} = \boldsymbol{F}^{-1}\boldsymbol{P} = \boldsymbol{S}^{\mathrm{T}}, \qquad S_{AB} = JF^{-1}_{Aa}\sigma_{ab}F^{-1}_{Bb} = F^{-1}_{Aa}P_{aB}. \tag{4.14}$$

$\boldsymbol{S}$ has no direct physical interpretation in terms of tractions (it is the "force" $\boldsymbol{F}^{-1}\,\mathrm{d}\boldsymbol{f}$ pulled back to the reference configuration, per unit reference area), but it is the natural stress for constitutive equations written in the reference configuration. It can be seen as $\boldsymbol{P}$ "symmetrised" by removing the spatial leg with $\boldsymbol{F}^{-1}$.

In class

Why $\boldsymbol{P}$ and $\boldsymbol{S}$? Gabriele Piola, the Italian mathematician who introduced them (the street of the Politecnico, Via Piola, is named after him), called them *primo* and *secondo*: first and second. The second could not be called "F", which is taken by the deformation gradient, hence $\boldsymbol{S}$.

Relations between stress tensors (stress table)

Each entry gives the tensor of its row in terms of the tensor of its column:

	$\boldsymbol{\sigma}$	$\boldsymbol{\tau}$	$\boldsymbol{P}$	$\boldsymbol{S}$
$\boldsymbol{\sigma} =$	$\boldsymbol{\sigma}$	$J^{-1}\boldsymbol{\tau}$	$J^{-1}\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}}$	$J^{-1}\boldsymbol{F}\boldsymbol{S}\boldsymbol{F}^{\mathrm{T}}$
$\boldsymbol{\tau} =$	$J\boldsymbol{\sigma}$	$\boldsymbol{\tau}$	$\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}}$	$\boldsymbol{F}\boldsymbol{S}\boldsymbol{F}^{\mathrm{T}}$
$\boldsymbol{P} =$	$J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}$	$\boldsymbol{\tau}\boldsymbol{F}^{-\mathrm{T}}$	$\boldsymbol{P}$	$\boldsymbol{F}\boldsymbol{S}$
$\boldsymbol{S} =$	$J\boldsymbol{F}^{-1}\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}$	$\boldsymbol{F}^{-1}\boldsymbol{\tau}\boldsymbol{F}^{-\mathrm{T}}$	$\boldsymbol{F}^{-1}\boldsymbol{P}$	$\boldsymbol{S}$

The table is easy to memorise: to pass from the stress of a row to that of a column, read the entry at their intersection.

Remark 4.1 (Small deformations). When the displacement gradient is small, $\boldsymbol{F} \approx \boldsymbol{I}$, $J \approx 1$ and all four stress tensors coincide to first order with the Cauchy stress of the linear theory. Their differences are a purely geometrical effect of finite deformation.

Example 4.1 (Uniaxial tension of a bar)**.** A bar of reference cross section A_0 is stretched uniformly, $[\boldsymbol{F}] = \mathrm{diag}(\lambda, \lambda_t, \lambda_t)$, by an axial force f; the current cross section is $A = \lambda_t^2 A_0$ and $J = \lambda\lambda_t^2$. The only non-zero Cauchy stress component is the true stress $\sigma_{11} = f/A$. Then

$$P_{11} = J\sigma_{11}F^{-1}_{11} = \lambda\lambda_t^2\frac{f}{\lambda_t^2 A_0}\frac{1}{\lambda} = \frac{f}{A_0}, \qquad S_{11} = F^{-1}_{11}P_{11} = \frac{f}{\lambda A_0}, \qquad \tau_{11} = J\sigma_{11} = \lambda\frac{f}{A_0}.$$

P_{11} is the *nominal* (engineering) stress measured in a tensile test: force over initial area.

4.6 Exercises

Exercise 4.1. Derive all the entries of the stress table from the definitions of $\boldsymbol{\tau}$, $\boldsymbol{P}$ and $\boldsymbol{S}$.

▷ Solution on p. 140

Exercise 4.2. Show that Eq. (4.7) is equivalent to the symmetry of $\boldsymbol{\sigma}$, and that $\boldsymbol{S}$ is symmetric if and only if $\boldsymbol{\sigma}$ is.

▷ Solution on p. 140

Exercise 4.3. For the simple shear $\boldsymbol{F} = \boldsymbol{I} + \gamma\, \boldsymbol{e}_1 \otimes \boldsymbol{E}_2$, compute $\boldsymbol{P}$, $\boldsymbol{S}$ and $\boldsymbol{\tau}$ for (a) the shear stress $\boldsymbol{\sigma} = s(\boldsymbol{e}_1 \otimes \boldsymbol{e}_2 + \boldsymbol{e}_2 \otimes \boldsymbol{e}_1)$ and (b) the uniaxial stress $\boldsymbol{\sigma} = \sigma_0\, \boldsymbol{e}_2 \otimes \boldsymbol{e}_2$. In which case is $\boldsymbol{P}$ symmetric?

▷ Solution on p. 140

Exercise 4.4. Given $\boldsymbol{\sigma}$, find the planes of maximum shear stress and show that $\tau_{\max} = \frac{1}{2}(\sigma_1 - \sigma_3)$, with $\sigma_1 \geq \sigma_2 \geq \sigma_3$.

▷ Solution on p. 141

5

Balance Principles

In class

So far we have described two different worlds. Kinematics required only geometry: motion, rotation, stretches, things we can see. Stress describes the internal forces. The balance principles put the two worlds together. The lecturer listed on the blackboard the principles to be explored:

1. conservation of mass (fundamental in fluid mechanics, usually trivial in solid mechanics, but not for growing biological tissues);
2. balance of linear and angular momentum (equations of motion, or of equilibrium in the static case);
3. balance of mechanical energy, written in terms of power (the theorem of power expended), which is the basis of the numerical methods;
4. heat and the principles of thermodynamics (zeroth, first and second), needed to formulate realistic constitutive equations.

This chapter covers all four: the first three were presented on the third day, the fourth (together with work-conjugate pairs and the principle of virtual work) on the fourth day.

5.1 Systems and mass

Mass is a continuously distributed physical property that measures the amount of material in a body: a positive scalar measure, in general invariant during the motion. It is an *extensive* quantity, related to (but not simply proportional to) the volume.

A given quantity of matter defines a *system*; everything else is the *surroundings*. The two are separated by the *boundary* (the system wall), through which they interact. Depending on what can cross the boundary:

- a **closed system** (control mass) is a fixed amount of mass in a region Ω whose boundary $\partial\Omega$ moves with time; only energy (work, heat) can cross the boundary;
- an *isolated system* is a closed system whose boundary does not allow any exchange of energy (mechanically and thermally insulated);
- an **open system** (control volume) is a fixed region Ω_c whose boundary does not depend on time; both energy and mass can cross it (fluids flowing through a pipe).

In this course we deal with solids and consider closed systems: no mass exchange, but exchange of work (tractions on the boundary, body forces such as gravity) and heat.

5.2 Conservation of mass

Assuming that there are no mass sources or sinks, the mass of a closed system is conserved, and so is the mass of any part of it:

$$m(\Omega_0) = m(\Omega) = \text{const} > 0, \qquad \mathrm{d}m(\boldsymbol{X}) = \mathrm{d}m(\boldsymbol{x}, t), \qquad \frac{\mathrm{D}m}{\mathrm{D}t} = 0. \tag{5.1}$$

Definition 5.1 (Mass density)**.** The reference and spatial **mass densities** are

$$\rho_0(\boldsymbol{X}) = \lim_{\Delta V \to 0} \frac{\Delta m}{\Delta V}, \qquad \rho(\boldsymbol{x}, t) = \lim_{\Delta v \to 0} \frac{\Delta m}{\Delta v}, \tag{5.2}$$

so that $\mathrm{d}m = \rho_0 \, \mathrm{d}V = \rho \, \mathrm{d}v$, with $\rho_0 > 0$ and $\rho > 0$.

ΔV is an idealised limit: a volume small with respect to the body but large enough to provide representative averages of the matter (excluding the atomistic scale). The lecturer's example: in a bucket of stones, a too small volume contains either a stone or nothing, and the density is not defined; the size of the averaging volume must be large with respect to the constituents.

5.2.1 Continuity equation

From $\rho_0 \, \mathrm{d}V = \rho \, \mathrm{d}v$ and $\mathrm{d}v = J \, \mathrm{d}V$,

Continuity equation (material description)

$$\rho_0(\boldsymbol{X}) = \rho(\boldsymbol{\chi}(\boldsymbol{X}, t), t) \, J(\boldsymbol{X}, t), \qquad \frac{\rho}{\rho_0} = \frac{\mathrm{d}V}{\mathrm{d}v} = J^{-1}. \tag{5.3}$$

This is the *local form* of the continuity equation (the name comes from fluid mechanics): it involves differentials. The density can change only if the volume changes. The *integral form* is

$$m = \int_{\Omega_0} \rho_0(\boldsymbol{X}) \, \mathrm{d}V = \int_{\Omega} \rho(\boldsymbol{x}, t) \, \mathrm{d}v > 0, \qquad \frac{\mathrm{D}m}{\mathrm{D}t} = \frac{\mathrm{D}}{\mathrm{D}t} \int_{\Omega} \rho(\boldsymbol{x}, t) \, \mathrm{d}v = 0. \tag{5.4}$$

The local form can also be recovered from the integral one: changing variables in the spatial integral, $\int_{\Omega_0} [\rho_0(\boldsymbol{X}) - \rho(\boldsymbol{\chi}(\boldsymbol{X}, t), t) J(\boldsymbol{X}, t)] \, \mathrm{d}V = 0$, and since this holds for every part of the body the integrand must vanish.

Since ρ_0 does not depend on time, $\dot{\rho}_0 = 0$, i.e.,

$$\frac{\mathrm{D}(\rho J)}{\mathrm{D}t} = \dot{\rho} J + \rho \dot{J} = 0. \tag{5.5}$$

Using $\dot{J} = J \operatorname{div} \boldsymbol{v}$ (Eq. (3.73)) and $J > 0$:

Continuity equation (spatial description)

$$\dot{\rho} + \rho \operatorname{div} \boldsymbol{v} = 0, \qquad \frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \boldsymbol{v}) = 0. \tag{5.6}$$

The second form follows from $\dot{\rho} = \partial \rho / \partial t + \operatorname{grad} \rho \cdot \boldsymbol{v}$ and the identity $\operatorname{div}(\rho \boldsymbol{v}) = \operatorname{grad} \rho \cdot \boldsymbol{v} + \rho \operatorname{div} \boldsymbol{v}$ (in index notation $(\rho v_a)_{,a} = \rho_{,a} v_a + \rho v_{a,a}$). It is the appropriate description in fluid dynamics: the local rate of change of density equals minus the divergence of the linear

momentum density $\rho\boldsymbol{v}$.

Example 5.1 (Slide 154)**.** Consider the plane velocity field $v_1 = (\alpha x_1 - \beta x_2)t$, $v_2 = \beta x_1 - \alpha x_2$, $v_3 = 0$ $(\alpha, \beta > 0)$, and assume $\operatorname{grad}\rho = \mathbf{0}$. Then $\operatorname{div}\boldsymbol{v} = \alpha t - \alpha = \alpha(t-1)$, and the continuity equation becomes

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} + \rho\,\alpha(t-1) = 0 \quad \Longrightarrow \quad \rho(t) = \rho_0 \exp\!\Big(-\tfrac{1}{2}\alpha t^2 + \alpha t\Big).$$

The motion is isochoric ($\operatorname{div}\boldsymbol{v} = 0$ for all t, $\rho = \rho_0$) if and only if $\alpha = 0$.

Example 5.2 (Slide 156)**.** For the uniform expansion $\boldsymbol{x} = [1+\alpha(t)t]\boldsymbol{X}$, $\boldsymbol{F} = [1+\alpha(t)t]\boldsymbol{I}$, $J = [1+\alpha(t)t]^3$ and $\rho = \rho_0 J^{-1} = \rho_0/[1+\alpha(t)t]^3$. For the motion $x_1 = X_1 + \alpha(t)t^2 X_2$, $x_2 = X_2 - \alpha(t)t^2 X_1$, $x_3 = X_3$, $J = 1 + \alpha^2(t)t^4$ and $\rho = \rho_0/(1+\alpha^2 t^4)$.

Complement Open systems

For a control volume Ω_c fixed in space, integration and time differentiation commute. Integrating $\partial\rho/\partial t = -\operatorname{div}(\rho\boldsymbol{v})$ and using the divergence theorem,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_c} \rho\,\mathrm{d}v = -\int_{\partial\Omega_c} \rho\boldsymbol{v}\cdot\boldsymbol{n}\,\mathrm{d}s :$$

the mass inside changes only because of the flux of mass $\rho\boldsymbol{v}$ across the control surface.

5.3 Reynolds' transport theorem

We want the material time derivative of a quantity integrated over the current volume,

$$\frac{\mathrm{D}}{\mathrm{D}t}\int_{\Omega} \Phi(\boldsymbol{x},t)\,\mathrm{d}v.$$

Two things change with time: the integrand and the domain of integration. The trick is to go back to the reference configuration, where the domain is fixed, differentiate there, and come back:

$$\begin{aligned}\frac{\mathrm{D}}{\mathrm{D}t}\int_{\Omega} \Phi(\boldsymbol{x},t)\,\mathrm{d}v &= \frac{\mathrm{D}}{\mathrm{D}t}\int_{\Omega_0} \Phi\big(\boldsymbol{\chi}(\boldsymbol{X},t),t\big)J(\boldsymbol{X},t)\,\mathrm{d}V = \int_{\Omega_0}\big(\dot\Phi J + \Phi\dot J\big)\,\mathrm{d}V \\ &= \int_{\Omega_0}\big(\dot\Phi + \Phi\operatorname{div}\boldsymbol{v}\big)J\,\mathrm{d}V = \int_{\Omega}\big(\dot\Phi + \Phi\operatorname{div}\boldsymbol{v}\big)\,\mathrm{d}v,\end{aligned}$$

where we used Eq. (3.73). Expanding the material derivative, $\dot\Phi = \partial\Phi/\partial t + \operatorname{grad}\Phi\cdot\boldsymbol{v}$, and using again $\operatorname{grad}\Phi\cdot\boldsymbol{v} + \Phi\operatorname{div}\boldsymbol{v} = \operatorname{div}(\Phi\boldsymbol{v})$ and the divergence theorem:

Reynolds' transport theorem

$$\frac{\mathrm{D}}{\mathrm{D}t}\int_{\Omega}\Phi\,\mathrm{d}v = \int_{\Omega}\Big(\dot\Phi + \Phi\operatorname{div}\boldsymbol{v}\Big)\,\mathrm{d}v = \int_{\Omega}\frac{\partial\Phi}{\partial t}\,\mathrm{d}v + \int_{\partial\Omega}\Phi\,\boldsymbol{v}\cdot\boldsymbol{n}\,\mathrm{d}s. \tag{5.7}$$

The first term is the local time rate of change inside the region; the second is the outward normal flux of $\Phi\boldsymbol{v}$ across the boundary, i.e., the rate of transport of Φ due to the motion of the region. The same holds for vector- and tensor-valued integrands.

Quantities per unit mass. If $\Phi = \rho\,\phi$, with ϕ a quantity per unit mass, it is simpler to integrate over the mass: $\rho\,\mathrm{d}v = \mathrm{d}m$ does not change in time, so

$$\frac{\mathrm{D}}{\mathrm{D}t}\int_\Omega \rho\,\phi\,\mathrm{d}v = \int_\Omega \rho\,\dot{\phi}\,\mathrm{d}v. \tag{5.8}$$

The same result follows from Eq. (5.7) and the continuity equation, since $\dot{\overline{\rho\phi}} + \rho\phi\,\mathrm{div}\,\boldsymbol{v} = \rho\dot{\phi} + \phi(\dot{\rho} + \rho\,\mathrm{div}\,\boldsymbol{v}) = \rho\dot{\phi}$. Slide 158 shows how this avoids computing the Jacobian.

5.4 Balance of linear and angular momentum

Now the motion is referred to a mass, not only to a geometric change.

Definition 5.2 (Linear and angular momentum)**.** The total **linear momentum** and the total **angular momentum** with respect to a fixed point (pole) $\boldsymbol{x}_0$ are

$$\mathbf{L}(t) = \int_\Omega \rho\,\boldsymbol{v}\,\mathrm{d}v = \int_{\Omega_0} \rho_0 \boldsymbol{V}\,\mathrm{d}V, \tag{5.9}$$

$$\mathbf{J}(t) = \int_\Omega \boldsymbol{r}\times\rho\,\boldsymbol{v}\,\mathrm{d}v = \int_{\Omega_0} \boldsymbol{r}\times\rho_0\boldsymbol{V}\,\mathrm{d}V, \qquad \boldsymbol{r} = \boldsymbol{x} - \boldsymbol{x}_0. \tag{5.10}$$

($\mathbf{L}$ and $\mathbf{J}$ are printed upright to distinguish them from the velocity gradient $\boldsymbol{l}$ and the Jacobian J.)

Definition 5.3 (Balance of momentum)**.** The balance principles of linear and angular momentum (Cauchy and Euler's generalisation of Newton's laws) postulate that, for any part of the body,

$$\frac{\mathrm{D}}{\mathrm{D}t}\mathbf{L}(t) = \mathbf{F}(t), \qquad \frac{\mathrm{D}}{\mathrm{D}t}\mathbf{J}(t) = \mathbf{M}(t), \tag{5.11}$$

where $\mathbf{F}$ and $\mathbf{M}$ are the resultant external force and the resultant external moment about $\boldsymbol{x}_0$.

By Eq. (5.8), $\dot{\mathbf{L}} = \int_\Omega \rho\dot{\boldsymbol{v}}\,\mathrm{d}v$ and $\dot{\mathbf{J}} = \int_\Omega \boldsymbol{r}\times\rho\dot{\boldsymbol{v}}\,\mathrm{d}v$ (the term $\dot{\boldsymbol{r}}\times\rho\boldsymbol{v} = \boldsymbol{v}\times\rho\boldsymbol{v}$ vanishes): the external forces equal the inertia forces, the external moments equal the moment of the inertia forces.

Spatial description. For a non-polar continuum (no distributed couples), the external actions are the body forces $\boldsymbol{b}$ (per unit current volume, such as the self weight) and the surface tractions $\boldsymbol{t}$ (per unit current area):

$$\mathbf{F} = \int_{\partial\Omega} \boldsymbol{t}\,\mathrm{d}s + \int_\Omega \boldsymbol{b}\,\mathrm{d}v, \qquad \mathbf{M} = \int_{\partial\Omega} \boldsymbol{r}\times\boldsymbol{t}\,\mathrm{d}s + \int_\Omega \boldsymbol{r}\times\boldsymbol{b}\,\mathrm{d}v. \tag{5.12}$$

(Distributed couples lead to polar, or Cosserat, continua.) Using Cauchy's theorem $\boldsymbol{t} = \boldsymbol{\sigma}\boldsymbol{n}$ and the divergence theorem, $\int_{\partial\Omega}\boldsymbol{\sigma}\boldsymbol{n}\,\mathrm{d}s = \int_\Omega \mathrm{div}\,\boldsymbol{\sigma}\,\mathrm{d}v$, so that the balance of linear momentum becomes

$$\int_\Omega (\mathrm{div}\,\boldsymbol{\sigma} + \boldsymbol{b} - \rho\dot{\boldsymbol{v}})\,\mathrm{d}v = \mathbf{0}.$$

Since this holds for any part of the body:

Cauchy's equations of motion

$$\operatorname{div}\boldsymbol{\sigma} + \boldsymbol{b} = \rho\dot{\boldsymbol{v}}, \qquad \frac{\partial \sigma_{ab}}{\partial x_b} + b_a = \rho \dot{v}_a, \tag{5.13}$$

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\mathrm{T}}. \tag{5.14}$$

Equation (5.13) is *Cauchy's first equation of motion*, with $\dot{\boldsymbol{v}} = \partial \boldsymbol{v}/\partial t + \boldsymbol{l}\boldsymbol{v}$; in the static case ($\dot{\boldsymbol{v}} = \boldsymbol{0}$) it is the familiar equilibrium equation $\operatorname{div}\boldsymbol{\sigma} + \boldsymbol{b} = \boldsymbol{0}$. Equation (5.14), *Cauchy's second equation of motion*, follows from the balance of angular momentum (Holzapfel [3, Sect. 4.3]): locally, the balance of angular momentum is equivalent to the symmetry of the Cauchy stress. Globally, of course, the resultant moment must still be balanced when solving a structural problem.

In class

The lecturer derived the equation in the spatial configuration "because this is exactly the form you learnt in the basic courses", where the spatial and reference configurations coincide. In a polar continuum, with distributed couples, the symmetry of the stress does not hold and becomes part of a constitutive statement.

Material description. Introduce the body force $\boldsymbol{B}$ per unit reference volume and the first Piola–Kirchhoff traction $\boldsymbol{T} = \boldsymbol{P}\boldsymbol{N}$. Since $\boldsymbol{B}\,\mathrm{d}V = \boldsymbol{b}\,\mathrm{d}v$, $\boldsymbol{B} = J\boldsymbol{b}$; since $\boldsymbol{T}\,\mathrm{d}S = \boldsymbol{t}\,\mathrm{d}s$, the force is the same. The global balance laws read

$$\frac{\mathrm{D}}{\mathrm{D}t}\int_{\Omega_0} \rho_0 \boldsymbol{V}\,\mathrm{d}V = \int_{\partial\Omega_0} \boldsymbol{T}\,\mathrm{d}S + \int_{\Omega_0} \boldsymbol{B}\,\mathrm{d}V, \tag{5.15}$$

$$\frac{\mathrm{D}}{\mathrm{D}t}\int_{\Omega_0} \boldsymbol{r}\times\rho_0 \boldsymbol{V}\,\mathrm{d}V = \int_{\partial\Omega_0} \boldsymbol{r}\times\boldsymbol{T}\,\mathrm{d}S + \int_{\Omega_0} \boldsymbol{r}\times\boldsymbol{B}\,\mathrm{d}V. \tag{5.16}$$

To obtain the local form we need the relation between the material and spatial divergence of the stress.

Proposition 5.1 (Piola identity)**.**

$$\operatorname{Div}(J\boldsymbol{F}^{-\mathrm{T}}) = \boldsymbol{0}, \qquad \frac{\partial (JF^{-1}_{Aa})}{\partial X_A} = 0. \tag{5.17}$$

Proof. For any reference region, by the divergence theorem and Nanson's formula,

$$\int_{\Omega_0} \operatorname{Div}(J\boldsymbol{F}^{-\mathrm{T}})\,\mathrm{d}V = \int_{\partial\Omega_0} J\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{N}\,\mathrm{d}S = \int_{\partial\Omega} \boldsymbol{n}\,\mathrm{d}s = \int_{\partial\Omega} \boldsymbol{I}\boldsymbol{n}\,\mathrm{d}s = \int_{\Omega} \operatorname{div}\boldsymbol{I}\,\mathrm{d}v = \boldsymbol{0},$$

and the region is arbitrary. ∎

Then, with $\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}$,

$$\operatorname{Div}\boldsymbol{P} = \operatorname{Grad}\boldsymbol{\sigma} : \boldsymbol{F}^{-\mathrm{T}} J + \boldsymbol{\sigma}\,\operatorname{Div}(J\boldsymbol{F}^{-\mathrm{T}}) = J\operatorname{div}\boldsymbol{\sigma}, \qquad \frac{\partial P_{aA}}{\partial X_A} = J\frac{\partial \sigma_{ab}}{\partial x_b}, \tag{5.18}$$

where the first term uses the chain rule $\partial\sigma_{ab}/\partial X_A = (\partial\sigma_{ab}/\partial x_c)F_{cA}$ and $F_{cA}F^{-1}_{Ab} = \delta_{cb}$. Hence

$$\int_\Omega (\operatorname{div}\boldsymbol{\sigma} + \boldsymbol{b} - \rho\dot{\boldsymbol{v}})\,\mathrm{d}v = \int_{\Omega_0} (\operatorname{div}\boldsymbol{\sigma} + \boldsymbol{b} - \rho\dot{\boldsymbol{v}})J\,\mathrm{d}V = \int_{\Omega_0} (\operatorname{Div}\boldsymbol{P} + \boldsymbol{B} - \rho_0\dot{\boldsymbol{V}})\,\mathrm{d}V = \boldsymbol{0},$$

using $\rho J = \rho_0$ and $\boldsymbol{B} = J\boldsymbol{b}$, and locally:

Equation of motion in material description

$$\operatorname{Div}\boldsymbol{P} + \boldsymbol{B} = \rho_0\dot{\boldsymbol{V}}, \qquad \frac{\partial P_{aA}}{\partial X_A} + B_a = \rho_0\dot{V}_a; \qquad \boldsymbol{P}\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{F}\boldsymbol{P}^{\mathrm{T}}. \tag{5.19}$$

The form is the same as the spatial one, with $\boldsymbol{P}$, $\boldsymbol{B}$, ρ_0 in place of $\boldsymbol{\sigma}$, $\boldsymbol{b}$, ρ. Cauchy's equation of motion is in general nonlinear: the nonlinearities come from geometry (the kinematics of the body: the current configuration is unknown) and from the material ($\boldsymbol{\sigma}$ may depend nonlinearly on the deformation).

Example 5.3 (Slides 165–166)**.** Consider, for $t > 0$, the motion and the Cauchy stress

$$x_1 = e^t X_1 - e^{-t}X_2, \quad x_2 = e^t X_1 + e^{-t}X_2, \quad x_3 = X_3; \qquad [\boldsymbol{\sigma}] = \begin{bmatrix} x_1^2 & \alpha x_2 x_3^2 & 0 \\ \alpha x_2 x_3^2 & x_2^2 & 0 \\ 0 & 0 & \beta x_1^3 \end{bmatrix}.$$

Find the body forces that satisfy the equation of motion and the continuity equation, and the tractions acting on the tangent planes to the spheres $\Phi = x_1^2 + x_2^2 + x_3^2 = \text{const}$.
Kinematics.

$$[\boldsymbol{F}] = \begin{bmatrix} e^t & -e^{-t} & 0 \\ e^t & e^{-t} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad J = 2, \quad \rho = \frac{\rho_0}{2}, \quad \{\boldsymbol{v}\} = \begin{Bmatrix} x_2 \\ x_1 \\ 0 \end{Bmatrix}, \quad \{\dot{\boldsymbol{v}}\} = \begin{Bmatrix} x_1 \\ x_2 \\ 0 \end{Bmatrix},$$

where $v_1 = e^t X_1 + e^{-t}X_2 = x_2$, $v_2 = e^t X_1 - e^{-t}X_2 = x_1$, and $\dot{v}_1 = v_2 = x_1$, $\dot{v}_2 = v_1 = x_2$.
Body forces. From $\boldsymbol{b} = \rho\dot{\boldsymbol{v}} - \operatorname{div}\boldsymbol{\sigma}$, with $\operatorname{div}\boldsymbol{\sigma} = \{2x_1 + \alpha x_3^2,\ 2x_2,\ 0\}^{\mathrm{T}}$,

$$\{\boldsymbol{b}\} = \frac{\rho_0}{2}\begin{Bmatrix} x_1 \\ x_2 \\ 0 \end{Bmatrix} - \begin{Bmatrix} 2x_1 + \alpha x_3^2 \\ 2x_2 \\ 0 \end{Bmatrix}.$$

Tractions. The unit normal to the sphere is $\boldsymbol{n} = \operatorname{grad}\Phi/\left|\operatorname{grad}\Phi\right| = \boldsymbol{x}/\left|\boldsymbol{x}\right|$, and

$$\{\boldsymbol{t}\} = [\boldsymbol{\sigma}]\,\{\boldsymbol{n}\} = \frac{1}{|\boldsymbol{x}|}\begin{Bmatrix} x_1^3 + \alpha x_2^2 x_3^2 \\ \alpha x_1 x_2 x_3^2 + x_2^3 \\ \beta x_1^3 x_3 \end{Bmatrix}.$$

(The third component on the slide reads $\beta x_1^2 x_3$; the product $\sigma_{33}n_3$ gives $\beta x_1^3 x_3/\left|\boldsymbol{x}\right|$.)

5.5 Balance of mechanical energy

In class

In Italian this is the *teorema delle potenze* (theorem of power). Only mechanical power is considered here; thermal power will be added with thermodynamics, while electric, magnetic, chemical and nuclear energies are excluded. The balance of mechanical energy is a consequence of the balance of linear momentum.

Consider a body loaded by tractions $\boldsymbol{t}$ on the boundary and body forces $\boldsymbol{b}$, and in equilibrium with the stress $\boldsymbol{\sigma}$ (i.e., satisfying Eq. (5.13)). The power of a force is obtained by multiplying it by a velocity. Define:

1. the *external mechanical power*

$$\mathcal{P}_{\text{ext}}(t) = \int_{\partial\Omega} \boldsymbol{t}\cdot\boldsymbol{v}\,\mathrm{d}s + \int_{\Omega} \boldsymbol{b}\cdot\boldsymbol{v}\,\mathrm{d}v; \tag{5.20}$$

2. the *kinetic energy*

$$\mathcal{K}(t) = \int_{\Omega} \tfrac{1}{2}\rho\,\boldsymbol{v}\cdot\boldsymbol{v}\,\mathrm{d}v; \tag{5.21}$$

3. the *stress power* (rate of internal mechanical work)

$$\mathcal{P}_{\text{int}}(t) = \int_{\Omega} \boldsymbol{\sigma} : \boldsymbol{d}\,\mathrm{d}v. \tag{5.22}$$

In the stress power, $\boldsymbol{\sigma}:\boldsymbol{l} = \boldsymbol{\sigma}:(\boldsymbol{d}+\boldsymbol{w}) = \boldsymbol{\sigma}:\boldsymbol{d}$, because the contraction of a symmetric with a skew tensor vanishes (Section 1.10).

Theorem 5.2 (Theorem of power expended)**.** *For any velocity field compatible with the balance of linear momentum,*

$$\boxed{\frac{\mathrm{D}\mathcal{K}}{\mathrm{D}t} + \mathcal{P}_{\text{int}} = \mathcal{P}_{\text{ext}}}, \qquad \frac{\mathrm{D}}{\mathrm{D}t}\int_{\Omega} \tfrac{1}{2}\rho\,\boldsymbol{v}\cdot\boldsymbol{v}\,\mathrm{d}v + \int_{\Omega} \boldsymbol{\sigma}:\boldsymbol{d}\,\mathrm{d}v = \int_{\partial\Omega} \boldsymbol{t}\cdot\boldsymbol{v}\,\mathrm{d}s + \int_{\Omega} \boldsymbol{b}\cdot\boldsymbol{v}\,\mathrm{d}v. \tag{5.23}$$

Proof. By Cauchy's theorem and the divergence theorem,

$$\int_{\partial\Omega} \boldsymbol{t}\cdot\boldsymbol{v}\,\mathrm{d}s = \int_{\partial\Omega} \boldsymbol{\sigma}\boldsymbol{n}\cdot\boldsymbol{v}\,\mathrm{d}s = \int_{\partial\Omega} \boldsymbol{\sigma}^{\mathrm{T}}\boldsymbol{v}\cdot\boldsymbol{n}\,\mathrm{d}s = \int_{\Omega} \operatorname{div}(\boldsymbol{\sigma}\boldsymbol{v})\,\mathrm{d}v = \int_{\Omega} \left(\operatorname{div}\boldsymbol{\sigma}\cdot\boldsymbol{v} + \boldsymbol{\sigma}:\boldsymbol{l}\right)\mathrm{d}v,$$

where $\boldsymbol{\sigma}^{\mathrm{T}} = \boldsymbol{\sigma}$ and, in index notation, $(\sigma_{ab}v_a)_{,b} = \sigma_{ab,b}v_a + \sigma_{ab}v_{a,b}$. By Eq. (5.8), $\dot{\mathcal{K}} = \int_{\Omega}\rho\dot{\boldsymbol{v}}\cdot\boldsymbol{v}\,\mathrm{d}v$. Hence

$$\mathcal{P}_{\text{ext}} - \dot{\mathcal{K}} = \int_{\Omega} \left(\operatorname{div}\boldsymbol{\sigma} + \boldsymbol{b} - \rho\dot{\boldsymbol{v}}\right)\cdot\boldsymbol{v}\,\mathrm{d}v + \int_{\Omega} \boldsymbol{\sigma}:\boldsymbol{d}\,\mathrm{d}v = \mathcal{P}_{\text{int}},$$

since the first integral vanishes by the balance of linear momentum. ∎

In class

The trick of the proof on slide 169 is to add and subtract the rate of kinetic energy: $\mathcal{P}_{\text{ext}} + \dot{\mathcal{K}} - \dot{\mathcal{K}} = \mathcal{P}_{\text{int}} + \dot{\mathcal{K}}$. The terms that remain reproduce the linear momentum balance, and the statement becomes an identity. The velocity field does not have to be the actual velocity of the motion: it can be any (virtual) velocity field superposed on the body, and from this observation one obtains the principle of virtual power (and virtual work) used in the finite element method.

The power of the external forces is spent partly in changing the kinetic energy and partly in the stress power. Consequences:

- when the external power is zero, the mechanical energy balance describes free vibrations;
- when the rate of kinetic energy is zero, the problem is *quasi-static*, and $\mathcal{P}_{\text{int}} = \mathcal{P}_{\text{ext}}$;
- when the stress power is zero, the body undergoes a rigid-body motion ($\boldsymbol{d} = \boldsymbol{O}$);
- the theorem holds for any velocity field, for any magnitude of the actual displacements, and for any material type, since no constitutive assumption has been made (the slide specifies non-dissipative materials because thermal effects are excluded here).

Material description. Using $\boldsymbol{t}\,\mathrm{d}s = \boldsymbol{T}\,\mathrm{d}S$, $\boldsymbol{v} = \boldsymbol{V}$, $J\boldsymbol{b} = \boldsymbol{B}$, $\mathrm{d}v = J\,\mathrm{d}V$ and $\rho\,\mathrm{d}v = \rho_0\,\mathrm{d}V$, and, for the stress power, the symmetry of $\boldsymbol{\sigma}$ and Eq. (3.67),

$$\int_\Omega \boldsymbol{\sigma} : \boldsymbol{d}\,\mathrm{d}v = \int_{\Omega_0} J\boldsymbol{\sigma} : \boldsymbol{l}\,\mathrm{d}V = \int_{\Omega_0} J\boldsymbol{\sigma} : \dot{\boldsymbol{F}}\boldsymbol{F}^{-1}\,\mathrm{d}V = \int_{\Omega_0} J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}} : \dot{\boldsymbol{F}}\,\mathrm{d}V = \int_{\Omega_0} \boldsymbol{P} : \dot{\boldsymbol{F}}\,\mathrm{d}V,$$

the theorem becomes

Theorem of power expended (material description)

$$\frac{\mathrm{D}}{\mathrm{D}t}\int_{\Omega_0} \tfrac{1}{2}\rho_0 \boldsymbol{V}\cdot\boldsymbol{V}\,\mathrm{d}V + \int_{\Omega_0} \boldsymbol{P} : \dot{\boldsymbol{F}}\,\mathrm{d}V = \int_{\partial\Omega_0} \boldsymbol{T}\cdot\boldsymbol{V}\,\mathrm{d}S + \int_{\Omega_0} \boldsymbol{B}\cdot\boldsymbol{V}\,\mathrm{d}V. \tag{5.24}$$

The stress power per unit reference volume is $J\boldsymbol{\sigma} : \boldsymbol{d} = \boldsymbol{P} : \dot{\boldsymbol{F}}$: $\boldsymbol{P}$ is *work conjugate* to $\boldsymbol{F}$. Similarly, using $\dot{\boldsymbol{E}} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{d}\boldsymbol{F}$ (Eq. (3.72)), $J\boldsymbol{\sigma} : \boldsymbol{d} = \boldsymbol{\tau} : \boldsymbol{d} = \boldsymbol{S} : \dot{\boldsymbol{E}}$. These work-conjugate stress–strain pairs are studied systematically in Section 5.6.

In class

The third day ended here (slide 171). The lecturer was "too tired" to treat the conjugate stress–strain pairs properly, and postponed them to the next lecture (from slide 172), followed by thermodynamics, heat, and the constitutive relations (objectivity, hyperelasticity). She also recommended, as exercises, deriving the material form of the linear momentum balance and of the theorem of power expended step by step.

5.6 Work-conjugate stress–strain pairs

In class

The fourth day opened by completing the material form of the theorem of power expended (left as an exercise at the end of the previous lecture) and by deriving, on the blackboard and in index notation, the work conjugate pairs of slide 172.

The stress power can be written with any of the stress tensors of Chapter 4, provided each is paired with the proper rate:

Work-conjugate pairs

$$\mathcal{P}_{\text{int}}(t) = \int_{\Omega} \boldsymbol{\sigma} : \boldsymbol{d}\, \mathrm{d}v = \int_{\Omega_0} J\boldsymbol{\sigma} : \boldsymbol{d}\, \mathrm{d}V = \int_{\Omega_0} \boldsymbol{\tau} : \boldsymbol{d}\, \mathrm{d}V = \int_{\Omega_0} \boldsymbol{P} : \dot{\boldsymbol{F}}\, \mathrm{d}V = \int_{\Omega_0} \boldsymbol{S} : \dot{\boldsymbol{E}}\, \mathrm{d}V. \quad (5.25)$$

$(\boldsymbol{\sigma}, \boldsymbol{d})$, $(\boldsymbol{\tau}, \boldsymbol{d})$, $(\boldsymbol{P}, \dot{\boldsymbol{F}})$ and $(\boldsymbol{S}, \dot{\boldsymbol{E}})$ are *work-conjugate* (more precisely power-conjugate) pairs: their double contraction gives the stress power, per unit current volume for the first one and per unit reference volume for the others.

Proof. The first equalities follow from $\mathrm{d}v = J\,\mathrm{d}V$ and $\boldsymbol{\tau} = J\boldsymbol{\sigma}$. For $\boldsymbol{P}$: since $\boldsymbol{\sigma}$ is symmetric, $\boldsymbol{\sigma} : \boldsymbol{w} = 0$ and $\boldsymbol{\sigma} : \boldsymbol{d} = \boldsymbol{\sigma} : \boldsymbol{l}$; with $\boldsymbol{l} = \dot{\boldsymbol{F}}\boldsymbol{F}^{-1}$, in index notation,

$$J\sigma_{ab}\, \dot{F}_{aA} F^{-1}_{Ab} = (J\sigma_{ab} F^{-1}_{Ab}) \dot{F}_{aA} = P_{aA} \dot{F}_{aA} = \boldsymbol{P} : \dot{\boldsymbol{F}}.$$

For $\boldsymbol{S}$: from Eq. (3.72), $\dot{\boldsymbol{E}} = \boldsymbol{F}^{\mathrm{T}} \boldsymbol{d} \boldsymbol{F}$, so $\boldsymbol{d} = \boldsymbol{F}^{-\mathrm{T}} \dot{\boldsymbol{E}} \boldsymbol{F}^{-1}$ (the push-forward of $\dot{\boldsymbol{E}}$), and $J\boldsymbol{\sigma} = \boldsymbol{F}\boldsymbol{S}\boldsymbol{F}^{\mathrm{T}}$ (the push-forward of $\boldsymbol{S}$). Then

$$J\boldsymbol{\sigma} : \boldsymbol{d} = F_{aA} S_{AB} F_{bB}\, F^{-1}_{Ca} \dot{E}_{CD} F^{-1}_{Db} = \delta_{AC} \delta_{BD} S_{AB} \dot{E}_{CD} = S_{AB} \dot{E}_{AB} = \boldsymbol{S} : \dot{\boldsymbol{E}},$$

since $F_{aA} F^{-1}_{Ca} = \delta_{AC}$ and $F_{bB} F^{-1}_{Db} = \delta_{BD}$. In compact form, with $\boldsymbol{A} : (\boldsymbol{B}\boldsymbol{C}) = (\boldsymbol{B}^{\mathrm{T}}\boldsymbol{A}) : \boldsymbol{C} = (\boldsymbol{A}\boldsymbol{C}^{\mathrm{T}}) : \boldsymbol{B}$: $\boldsymbol{F}\boldsymbol{S}\boldsymbol{F}^{\mathrm{T}} : \boldsymbol{F}^{-\mathrm{T}} \dot{\boldsymbol{E}} \boldsymbol{F}^{-1} = \boldsymbol{S} : \dot{\boldsymbol{E}}$. ■

Note the general structure: a contravariant stress (push-forward or pull-back of $\boldsymbol{\tau}$) contracted with a covariant strain rate (pull-back or push-forward of $\boldsymbol{d}$) gives an invariant (Section 3.12).

Rotated quantities. The stress power is also invariant under a change of the reference system. With the rotation $\boldsymbol{R}$ of the polar decomposition, define the rotated stress and rate of deformation $\boldsymbol{\sigma}_r = \boldsymbol{R}^{\mathrm{T}} \boldsymbol{\sigma} \boldsymbol{R}$, $\boldsymbol{d}_r = \boldsymbol{R}^{\mathrm{T}} \boldsymbol{d} \boldsymbol{R}$; then $\boldsymbol{\sigma} : \boldsymbol{d} = \boldsymbol{\sigma}_r : \boldsymbol{d}_r$ and $\mathcal{P}_{\text{int}} = \int_{\Omega_0} J\boldsymbol{\sigma}_r : \boldsymbol{d}_r\, \mathrm{d}V$. Other conjugate pairs (e.g., the Biot stress, conjugate to $\dot{\boldsymbol{U}}$) are discussed in Holzapfel [3, Sect. 4.5].

> **In class**
>
> The table of conjugate pairs is the key to the constitutive relations: when we write a constitutive law, the stress measure must be associated with its own strain measure. In small strain the pair is $(\boldsymbol{\sigma}, \boldsymbol{\varepsilon})$ for work, and $(\boldsymbol{\sigma}, \dot{\boldsymbol{\varepsilon}})$ for power.

5.7 Principle of virtual work

> **In class**
>
> The principle of virtual work (slides 173–181) was not developed in class, because it is proved in the undergraduate courses and "it would take at least one hour". The lecturer stressed that it is the weak form of the linear momentum balance, that it is the form actually used to solve problems (finite elements), and that it uses the same conjugate pairs, with the rates replaced by virtual (incremental) quantities. This section develops it following the slides and Holzapfel [3, Sect. 8.1].

5.7.1 Virtual displacements and variations

At time t the current configuration is described by the displacement $\boldsymbol{u}(\boldsymbol{X},t) = \boldsymbol{x} - \boldsymbol{X}$. Consider a neighbouring displacement field $\boldsymbol{u} + \varepsilon\,\delta\boldsymbol{u}$, with ε a small parameter. The field $\delta\boldsymbol{u}$ is the **virtual displacement** (or first variation of $\boldsymbol{u}$): it is arbitrary, infinitesimal, *virtual* (it is not an actual change, it is performed at a fixed instant of time), and independent of $\boldsymbol{u}$. It can be expressed in material or spatial coordinates, $\delta\boldsymbol{u}(\boldsymbol{X}) = \delta\boldsymbol{u}(\boldsymbol{\chi}^{-1}(\boldsymbol{x}))$. Its properties are:

1. the variation of a derivative is the derivative of the variation, $\delta(\partial\boldsymbol{u}/\partial\boldsymbol{X}) = \partial(\delta\boldsymbol{u})/\partial\boldsymbol{X}$;
2. variation and definite integration commute;
3. the spatial and material gradients are related by $\operatorname{grad}\delta\boldsymbol{u} = \operatorname{Grad}\delta\boldsymbol{u}\,\boldsymbol{F}^{-1}$.

Definition 5.4 (First variation)**.** In the material description, the first variation of a smooth function $\mathcal{F}(\boldsymbol{u})$ is its Gateaux (directional) derivative in the direction $\delta\boldsymbol{u}$:

$$\delta\mathcal{F}(\boldsymbol{u},\delta\boldsymbol{u}) = D_{\delta\boldsymbol{u}}\mathcal{F}(\boldsymbol{u}) = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\mathcal{F}(\boldsymbol{u}+\varepsilon\delta\boldsymbol{u})\Big|_{\varepsilon=0}. \tag{5.26}$$

This is exactly the operation used to compute $\dot{\boldsymbol{F}}$ as a directional derivative in Section 3.16, with $\delta\boldsymbol{u}$ in place of $\boldsymbol{v}$. Hence

$$\delta\boldsymbol{F} = \operatorname{Grad}\delta\boldsymbol{u}, \qquad \delta F_{aA} = \frac{\partial\delta u_a}{\partial X_A}, \tag{5.27}$$

$$\delta\boldsymbol{F}^{-1} = -\boldsymbol{F}^{-1}\delta\boldsymbol{F}\boldsymbol{F}^{-1} = -\boldsymbol{F}^{-1}\operatorname{grad}\delta\boldsymbol{u}, \qquad \delta F^{-1}_{Aa} = -F^{-1}_{Ab}\frac{\partial\delta u_b}{\partial x_a}, \tag{5.28}$$

$$\delta\boldsymbol{E} = \tfrac{1}{2}(\delta\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} + \boldsymbol{F}^{\mathrm{T}}\delta\boldsymbol{F}) = \operatorname{sym}(\boldsymbol{F}^{\mathrm{T}}\operatorname{Grad}\delta\boldsymbol{u}). \tag{5.29}$$

In the spatial description the variation is computed as a Lie derivative (Section 3.16): pull back to the reference configuration, take the variation, push forward. For the Euler–Almansi strain, $\boldsymbol{\chi}_*^{-1}(\boldsymbol{e}) = \boldsymbol{E}$ and

$$\delta\boldsymbol{e} = \boldsymbol{F}^{-\mathrm{T}}\delta\boldsymbol{E}\,\boldsymbol{F}^{-1} = \operatorname{sym}(\operatorname{Grad}\delta\boldsymbol{u}\,\boldsymbol{F}^{-1}) = \operatorname{sym}(\operatorname{grad}\delta\boldsymbol{u}), \tag{5.30}$$

the spatial counterpart of $\boldsymbol{d} = \operatorname{sym}\operatorname{grad}\boldsymbol{v}$. For a scalar, $\delta f = \delta\mathcal{F}$.

5.7.2 Boundary value problem

The body occupies Ω at time t; its boundary is split into a *constrained* part $\partial\Omega_u$, where the displacement is assigned, $\boldsymbol{u} = \bar{\boldsymbol{u}}$ (Dirichlet, or essential, conditions), and a *loaded* part $\partial\Omega_\sigma$, where the traction is assigned, $\boldsymbol{t} = \boldsymbol{\sigma}\boldsymbol{n} = \bar{\boldsymbol{t}}$ (Neumann, or natural, conditions), with $\partial\Omega_u \cup \partial\Omega_\sigma = \partial\Omega$ and $\partial\Omega_u \cap \partial\Omega_\sigma = \emptyset$. Under body forces $\boldsymbol{b}$ and tractions $\bar{\boldsymbol{t}}$ the motion satisfies Cauchy's equation $\operatorname{div}\boldsymbol{\sigma} + \boldsymbol{b} = \rho\dot{\boldsymbol{v}}$ in Ω. An *admissible* variation (test function) $\boldsymbol{\eta}$ is a smooth vector field on Ω that vanishes on $\partial\Omega_u$.

Theorem 5.3 (Principle of virtual work, spatial form)**.** *A stress field $\boldsymbol{\sigma}$ satisfies the equation of motion with body forces $\boldsymbol{b}$ and prescribed tractions $\bar{\boldsymbol{t}}$ if and only if*

$$\int_\Omega [\boldsymbol{\sigma} : \operatorname{grad}\boldsymbol{\eta} - (\boldsymbol{b} - \rho\dot{\boldsymbol{v}})\cdot\boldsymbol{\eta}]\,\mathrm{d}v - \int_{\partial\Omega_\sigma}\bar{\boldsymbol{t}}\cdot\boldsymbol{\eta}\,\mathrm{d}s = 0 \qquad \forall\,\boldsymbol{\eta}\ \textit{admissible}. \tag{5.31}$$

Proof. Multiply Cauchy's equation by $\boldsymbol{\eta}$ and integrate: $\int_\Omega(-\operatorname{div}\boldsymbol{\sigma}-\boldsymbol{b}+\rho\dot{\boldsymbol{v}})\cdot\boldsymbol{\eta}\,\mathrm{d}v=0$. With the identity $\operatorname{div}\boldsymbol{\sigma}\cdot\boldsymbol{\eta}=\operatorname{div}(\boldsymbol{\sigma}\boldsymbol{\eta})-\boldsymbol{\sigma}:\operatorname{grad}\boldsymbol{\eta}$ (using $\boldsymbol{\sigma}=\boldsymbol{\sigma}^{\mathrm{T}}$) and the divergence theorem,

$$\int_\Omega\left[\boldsymbol{\sigma}:\operatorname{grad}\boldsymbol{\eta}-(\boldsymbol{b}-\rho\dot{\boldsymbol{v}})\cdot\boldsymbol{\eta}\right]\mathrm{d}v-\int_{\partial\Omega}\boldsymbol{\sigma}\boldsymbol{n}\cdot\boldsymbol{\eta}\,\mathrm{d}s=0,$$

and the boundary integral reduces to $\partial\Omega_\sigma$ because $\boldsymbol{\eta}=\boldsymbol{0}$ on $\partial\Omega_u$. Conversely, if Eq. (5.31) holds for all admissible $\boldsymbol{\eta}$, reversing the steps gives $\int_\Omega(\operatorname{div}\boldsymbol{\sigma}+\boldsymbol{b}-\rho\dot{\boldsymbol{v}})\cdot\boldsymbol{\eta}\,\mathrm{d}v-\int_{\partial\Omega_\sigma}(\boldsymbol{\sigma}\boldsymbol{n}-\bar{\boldsymbol{t}})\cdot\boldsymbol{\eta}\,\mathrm{d}s=0$ for all $\boldsymbol{\eta}$; choosing $\boldsymbol{\eta}$ with support inside Ω gives the local equation, and then the traction condition follows. ∎

Equation (5.31) is the *weak form* of the initial boundary value problem, and the equation of motion is its Euler–Lagrange equation. Essential conditions are built into the space of admissible functions, natural conditions appear in the weak form. Choosing $\boldsymbol{\eta}=\delta\boldsymbol{u}$ and using the symmetry of $\boldsymbol{\sigma}$, $\boldsymbol{\sigma}:\operatorname{grad}\delta\boldsymbol{u}=\boldsymbol{\sigma}:\delta\boldsymbol{e}$:

$$\underbrace{\int_\Omega\boldsymbol{\sigma}:\delta\boldsymbol{e}\,\mathrm{d}v}_{\delta W_{\mathrm{int}}}=\underbrace{\int_\Omega(\boldsymbol{b}-\rho\dot{\boldsymbol{v}})\cdot\delta\boldsymbol{u}\,\mathrm{d}v+\int_{\partial\Omega_\sigma}\bar{\boldsymbol{t}}\cdot\delta\boldsymbol{u}\,\mathrm{d}s}_{\delta W_{\mathrm{ext}}}:\tag{5.32}$$

the internal virtual work equals the external virtual work (including the inertia forces), the analogue of $\mathcal{P}_{\mathrm{int}}+\dot{\mathcal{K}}=\mathcal{P}_{\mathrm{ext}}$ with velocities replaced by virtual displacements.

Theorem 5.4 (Principle of virtual work, material form)**.** *A stress field $\boldsymbol{P}$ satisfies $\operatorname{Div}\boldsymbol{P}+\boldsymbol{B}=\rho_0\dot{\boldsymbol{V}}$ with prescribed nominal tractions $\bar{\boldsymbol{T}}$ on $\partial\Omega_{0\sigma}$ if and only if*

$$\int_{\Omega_0}\left[\boldsymbol{P}:\operatorname{Grad}\boldsymbol{\eta}-(\boldsymbol{B}-\rho_0\dot{\boldsymbol{V}})\cdot\boldsymbol{\eta}\right]\mathrm{d}V-\int_{\partial\Omega_{0\sigma}}\bar{\boldsymbol{T}}\cdot\boldsymbol{\eta}\,\mathrm{d}S=0\qquad\forall\,\boldsymbol{\eta}\ \textit{admissible}.\tag{5.33}$$

The proof is identical, with $\operatorname{Div}\boldsymbol{P}\cdot\boldsymbol{\eta}=\operatorname{Div}(\boldsymbol{P}^{\mathrm{T}}\boldsymbol{\eta})-\boldsymbol{P}:\operatorname{Grad}\boldsymbol{\eta}$. With $\boldsymbol{\eta}=\delta\boldsymbol{u}$, $\boldsymbol{P}:\operatorname{Grad}\delta\boldsymbol{u}=\boldsymbol{P}:\delta\boldsymbol{F}=\boldsymbol{S}:\delta\boldsymbol{E}$:

$$\int_{\Omega_0}\boldsymbol{S}:\delta\boldsymbol{E}\,\mathrm{d}V=\int_{\Omega_0}\boldsymbol{P}:\delta\boldsymbol{F}\,\mathrm{d}V=\int_{\Omega_0}(\boldsymbol{B}-\rho_0\dot{\boldsymbol{V}})\cdot\delta\boldsymbol{u}\,\mathrm{d}V+\int_{\partial\Omega_{0\sigma}}\bar{\boldsymbol{T}}\cdot\delta\boldsymbol{u}\,\mathrm{d}S.\tag{5.34}$$

The pairs $(\boldsymbol{\sigma},\delta\boldsymbol{e})$, $(\boldsymbol{P},\delta\boldsymbol{F})$ and $(\boldsymbol{S},\delta\boldsymbol{E})$ are the virtual counterparts of the power-conjugate pairs.

5.8 Continuum thermodynamics: heat and internal energy

In class

Kinematics (six or nine strain components related to three displacement components) and balance of linear and angular momentum (three equations for six or nine stress components) do not give enough equations: six (symmetric formulation, $\boldsymbol{\sigma}$–$\boldsymbol{\varepsilon}$ or $\boldsymbol{S}$–$\boldsymbol{E}$) or nine (non-symmetric, $\boldsymbol{P}$–$\boldsymbol{F}$) equations are missing. They are the *constitutive equations*, which cannot be derived from theory alone and must refer to experimental tests. A purely phenomenological fit of the data of one experiment often fails to predict a different experiment; constitutive equations should instead be built on strong physical principles, which account for the energy stored and dissipated by the material. This is why we need thermodynamics.

Mechanics alone cannot describe all processes when thermal and mechanical energy interact. A thermodynamic *equilibrium state* is a state of balance with *uniform temperature* and *zero (average) velocity*; thermodynamics studies equilibrium and non-equilibrium states. It uses:

- *state variables*: quantities characterising the state of the system (e.g., six strain components and the temperature);
- *functions of state*: functions of the state variables;
- *equations of state* (constitutive equations), relating the state functions.

The mean velocity field $\boldsymbol{v}$ is the average velocity of the particles in the neighbourhood of a point. It does not account for the thermal fluctuations of the particles (atoms) around their mean motion, which carry a non-negligible kinetic energy, known as *heat*. Heat is the form of (thermal) energy transferred between a system and its surroundings by virtue of a temperature gradient; the word is often used with the meaning of heat transfer.

5.8.1 Thermal power and heat flux

Definition 5.5 (Thermal power)**.** The **thermal power** (rate of thermal work) entering a body is

$$\mathcal{Q}(t) = \int_{\partial\Omega} q_n \,\mathrm{d}s + \int_{\Omega} r \,\mathrm{d}v = \int_{\partial\Omega_0} Q_N \,\mathrm{d}S + \int_{\Omega_0} R \,\mathrm{d}V, \tag{5.35}$$

where q_n and Q_N are the heat fluxes per unit current or reference area and unit time entering the body across its boundary, and r, R the heat sources per unit current or reference volume and unit time (heat generated inside, e.g., by radiation).

Theorem 5.5 (Stokes' heat flux theorem)**.** *The scalar fluxes are linear functions of the outward normal: there exist vector fields $\boldsymbol{q}(\boldsymbol{x},t)$ and $\boldsymbol{Q}(\boldsymbol{X},t)$ such that*

$$q_n(\boldsymbol{x},t,\boldsymbol{n}) = -\boldsymbol{q}(\boldsymbol{x},t)\cdot\boldsymbol{n}, \qquad Q_N(\boldsymbol{X},t,\boldsymbol{N}) = -\boldsymbol{Q}(\boldsymbol{X},t)\cdot\boldsymbol{N}. \tag{5.36}$$

This is the thermal counterpart of Cauchy's stress theorem; the minus sign makes $q_n > 0$ when the heat enters the body (the normal is outward). $\boldsymbol{q}$ is the *Cauchy (true) heat flux*, $\boldsymbol{Q}$ the *Piola–Kirchhoff (nominal) heat flux*; both are contravariant vectors. ($\boldsymbol{Q}$ here is a vector, not the orthogonal tensor of Chapter 6.) Since the heat crossing a surface element is the same, Nanson's formula gives

$$\int_{\partial\Omega_0} \boldsymbol{Q}\cdot\boldsymbol{N}\,\mathrm{d}S = \int_{\partial\Omega} \boldsymbol{q}\cdot\boldsymbol{n}\,\mathrm{d}s = \int_{\partial\Omega_0} \boldsymbol{q}\cdot J\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{N}\,\mathrm{d}S = \int_{\partial\Omega_0} J\boldsymbol{F}^{-1}\boldsymbol{q}\cdot\boldsymbol{N}\,\mathrm{d}S,$$

Piola–Kirchhoff heat flux

$$\boldsymbol{Q} = J\boldsymbol{F}^{-1}\boldsymbol{q}, \qquad Q_A = JF^{-1}_{Aa}q_a, \tag{5.37}$$

a Piola transformation (pull-back of the contravariant vector $J\boldsymbol{q}$). By the divergence theorem,

$$\mathcal{Q} = -\int_{\Omega} (\operatorname{div}\boldsymbol{q} - r)\,\mathrm{d}v = -\int_{\Omega_0} (\operatorname{Div}\boldsymbol{Q} - R)\,\mathrm{d}V. \tag{5.38}$$

5.8.2 Internal energy

Empirically, when a part $\mathcal{A}$ of the body is taken from a state 1 to a state 2 along a path Γ in the space of states (a *process*), the integral of the sum of stress power and thermal power,

$$\int_\Gamma [\mathcal{P}_{\text{int}}(\mathcal{A},t) + \mathcal{Q}(\mathcal{A},t)]\,\mathrm{d}t,$$

is independent of the path, and vanishes on a closed cycle. This suggests the existence of a function of state, the **internal energy** $\mathcal{E}$, defined up to an inessential additive constant through its rate:

$$\frac{\mathrm{D}\mathcal{E}}{\mathrm{D}t} = \mathcal{P}_{\text{int}} + \mathcal{Q}. \tag{5.39}$$

The internal energy accounts for all the microscopic forms of energy of the system. It is an extensive quantity, defined through densities per unit current volume $e_c(\boldsymbol{x},t)$ or reference volume $e(\boldsymbol{X},t)$:

$$\mathcal{E} = \int_\Omega e_c\,\mathrm{d}v = \int_{\Omega_0} e\,\mathrm{d}V, \qquad e(\boldsymbol{X},t) = J\,e_c(\boldsymbol{\chi}(\boldsymbol{X},t),t). \tag{5.40}$$

Local forms. In the spatial description, Eq. (5.39) reads $\frac{\mathrm{D}}{\mathrm{D}t}\int_\Omega e_c\,\mathrm{d}v = \int_\Omega(\boldsymbol{\sigma}:\boldsymbol{d} - \operatorname{div}\boldsymbol{q} + r)\,\mathrm{d}v$; with Reynolds' theorem Eq. (5.7) and the arbitrariness of the part,

$$\dot{e}_c + e_c \operatorname{div}\boldsymbol{v} = \frac{\partial e_c}{\partial t} + \operatorname{div}(e_c\boldsymbol{v}) = \boldsymbol{\sigma}:\boldsymbol{d} - \operatorname{div}\boldsymbol{q} + r. \tag{5.41}$$

In the material description the domain is fixed, and

Internal energy balance (local, material)

$$\dot{e} = \boldsymbol{P}:\dot{\boldsymbol{F}} - \operatorname{Div}\boldsymbol{Q} + R, \qquad \dot{e} = P_{aA}\dot{F}_{aA} - \frac{\partial Q_A}{\partial X_A} + R. \tag{5.42}$$

5.8.3 Balance of energy: the first law

Using Eq. (5.39) to write $\mathcal{P}_{\text{int}} = \dot{\mathcal{E}} - \mathcal{Q}$ in the theorem of power expended Eq. (5.23) gives

First law of thermodynamics

$$\frac{\mathrm{D}\mathcal{K}}{\mathrm{D}t} + \frac{\mathrm{D}\mathcal{E}}{\mathrm{D}t} = \mathcal{P}_{\text{ext}} + \mathcal{Q}. \tag{5.43}$$

The external mechanical power and the thermal power supplied to any part of the body are exactly converted into kinetic energy or internal energy: this is the *first law of thermodynamics*, or balance of energy. In global form,

$$\frac{\mathrm{D}}{\mathrm{D}t}\int_\Omega \left(\tfrac{1}{2}\rho\boldsymbol{v}\cdot\boldsymbol{v} + e_c\right)\mathrm{d}v = \int_{\partial\Omega}(\boldsymbol{t}\cdot\boldsymbol{v} + q_n)\,\mathrm{d}s + \int_\Omega(\boldsymbol{b}\cdot\boldsymbol{v} + r)\,\mathrm{d}v, \tag{5.44}$$

$$\frac{\mathrm{D}}{\mathrm{D}t}\int_{\Omega_0} \left(\tfrac{1}{2}\rho_0\boldsymbol{V}\cdot\boldsymbol{V} + e\right)\mathrm{d}V = \int_{\partial\Omega_0}(\boldsymbol{T}\cdot\boldsymbol{V} + Q_N)\,\mathrm{d}S + \int_{\Omega_0}(\boldsymbol{B}\cdot\boldsymbol{V} + R)\,\mathrm{d}V. \tag{5.45}$$

The local forms coincide with the internal energy balance above (Exercise 5.6).

In class

The first law gives the heat "the same dignity" as the external power: both can increase the internal energy or change the kinetic energy. But it does not tell the whole story: heat can only flow from hotter to colder regions, while external power can be reversed at will. Mechanical work can be completely transformed into heat (by friction), not vice versa. This irreversibility is the content of the second law.

Mechanical energy only. If no heat is involved, $\dot{\mathcal{E}} = \mathcal{P}_{\text{int}}$ and $\dot{\mathcal{K}} + \dot{\mathcal{E}} = \mathcal{P}_{\text{ext}}$: the total mechanical energy (kinetic plus internal) changes only because of the power of the external forces.

Conservative systems. Define the total potential energy $\Pi = \Pi_{\text{int}} + \Pi_{\text{ext}}$ as the sum of the strain energy and the potential energy of the loads. A mechanical system is *conservative* if both powers derive from potentials, $\mathcal{P}_{\text{int}} = \dot{\Pi}_{\text{int}}$, $\mathcal{P}_{\text{ext}} = -\dot{\Pi}_{\text{ext}}$; then

$$\Pi_{\text{int}} + \Pi_{\text{ext}} + \mathcal{K} = \text{const.} \tag{5.46}$$

Friction, viscosity and plasticity dissipate energy into non-mechanical forms and make the system non-conservative.

Remark 5.1 (Open systems). Slides on the energy and momentum balance of open systems (control volumes) were skipped: as for the mass (Section 5.2), the balance of a fixed control volume contains additional flux terms of the transported quantities across the control surface (Reynolds' theorem). They are not needed for solids.

5.9 Temperature, entropy and the second law

5.9.1 Zeroth law and empirical temperature

A system has *intensive* properties (stress, strain, temperature), which do not change if the system is doubled, and *extensive* properties (mass, internal energy, entropy), which double. A system is in uniform thermodynamic equilibrium if all its properties are independent of time and the intensive ones also of position. Two systems are in *thermal equilibrium* with each other if they remain in thermodynamic equilibrium when placed in full interaction.

Definition 5.6 (Zeroth law of thermodynamics)**.** If two systems A and B are in thermal equilibrium with a third system C, then A is in thermal equilibrium with B.

The third system is a *thermometer*: a system characterised by a single state variable ϑ (e.g., the length of a column of mercury in a sealed glass tube). In contact with A and B it reaches the values ϑ_A and ϑ_B, and A and B are in thermal equilibrium if and only if $\vartheta_A = \vartheta_B$. The state variable ϑ is the *empirical temperature*, an intensive property. It is not unique (Celsius, Fahrenheit, or any one-to-one function of them) and can be negative.

5.9.2 Absolute temperature and entropy

The first law governs the energy transfer but is insensitive to its direction. Experience shows irreversibility: heat flows from warmer to colder regions, and work is turned into heat by friction, not vice versa. In Carathéodory's formulation: near any state of a thermodynamic

system there are states that cannot be reached from it by reversible adiabatic processes. To express this we introduce a new extensive variable, the *entropy*, related to the heat through a multiplicative factor, the **absolute temperature** $\Theta(\boldsymbol{x},t)>0$ (measured in kelvin).

Definition 5.7 (Entropy)**.** The entropy (a measure of disorder) is described by densities per unit current volume $\eta_c(\boldsymbol{x},t)$ and reference volume $\eta(\boldsymbol{X},t)=J\,\eta_c(\boldsymbol{\chi}(\boldsymbol{X},t),t)$, and the total entropy is

$$\mathcal{S}(t)=\int_\Omega \eta_c\,\mathrm{d}v=\int_{\Omega_0}\eta\,\mathrm{d}V. \tag{5.47}$$

The rate of entropy input is

$$\tilde{\mathcal{Q}}(t)=-\int_{\partial\Omega}\boldsymbol{h}\cdot\boldsymbol{n}\,\mathrm{d}s+\int_\Omega\tilde{r}\,\mathrm{d}v=-\int_{\partial\Omega_0}\boldsymbol{H}\cdot\boldsymbol{N}\,\mathrm{d}S+\int_{\Omega_0}\tilde{R}\,\mathrm{d}V, \tag{5.48}$$

with $\boldsymbol{h}$, $\boldsymbol{H}$ the Cauchy and Piola–Kirchhoff entropy fluxes and $\tilde{r}$, $\tilde{R}$ the entropy sources per unit time.

Definition 5.8 (Second law of thermodynamics)**.** The *total entropy production* per unit time is never negative:

$$\Gamma(t)=\frac{\mathrm{D}\mathcal{S}}{\mathrm{D}t}-\tilde{\mathcal{Q}}\geq 0,\qquad \Gamma(t)=\frac{\mathrm{D}}{\mathrm{D}t}\int_\Omega\eta_c\,\mathrm{d}v+\int_{\partial\Omega}\boldsymbol{h}\cdot\boldsymbol{n}\,\mathrm{d}s-\int_\Omega\tilde{r}\,\mathrm{d}v\geq 0. \tag{5.49}$$

The second law is *not* a balance principle: it is an inequality that fixes the direction of the energy transfer, i.e., postulates the irreversibility of thermodynamic processes. Reversible processes produce no entropy ($\Gamma=0$); irreversible processes (almost all real ones) increase it ($\Gamma>0$).

5.9.3 Clausius–Duhem and Clausius–Planck inequalities

The link between entropy and heat is the postulate

$$\boldsymbol{h}=\frac{\boldsymbol{q}}{\Theta},\quad \tilde{r}=\frac{r}{\Theta},\qquad \boldsymbol{H}=\frac{\boldsymbol{Q}}{\Theta},\quad \tilde{R}=\frac{R}{\Theta}: \tag{5.50}$$

since $\Theta>0$, entropy flows in the same direction as heat. (This postulate is not valid, e.g., , for the thermodynamics of diffusion.) The second law becomes the **Clausius–Duhem inequality**:

$$\Gamma=\frac{\mathrm{D}}{\mathrm{D}t}\int_\Omega\eta_c\,\mathrm{d}v+\int_{\partial\Omega}\frac{\boldsymbol{q}}{\Theta}\cdot\boldsymbol{n}\,\mathrm{d}s-\int_\Omega\frac{r}{\Theta}\,\mathrm{d}v\geq 0, \tag{5.51}$$

or, equivalently, in the reference configuration,

$$\Gamma=\frac{\mathrm{D}}{\mathrm{D}t}\int_{\Omega_0}\eta\,\mathrm{d}V+\int_{\partial\Omega_0}\frac{\boldsymbol{Q}}{\Theta}\cdot\boldsymbol{N}\,\mathrm{d}S-\int_{\Omega_0}\frac{R}{\Theta}\,\mathrm{d}V\geq 0. \tag{5.52}$$

Writing $\Gamma=\int_{\Omega_0}\gamma\,\mathrm{d}V$ and using the divergence theorem, the local forms of the entropy production per unit volume are

$$\gamma=\dot{\eta}-\frac{R}{\Theta}+\mathrm{Div}\Big(\frac{\boldsymbol{Q}}{\Theta}\Big)\geq 0,\qquad \gamma_c=\dot{\eta}_c+\eta_c\,\mathrm{div}\,\boldsymbol{v}-\frac{r}{\Theta}+\mathrm{div}\Big(\frac{\boldsymbol{q}}{\Theta}\Big)\geq 0. \tag{5.53}$$

With $\operatorname{Div}(\boldsymbol{Q}/\Theta) = \frac{1}{\Theta}\operatorname{Div}\boldsymbol{Q} - \frac{1}{\Theta^2}\boldsymbol{Q}\cdot\operatorname{Grad}\Theta$ (in components, $(Q_A/\Theta)_{,A} = Q_{A,A}/\Theta - Q_A\Theta_{,A}/\Theta^2$) and multiplying by $\Theta > 0$:

$$\Theta\gamma = \Theta\dot{\eta} - R + \operatorname{Div}\boldsymbol{Q} - \frac{1}{\Theta}\boldsymbol{Q}\cdot\operatorname{Grad}\Theta \geq 0. \tag{5.54}$$

Finally, eliminate the heat source with the energy balance Eq. (5.42), $\operatorname{Div}\boldsymbol{Q} - R = \boldsymbol{P} : \dot{\boldsymbol{F}} - \dot{e}$:

Clausius–Duhem inequality (local, material)

$$\Theta\gamma = \boldsymbol{P} : \dot{\boldsymbol{F}} - \dot{e} + \Theta\dot{\eta} - \frac{1}{\Theta}\boldsymbol{Q}\cdot\operatorname{Grad}\Theta \geq 0. \tag{5.55}$$

The last term is the entropy production by heat conduction. Heat flows against the temperature gradient, $\boldsymbol{Q}\cdot\operatorname{Grad}\Theta \leq 0$, so this term is never negative. Requiring the stronger condition that the remaining part be non-negative by itself gives the

Clausius–Planck inequality

$$\mathcal{D}_{\text{int}} = \boldsymbol{P} : \dot{\boldsymbol{F}} - \dot{e} + \Theta\dot{\eta} \geq 0, \qquad \mathcal{D}_{\text{int}} = P_{aA}\dot{F}_{aA} - \dot{e} + \Theta\dot{\eta} \geq 0, \tag{5.56}$$

where $\mathcal{D}_{\text{int}}$ is the **internal dissipation**: the stress power not converted into internal energy (net of the reversible thermal part $\Theta\dot{\eta}$) is dissipated. This is the form of the second law used to formulate constitutive equations (Chapter 7).

In class

In class the derivation was done on the blackboard in the material description; the spatial form, slightly more involved because of the time derivative of an integral over a moving domain, was left as an exercise (Exercise 5.7). The sign of the conduction term was justified by the fact that heat flows against the temperature gradient. Strictly speaking, the Clausius–Planck inequality is a *stronger* requirement than Clausius–Duhem; it is the one usually adopted (Holzapfel [3, Sect. 4.7]).

5.10 Exercises

Exercise 5.1. Derive $\dot{\rho} + \rho\operatorname{div}\boldsymbol{v} = 0$ directly from the integral form of the continuity equation using Reynolds' theorem.

▷ Solution on p. 141

Exercise 5.2. For the expanding hollow sphere of Example 3.6, verify that the spatial velocity satisfies $\operatorname{div}\boldsymbol{v} = 0$.

▷ Solution on p. 142

Exercise 5.3. Prove the Piola identity in index notation, using $\partial J/\partial F_{aA} = JF^{-1}_{Aa}$ and the symmetry of second derivatives of $\boldsymbol{\chi}$.

▷ Solution on p. 142

Exercise 5.4. Derive the balance of angular momentum in local form and show that it is equivalent to $\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\mathrm{T}}$, and to $\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{F}\boldsymbol{P}^{\mathrm{T}}$ in the material description.

▷ Solution on p. 142

Exercise 5.5. Prove Eq. (5.24) directly in the material description, using Eq. (5.19) and the material divergence theorem. Show that $\boldsymbol{P} : \dot{\boldsymbol{F}} = \boldsymbol{S} : \dot{\boldsymbol{E}}$.

▷ Solution on p. 142

Exercise 5.6. Starting from the global first law in the material and in the spatial description, derive the local forms of the energy balance, $\dot{e} = \boldsymbol{P} : \dot{\boldsymbol{F}} - \operatorname{Div} \boldsymbol{Q} + R$ and $\dot{e}_c + e_c \operatorname{div} \boldsymbol{v} = \boldsymbol{\sigma} : \boldsymbol{d} - \operatorname{div} \boldsymbol{q} + r$, and show that they are equivalent (slide 187).

▷ Solution on p. 143

Exercise 5.7. Derive the spatial local form of the Clausius–Duhem inequality,

$$\Theta \gamma_c = \boldsymbol{\sigma} : \boldsymbol{d} - \dot{e}_c - e_c \operatorname{div} \boldsymbol{v} + \Theta\big(\dot{\eta}_c + \eta_c \operatorname{div} \boldsymbol{v}\big) - \frac{1}{\Theta} \boldsymbol{q} \cdot \operatorname{grad} \Theta \geq 0,$$

and check that it coincides with J^{-1} times its material counterpart Eq. (5.55).

▷ Solution on p. 143

Exercise 5.8. Show that $\delta \boldsymbol{E} = \operatorname{sym}(\boldsymbol{F}^{\mathrm{T}} \operatorname{Grad} \delta \boldsymbol{u})$ and that $\boldsymbol{S} : \delta \boldsymbol{E} = \boldsymbol{P} : \delta \boldsymbol{F} = J \boldsymbol{\sigma} : \delta \boldsymbol{e}$. Deduce that the internal virtual work has the same value in the three descriptions.

▷ Solution on p. 144

Exercise 5.9. A bar of length L, cross section A and Young's modulus E (small strain), clamped at $x = 0$, is loaded by a uniform axial body force b per unit volume and by an end force F at $x = L$. Using the principle of virtual work, compute the Galerkin approximations with the one-term trial function $u = a_1 x$ and with the two-term trial function $u = a_1 x + a_2 x^2$ (test functions of the same form), and compare them with the exact solution.

▷ Solution on p. 144

Exercise 5.10. Using the Clausius–Duhem inequality, show that in a rigid heat conductor at rest ($\dot{\boldsymbol{F}} = \boldsymbol{O}$) obeying Fourier's law $\boldsymbol{q} = -k \operatorname{grad} \Theta$ one must have $k \geq 0$, and that heat flows from hot to cold. Write the heat equation for the case $e = c\,\Theta$ with constant c and k.

▷ Solution on p. 144

6

Objectivity

In class

In class this file was presented *after* the constitutive theory of Chapter 7 (the lecturer "took the wrong file" first), in the last part of the fourth day and "in a very compact way", leaving to the students the task of going through the slides in detail. The book follows the logical order of the slide files: first objectivity, a requirement that every constitutive equation must meet, then the general constitutive framework. All the derivations skipped in class are given here, and most of them are also proposed as exercises.

When we define a constitutive relation, for instance a function $\boldsymbol{g}$ giving the Cauchy stress in terms of the deformation gradient, $\boldsymbol{\sigma} = \boldsymbol{g}(\boldsymbol{F})$, we must make sure that the result of the law *does not depend on the observer*. Two observers looking at the same body must find the same physical response, even if they move one with respect to the other. This requirement is called **material frame indifference**, or **objectivity**. A physical quantity that does not depend on the observer is called *objective* (or frame-indifferent).

6.1 Observers and Euclidean transformations

An **observer** is someone able to measure distances and lapses of time: an observer is equipped with a ruler and a clock. An *event* is characterised by a position and an instant, $(\boldsymbol{x}, t)$. Given two events $(\boldsymbol{x}, t)$ and $(\boldsymbol{x}_0, t_0)$, for example a light flashing first in one place and then in another, the observer measures the distance $|\boldsymbol{x} - \boldsymbol{x}_0|$ between the two places and the time lapse $t - t_0$ between the two instants.

A second observer records the same two events as $(\boldsymbol{x}^+, t^+)$ and $(\boldsymbol{x}_0^+, t_0^+)$ (Fig. 6.1). We require that the two observers measure the same distances and the same time lapses:

$$\left|\boldsymbol{x}^+ - \boldsymbol{x}_0^+\right| = |\boldsymbol{x} - \boldsymbol{x}_0|\,, \qquad t^+ - t_0^+ = t - t_0. \tag{6.1}$$

The second condition is satisfied if the two clocks differ by a constant shift, $t^+ = t + \alpha$. The first one is satisfied if and only if the two vectors are related by an orthogonal tensor, $\boldsymbol{x}^+ - \boldsymbol{x}_0^+ = \boldsymbol{Q}(t)(\boldsymbol{x} - \boldsymbol{x}_0)$, because orthogonal tensors, and only them, preserve the length of every vector (Section 1.9). From $\boldsymbol{Q}\boldsymbol{Q}^{\mathrm{T}} = \boldsymbol{I}$, $(\det \boldsymbol{Q})^2 = 1$, hence $\det \boldsymbol{Q} = \pm 1$; the value -1 corresponds to a reflection, which would transform a right-handed frame into a left-handed one, and is excluded: $\boldsymbol{Q}$ is a *proper* orthogonal tensor (a rotation), $\det \boldsymbol{Q} = +1$.

Definition 6.1 (Euclidean transformation)**.** A change of observer is the one-to-one mapping

$$\boldsymbol{x}^+ = \boldsymbol{c}(t) + \boldsymbol{Q}(t)\,\boldsymbol{x}, \qquad t^+ = t + \alpha, \tag{6.2}$$

with $\boldsymbol{c}(t) = \boldsymbol{x}_0^+ - \boldsymbol{Q}(t)\boldsymbol{x}_0$ a vector (relative translation), $\boldsymbol{Q}(t)$ a proper orthogonal tensor (relative rotation) and α a constant. It is called a **Euclidean transformation**: the two observers move one with respect to the other by a time-dependent *rigid-body motion*, a rotation plus a translation.

Note that $\boldsymbol{c}$ and $\boldsymbol{Q}$ are, in general, functions of time. If they are, the second observer is *accelerated* with respect to the first one; distances are still preserved, but, as we shall see, rates are not transformed in a simple way. In what follows we set $\alpha = 0$, as the shift of time origin has no consequences.

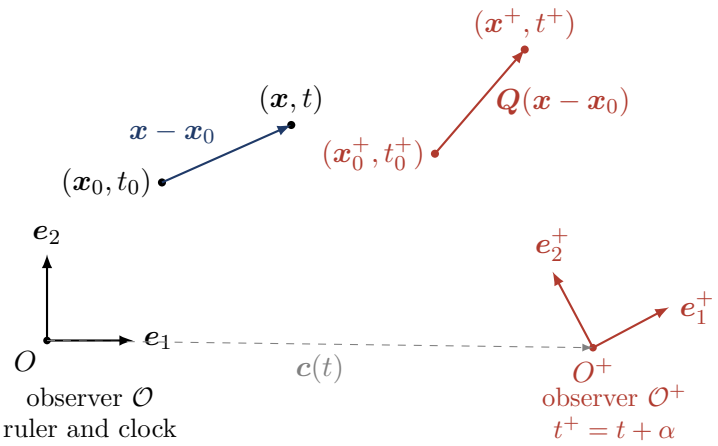


Figure 6.1: Two observers record the same pair of events; they measure the same distances and time lapses if and only if they are related by a Euclidean transformation (slides 198–200).

6.2 Objective spatial fields

Vectors. Let $\boldsymbol{u} = \boldsymbol{y} - \boldsymbol{x}$ be the vector joining two points recorded by the first observer. The second observer records

$$\boldsymbol{u}^+ = \boldsymbol{y}^+ - \boldsymbol{x}^+ = \boldsymbol{c} + \boldsymbol{Q}\boldsymbol{y} - \boldsymbol{c} - \boldsymbol{Q}\boldsymbol{x} = \boldsymbol{Q}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{Q}\boldsymbol{u}.$$

The translation disappears and $\boldsymbol{u}^+$ has the same length as $\boldsymbol{u}$: the two observers see the same vector, only described from a rotated frame. A spatial vector field that transforms according to this rule is objective.

Second-order tensors. A spatial tensor $\boldsymbol{A}$ maps spatial vectors into spatial vectors. Write it as a sum of dyads, $\boldsymbol{A} = \sum \boldsymbol{u} \otimes \boldsymbol{v}$, with $\boldsymbol{u}, \boldsymbol{v}$ objective vectors. Then, using $(\boldsymbol{Q}\boldsymbol{u}) \otimes (\boldsymbol{Q}\boldsymbol{v}) = \boldsymbol{Q}(\boldsymbol{u} \otimes \boldsymbol{v})\boldsymbol{Q}^{\mathrm{T}}$,

$$\boldsymbol{A}^+ = \sum \boldsymbol{u}^+ \otimes \boldsymbol{v}^+ = \sum \boldsymbol{Q}\boldsymbol{u} \otimes \boldsymbol{Q}\boldsymbol{v} = \boldsymbol{Q}\Big(\sum \boldsymbol{u} \otimes \boldsymbol{v}\Big)\boldsymbol{Q}^{\mathrm{T}} = \boldsymbol{Q}\boldsymbol{A}\boldsymbol{Q}^{\mathrm{T}}.$$

Equivalently, $\boldsymbol{A}$ is objective if $\boldsymbol{w} = \boldsymbol{A}\boldsymbol{u}$ implies $\boldsymbol{w}^+ = \boldsymbol{A}^+\boldsymbol{u}^+$ for all objective $\boldsymbol{u}, \boldsymbol{w}$: $\boldsymbol{Q}\boldsymbol{A}\boldsymbol{u} = \boldsymbol{A}^+\boldsymbol{Q}\boldsymbol{u}$ for all $\boldsymbol{u}$, i.e., $\boldsymbol{A}^+ = \boldsymbol{Q}\boldsymbol{A}\boldsymbol{Q}^{\mathrm{T}}$.

Objective spatial fields

Under the Euclidean transformation Eq. (6.2), a spatial field is objective if

$$\Phi^+(\boldsymbol{x}^+,t^+)=\Phi(\boldsymbol{x},t),\quad \boldsymbol{u}^+(\boldsymbol{x}^+,t^+)=\boldsymbol{Q}(t)\boldsymbol{u}(\boldsymbol{x},t),\quad \boldsymbol{A}^+(\boldsymbol{x}^+,t^+)=\boldsymbol{Q}(t)\boldsymbol{A}(\boldsymbol{x},t)\boldsymbol{Q}^{\mathrm{T}}(t), \tag{6.3}$$

for a scalar, a vector and a second-order tensor field, respectively.

A scalar is objective by nature only if it is a genuine physical scalar (temperature, mass density, a distance): it has no components to transform. Not every scalar *built* from vectors is objective, however: $|\boldsymbol{v}|$ is not, because $\boldsymbol{v}$ is not (Section 6.3).

6.3 Velocity and acceleration are not objective

Apply the change of observer to the motion: if $\boldsymbol{x}=\boldsymbol{\chi}(\boldsymbol{X},t)$ is the motion recorded by the first observer, the second one records

$$\boldsymbol{x}^+=\boldsymbol{\chi}^+(\boldsymbol{X},t^+)=\boldsymbol{c}(t)+\boldsymbol{Q}(t)\boldsymbol{\chi}(\boldsymbol{X},t). \tag{6.4}$$

The spin of the observer. Differentiating $\boldsymbol{Q}\boldsymbol{Q}^{\mathrm{T}}=\boldsymbol{I}$ in time gives $\dot{\boldsymbol{Q}}\boldsymbol{Q}^{\mathrm{T}}+\boldsymbol{Q}\dot{\boldsymbol{Q}}^{\mathrm{T}}=\boldsymbol{O}$, hence the tensor

$$\boldsymbol{\Omega}=\dot{\boldsymbol{Q}}\boldsymbol{Q}^{\mathrm{T}}=-\boldsymbol{Q}\dot{\boldsymbol{Q}}^{\mathrm{T}}=-\boldsymbol{\Omega}^{\mathrm{T}} \tag{6.5}$$

is skew: it is the angular velocity tensor of the second observer relative to the first one, $\dot{\boldsymbol{Q}}=\boldsymbol{\Omega}\boldsymbol{Q}$, the same kind of tensor as the spin $\boldsymbol{w}$ (Section 3.16). Its axial vector $\boldsymbol{\omega}$ satisfies $\boldsymbol{\Omega}\boldsymbol{u}=\boldsymbol{\omega}\times\boldsymbol{u}$. Differentiating once more, $\ddot{\boldsymbol{Q}}=\dot{\boldsymbol{\Omega}}\boldsymbol{Q}+\boldsymbol{\Omega}\dot{\boldsymbol{Q}}=(\dot{\boldsymbol{\Omega}}+\boldsymbol{\Omega}^2)\boldsymbol{Q}$.

Velocity. The material time derivative of Eq. (6.4) gives

$$\boldsymbol{v}^+=\dot{\boldsymbol{c}}+\dot{\boldsymbol{Q}}\boldsymbol{x}+\boldsymbol{Q}\boldsymbol{v}=\dot{\boldsymbol{c}}+\boldsymbol{\Omega}\boldsymbol{Q}\boldsymbol{x}+\boldsymbol{Q}\boldsymbol{v}.$$

Inverting Eq. (6.4), $\boldsymbol{Q}\boldsymbol{x}=\boldsymbol{x}^+-\boldsymbol{c}$, so

$$\boldsymbol{v}^+=\boldsymbol{Q}\boldsymbol{v}+\dot{\boldsymbol{c}}+\boldsymbol{\Omega}(\boldsymbol{x}^+-\boldsymbol{c}). \tag{6.6}$$

The velocity is *not* an objective vector: besides the rotated velocity $\boldsymbol{Q}\boldsymbol{v}$, the second observer sees the translational velocity $\dot{\boldsymbol{c}}$ and the rotational velocity $\boldsymbol{\omega}\times(\boldsymbol{x}^+-\boldsymbol{c})$ of the relative motion of the frames. The velocity is objective only if $\dot{\boldsymbol{c}}=\boldsymbol{0}$ and $\boldsymbol{\Omega}=\boldsymbol{O}$, i.e., if the Euclidean transformation is *time independent*.

Acceleration. Differentiating Eq. (6.6) again, with $\dot{\boldsymbol{x}}^+=\boldsymbol{v}^+$ and $\dot{\boldsymbol{Q}}\boldsymbol{v}=\boldsymbol{\Omega}\boldsymbol{Q}\boldsymbol{v}=\boldsymbol{\Omega}(\boldsymbol{v}^+-\dot{\boldsymbol{c}}-\boldsymbol{\Omega}(\boldsymbol{x}^+-\boldsymbol{c}))$:

$$\boldsymbol{a}^+=\boldsymbol{Q}\boldsymbol{a}+\dot{\boldsymbol{Q}}\boldsymbol{v}+\ddot{\boldsymbol{c}}+\dot{\boldsymbol{\Omega}}(\boldsymbol{x}^+-\boldsymbol{c})+\boldsymbol{\Omega}(\boldsymbol{v}^+-\dot{\boldsymbol{c}}),$$

Transformation of the acceleration

$$\boldsymbol{a}^+=\boldsymbol{Q}\boldsymbol{a}+\ddot{\boldsymbol{c}}+\underbrace{\dot{\boldsymbol{\Omega}}(\boldsymbol{x}^+-\boldsymbol{c})}_{\text{Euler}}\underbrace{-\boldsymbol{\Omega}^2(\boldsymbol{x}^+-\boldsymbol{c})}_{\text{centrifugal}}+\underbrace{2\boldsymbol{\Omega}(\boldsymbol{v}^+-\dot{\boldsymbol{c}})}_{\text{Coriolis}}. \tag{6.7}$$

The extra terms are the translational acceleration $\ddot{\boldsymbol{c}}$ and the well-known Euler, centrifugal and Coriolis accelerations. With $\boldsymbol{\Omega}\boldsymbol{u}=\boldsymbol{\omega}\times\boldsymbol{u}$: $\dot{\boldsymbol{\Omega}}(\boldsymbol{x}^+-\boldsymbol{c})=\dot{\boldsymbol{\omega}}\times(\boldsymbol{x}^+-\boldsymbol{c})$, $-\boldsymbol{\Omega}^2(\boldsymbol{x}^+-\boldsymbol{c})=$

$-\boldsymbol{\omega}\times(\boldsymbol{\omega}\times(\boldsymbol{x}^+-\boldsymbol{c}))$, $2\boldsymbol{\Omega}(\boldsymbol{v}^+-\dot{\boldsymbol{c}})=2\boldsymbol{\omega}\times(\boldsymbol{v}^+-\dot{\boldsymbol{c}})$. (Writing the last term with the relative velocity $\boldsymbol{Q}\boldsymbol{v}=\boldsymbol{v}^+-\dot{\boldsymbol{c}}-\boldsymbol{\Omega}(\boldsymbol{x}^+-\boldsymbol{c})$ changes the sign of the centrifugal term, which is how it often appears in dynamics textbooks; see Exercise 6.2.)

The acceleration is objective if and only if $\ddot{\boldsymbol{c}}=\boldsymbol{0}$ and $\boldsymbol{\Omega}=\boldsymbol{O}$, i.e., for a *Galilean transformation*: $\boldsymbol{c}(t)=\boldsymbol{c}_0+\boldsymbol{v}_0 t$ and $\boldsymbol{Q}$ constant. Newton's laws are invariant only under Galilean transformations; this is the reason why inertial frames are privileged in dynamics.

In class

"Look at what a big mess comes out": the lecturer only showed the result Eq. (6.7) and suggested deriving it as an exercise (Exercise 6.2). The important message was a warning for the constitutive theory: *rates introduce extra terms*. Constitutive relations written in rate form must be checked "one hundred times" for objectivity, and it is safer to write them in terms of total quantities whenever possible.

6.4 Superimposed rigid-body motions

The previous sections considered *one* motion seen by *two* observers. By the *principle of relativity*, the description of a single motion by two different observers is equivalent to the description, by a single observer, of two different motions that differ by a rigid-body motion (Fig. 6.2). The second motion $\boldsymbol{\chi}^+=\boldsymbol{c}+\boldsymbol{Q}\boldsymbol{\chi}$ is obtained by *superimposing* a rigid-body motion on the first one, and all the results of the previous sections hold unchanged. This is the point of view of Holzapfel [3, Sect. 5.2], and the one we adopt from now on: a quantity is objective if it transforms according to Eq. (6.3) under every superimposed rigid-body motion.

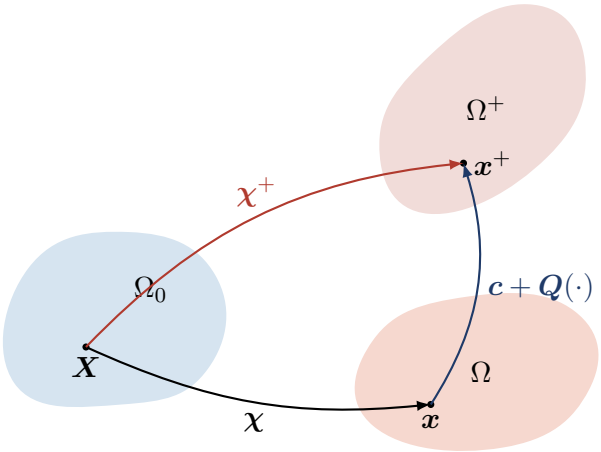


Figure 6.2: A rigid-body motion superimposed on the current configuration: $\boldsymbol{\chi}^+=\boldsymbol{c}+\boldsymbol{Q}\boldsymbol{\chi}$. The reference configuration is untouched.

Material and two-point fields. The superimposed motion acts on the current configuration only: the reference configuration, and therefore every *material* field, is unaffected,

$$\Phi^+=\Phi,\qquad \boldsymbol{U}^+=\boldsymbol{U},\qquad \boldsymbol{A}^+=\boldsymbol{A}\qquad \text{(material scalar, vector, tensor).}\tag{6.8}$$

A *two-point* tensor has one leg in each configuration; only the spatial leg is rotated. Two-point tensors transform as spatial vectors: $\boldsymbol{A}^+=\boldsymbol{Q}\boldsymbol{A}$.

6.5 Objectivity of kinematic quantities

Deformation gradient. Taking the material gradient of Eq. (6.4), with $\boldsymbol{c}$ and $\boldsymbol{Q}$ independent of $\boldsymbol{X}$,

$$\boldsymbol{F}^+ = \operatorname{Grad} \boldsymbol{\chi}^+ = \boldsymbol{Q}\boldsymbol{F}, \qquad F^+_{aA} = Q_{ab}F_{bA}, \qquad J^+ = \det \boldsymbol{Q} \det \boldsymbol{F} = J. \tag{6.9}$$

$\boldsymbol{F}$ is an objective two-point tensor: $\boldsymbol{Q}$ acts only on its spatial index a.

Polar decomposition. From $\boldsymbol{F}^+ = \boldsymbol{R}^+\boldsymbol{U}^+ = \boldsymbol{Q}\boldsymbol{R}\boldsymbol{U}$, where $\boldsymbol{Q}\boldsymbol{R}$ is a rotation and $\boldsymbol{U}$ is symmetric positive definite, the uniqueness of the polar decomposition gives

$$\boldsymbol{R}^+ = \boldsymbol{Q}\boldsymbol{R}, \qquad \boldsymbol{U}^+ = \boldsymbol{U}. \tag{6.10}$$

Similarly, $\boldsymbol{F}^+ = \boldsymbol{V}^+\boldsymbol{R}^+ = \boldsymbol{Q}\boldsymbol{V}\boldsymbol{R} = (\boldsymbol{Q}\boldsymbol{V}\boldsymbol{Q}^{\mathrm{T}})(\boldsymbol{Q}\boldsymbol{R})$ gives $\boldsymbol{V}^+ = \boldsymbol{Q}\boldsymbol{V}\boldsymbol{Q}^{\mathrm{T}}$. $\boldsymbol{U}$ is a material tensor (unaffected), $\boldsymbol{V}$ is an objective spatial tensor, $\boldsymbol{R}$ is a two-point tensor.

Strain tensors. Directly from Eq. (6.9) (Exercise 6.1):

$$\boldsymbol{C}^+ = \boldsymbol{C}, \quad \boldsymbol{E}^+ = \boldsymbol{E}, \qquad \boldsymbol{b}^+ = \boldsymbol{Q}\boldsymbol{b}\boldsymbol{Q}^{\mathrm{T}}, \quad \boldsymbol{e}^+ = \boldsymbol{Q}\boldsymbol{e}\boldsymbol{Q}^{\mathrm{T}}, \tag{6.11}$$

where $2\boldsymbol{e} = \boldsymbol{I} - \boldsymbol{b}^{-1} = \boldsymbol{I} - \boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{-1}$ is the Euler–Almansi strain. (Slide 214 writes $\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}$; the correct product is $\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{-1}$.) More generally, every Lagrangian strain of the Seth–Hill family (Section 3.15) is unaffected and every Eulerian one is objective.

Velocity gradient, rate of deformation and spin. With $\dot{\boldsymbol{F}}^+ = \dot{\boldsymbol{Q}}\boldsymbol{F} + \boldsymbol{Q}\dot{\boldsymbol{F}}$ and Eq. (3.67),

$$\boldsymbol{l}^+ = \dot{\boldsymbol{F}}^+\boldsymbol{F}^{+-1} = (\dot{\boldsymbol{Q}}\boldsymbol{F} + \boldsymbol{Q}\dot{\boldsymbol{F}})\boldsymbol{F}^{-1}\boldsymbol{Q}^{\mathrm{T}} = \boldsymbol{\Omega} + \boldsymbol{Q}\boldsymbol{l}\boldsymbol{Q}^{\mathrm{T}}. \tag{6.12}$$

Taking the symmetric and skew parts, since $\boldsymbol{\Omega}$ is skew,

$$\boldsymbol{d}^+ = \boldsymbol{Q}\boldsymbol{d}\boldsymbol{Q}^{\mathrm{T}}, \qquad \boldsymbol{w}^+ = \boldsymbol{\Omega} + \boldsymbol{Q}\boldsymbol{w}\boldsymbol{Q}^{\mathrm{T}}. \tag{6.13}$$

The rate of deformation is objective; the velocity gradient and the spin are not. This is physically sound: a rigid rotation does not stretch, but it does spin.

6.6 Objectivity of stress tensors

The basic *assumption* is that forces are objective: two observers see the same force acting on the same surface element, described in rotated frames. Then the traction and the unit normal are objective vectors, $\boldsymbol{t}^+ = \boldsymbol{Q}\boldsymbol{t}$, $\boldsymbol{n}^+ = \boldsymbol{Q}\boldsymbol{n}$. From Cauchy's theorem in the two descriptions, $\boldsymbol{t}^+ = \boldsymbol{\sigma}^+\boldsymbol{n}^+$:

$$\boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{n} = \boldsymbol{\sigma}^+\boldsymbol{Q}\boldsymbol{n} \quad \forall \boldsymbol{n} \qquad \Longrightarrow \qquad \boldsymbol{\sigma}^+ = \boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{Q}^{\mathrm{T}}.$$

The Cauchy stress is an objective spatial tensor, and so is the Kirchhoff stress, $\boldsymbol{\tau}^+ = J^+\boldsymbol{\sigma}^+ = \boldsymbol{Q}\boldsymbol{\tau}\boldsymbol{Q}^{\mathrm{T}}$. For the first Piola–Kirchhoff stress, $\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}$ (Eq. (4.6)):

$$\boldsymbol{P}^+ = J^+\boldsymbol{\sigma}^+\boldsymbol{F}^{+\,-\mathrm{T}} = J\boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{F}^{-\mathrm{T}} = \boldsymbol{Q}\boldsymbol{P},$$

(using $(\boldsymbol{QF})^{-\mathrm{T}} = \boldsymbol{Q}^{-\mathrm{T}}\boldsymbol{F}^{-\mathrm{T}} = \boldsymbol{QF}^{-\mathrm{T}}$): $\boldsymbol{P}$ is a two-point tensor, objective like $\boldsymbol{F}$. Finally, $\boldsymbol{S}^{+} = \boldsymbol{F}^{+-1}\boldsymbol{P}^{+} = \boldsymbol{F}^{-1}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{QP} = \boldsymbol{S}$: the second Piola–Kirchhoff stress is a material tensor and is unaffected (Exercise 6.4).

In class

The lecturer derived on the blackboard $\boldsymbol{\sigma}^{+}$ and $\boldsymbol{P}^{+}$ (the latter in a rather laborious way, multiplying $\boldsymbol{PF}^{\mathrm{T}} = J\boldsymbol{\sigma}$ by $\boldsymbol{F}^{-\mathrm{T}}$), and left $\boldsymbol{S}^{+} = \boldsymbol{S}$ to the students: "you should say it immediately on the basis of the knowledge that you have, because $\boldsymbol{S}$ is a material stress".

6.7 Rates of objective fields

The material time derivative of an objective field is, in general, *not* objective. For a vector, $\boldsymbol{u}^{+} = \boldsymbol{Qu}$ gives

$$\dot{\boldsymbol{u}}^{+} = \boldsymbol{Q}\dot{\boldsymbol{u}} + \dot{\boldsymbol{Q}}\boldsymbol{u} = \boldsymbol{Q}\dot{\boldsymbol{u}} + \boldsymbol{\Omega}\boldsymbol{u}^{+},$$

and for a tensor, $\boldsymbol{A}^{+} = \boldsymbol{QAQ}^{\mathrm{T}}$ gives

$$\dot{\boldsymbol{A}}^{+} = \boldsymbol{Q}\dot{\boldsymbol{A}}\boldsymbol{Q}^{\mathrm{T}} + \boldsymbol{\Omega}\boldsymbol{A}^{+} - \boldsymbol{A}^{+}\boldsymbol{\Omega}. \tag{6.14}$$

For instance, $\dot{\boldsymbol{\sigma}}$ is not objective. This matters in rate constitutive equations (hypoelasticity, rate formulations of plasticity for large strains), where the stress rate must be replaced by an *objective rate*.

Complement (objective stress rates, Holzapfel Sect. 5.3)

Combining Eq. (6.14) with the transformation rules of $\boldsymbol{l}$ and $\boldsymbol{w}$, objective rates of an objective tensor $\boldsymbol{\sigma}$ are obtained by subtracting the terms generated by the rotation of the frame:

- *Jaumann (co-rotational) rate*: $\overset{\triangledown}{\boldsymbol{\sigma}} = \dot{\boldsymbol{\sigma}} - \boldsymbol{w\sigma} + \boldsymbol{\sigma w}$;
- *Green–Naghdi rate*: as Jaumann's with $\boldsymbol{w}$ replaced by the rate of rotation $\boldsymbol{\Omega}_R = \dot{\boldsymbol{R}}\boldsymbol{R}^{\mathrm{T}}$, which transforms as $\boldsymbol{\Omega}_R^{+} = \boldsymbol{\Omega} + \boldsymbol{Q}\boldsymbol{\Omega}_R\boldsymbol{Q}^{\mathrm{T}}$;
- *Oldroyd rate*, the Lie derivative of $\boldsymbol{\sigma}$ (pull back with $\boldsymbol{F}$, differentiate, push forward): $\overset{\circ}{\boldsymbol{\sigma}} = \boldsymbol{F}\Big(\frac{\mathrm{D}}{\mathrm{D}t}\big(\boldsymbol{F}^{-1}\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}\big)\Big)\boldsymbol{F}^{\mathrm{T}} = \dot{\boldsymbol{\sigma}} - \boldsymbol{l\sigma} - \boldsymbol{\sigma l}^{\mathrm{T}}$;
- *Truesdell rate*, the Piola transform of the Lie derivative: $\overset{\circ}{\boldsymbol{\sigma}} + \boldsymbol{\sigma}\,\mathrm{tr}\,\boldsymbol{l} = J^{-1}\boldsymbol{F}\dot{\boldsymbol{S}}\boldsymbol{F}^{\mathrm{T}}$.

The Oldroyd and Truesdell rates are objective because $\boldsymbol{F}^{-1}\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}$ and $\boldsymbol{S}$ are material tensors, hence unaffected by the superimposed motion, and their push-forward with $\boldsymbol{F}^{+} = \boldsymbol{QF}$ produces the factors $\boldsymbol{Q}(\cdot)\boldsymbol{Q}^{\mathrm{T}}$. For the Jaumann rate, see Exercise 6.5.

6.8 Objective constitutive equations

Principle of material frame indifference

Constitutive equations must be invariant under changes of observer (equivalently, under superimposed rigid-body motions): if the constitutive law gives the response to a motion $\boldsymbol{\chi}$, it must give the correspondingly transformed response to the motion $\boldsymbol{\chi}^{+} = \boldsymbol{c} + \boldsymbol{Q}\boldsymbol{\chi}$, for every $\boldsymbol{c}(t)$ and every rotation $\boldsymbol{Q}(t)$.

Cauchy-elastic materials. Consider a homogeneous material in which the Cauchy stress depends only on the current deformation gradient,

$$\boldsymbol{\sigma} = \boldsymbol{g}(\boldsymbol{F}), \tag{6.15}$$

(a dependence on the position $\boldsymbol{X}$ is dropped for homogeneous materials). The response to the motion $\boldsymbol{\chi}^+$ is $\boldsymbol{\sigma}^+ = \boldsymbol{g}(\boldsymbol{F}^+) = \boldsymbol{g}(\boldsymbol{Q}\boldsymbol{F})$, and objectivity requires $\boldsymbol{\sigma}^+ = \boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{Q}^{\mathrm{T}} = \boldsymbol{Q}\boldsymbol{g}(\boldsymbol{F})\boldsymbol{Q}^{\mathrm{T}}$:

$$\boldsymbol{g}(\boldsymbol{Q}\boldsymbol{F}) = \boldsymbol{Q}\,\boldsymbol{g}(\boldsymbol{F})\,\boldsymbol{Q}^{\mathrm{T}} \qquad \forall \boldsymbol{F},\ \det \boldsymbol{F} > 0,\ \forall \boldsymbol{Q} \text{ proper orthogonal.} \tag{6.16}$$

This is a restriction on the function $\boldsymbol{g}$, not on $\boldsymbol{F}$.

Reduced form. Since Eq. (6.16) holds for every rotation, choose the particular one $\boldsymbol{Q} = \boldsymbol{R}^{\mathrm{T}}$, with $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U}$:

$$\boldsymbol{g}(\boldsymbol{R}^{\mathrm{T}}\boldsymbol{R}\boldsymbol{U}) = \boldsymbol{g}(\boldsymbol{U}) = \boldsymbol{R}^{\mathrm{T}}\boldsymbol{g}(\boldsymbol{F})\boldsymbol{R}.$$

Hence

Reduced form (Cauchy stress)

$$\boldsymbol{\sigma} = \boldsymbol{g}(\boldsymbol{F}) = \boldsymbol{R}\,\boldsymbol{g}(\boldsymbol{U})\,\boldsymbol{R}^{\mathrm{T}}. \tag{6.17}$$

The condition is also sufficient: if $\boldsymbol{g}(\boldsymbol{F}) = \boldsymbol{R}\boldsymbol{g}(\boldsymbol{U})\boldsymbol{R}^{\mathrm{T}}$, then $\boldsymbol{g}(\boldsymbol{Q}\boldsymbol{F}) = (\boldsymbol{Q}\boldsymbol{R})\boldsymbol{g}(\boldsymbol{U})(\boldsymbol{Q}\boldsymbol{R})^{\mathrm{T}} = \boldsymbol{Q}\boldsymbol{g}(\boldsymbol{F})\boldsymbol{Q}^{\mathrm{T}}$, because $\boldsymbol{Q}\boldsymbol{F}$ has rotation $\boldsymbol{Q}\boldsymbol{R}$ and the same stretch $\boldsymbol{U}$. The constitutive function may be prescribed arbitrarily on the *stretches* $\boldsymbol{U}$ (the deformation), and the rotation $\boldsymbol{R}$ enters only to rotate the result: a rigid rotation does not produce stress.

Piola–Kirchhoff stresses. The first Piola–Kirchhoff stress is

$$\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}} = \det \boldsymbol{F}\,\boldsymbol{g}(\boldsymbol{F})\boldsymbol{F}^{-\mathrm{T}} := \boldsymbol{G}(\boldsymbol{F}).$$

By $\boldsymbol{P}^+ = \boldsymbol{Q}\boldsymbol{P}$ and $\boldsymbol{P}^+ = \boldsymbol{G}(\boldsymbol{F}^+)$, objectivity requires

$$\boldsymbol{Q}\boldsymbol{G}(\boldsymbol{F}) = \boldsymbol{G}(\boldsymbol{Q}\boldsymbol{F}) \quad \forall \boldsymbol{Q} \qquad \Longleftrightarrow \qquad \boldsymbol{G}(\boldsymbol{F}) = \boldsymbol{R}\,\boldsymbol{G}(\boldsymbol{U}), \tag{6.18}$$

again by the choice $\boldsymbol{Q} = \boldsymbol{R}^{\mathrm{T}}$. For the second Piola–Kirchhoff stress, $\boldsymbol{S} = J\boldsymbol{F}^{-1}\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}$, and with $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U}$ and Eq. (6.17),

$$\begin{aligned}\boldsymbol{S} = \det \boldsymbol{U}\,\boldsymbol{U}^{-1}\boldsymbol{R}^{\mathrm{T}}\boldsymbol{\sigma}\boldsymbol{R}\,\boldsymbol{U}^{-1} &= \det \boldsymbol{U}\,\boldsymbol{U}^{-1}\boldsymbol{g}(\boldsymbol{U})\boldsymbol{U}^{-1} \\ &= \det \boldsymbol{C}^{1/2}\,\boldsymbol{C}^{-1/2}\boldsymbol{g}(\boldsymbol{C}^{1/2})\boldsymbol{C}^{-1/2} := \boldsymbol{H}(\boldsymbol{C}).\end{aligned} \tag{6.19}$$

Both $\boldsymbol{S}$ and $\boldsymbol{C}$ are unaffected by superimposed rigid motions, so *any* relation $\boldsymbol{S} = \boldsymbol{H}(\boldsymbol{C})$, or $\boldsymbol{S} = \tilde{\boldsymbol{H}}(\boldsymbol{E})$, is automatically objective (Exercise 6.6). This is the main reason why constitutive equations at finite strain are preferably written in the material description, in terms of material stress and strain measures.

Example 6.1 (Saint Venant–Kirchhoff versus a naive extension)**.** The *Saint Venant–Kirchhoff* model $\boldsymbol{S} = \lambda(\operatorname{tr}\boldsymbol{E})\boldsymbol{I} + 2\mu\boldsymbol{E}$ is objective, since it relates two material tensors. On the contrary, the naive extension of Hooke's law $\boldsymbol{\sigma} = \lambda \operatorname{tr}(\tilde{\boldsymbol{\varepsilon}})\boldsymbol{I} + 2\mu\tilde{\boldsymbol{\varepsilon}}$ with $\tilde{\boldsymbol{\varepsilon}} = \operatorname{sym}\boldsymbol{F} - \boldsymbol{I}$

(the small-strain tensor written with $\boldsymbol{F}$ in place of $\boldsymbol{I} + \operatorname{Grad} \boldsymbol{u}$) is *not*: a pure rotation $\boldsymbol{F} = \boldsymbol{Q}$ produces $\tilde{\boldsymbol{\varepsilon}} = \operatorname{sym} \boldsymbol{Q} - \boldsymbol{I} \neq \boldsymbol{O}$, i.e., a stress without deformation (Exercise 6.7).

Table 6.1: Behaviour of the main fields under a superimposed rigid-body motion $\boldsymbol{\chi}^+ = \boldsymbol{c} + \boldsymbol{Q}\boldsymbol{\chi}$ (slide 220).

Type	Rule	Examples
spatial scalar	$\Phi^+ = \Phi$	ρ, Θ, J, e_c, η_c
spatial vector	$\boldsymbol{u}^+ = \boldsymbol{Q}\boldsymbol{u}$	$\boldsymbol{t}$, $\boldsymbol{T}$, $\boldsymbol{n}$, $\boldsymbol{q}$, $\mathrm{d}\boldsymbol{x}$
spatial tensor	$\boldsymbol{A}^+ = \boldsymbol{Q}\boldsymbol{A}\boldsymbol{Q}^{\mathrm{T}}$	$\boldsymbol{\sigma}$, $\boldsymbol{\tau}$, $\boldsymbol{b}$, $\boldsymbol{V}$, $\boldsymbol{e}$, $\boldsymbol{d}$
two-point tensor	$\boldsymbol{A}^+ = \boldsymbol{Q}\boldsymbol{A}$	$\boldsymbol{F}$, $\boldsymbol{R}$, $\boldsymbol{P}$
material vector	$\boldsymbol{U}^+ = \boldsymbol{U}$	$\boldsymbol{N}$, $\mathrm{d}\boldsymbol{X}$, $\operatorname{Grad}\Theta$, $\boldsymbol{Q}$ (heat flux)
material tensor	$\boldsymbol{A}^+ = \boldsymbol{A}$	$\boldsymbol{C}$, $\boldsymbol{U}$, $\boldsymbol{E}$, $\boldsymbol{S}$
not objective	$\boldsymbol{v}^+$, $\boldsymbol{a}^+$, $\boldsymbol{l}^+$, $\boldsymbol{w}^+$, $\dot{\boldsymbol{\sigma}}^+$	see Eqs. (6.6), (6.7), (6.13) and (6.14)

In class

The reduced forms (slides 218–219) were only sketched in class, with the advice to "try to do these things by following the relations on page 219". The lecturer underlined the idea behind them: in $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U}$ only $\boldsymbol{U}$ is responsible for the deformation, while $\boldsymbol{R}$ is a rigid rotation and must not appear in the constitutive function. The table of slide 220 "is useful to be consulted".

6.9 Isotropy

In class

At the end of the lecture the lecturer announced a second property of constitutive equations, to be exploited in the next lecture: besides being independent of the observer, the response may be independent of the *direction* in the material (a property measured along the axis x_1 is the same along x_2). These two slides (221–223) will be moved to the hyperelasticity file. They are anticipated here because they complete the discussion of the reduced forms.

Objectivity concerns a rotation of the *current* configuration (or of the observer). *Material symmetry* concerns a rotation of the *reference* configuration, $\boldsymbol{F} \mapsto \boldsymbol{F}\boldsymbol{Q}$: a material is isotropic if its response is the same for every rotated reference configuration,

$$\boldsymbol{g}(\boldsymbol{F}\boldsymbol{Q}) = \boldsymbol{g}(\boldsymbol{F}) \qquad \forall \boldsymbol{Q} \text{ proper orthogonal.} \tag{6.20}$$

With $\boldsymbol{F} = \boldsymbol{V}\boldsymbol{R}$ and $\boldsymbol{Q} = \boldsymbol{R}^{\mathrm{T}}$, isotropy gives $\boldsymbol{g}(\boldsymbol{F}) = \boldsymbol{g}(\boldsymbol{V})$: the stress depends on $\boldsymbol{F}$ only through $\boldsymbol{V} = \boldsymbol{b}^{1/2}$, i.e.,

$$\boldsymbol{\sigma} = \boldsymbol{h}(\boldsymbol{b}), \qquad \boldsymbol{b} = \boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}. \tag{6.21}$$

Objectivity then requires, with $\boldsymbol{b}^+ = \boldsymbol{Q}\boldsymbol{b}\boldsymbol{Q}^{\mathrm{T}}$ and $\boldsymbol{\sigma}^+ = \boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{Q}^{\mathrm{T}}$,

Isotropic Cauchy-elastic material

$$\boldsymbol{h}(\boldsymbol{Q}\boldsymbol{b}\boldsymbol{Q}^{\mathrm{T}}) = \boldsymbol{Q}\,\boldsymbol{h}(\boldsymbol{b})\,\boldsymbol{Q}^{\mathrm{T}} \qquad \forall \boldsymbol{Q} \text{ orthogonal,} \tag{6.22}$$

i.e., $\boldsymbol{h}$ is an *isotropic tensor function*. The material has no preferred directions and its response is not affected by the choice of the reference configuration (slide 221). Note that, for a function of $\boldsymbol{b}$, the requirement of objectivity and that of isotropy of the function coincide. An immediate consequence is that $\boldsymbol{\sigma}$ and $\boldsymbol{b}$ are *coaxial*: take $\boldsymbol{Q}$ the reflection $\boldsymbol{I} - 2\boldsymbol{n}_i \otimes \boldsymbol{n}_i$ through the plane orthogonal to an eigenvector $\boldsymbol{n}_i$ of $\boldsymbol{b}$; it leaves $\boldsymbol{b}$ unchanged, so $\boldsymbol{h}(\boldsymbol{b}) = \boldsymbol{Q}\boldsymbol{h}(\boldsymbol{b})\boldsymbol{Q}^{\mathrm{T}}$, which forces $\boldsymbol{h}(\boldsymbol{b})\boldsymbol{n}_i \parallel \boldsymbol{n}_i$.

Theorem 6.1 (Rivlin–Ericksen representation)**.** *A tensor function $\boldsymbol{h}$ of a symmetric tensor $\boldsymbol{b}$ is isotropic if and only if it can be written as*

$$\boldsymbol{\sigma} = \boldsymbol{h}(\boldsymbol{b}) = \alpha_0 \boldsymbol{I} + \alpha_1 \boldsymbol{b} + \alpha_2 \boldsymbol{b}^2, \tag{6.23}$$

with scalar coefficients $\alpha_i = \alpha_i(I_1, I_2, I_3)$ that are functions of the principal invariants of $\boldsymbol{b}$, $I_1 = \operatorname{tr} \boldsymbol{b}$, $I_2 = \frac{1}{2}\left[(\operatorname{tr} \boldsymbol{b})^2 - \operatorname{tr}(\boldsymbol{b}^2)\right]$, $I_3 = \det \boldsymbol{b}$.

Using the Cayley–Hamilton theorem, $\boldsymbol{b}^2 = I_1 \boldsymbol{b} - I_2 \boldsymbol{I} + I_3 \boldsymbol{b}^{-1}$, the representation can be written equivalently as

$$\boldsymbol{\sigma} = \beta_0 \boldsymbol{I} + \beta_1 \boldsymbol{b} + \beta_{-1} \boldsymbol{b}^{-1}, \qquad \beta_0 = \alpha_0 - I_2 \alpha_2, \quad \beta_1 = \alpha_1 + I_1 \alpha_2, \quad \beta_{-1} = I_3 \alpha_2. \tag{6.24}$$

In the material description, from $\boldsymbol{S} = J \boldsymbol{F}^{-1} \boldsymbol{\sigma} \boldsymbol{F}^{-\mathrm{T}}$ and $\boldsymbol{F}^{-1} \boldsymbol{b}^n \boldsymbol{F}^{-\mathrm{T}} = \boldsymbol{C}^{n-1}$, the isotropic response is $\boldsymbol{S} = J(\beta_0 \boldsymbol{C}^{-1} + \beta_1 \boldsymbol{I} + \beta_{-1} \boldsymbol{C}^{-2})$, with the same coefficients (the invariants of $\boldsymbol{C}$ and $\boldsymbol{b}$ coincide).

Incompressible materials. If the material is incompressible, $J = 1$, the stress is determined by the deformation only up to an arbitrary hydrostatic pressure p, a Lagrange multiplier to be found from the balance equations and boundary conditions:

$$\boldsymbol{\sigma} = -p \boldsymbol{I} + \boldsymbol{h}(\boldsymbol{b}), \qquad \boldsymbol{S} = -p\, \boldsymbol{C}^{-1} + \boldsymbol{H}(\boldsymbol{C}). \tag{6.25}$$

(Slide 223 writes $-p\boldsymbol{C}$; the correct term is the pull-back of $-p\boldsymbol{I}$, $\boldsymbol{F}^{-1}(-p\boldsymbol{I})\boldsymbol{F}^{-\mathrm{T}} = -p\boldsymbol{C}^{-1}$.) These forms are developed in the chapter on hyperelasticity.

6.10 Exercises

Exercise 6.1. Prove Eq. (6.11): $\boldsymbol{C}^+ = \boldsymbol{C}$, $\boldsymbol{E}^+ = \boldsymbol{E}$, $\boldsymbol{b}^+ = \boldsymbol{Q}\boldsymbol{b}\boldsymbol{Q}^{\mathrm{T}}$, $\boldsymbol{e}^+ = \boldsymbol{Q}\boldsymbol{e}\boldsymbol{Q}^{\mathrm{T}}$. Show also that the principal stretches are unaffected, while the principal spatial directions are rotated.

▷ Solution on p. 145

Exercise 6.2. Derive Eq. (6.7) step by step. Rewrite it in terms of the relative velocity $\boldsymbol{Q}\boldsymbol{v}$ and the angular velocity vector $\boldsymbol{\omega}$, and identify the classical Euler, centrifugal and Coriolis accelerations. Show that the acceleration is objective for Galilean transformations.

▷ Solution on p. 145

Exercise 6.3. Show that $\boldsymbol{\Omega} = \dot{\boldsymbol{Q}}\boldsymbol{Q}^{\mathrm{T}}$ is skew, and derive Eq. (6.12) and Eq. (6.13). For the rigid rotation $\boldsymbol{\chi}(\boldsymbol{X}, t) = \boldsymbol{Q}_0(t)\boldsymbol{X}$ superimposed on the identity, compute $\boldsymbol{l}$, $\boldsymbol{d}$, $\boldsymbol{w}$ and comment.

▷ Solution on p. 145

Exercise 6.4. Show that $\boldsymbol{\tau}^{+} = \boldsymbol{Q}\boldsymbol{\tau}\boldsymbol{Q}^{\mathrm{T}}$, $\boldsymbol{P}^{+} = \boldsymbol{Q}\boldsymbol{P}$, $\boldsymbol{S}^{+} = \boldsymbol{S}$, and that the stress power per unit reference volume, $\boldsymbol{P} : \dot{\boldsymbol{F}}$, is unaffected by superimposed rigid-body motions even though $\dot{\boldsymbol{F}}$ is not objective.

▷ Solution on p. 146

Exercise 6.5. Show that $\dot{\boldsymbol{\sigma}}$ is not objective and that the Jaumann rate $\overset{\triangledown}{\boldsymbol{\sigma}} = \dot{\boldsymbol{\sigma}} - \boldsymbol{w}\boldsymbol{\sigma} + \boldsymbol{\sigma}\boldsymbol{w}$ and the Oldroyd rate $\overset{\circ}{\boldsymbol{\sigma}} = \dot{\boldsymbol{\sigma}} - \boldsymbol{l}\boldsymbol{\sigma} - \boldsymbol{\sigma}\boldsymbol{l}^{\mathrm{T}}$ are objective.

▷ Solution on p. 146

Exercise 6.6. Starting from $\boldsymbol{\sigma} = \boldsymbol{g}(\boldsymbol{F})$, derive the reduced forms $\boldsymbol{P} = \boldsymbol{R}\boldsymbol{G}(\boldsymbol{U})$ and $\boldsymbol{S} = \boldsymbol{H}(\boldsymbol{C})$ of slide 219. Conversely, show that for an arbitrary function $\boldsymbol{H}$ the law $\boldsymbol{\sigma} = J^{-1}\boldsymbol{F}\boldsymbol{H}(\boldsymbol{C})\boldsymbol{F}^{\mathrm{T}}$ satisfies Eq. (6.16).

▷ Solution on p. 146

Exercise 6.7. Check the objectivity of the following constitutive laws: (a) $\boldsymbol{S} = \lambda(\operatorname{tr}\boldsymbol{E})\boldsymbol{I} + 2\mu\boldsymbol{E}$; (b) $\boldsymbol{\sigma} = \lambda \operatorname{tr}(\tilde{\boldsymbol{\varepsilon}})\boldsymbol{I} + 2\mu\tilde{\boldsymbol{\varepsilon}}$ with $\tilde{\boldsymbol{\varepsilon}} = \operatorname{sym}\boldsymbol{F} - \boldsymbol{I}$; (c) $\boldsymbol{\sigma} = \mu(\boldsymbol{b} - \boldsymbol{I})$; (d) $\boldsymbol{\sigma} = \mu(\boldsymbol{C} - \boldsymbol{I})$; (e) $\boldsymbol{\sigma} = 2\eta\,\boldsymbol{d}$ and $\boldsymbol{\sigma} = 2\eta\,\boldsymbol{l}$ (viscous fluids).

▷ Solution on p. 146

Exercise 6.8. Show that $\boldsymbol{\sigma} = \beta_0\boldsymbol{I} + \beta_1\boldsymbol{b} + \beta_{-1}\boldsymbol{b}^{-1}$ satisfies Eq. (6.22), derive the relation between the coefficients α_i and β_i, and compute the corresponding second Piola–Kirchhoff stress. For an incompressible neo-Hookean material, $\boldsymbol{\sigma} = -p\boldsymbol{I} + \mu\boldsymbol{b}$, compute $\boldsymbol{\sigma}$ in uniaxial tension $\boldsymbol{F} = \operatorname{diag}(\lambda, \lambda^{-1/2}, \lambda^{-1/2})$ with traction-free lateral surfaces.

▷ Solution on p. 147

Part III

Constitutive Theory

7

Constitutive Theory

In class

"With all this information that we got from kinematics, from statics and from thermodynamics, we are finally able to formulate and describe a constitutive relationship." The constitutive theory was presented on the fourth day, right after the second law and before objectivity (Chapter 6). The lecturer did not prove the Coleman–Noll results ("I can prove it, but it takes too much time") and invited the students to read the original paper [1], warning that, written in 1963, it does not use the modern notation. The proof is given here in full, in the notation of the course. The fluxes and kinetic relations (slide 233) were skipped for lack of time and are also developed here.

7.1 Why constitutive equations

The balance principles of Chapter 5 hold for every continuum, whatever it is made of. They are not sufficient to determine the motion. Count, in the material description, the unknown fields of a thermomechanical problem: the motion $\boldsymbol{\chi}$ (3 components), the stress $\boldsymbol{P}$ (9 components, of which 6 are independent because of the balance of angular momentum), the temperature Θ, the internal energy e, the entropy η and the heat flux $\boldsymbol{Q}$ (3 components): 14 scalar unknowns once $\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{F}\boldsymbol{P}^{\mathrm{T}}$ is used. The balance equations provide only

$$\operatorname{Div}\boldsymbol{P} + \boldsymbol{B} = \rho_0\dot{\boldsymbol{V}} \quad (3), \qquad \dot{e} = \boldsymbol{P} : \dot{\boldsymbol{F}} - \operatorname{Div}\boldsymbol{Q} + R \quad (1),$$

while the conservation of mass only gives ρ and the second law is an inequality. The missing equations are **constitutive equations**, which characterise the specific material. They link the stress to the strain (and possibly to the strain rate, the temperature, the history of the deformation), and the heat flux to the temperature field. Constitutive equations may be proposed empirically, from the results of experiments, but they cannot be arbitrary: they must satisfy a set of general principles, among which

- *determinism*: the state at a point is determined by the history of the motion and of the temperature of the body;
- *local action*: the response at a particle depends only on the motion of an arbitrarily small neighbourhood of the particle;
- *objectivity*: the response does not depend on the observer (Chapter 6);
- *thermodynamic consistency*: the response satisfies the second law for every process (Section 7.4).

Materials for which the response at $\boldsymbol{X}$ depends only on $\boldsymbol{F}$ (and not on its gradients) are called *simple materials*: they do not show any particular microstructure.

7.2 Classes of simple materials

We consider homogeneous materials (no dependence on the position $\boldsymbol{X}$) in the material description, and classify them according to the variables that affect the stress response (Table 7.1).

Table 7.1: Simple material behaviours (slide 225).

Material	Stress function
Elastic isothermal	$\boldsymbol{P} = \boldsymbol{P}(\boldsymbol{F})$
Elastic	$\boldsymbol{P} = \boldsymbol{P}(\boldsymbol{F}, \Theta)$
Rate-independent	$\boldsymbol{P} = \boldsymbol{P}(\boldsymbol{F}, \Theta, \boldsymbol{\xi})$
Viscous	$\boldsymbol{P} = \boldsymbol{P}(\dot{\boldsymbol{F}}; \boldsymbol{F}, \Theta, \boldsymbol{\xi})$
Visco-elastic	$\boldsymbol{P} = \boldsymbol{P}(\dot{\boldsymbol{F}}; \boldsymbol{F}, \Theta)$
Visco-elastic isothermal	$\boldsymbol{P} = \boldsymbol{P}(\dot{\boldsymbol{F}}; \boldsymbol{F})$

- *Elastic*: the stress in the current configuration depends only on the current value of the deformation, measured by $\boldsymbol{F}$ with respect to the reference configuration (and on the temperature, if thermal effects are considered). The absolute temperature Θ is the proper variable.

- *Rate-independent*: the response does not depend on time, but depends on the *history* of loading. In a metal loaded beyond yield, unloaded and reloaded, the same stress corresponds to different strains: the stress is not a function of the strain alone. The history is accounted for by a collection $\boldsymbol{\xi}$ of **internal variables** (scalars, vectors or tensors, depending on the model, e.g., the accumulated plastic strain in plasticity).

- *Viscous*: the response depends also on the rate of deformation $\dot{\boldsymbol{F}}$. The semicolon in $\boldsymbol{P}(\dot{\boldsymbol{F}}; \boldsymbol{F}, \Theta, \boldsymbol{\xi})$ separates the dependence on the rate from the dependence on the state: when $\dot{\boldsymbol{F}} = \boldsymbol{O}$ the material behaves as a rate-independent one. Creep and relaxation are typical viscous phenomena.

- *Visco-elastic*: viscous, without internal variables.

The collection $\boldsymbol{\xi}$ of internal variables depends on the class of materials and cannot be specified universally. Internal variables are also called *hidden* variables: they cannot be measured or seen from outside, their existence is postulated.

Remark 7.1 (Entropy instead of temperature). In what follows we use the entropy η, rather than the temperature Θ, as independent thermal variable, because the Coleman–Noll theory was developed in these variables. The temperature is easier to measure, but the two are conjugate, and the thermodynamic potentials of Section 7.5 allow us to switch from one to the other.

7.3 Viscosity: equilibrium and viscous stress

Definition 7.1 (Viscous material)**.** A material is *viscous* if its response depends on the rate of deformation:

$$\boldsymbol{P} = \boldsymbol{P}(\dot{\boldsymbol{F}}; \boldsymbol{F}, \eta, \boldsymbol{\xi}). \tag{7.1}$$

The **equilibrium stress** (or elastic stress) is the stress for $\dot{\boldsymbol{F}} = \boldsymbol{O}$, and the **viscous stress** is the difference:

$$\boldsymbol{P}^e(\boldsymbol{F}, \eta, \boldsymbol{\xi}) = \boldsymbol{P}(\dot{\boldsymbol{F}} = \boldsymbol{O}; \boldsymbol{F}, \eta, \boldsymbol{\xi}), \qquad \boldsymbol{P}^v(\dot{\boldsymbol{F}}; \boldsymbol{F}, \eta, \boldsymbol{\xi}) = \boldsymbol{P} - \boldsymbol{P}^e. \tag{7.2}$$

The viscous stress vanishes when the deformation stops changing. It is assumed to satisfy the regularity condition

$$\boldsymbol{P}^v : \dot{\boldsymbol{F}} = o\big(\big|\dot{\boldsymbol{F}}\big|\big) \quad \text{as } \dot{\boldsymbol{F}} \to \boldsymbol{O}, \tag{7.3}$$

i.e., the viscous power vanishes faster than the rate itself: in particular, $\boldsymbol{P}^v \to \boldsymbol{O}$ when $\dot{\boldsymbol{F}} \to \boldsymbol{O}$. The same decomposition holds for the Cauchy stress,

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^e + \boldsymbol{\sigma}^v = J^{-1}\boldsymbol{P}^e\boldsymbol{F}^{\mathrm{T}} + J^{-1}\boldsymbol{P}^v\boldsymbol{F}^{\mathrm{T}}. \tag{7.4}$$

Internal variables and the thermodynamic state. The evolution of a dissipative system is regarded as a sequence of fictitious, or *constrained*, equilibrium states: going from the reference configuration Ω_0 to the current configuration Ω, the body passes through intermediate configurations, and in each of them the internal variables are frozen at their current values and the system is assumed to be in equilibrium. The local thermodynamic state of an infinitesimal neighbourhood of a particle is then defined by a finite set of variables, which include $\boldsymbol{\xi}$.

7.4 The Coleman–Noll procedure

Thermodynamics restricts the constitutive relations that we can write: instead of any connection between stress and strain, only those consistent with the balance of energy and with the dissipation inequality are admissible. The systematic method to obtain these restrictions is due to Coleman and Noll [1]. We first present the original argument, then its extension to internal variables given in the slides.

7.4.1 Thermodynamic processes

A *thermodynamic process* is a set of fields over the body and the time interval: the motion $\boldsymbol{\chi}$, the stress $\boldsymbol{P}$, the body force $\boldsymbol{B}$, the internal energy e, the heat flux $\boldsymbol{Q}$, the heat source R, the entropy η and the absolute temperature $\Theta > 0$, such that the balance of linear momentum and the balance of energy hold for every part of the body:

$$\operatorname{Div}\boldsymbol{P} + \boldsymbol{B} = \rho_0\ddot{\boldsymbol{\chi}}, \qquad \boldsymbol{P} : \dot{\boldsymbol{F}} - \operatorname{Div}\boldsymbol{Q} + R = \dot{e}. \tag{7.5}$$

The key observation is that the two balance equations do not restrict $\boldsymbol{\chi}$, $\boldsymbol{P}$, e, $\boldsymbol{Q}$, η, Θ: given these six fields, Eq. (7.5) *defines* the body force $\boldsymbol{B}$ and the heat supply R needed to sustain the process. Body force and heat supply are not regarded as given a priori, but as assignable

at will by the external world (Coleman and Noll are "not disturbed by the fact that it might be difficult to control them experimentally").

7.4.2 The original assumptions

A material is defined by *constitutive assumptions*, which restrict the processes that are admissible in a body made of that material. The *elastic material with heat conduction and viscosity* of the paper is defined by the response functions

$$e = \hat{e}(\boldsymbol{F}, \eta) \qquad \text{(caloric equation of state)}, \tag{7.6}$$

$$\Theta = \hat{\Theta}(\boldsymbol{F}, \eta) \qquad \text{(temperature equation)}, \tag{7.7}$$

$$\boldsymbol{P} = \hat{\boldsymbol{P}}^e(\boldsymbol{F}, \eta) + \hat{\boldsymbol{P}}^v(\boldsymbol{F}, \eta)[\dot{\boldsymbol{F}}] \qquad \text{(elastic + viscous stress)}, \tag{7.8}$$

$$\boldsymbol{Q} = \hat{\boldsymbol{Q}}(\boldsymbol{F}, \eta, \operatorname{Grad}\Theta) \qquad \text{(heat conduction)}, \tag{7.9}$$

where $\hat{\boldsymbol{P}}^v(\boldsymbol{F}, \eta)[\cdot]$ is linear in $\dot{\boldsymbol{F}}$ (the paper notes that it suffices that $\hat{\boldsymbol{P}}^v \to \boldsymbol{O}$ as $\dot{\boldsymbol{F}} \to \boldsymbol{O}$). A process is *admissible* if these relations hold at every point and time. To build an admissible process it is sufficient to prescribe the motion $\boldsymbol{\chi}(\boldsymbol{X}, t)$ and the entropy $\eta(\boldsymbol{X}, t)$ arbitrarily: then $\boldsymbol{F}$ and $\dot{\boldsymbol{F}}$ follow, Eqs. (7.6) to (7.8) give e, Θ and $\boldsymbol{P}$, Eq. (7.9) gives $\boldsymbol{Q}$, and finally $\boldsymbol{B}$ and R are computed from Eq. (7.5).

Coleman–Noll postulate

The Clausius–Duhem inequality Eq. (5.55) must hold for *every* admissible process:

$$\Theta\gamma = \boldsymbol{P} : \dot{\boldsymbol{F}} - \dot{e} + \Theta\dot{\eta} - \frac{1}{\Theta}\boldsymbol{Q} \cdot \operatorname{Grad}\Theta \geq 0.$$

It is not a restriction on the processes, but on the response functions.

7.4.3 Consequences

By the chain rule, $\dot{e} = \frac{\partial \hat{e}}{\partial \boldsymbol{F}} : \dot{\boldsymbol{F}} + \frac{\partial \hat{e}}{\partial \eta}\dot{\eta}$, and the inequality becomes

$$\Big(\hat{\Theta} - \frac{\partial \hat{e}}{\partial \eta}\Big)\dot{\eta} + \Big(\hat{\boldsymbol{P}}^e - \frac{\partial \hat{e}}{\partial \boldsymbol{F}}\Big) : \dot{\boldsymbol{F}} + \hat{\boldsymbol{P}}^v[\dot{\boldsymbol{F}}] : \dot{\boldsymbol{F}} - \frac{1}{\Theta}\hat{\boldsymbol{Q}} \cdot \operatorname{Grad}\Theta \geq 0. \tag{7.10}$$

Step 1: temperature relation. Consider processes with *homogeneous* deformation and entropy: $\boldsymbol{F}$ and η depend only on time. By Eq. (7.7) the temperature is homogeneous too, and $\operatorname{Grad}\Theta = \boldsymbol{0}$. Since $\boldsymbol{F}(t)$ and $\eta(t)$ are arbitrary functions, at a given instant the values of $\boldsymbol{F}$, $\dot{\boldsymbol{F}}$, η, $\dot{\eta}$ can be assigned independently. Taking $\dot{\boldsymbol{F}} = \boldsymbol{O}$, Eq. (7.10) reduces to $(\hat{\Theta} - \partial\hat{e}/\partial\eta)\dot{\eta} \geq 0$ for every $\dot{\eta}$, positive or negative, which is possible only if

$$\Theta = \hat{\Theta}(\boldsymbol{F}, \eta) = \frac{\partial \hat{e}}{\partial \eta}(\boldsymbol{F}, \eta). \tag{7.11}$$

Step 2: stress relation and dissipation inequality. Still in homogeneous processes, take $\dot{\eta} = 0$ and replace $\dot{\boldsymbol{F}}$ by $\alpha\dot{\boldsymbol{F}}$, with α an arbitrary real number. By the linearity of $\hat{\boldsymbol{P}}^v$,

$$\alpha\Big(\hat{\boldsymbol{P}}^e - \frac{\partial \hat{e}}{\partial \boldsymbol{F}}\Big) : \dot{\boldsymbol{F}} + \alpha^2\, \hat{\boldsymbol{P}}^v[\dot{\boldsymbol{F}}] : \dot{\boldsymbol{F}} \geq 0 \qquad \forall \alpha \in \mathbb{R}.$$

A polynomial $a\alpha + b\alpha^2$ is non-negative for every α if and only if $a = 0$ and $b \geq 0$. Since $\dot{\boldsymbol{F}}$ is arbitrary,

$$\boldsymbol{P}^e = \hat{\boldsymbol{P}}^e(\boldsymbol{F}, \eta) = \frac{\partial \hat{e}}{\partial \boldsymbol{F}}(\boldsymbol{F}, \eta), \qquad \hat{\boldsymbol{P}}^v(\boldsymbol{F}, \eta)[\dot{\boldsymbol{F}}] : \dot{\boldsymbol{F}} \geq 0 \quad \forall \dot{\boldsymbol{F}}. \tag{7.12}$$

The first is the *stress relation*: the equilibrium stress derives from the internal energy as from a potential. The second is the *dissipation inequality*: the viscous stress never produces power.

Step 3: heat conduction inequality. Suppose that the temperature relation Eq. (7.11) can be inverted to give $\eta = \tilde{\eta}(\boldsymbol{F}, \Theta)$; this is the case, in particular, if $\hat{e}$ is a convex function of η. Consider now processes with *constant, homogeneous* deformation, $\dot{\boldsymbol{F}} = \boldsymbol{O}$, and time-independent, non-uniform temperature $\Theta(\boldsymbol{X})$; the entropy $\eta(\boldsymbol{X}) = \tilde{\eta}(\boldsymbol{F}, \Theta(\boldsymbol{X}))$ is then time independent, and the process is admissible. For such processes Eq. (7.10) reduces to

$$\hat{\boldsymbol{Q}}(\boldsymbol{F}, \eta, \operatorname{Grad}\Theta) \cdot \operatorname{Grad}\Theta \leq 0. \tag{7.13}$$

Since $\Theta(\boldsymbol{X})$ can be chosen so that, at a given point, Θ and $\operatorname{Grad}\Theta$ take arbitrary values, this *heat conduction inequality* must hold for every vector $\boldsymbol{g}$ in place of $\operatorname{Grad}\Theta$: heat flows against the temperature gradient. Moreover, the function $\varphi(\boldsymbol{g}) = \boldsymbol{g} \cdot \hat{\boldsymbol{Q}}(\boldsymbol{F}, \eta, \boldsymbol{g})$ has a maximum, equal to 0, at $\boldsymbol{g} = \boldsymbol{0}$, so its gradient vanishes there: $\hat{\boldsymbol{Q}}(\boldsymbol{F}, \eta, \boldsymbol{0}) = \boldsymbol{0}$. There is no heat flux without a temperature gradient (no "piezo-caloric" effect).

Sufficiency. With Eqs. (7.11) and (7.12), the entropy production Eq. (7.10) reduces to

$$\Theta\gamma = \hat{\boldsymbol{P}}^v[\dot{\boldsymbol{F}}] : \dot{\boldsymbol{F}} - \frac{1}{\Theta}\hat{\boldsymbol{Q}} \cdot \operatorname{Grad}\Theta \geq 0, \tag{7.14}$$

which is non-negative whenever the dissipation and heat conduction inequalities hold. Hence:

Theorem 7.1 (Coleman–Noll, 1963)**.** *The Clausius–Duhem inequality holds for every admissible process of an elastic material with heat conduction and viscosity if and only if*

1. *the temperature is* $\Theta = \partial\hat{e}/\partial\eta$*;*
2. *the equilibrium stress is* $\boldsymbol{P}^e = \partial\hat{e}/\partial\boldsymbol{F}$*;*
3. *the viscous stress satisfies* $\hat{\boldsymbol{P}}^v[\dot{\boldsymbol{F}}] : \dot{\boldsymbol{F}} \geq 0$ *for every* $\dot{\boldsymbol{F}}$*;*
4. *the heat flux satisfies* $\hat{\boldsymbol{Q}}(\boldsymbol{F}, \eta, \boldsymbol{g}) \cdot \boldsymbol{g} \leq 0$ *for every* $\boldsymbol{g}$*.*

The material is therefore determined by three response functions only: $\hat{e}$, $\hat{\boldsymbol{P}}^v$ *and* $\hat{\boldsymbol{Q}}$*.*

Remark 7.2. As the paper stresses, each of these results is by itself a familiar formula. What is new is the logical connection: they follow from the second law, assumed to hold for all processes, rather than being postulated one by one. The method, the *Coleman–Noll procedure*, is to expand the dissipation inequality with the chain rule and to exploit the arbitrariness of the rates that appear linearly.

Example 7.1 (Linearly viscous fluid with heat conduction)**.** For a fluid (isotropy group equal to the whole unimodular group) the internal energy depends on $\boldsymbol{F}$ only through J, and the paper shows that the equilibrium stress is a pressure, the viscous stress is linear

and isotropic in $\boldsymbol{d}$, and the heat flux is of Fourier's type:

$$\boldsymbol{\sigma} = -p\boldsymbol{I} + \lambda(\operatorname{tr}\boldsymbol{d})\boldsymbol{I} + 2\mu\boldsymbol{d}, \qquad \boldsymbol{q} = -k\operatorname{grad}\Theta, \tag{7.15}$$

where the pressure is given by the same formula as for the inviscid fluid, $p = -\frac{\mathrm{d}\hat{e}}{\mathrm{d}J}$ (with e per unit reference volume: from $\boldsymbol{P}^e = \hat{e}'(J)\,J\boldsymbol{F}^{-\mathrm{T}}$, $\boldsymbol{\sigma}^e = J^{-1}\boldsymbol{P}^e\boldsymbol{F}^{\mathrm{T}} = \hat{e}'(J)\boldsymbol{I}$). The spatial form of the dissipation, $J^{-1}\boldsymbol{P}^v\!:\!\dot{\boldsymbol{F}} = \boldsymbol{\sigma}^v\!:\!\boldsymbol{l} = \boldsymbol{\sigma}^v\!:\!\boldsymbol{d}$, gives, with $\boldsymbol{d} = \frac{1}{3}(\operatorname{tr}\boldsymbol{d})\boldsymbol{I} + \operatorname{dev}\boldsymbol{d}$,

$$\boldsymbol{\sigma}^v : \boldsymbol{d} = \lambda(\operatorname{tr}\boldsymbol{d})^2 + 2\mu\,\boldsymbol{d} : \boldsymbol{d} = \Big(\lambda + \tfrac{2}{3}\mu\Big)(\operatorname{tr}\boldsymbol{d})^2 + 2\mu\,\operatorname{dev}\boldsymbol{d} : \operatorname{dev}\boldsymbol{d} \geq 0 \quad \forall\boldsymbol{d}, \tag{7.16}$$

which is equivalent to $\mu \geq 0$ and $3\lambda + 2\mu \geq 0$: the shear viscosity and the bulk viscosity $\lambda + \frac{2}{3}\mu$ must be non-negative. The heat conduction inequality gives $-\boldsymbol{q}\cdot\operatorname{grad}\Theta = k\,|\operatorname{grad}\Theta|^2 \geq 0$, i.e., $k \geq 0$ (Exercise 7.4).

7.4.4 Extension: rates, gradients and internal variables

The slides (226–229) present the procedure in a more general setting. The local thermodynamic state is assumed to be defined by

$$\boldsymbol{F},\ \operatorname{Grad}\boldsymbol{F},\dots,\ \dot{\boldsymbol{F}},\dots,\ \eta,\ \operatorname{Grad}\eta,\dots,\ \dot{\eta},\dots,\ \boldsymbol{\xi},$$

and the internal energy and the temperature are assumed to depend on all of them. The equilibrium stress, the viscous stress and the *thermodynamic forces* conjugate to the internal variables are *defined* as

$$\boldsymbol{P}^e = \frac{\partial e}{\partial \boldsymbol{F}}, \qquad \boldsymbol{P}^v = \boldsymbol{P} - \boldsymbol{P}^e, \qquad \boldsymbol{\Xi} = -\frac{\partial e}{\partial \boldsymbol{\xi}}. \tag{7.17}$$

By the chain rule,

$$\dot{e} = \frac{\partial e}{\partial \boldsymbol{F}}\!:\!\dot{\boldsymbol{F}} + \frac{\partial e}{\partial \operatorname{Grad}\boldsymbol{F}}\vdots\operatorname{Grad}\dot{\boldsymbol{F}} + \frac{\partial e}{\partial \dot{\boldsymbol{F}}}\!:\!\ddot{\boldsymbol{F}} + \dots + \frac{\partial e}{\partial \eta}\dot{\eta} + \frac{\partial e}{\partial \operatorname{Grad}\eta}\cdot\operatorname{Grad}\dot{\eta} + \frac{\partial e}{\partial \dot{\eta}}\ddot{\eta} + \dots + \frac{\partial e}{\partial \boldsymbol{\xi}}\cdot\dot{\boldsymbol{\xi}},$$

and the Clausius–Duhem inequality becomes

$$\begin{aligned}\boldsymbol{P}^v : \dot{\boldsymbol{F}} + \boldsymbol{\Xi}\cdot\dot{\boldsymbol{\xi}} + \Big(\Theta - \frac{\partial e}{\partial \eta}\Big)\dot{\eta} - \frac{1}{\Theta}\boldsymbol{Q}\cdot\operatorname{Grad}\Theta \\ - \frac{\partial e}{\partial \operatorname{Grad}\boldsymbol{F}}\vdots\operatorname{Grad}\dot{\boldsymbol{F}} - \frac{\partial e}{\partial \dot{\boldsymbol{F}}}\!:\!\ddot{\boldsymbol{F}} - \frac{\partial e}{\partial \operatorname{Grad}\eta}\cdot\operatorname{Grad}\dot{\eta} - \frac{\partial e}{\partial \dot{\eta}}\ddot{\eta} - \dots \geq 0.\end{aligned}$$

Since $\boldsymbol{\chi}$ and η can be chosen arbitrarily, at a given point and instant the higher derivatives $\ddot{\boldsymbol{F}}$, $\operatorname{Grad}\dot{\boldsymbol{F}}$, $\ddot{\eta}$, $\operatorname{Grad}\dot{\eta},\dots$ can be assigned arbitrarily, independently of the state. They appear linearly and only in the last line (the response functions Θ, $\boldsymbol{P}$, $\boldsymbol{Q}$ and the evolution of $\boldsymbol{\xi}$ do not depend on them), so their coefficients must vanish: e does not depend on the rates and gradients. Similarly, assuming that Θ, $\boldsymbol{P}^v$, $\boldsymbol{Q}$ and $\dot{\boldsymbol{\xi}}$ do not depend on $\dot{\eta}$, the arbitrariness of $\dot{\eta}$ gives the temperature relation. The results are summarised in

Coleman–Noll: results (slide 229)

$$e = e(\boldsymbol{F}, \eta, \boldsymbol{\xi}), \qquad \Theta = \frac{\partial e}{\partial \eta}(\boldsymbol{F}, \eta, \boldsymbol{\xi}), \tag{7.18}$$

and the dissipation inequality

$$\boldsymbol{P}^v : \dot{\boldsymbol{F}} + \boldsymbol{\Xi} \cdot \dot{\boldsymbol{\xi}} - \frac{1}{\Theta} \boldsymbol{Q} \cdot \operatorname{Grad} \Theta \geq 0. \tag{7.19}$$

The three sources of dissipation are the viscous stress, the evolution of the internal variables and heat conduction. The elastic part of the stress disappears from the dissipation, because it is the derivative of e with respect to $\boldsymbol{F}$. Note that, differently from $\dot{\boldsymbol{F}}$, the rate $\dot{\boldsymbol{\xi}}$ cannot be chosen at will: it is given by an evolution law (a kinetic relation, Section 7.6); for this reason the term $\boldsymbol{\Xi} \cdot \dot{\boldsymbol{\xi}}$ remains in the dissipation.

7.5 Thermodynamic potentials

The internal energy $e(\boldsymbol{F}, \eta, \boldsymbol{\xi})$ is a *thermodynamic potential*: it generates the equilibrium stress, the temperature and the thermodynamic forces. However, its natural variable η cannot be measured directly, and it is more useful in practice to work with alternative potentials, obtained by **Legendre transforms** that exchange a variable with its conjugate.

Complement (the Legendre transform)

Given a function $f(x)$, its Legendre transform is $f^*(y) = \sup_x \big[xy - f(x)\big]$. If f is convex and differentiable, the supremum is attained where $y = f'(x)$, and f^* is convex with $(f^*)'(y) = x$: the roles of x and $y = f'(x)$ are exchanged, and $f(x) + f^*(y) = xy$ at corresponding points. With inf and a change of sign, $-f^*(y) = \inf_x \big[f(x) - xy\big]$. The slides use the infimum or the supremum (rather than the minimum or maximum) because the transform does not require a stationary point to exist.

Helmholtz free energy. Replace the entropy with the temperature:

$$A(\boldsymbol{F}, \Theta, \boldsymbol{\xi}) = \inf_{\eta} \big[e(\boldsymbol{F}, \eta, \boldsymbol{\xi}) - \Theta\eta\big]. \tag{7.20}$$

For e convex in η, the infimum is attained where $\Theta = \partial e / \partial \eta$, the temperature relation, and $A = e - \Theta\eta$ at that point. Differentiating, $\mathrm{d}A = \frac{\partial e}{\partial \boldsymbol{F}} : \mathrm{d}\boldsymbol{F} + \frac{\partial e}{\partial \eta}\,\mathrm{d}\eta + \frac{\partial e}{\partial \boldsymbol{\xi}} \cdot \mathrm{d}\boldsymbol{\xi} - \Theta\,\mathrm{d}\eta - \eta\,\mathrm{d}\Theta$, and the $\mathrm{d}\eta$ terms cancel:

Helmholtz free energy

$$\eta = -\frac{\partial A}{\partial \Theta}, \qquad \boldsymbol{P}^e = \frac{\partial A}{\partial \boldsymbol{F}}, \qquad \boldsymbol{\Xi} = -\frac{\partial A}{\partial \boldsymbol{\xi}}. \tag{7.21}$$

It is the most used potential, because it uses the variables we are familiar with. In terms of A, the Clausius–Planck inequality Eq. (5.56) reads (Exercise 7.1)

$$\mathcal{D}_{\text{int}} = \boldsymbol{P} : \dot{\boldsymbol{F}} - \dot{A} - \eta\dot{\Theta} \geq 0, \tag{7.22}$$

Holzapfel's form [3, Sect. 4.7], with Ψ in place of A. In isothermal conditions $\mathcal{D}_{\text{int}} = \boldsymbol{P} : \dot{\boldsymbol{F}} - \dot{A} \geq 0$.

Enthalpy. Replace the deformation with the equilibrium stress:

$$H(\boldsymbol{P}^e, \eta, \boldsymbol{\xi}) = \inf_{\boldsymbol{F}} \big[e(\boldsymbol{F}, \eta, \boldsymbol{\xi}) - \boldsymbol{P}^e : \boldsymbol{F} \big], \qquad \Theta = \frac{\partial H}{\partial \eta}, \quad \boldsymbol{F} = -\frac{\partial H}{\partial \boldsymbol{P}^e}, \quad \boldsymbol{\Xi} = -\frac{\partial H}{\partial \boldsymbol{\xi}}. \tag{7.23}$$

Gibbs free energy. Replace both, starting from the Helmholtz energy:

$$G(\boldsymbol{P}^e, \Theta, \boldsymbol{\xi}) = \sup_{\boldsymbol{F}} \big[\boldsymbol{P}^e : \boldsymbol{F} - A(\boldsymbol{F}, \Theta, \boldsymbol{\xi}) \big], \qquad \eta = \frac{\partial G}{\partial \Theta}, \quad \boldsymbol{F} = \frac{\partial G}{\partial \boldsymbol{P}^e}, \quad \boldsymbol{\Xi} = \frac{\partial G}{\partial \boldsymbol{\xi}}. \tag{7.24}$$

At the supremum $\boldsymbol{P}^e = \partial A / \partial \boldsymbol{F}$ and $G = \boldsymbol{P}^e : \boldsymbol{F} - A$, so $\partial G / \partial \boldsymbol{\xi} = -\partial A / \partial \boldsymbol{\xi} = +\boldsymbol{\Xi}$. (Slide 232 writes $\boldsymbol{\Xi} = -\partial G / \partial \boldsymbol{\xi}$: with the sign convention $G = \sup[\boldsymbol{P}^e : \boldsymbol{F} - A]$ used there, the correct sign is $+$; the sign would be $-$ with the alternative convention $G = A - \boldsymbol{P}^e : \boldsymbol{F}$, see Exercise 7.2.)

Remark 7.3. The transforms in $\boldsymbol{F}$ require A to be convex in $\boldsymbol{F}$, which is not the case at finite strains for most materials (convexity in $\boldsymbol{F}$ is incompatible with the growth condition $A \to \infty$ as $J \to 0^+$ and would exclude physically reasonable phenomena such as buckling). For this reason, enthalpy and Gibbs energy are rarely used in finite elasticity, while the Helmholtz free energy is the basic potential of hyperelasticity.

Link with small strain theory. For a one-dimensional hyperelastic material with a monotone stress–strain curve $\sigma = \sigma(\varepsilon)$, the *strain energy density* $\omega(\varepsilon) = \int_0^\varepsilon \sigma \, \mathrm{d}\varepsilon$ and the *complementary energy density* $\gamma(\sigma) = \int_0^\sigma \varepsilon \, \mathrm{d}\sigma$ are areas under and to the left of the curve (Fig. 7.1). They are Legendre transforms of each other, $\omega(\varepsilon) + \gamma(\sigma) = \sigma\varepsilon$, with $\sigma = \frac{\mathrm{d}\omega}{\mathrm{d}\varepsilon}$ and $\varepsilon = \frac{\mathrm{d}\gamma}{\mathrm{d}\sigma}$. In isothermal conditions and without internal variables, the Helmholtz free energy coincides with the strain energy density, and the Gibbs energy with the complementary energy density. For a linear material they are numerically equal, $\omega = \gamma = \frac{1}{2}\sigma\varepsilon$; in general they are not.

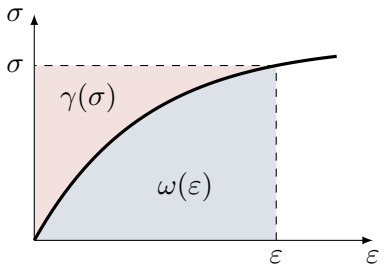


Figure 7.1: Strain energy density ω (area under the curve, Helmholtz energy) and complementary energy density γ (area to the left of the curve, Gibbs energy): $\omega + \gamma = \sigma\varepsilon$ (slide 232).

In class

"These are the things that remain in your minds when you are studying": the lecturer suggested remembering the potentials through the one-dimensional picture, with the notation of her undergraduate course, ω for the strain energy density and γ for the complementary energy density.

7.6 Thermodynamic forces, fluxes and kinetic relations

In class

This part (slide 233) was skipped ("we are already late enough"). The lecturer only remarked that the internal variables need an evolution law, a *kinetic relation*, to complete the description: in plasticity, for instance, one internal variable can be the accumulated plastic strain, a scalar, whose evolution must be given in rate form.

The dissipation inequality Eq. (7.19) is a sum of products of pairs of quantities. Collect *thermodynamic forces* and corresponding *fluxes* in two arrays:

$$\boldsymbol{X} = \begin{bmatrix} \dot{\boldsymbol{F}} \\ \boldsymbol{\Xi} \\ -\operatorname{Grad}\Theta/\Theta \end{bmatrix}, \qquad \boldsymbol{J} = \begin{bmatrix} \boldsymbol{P}^v \\ \dot{\boldsymbol{\xi}} \\ \boldsymbol{Q} \end{bmatrix}, \qquad \boldsymbol{X}\cdot\boldsymbol{J} = \dot{\boldsymbol{F}}:\boldsymbol{P}^v + \boldsymbol{\Xi}\cdot\dot{\boldsymbol{\xi}} - \frac{1}{\Theta}\operatorname{Grad}\Theta\cdot\boldsymbol{Q} \geq 0. \tag{7.25}$$

(Here $\boldsymbol{J}$ denotes the array of fluxes, not to be confused with the Jacobian J.) If the fluxes are determined only by the local thermodynamic forces and the local thermodynamic state, the **kinetic relations** take the form

$$\boldsymbol{J} = \boldsymbol{J}(\boldsymbol{X}; \boldsymbol{F}, \Theta, \boldsymbol{\xi}). \tag{7.26}$$

They include the viscous law, the evolution laws of the internal variables and the heat conduction law, possibly fully coupled (a temperature gradient may produce a viscous stress, a strain rate may produce a heat flux).

Onsager reciprocity. Close to equilibrium the fluxes vary linearly with all the forces, $\boldsymbol{J} = \mathsf{L}\boldsymbol{X}$. On the basis of an extension of the theory of fluctuations (Onsager, 1931), it may be accepted as an experimentally founded law of nature that the matrix of coefficients is symmetric. In the nonlinear setting the slides assume the **Onsager reciprocity relations**

$$\frac{\partial J_i}{\partial X_j} = \frac{\partial J_j}{\partial X_i}, \tag{7.27}$$

which are the integrability conditions for the existence of a scalar **kinetic potential** (dissipation potential) ψ such that

Kinetic potential

$$\boldsymbol{J} = \frac{\partial \psi}{\partial \boldsymbol{X}}(\boldsymbol{X}; \boldsymbol{F}, \Theta, \boldsymbol{\xi}): \qquad \boldsymbol{P}^v = \frac{\partial \psi}{\partial \dot{\boldsymbol{F}}}, \quad \dot{\boldsymbol{\xi}} = \frac{\partial \psi}{\partial \boldsymbol{\Xi}}, \quad \boldsymbol{Q} = \frac{\partial \psi}{\partial(-\operatorname{Grad}\Theta/\Theta)}. \tag{7.28}$$

All the fluxes are derivatives of one scalar function; this describes fully nonlinear and coupled strain rate, viscosity and heat conduction relations. A sufficient condition for the dissipation inequality is that ψ be convex in $\boldsymbol{X}$, with $\psi(\boldsymbol{0}) = 0$ and a minimum at $\boldsymbol{X} = \boldsymbol{0}$: by convexity $\psi(\boldsymbol{0}) \geq \psi(\boldsymbol{X}) + \frac{\partial \psi}{\partial \boldsymbol{X}} \cdot (\boldsymbol{0} - \boldsymbol{X})$, hence

$$\boldsymbol{X}\cdot\boldsymbol{J} = \boldsymbol{X}\cdot\frac{\partial \psi}{\partial \boldsymbol{X}} \geq \psi(\boldsymbol{X}) - \psi(\boldsymbol{0}) \geq 0. \tag{7.29}$$

Examples:

- the quadratic potential $\psi = \frac{1}{2}\boldsymbol{X}\cdot\mathsf{L}\boldsymbol{X}$ with L symmetric positive semi-definite gives

linear, coupled kinetic relations (Onsager);

- uncoupled, isotropic Newtonian viscosity and Fourier conduction (spatial form): $\psi = \frac{\lambda}{2}(\operatorname{tr}\boldsymbol{d})^2 + \mu\,\boldsymbol{d}:\boldsymbol{d} + \frac{k\Theta}{2}\,|\operatorname{grad}\Theta/\Theta|^2$, which gives back Example 7.1 (Exercise 7.5);
- in rate-independent plasticity ψ is positively homogeneous of degree one in $\boldsymbol{\Xi}$, hence not differentiable at the origin; the derivative is replaced by the subdifferential, and the elastic domain and the flow rule follow. This is the subject of the chapter on plasticity.

7.7 Exercises

Exercise 7.1. Starting from $A = e - \Theta\eta$, derive Eq. (7.21) and the Clausius–Planck inequality in the form Eq. (7.22). Show that, with $A = A(\boldsymbol{F}, \Theta, \boldsymbol{\xi})$, the Coleman–Noll argument applied to Eq. (7.22) gives directly $\boldsymbol{P}^e = \partial A/\partial \boldsymbol{F}$, $\eta = -\partial A/\partial\Theta$ and $\mathcal{D}_{\text{int}} = \boldsymbol{P}^v : \dot{\boldsymbol{F}} + \boldsymbol{\Xi}\cdot\dot{\boldsymbol{\xi}} \geq 0$.

▷ Solution on p. 147

Exercise 7.2. Derive the relations of the enthalpy $H(\boldsymbol{P}^e, \eta, \boldsymbol{\xi})$ and of the Gibbs energy $G(\boldsymbol{P}^e, \Theta, \boldsymbol{\xi})$, and verify the sign of $\partial G/\partial\boldsymbol{\xi}$. What changes with the convention $G = A - \boldsymbol{P}^e : \boldsymbol{F}$ (inf over $\boldsymbol{F}$)?

▷ Solution on p. 147

Exercise 7.3. One-dimensional nonlinear elasticity: for $\sigma = E_0\varepsilon/(1 + \varepsilon/\varepsilon_0)$ $(\varepsilon \geq 0)$, compute the strain energy density $\omega(\varepsilon)$ and the complementary energy density $\gamma(\sigma)$, check that $\omega + \gamma = \sigma\varepsilon$ and $\frac{\mathrm{d}\gamma}{\mathrm{d}\sigma} = \varepsilon$, and compare ω and γ with their linear counterparts.

▷ Solution on p. 148

Exercise 7.4. For the linearly viscous fluid with Fourier conduction of Example 7.1, prove that $\lambda(\operatorname{tr}\boldsymbol{d})^2 + 2\mu\,\boldsymbol{d}:\boldsymbol{d} \geq 0$ for every symmetric $\boldsymbol{d}$ if and only if $\mu \geq 0$ and $3\lambda + 2\mu \geq 0$, and that the heat conduction inequality requires $k \geq 0$. Show that the slides' condition $\boldsymbol{P}^v : \dot{\boldsymbol{F}} \geq 0$ coincides with $J\boldsymbol{\sigma}^v : \boldsymbol{d} \geq 0$.

▷ Solution on p. 148

Exercise 7.5. Verify that the kinetic potential $\psi = \frac{\lambda}{2}(\operatorname{tr}\boldsymbol{d})^2 + \mu\,\boldsymbol{d}:\boldsymbol{d} + \frac{k\Theta}{2}\,|\operatorname{grad}\Theta/\Theta|^2$ generates the viscous stress and the heat flux of Example 7.1, and that it is convex if and only if the conditions of Exercise 7.4 hold.

▷ Solution on p. 148

Exercise 7.6. Kelvin–Voigt and Maxwell models (one-dimensional, small strain, isothermal). (a) For $\sigma = E\varepsilon + \eta_v\dot{\varepsilon}$, identify the equilibrium and viscous stresses, the free energy and the dissipation, and give the condition on η_v. (b) For a Maxwell element (spring E and dashpot η_v in series), take the viscous strain $\xi = \varepsilon^v$ as internal variable, with $A = \frac{1}{2}E(\varepsilon - \xi)^2$. Compute the thermodynamic force Ξ, the dissipation, and show that the evolution law $\eta_v\dot{\xi} = \Xi$ derives from a kinetic potential and satisfies the second law.

▷ Solution on p. 149

Exercise 7.7. Thermoelastic material (no viscosity, no internal variables) with $A = A(\boldsymbol{F}, \Theta)$. Show that the energy balance reduces to $\Theta\dot{\eta} = R - \operatorname{Div}\boldsymbol{Q}$, and that, with the specific heat $c = -\Theta\,\partial^2 A/\partial\Theta^2$, it becomes the *coupled heat equation*

$$c\,\dot{\Theta} = R - \operatorname{Div}\boldsymbol{Q} + \Theta\frac{\partial^2 A}{\partial\Theta\,\partial\boldsymbol{F}} : \dot{\boldsymbol{F}}.$$

Interpret the last term (thermoelastic coupling).

▷ Solution on p. 149

A

Review of Small-Strain Linear Isotropic Elasticity

In class

This appendix is the English rewriting of a recorded lecture in Italian by the same lecturer, belonging to her undergraduate course on the mechanics of structures, which was provided together with the material of the first day. It recalls the linear, small-strain theory that the nonlinear theory of this course generalises: strain energy, Hooke's law, elastic constants and their bounds, bulk modulus and anisotropy. The notation is that of Chapter 1: $\boldsymbol{\varepsilon}$ is the infinitesimal strain tensor, $\boldsymbol{\sigma}$ the stress tensor, both symmetric.

A.1 Strain energy of a linear isotropic material

For a linear elastic *isotropic* material, the strain energy density ω (energy per unit volume) can depend on the strain only through invariants of $\boldsymbol{\varepsilon}$ (isotropy). Moreover, the energy of a linear material is a *quadratic form* of the strains. Of the three invariants, I_1 is linear in the strains, I_2 quadratic and I_3 cubic; hence I_3 cannot appear, and the only admissible terms are I_2 and I_1^2. The most commonly used expression is

Strain energy, linear isotropic elasticity

$$\omega(\boldsymbol{\varepsilon}) = \tfrac{1}{2}(\lambda + 2\mu)\, I_1^2 - 2\mu\, I_2, \tag{A.1}$$

where, from (1.69)–(1.70),

$$I_1 = \operatorname{tr}\boldsymbol{\varepsilon} = \varepsilon_{kk} = \varepsilon_{11} + \varepsilon_{22} + \varepsilon_{33}, \tag{A.2}$$

$$I_2 = (\varepsilon_{11}\varepsilon_{22} - \varepsilon_{21}\varepsilon_{12}) + (\varepsilon_{22}\varepsilon_{33} - \varepsilon_{32}\varepsilon_{23}) + (\varepsilon_{33}\varepsilon_{11} - \varepsilon_{31}\varepsilon_{13}), \tag{A.3}$$

the repeated index k in ε_{kk} implying the sum.

Lamé constants. The constants λ and μ are the *Lamé constants*, after G. Lamé, who proposed this form of the elastic law. It is not the only possible parametrisation (other pairs of constants can be used), but it is the most widely used.

- μ coincides exactly with the *shear modulus* G of the material: in some formulations G appears in place of μ. It is a quantity we certainly know how to measure, and is given in MPa or GPa depending on the material.
- λ is also dimensionally an elastic modulus (MPa or GPa), but its physical meaning is not as immediate as that of G and of Young's modulus E: it is not directly related to a simple experimental test. It is nevertheless convenient to write the constitutive law

with λ added to 2μ.

Why 2μ. The factor 2 appears because the law is written with the *tensorial* shear strains ε_{ij} $(i \neq j)$, whereas engineers like to relate shear stresses to the *engineering* shear strains (angular distortions) $\gamma_{ij} = \varepsilon_{ij} + \varepsilon_{ji} = 2\varepsilon_{ij}$, for which, in the linear range, $\tau = G\gamma$.

A.2 Hooke's law

The stress follows by differentiating the energy, $\sigma_{ij} = \partial\omega/\partial\varepsilon_{ij}$. Since $\omega = \omega(I_1, I_2)$, the chain rule gives

$$\frac{\partial\omega}{\partial\varepsilon_{ij}} = \frac{\partial\omega}{\partial I_1}\frac{\partial I_1}{\partial\varepsilon_{ij}} + \frac{\partial\omega}{\partial I_2}\frac{\partial I_2}{\partial\varepsilon_{ij}}, \tag{A.4}$$

so four derivatives are needed.

Derivative of I_1. For a term of the sum $\varepsilon_{kk} = \varepsilon_{11} + \varepsilon_{22} + \varepsilon_{33}$ to have a non-zero derivative with respect to ε_{ij}, i and j must be equal (and equal to k). Formally,

$$\frac{\partial I_1}{\partial\varepsilon_{ij}} = \frac{\partial\varepsilon_{kk}}{\partial\varepsilon_{ij}} = \delta_{ki}\delta_{kj} = \delta_{ij} : \tag{A.5}$$

the first delta is 1 only if $k = i$, the second only if $k = j$; the product of two unit matrices sharing the index k is again the unit matrix. Check directly: differentiating $\varepsilon_{11} + \varepsilon_{22} + \varepsilon_{33}$ with respect to ε_{11} gives 1, with respect to ε_{12} gives 0: the result is the identity matrix.

Derivative of I_2. Separate the diagonal and off-diagonal components. With respect to ε_{11}:

$$\frac{\partial I_2}{\partial\varepsilon_{11}} = \varepsilon_{22} + \varepsilon_{33} = (\varepsilon_{11} + \varepsilon_{22} + \varepsilon_{33}) - \varepsilon_{11} = I_1 - \varepsilon_{11}, \tag{A.6}$$

where we added and subtracted the component with respect to which we differentiate. The rule holds for any diagonal term: $\partial I_2/\partial\varepsilon_{ii} = I_1 - \varepsilon_{ii}$ (no sum). With respect to an off-diagonal term, e.g., ε_{12}, only $-\varepsilon_{21}\varepsilon_{12}$ contributes: $\partial I_2/\partial\varepsilon_{12} = -\varepsilon_{21}$; in general $\partial I_2/\partial\varepsilon_{ij} = -\varepsilon_{ji}$ for $i \neq j$. The two cases are unified as

$$\frac{\partial I_2}{\partial\varepsilon_{ij}} = I_1\,\delta_{ij} - \varepsilon_{ji} = I_1\,\delta_{ij} - \varepsilon_{ij}, \tag{A.7}$$

since $\boldsymbol{\varepsilon}$ is symmetric.[1]

Derivatives of ω. From (A.1), $\partial\omega/\partial I_1 = (\lambda + 2\mu)I_1$ (the derivative of I_1^2 is $2I_1$, times the coefficient $\frac{1}{2}(\lambda + 2\mu)$) and $\partial\omega/\partial I_2 = -2\mu$. Therefore

$$\sigma_{ij} = \frac{\partial\omega}{\partial\varepsilon_{ij}} = (\lambda + 2\mu)I_1\delta_{ij} - 2\mu(I_1\delta_{ij} - \varepsilon_{ij}). \tag{A.8}$$

The terms $\pm 2\mu I_1\delta_{ij}$ cancel, and we obtain

[1] Strictly, the components of a symmetric tensor are not independent; the derivatives are computed treating the nine components as independent and the result is symmetric. This is consistent with the tensor result $\partial I_2/\partial\boldsymbol{A} = I_1\boldsymbol{I} - \boldsymbol{A}^{\mathrm{T}}$.

Hooke's law (Lamé form)

$$\sigma_{ij} = \lambda\, \varepsilon_{kk}\, \delta_{ij} + 2\mu\, \varepsilon_{ij}, \qquad \boldsymbol{\sigma} = \lambda\, (\operatorname{tr} \boldsymbol{\varepsilon})\, \boldsymbol{I} + 2\mu\, \boldsymbol{\varepsilon}. \tag{A.9}$$

The first term, multiplied by δ_{ij}, is present only in the components with equal indices (normal stresses); the second holds for all the components of the matrix.

This is *Hooke's law*. Hooke performed experiments on springs and found that the force in a spring is the spring constant times the elongation: a linear law. Strictly speaking, Hooke's law is the concept of linearity, and applies to any linear elastic material, isotropic or not; but what is commonly called Hooke's law in our profession is (A.9), for isotropic materials. In terms of the fourth-order tensors of Section 1.13, $\mathbb{C} = \lambda\, \boldsymbol{I} \otimes \boldsymbol{I} + 2\mu\, \mathbb{I}_{\mathrm{s}}$.

A.2.1 Matrix (Voigt) form

Ordering the components as $\{\sigma_{11}, \sigma_{22}, \sigma_{33}, \sigma_{12}, \sigma_{23}, \sigma_{31}\}$ (the shear stresses are often denoted $\tau_{12}, \tau_{23}, \tau_{31}$), (A.9) reads

$$\begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \tau_{12} \\ \tau_{23} \\ \tau_{31} \end{Bmatrix} = \begin{bmatrix} \lambda+2\mu & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda+2\mu & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda+2\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & 2\mu & 0 & 0 \\ 0 & 0 & 0 & 0 & 2\mu & 0 \\ 0 & 0 & 0 & 0 & 0 & 2\mu \end{bmatrix} \begin{Bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{12} \\ \varepsilon_{23} \\ \varepsilon_{31} \end{Bmatrix}. \tag{A.10}$$

Each component with distinct indices, e.g., σ_{12}, also represents the equal component σ_{21}: the equal terms have been compacted. Every stress component is linked to the strain component with the same indices by 2μ on the diagonal; the normal stresses also receive λ times each normal strain. The two off-diagonal 3×3 blocks are null, and the lower-right block is diagonal: shear stresses are linked exclusively to the shear strains with the same indices.

In the engineering literature the strain is commonly stored as $\{\varepsilon_{11}, \varepsilon_{22}, \varepsilon_{33}, \gamma_{12}, \gamma_{23}, \gamma_{31}\}$. With the γ's instead of the ε's, the upper part of the matrix is unchanged, but the lower diagonal block contains $\mu = G$ instead of 2μ, recovering $\tau_{ij} = G\gamma_{ij}$. The meaning of the two forms is exactly the same, but attention must be paid to which one is being used.

A.3 Volumetric strain

Definition A.1 (Volumetric strain)**.** The *volumetric strain* θ of a small portion of material is the ratio between the change of volume and the initial volume,

$$\theta = \frac{\Delta V}{V_0} = \frac{V - V_0}{V_0}. \tag{A.11}$$

Under the small strain hypothesis it has a very simple expression: the first invariant of the small strain tensor.

Consider a small cube of material, with edges $\Delta x_1, \Delta x_2, \Delta x_3$ aligned with the Cartesian axes, and initial volume $V_0 = \Delta x_1 \Delta x_2 \Delta x_3$. After the deformation the edges change length; by the definition of normal strain, $\varepsilon_{11} = (\Delta \tilde{x}_1 - \Delta x_1)/\Delta x_1$, so the new length is $\Delta \tilde{x}_1 = (1 + \varepsilon_{11})\Delta x_1$, and similarly for the other edges. (Shear strains change the angles but, to first

order, not the volume.) The current volume is

$$V = \Delta\tilde{x}_1 \Delta\tilde{x}_2 \Delta\tilde{x}_3 = (1+\varepsilon_{11})(1+\varepsilon_{22})(1+\varepsilon_{33})\, V_0. \tag{A.12}$$

Expanding the product: the product of the three 1's gives 1; each strain times two 1's gives the linear terms $\varepsilon_{11}+\varepsilon_{22}+\varepsilon_{33}$; then there are quadratic terms ($\varepsilon_{11}\varepsilon_{22}$, ...) and one cubic term ($\varepsilon_{11}\varepsilon_{22}\varepsilon_{33}$). For small strains the quadratic and cubic terms are negligible:

$$V \simeq (1+\varepsilon_{11}+\varepsilon_{22}+\varepsilon_{33})\, V_0, \qquad \theta = \frac{V-V_0}{V_0} \simeq \varepsilon_{11}+\varepsilon_{22}+\varepsilon_{33} = \varepsilon_{kk} = I_1(\boldsymbol{\varepsilon}). \tag{A.13}$$

Whatever the initial volume, its value drops out, and the volumetric strain is the *sum* of the normal strains. It may look strange that a volume change, which is a product, becomes a sum: this happens because we are linearising, having neglected the higher-order terms. In finite strain this will no longer be true (the volume ratio will be the determinant of the deformation gradient, Chapter 3).

A.4 Hooke's law in terms of E and ν

It is often preferable to write the constitutive law in terms of the elastic constants determined by experimental tests: Young's modulus E and Poisson's ratio ν. One has (proof in Appendix A.6)

$$\mu = G = \frac{E}{2(1+\nu)}, \qquad \lambda = \frac{\nu E}{(1+\nu)(1-2\nu)}. \tag{A.14}$$

Substituting in (A.9):

$$\sigma_{11} = \frac{E}{(1+\nu)(1-2\nu)}\big[(1-\nu)\varepsilon_{11} + \nu(\varepsilon_{22}+\varepsilon_{33})\big], \qquad \tau_{12} = G\,\gamma_{12}, \tag{A.15}$$

and analogous expressions, with the appropriate rotation of the indices, for the other components. This is the *direct* law in terms of the fundamental elastic constants, which have a precise physical meaning; in the Lamé form μ and G are identical, but the physical meaning of λ, which depends on both E and ν in a non-simple way, is not evident.

> **In class**
>
> "Good professors do not know this expression by heart; I do not either and I always have to look it up, because it is never used. I will never ask you to know it by memory." What must be known is the Lamé form (A.9) and the inverse law below.

A.4.1 Inverse law

In practice it is extremely convenient to have the *inverse* constitutive law, $\boldsymbol{\varepsilon} = \mathbb{C}^{-1}\boldsymbol{\sigma}$ in matrix form. The constitutive matrix must be invertible: if the law is linear and we can express the stress as a function of the strain, we must also be able to express the strain as a function of the stress. Inverting the matrix (A.10) explicitly is possible but tedious; the result in terms of E and ν, however, is very simple:

Inverse Hooke's law

$$\varepsilon_{11} = \frac{\sigma_{11}}{E} - \frac{\nu}{E}(\sigma_{22} + \sigma_{33}), \qquad \gamma_{12} = \frac{\tau_{12}}{G}, \qquad \text{(and cyclic permutations)}. \tag{A.16}$$

This relation is so simple that it cannot be forgotten: each normal strain depends on all the normal stresses, on the direct one only through E, on the other two also through Poisson's ratio. In summary: in the direct form it is easier to remember the law with the Lamé constants; in the inverse form it is easier to remember it with the fundamental constants E, ν, G (whereas inverting the "clean" Lamé matrix produces a messy result).

A.5 Bounds on the elastic constants

For the constitutive law to make sense, the constitutive matrix must be invertible, and more precisely *positive definite*:

$$\boldsymbol{\varepsilon} \cdot \mathbb{C}\boldsymbol{\varepsilon} > 0 \quad \forall\, \boldsymbol{\varepsilon} \neq \mathbf{0} \qquad \Big(\text{in tensor form:} \quad \omega = \tfrac{1}{2}\, \mathbb{C}_{ijhk}\, \varepsilon_{ij} \varepsilon_{hk} > 0\Big). \tag{A.17}$$

This guarantees the existence of a strain energy with all the regularity required: the energy represents the area below the stress–strain curve, and must be positive. Positive definiteness means that all the eigenvalues of the matrix are positive, which is obtained by imposing that the determinants of all the *principal* submatrices are positive (Sylvester's criterion).

Reduction to principal axes. Working with $\mathbb{C}$ produces complicated expressions; it is easier to use the inverse law, since the energy can equally be written as $\omega = \frac{1}{2}\, \boldsymbol{\sigma} \cdot \mathbb{C}^{-1} \boldsymbol{\sigma}$ and $\mathbb{C}^{-1}$ is positive definite if and only if $\mathbb{C}$ is. The lower (shear) part of $\mathbb{C}^{-1}$ is diagonal with entries $1/G$: it is positive definite if $G = \mu > 0$, which is certainly the case. For the upper part, we exploit isotropy: the law has the same form in every reference system, and for an isotropic material the principal directions of stress and strain *coincide* (solving the eigenvalue problems of stress and strain one finds the same eigenvectors; this is not true for anisotropic materials). In the principal system the shear components vanish, and

$$\begin{Bmatrix} \varepsilon_{\mathrm{I}} \\ \varepsilon_{\mathrm{II}} \\ \varepsilon_{\mathrm{III}} \end{Bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu \\ -\nu & 1 & -\nu \\ -\nu & -\nu & 1 \end{bmatrix} \begin{Bmatrix} \sigma_{\mathrm{I}} \\ \sigma_{\mathrm{II}} \\ \sigma_{\mathrm{III}} \end{Bmatrix}, \tag{A.18}$$

i.e. $\varepsilon_{\mathrm{I}} = \sigma_{\mathrm{I}}/E - \nu(\sigma_{\mathrm{II}} + \sigma_{\mathrm{III}})/E$ and cyclic permutations. We now impose that the determinants of all principal submatrices of this 3×3 matrix be strictly positive.

1. 1×1: $1/E > 0$, i.e. $E > 0$. It cannot be zero (it would compromise the matrix). We already know it from experiments: in a uniaxial test the stress–strain line must have a positive slope, otherwise the material makes no sense.

2. 2×2 (by symmetry any principal one): $\frac{1}{E^2}(1 - \nu^2) > 0$. Since $E^2 > 0$, we need $(1 - \nu)(1 + \nu) > 0$, i.e.

$$-1 < \nu < 1, \tag{A.19}$$

with strict inequalities.

3. 3×3: by the rule of Sarrus,

$$\frac{1}{E^3}(1 - 2\nu^3 - 3\nu^2) = \frac{1}{E^3}(1+\nu)^2(1-2\nu) > 0. \tag{A.20}$$

(Check the factorisation: $(1 + 2\nu + \nu^2)(1 - 2\nu) = 1 - 3\nu^2 - 2\nu^3$.) Since $E^3 > 0$ and $(1 + \nu)^2 > 0$ by the previous condition, the sign is determined by $1 - 2\nu > 0$, i.e. $\nu < \frac{1}{2}$.

Combining the conditions:

Bounds for linear isotropic elasticity

$$E > 0, \qquad G = \mu > 0, \qquad -1 < \nu < \tfrac{1}{2}. \tag{A.21}$$

These bounds hold *only* for linear elastic isotropic materials. They do not hold for anisotropic materials, nor outside linearity (large strains). They provide a test: measure Poisson's ratio; if it falls in this range, Hooke's law can be applied; otherwise, forget about linear isotropic elasticity.

Auxetic materials. Materials with $\nu < 0$ are called *auxetic*: pulling a specimen, it elongates and at the same time expands laterally. This happens only if the material has an internal microstructure activated by the traction; it cannot happen in a continuum without microstructure such as the one considered here. Auxetic materials do exist in nature (some biological materials) and are nowadays built on purpose as microstructured *metamaterials*. Construction materials (for metallic structures, mechanical, aerospace and civil engineering) have, in general, a positive Poisson's ratio, and the bound is usually stated as $0 \le \nu < \frac{1}{2}$. The upper limit $\nu = \frac{1}{2}$ (incompressibility) is *not* acceptable in linear elasticity, because it would make the constitutive matrix singular, as just seen.

A.6 Lamé constants from the uniaxial test

To prove (A.14), apply (A.9) to a uniaxial tension test along axis 1: the bar elongates and contracts laterally.

- Stress: $\sigma_{11} = \sigma$, $\sigma_{22} = \sigma_{33} = 0$, and all shear components $\sigma_{ij} = 0$, $i \neq j$.
- Strain: $\varepsilon_{11} = \varepsilon$ and, by the definition of Poisson's ratio (transverse strains), $\varepsilon_{22} = \varepsilon_{33} = -\nu\varepsilon$.

Then $I_1 = \varepsilon - 2\nu\varepsilon = (1 - 2\nu)\varepsilon$, and

$$\sigma_{11} = \sigma = \lambda(1-2\nu)\varepsilon + 2\mu\varepsilon = \big[\lambda(1-2\nu) + 2\mu\big]\varepsilon, \tag{A.22}$$

$$\sigma_{22} = 0 = \lambda(1-2\nu)\varepsilon - 2\mu\nu\varepsilon \quad \Longrightarrow \quad \lambda(1-2\nu) = 2\mu\nu. \tag{A.23}$$

(The equation for σ_{33} is identical to the second one.) In a uniaxial test $\sigma = E\varepsilon$, so the first equation gives $E = \lambda(1 - 2\nu) + 2\mu$; substituting the second,

$$E = 2\mu\nu + 2\mu = 2\mu(1+\nu) \quad \Longrightarrow \quad \mu = G = \frac{E}{2(1+\nu)}. \tag{A.24}$$

Then, from $\lambda(1-2\nu) = 2\mu\nu = \nu E/(1+\nu)$,

$$\lambda = \frac{\nu E}{(1+\nu)(1-2\nu)}. \tag{A.25}$$

The relation between λ and E, ν is thus not just a datum but can be proved through the uniaxial tension test.

Range of G. With $G = E/[2(1+\nu)]$ and the engineering range $0 \le \nu < \frac{1}{2}$: for $\nu \to \frac{1}{2}$, $G \to E/3$; for $\nu = 0$, $G = E/2$. Hence, for engineering materials,

$$\frac{E}{3} < G \le \frac{E}{2}, \tag{A.26}$$

i.e. the shear modulus is between one third and one half of Young's modulus.

A.7 Bulk modulus

For an isotropic material the volumetric strain is the same in every reference system: $\theta = \varepsilon_{11} + \varepsilon_{22} + \varepsilon_{33} = \varepsilon_{\mathrm{I}} + \varepsilon_{\mathrm{II}} + \varepsilon_{\mathrm{III}}$, where the principal strain system coincides with the principal stress system. Summing the three inverse laws in the principal system,

$$\theta = \frac{\sigma_{\mathrm{I}} + \sigma_{\mathrm{II}} + \sigma_{\mathrm{III}}}{E} - \frac{\nu}{E}\,2(\sigma_{\mathrm{I}} + \sigma_{\mathrm{II}} + \sigma_{\mathrm{III}}) = \frac{1-2\nu}{E}\,(\sigma_{\mathrm{I}} + \sigma_{\mathrm{II}} + \sigma_{\mathrm{III}}), \tag{A.27}$$

since each principal stress appears twice in the negative terms. The sum of the principal stresses is the first stress invariant, equal to three times the isotropic (mean) stress p (Example 1.3): $\sigma_{\mathrm{I}} + \sigma_{\mathrm{II}} + \sigma_{\mathrm{III}} = 3p$. Hence the linear relation between volumetric strain and isotropic stress

Bulk modulus

$$\theta = \frac{3(1-2\nu)}{E}\,p, \qquad p = K\,\theta, \qquad K = \frac{E}{3(1-2\nu)}. \tag{A.28}$$

K is the *bulk modulus* (modulus of volumetric compressibility). More generally, the bulk modulus relates the change of pressure to the change of volume; for small changes it is a constant, $K = \Delta p/\theta$. The expression in terms of E and ν, however, holds only for linear isotropic materials, as a consequence of linearity (the law is linear), of small strains (the volumetric strain is linear) and of isotropy. Note that $K \to \infty$ as $\nu \to \frac{1}{2}$: an incompressible material. In terms of the Lamé constants, $K = \lambda + \frac{2}{3}\mu$.

A.8 Anisotropic and orthotropic materials

Isotropy does not hold for all materials: in experimental tests a material does not necessarily respond with the same elastic modulus and the same Poisson's ratio in all directions. Materials that are not isotropic are called *anisotropic*. Within this category there are materials that still show some symmetries, though not with respect to all the directions of space.

Definition A.2 (Orthotropic material)**.** A material is *orthotropic* if three mutually orthogonal planes exist with respect to which the material behaviour is symmetric. The

three directions normal to these planes are the *principal directions of orthotropy.*

Pulling along one principal direction or along another, the material responds differently. Performing three uniaxial tests on specimens cut along the three (mutually orthogonal) principal directions, one measures three Young's moduli E_1, E_2, E_3 and six Poisson's ratios $\nu_{12}, \nu_{13}, \nu_{21}, \nu_{23}, \nu_{31}, \nu_{32}$; three shear tests give three shear moduli G_{12}, G_{23}, G_{31}: $3+6+3=12$ coefficients. In the principal axes of orthotropy the inverse law reads

$$\begin{Bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \gamma_{12} \\ \gamma_{23} \\ \gamma_{31} \end{Bmatrix} = \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{21}}{E_2} & -\frac{\nu_{31}}{E_3} & 0 & 0 & 0 \\ -\frac{\nu_{12}}{E_1} & \frac{1}{E_2} & -\frac{\nu_{32}}{E_3} & 0 & 0 & 0 \\ -\frac{\nu_{13}}{E_1} & -\frac{\nu_{23}}{E_2} & \frac{1}{E_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{12}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{23}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{31}} \end{bmatrix} \begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \tau_{12} \\ \tau_{23} \\ \tau_{31} \end{Bmatrix}, \tag{A.29}$$

with null off-diagonal blocks and a diagonal shear block. Here ν_{ij} is the ratio of the transverse strain in direction j to the axial strain in direction i in a uniaxial test along i, so that column i collects the strains produced by σ_{ii} alone.

If the material is linear (the coefficients are constants), it is automatically *hyperelastic*: a strain energy exists and the compliance matrix is symmetric. The 12 coefficients are therefore not independent: symmetry imposes

$$\frac{\nu_{12}}{E_1} = \frac{\nu_{21}}{E_2}, \qquad \frac{\nu_{23}}{E_2} = \frac{\nu_{32}}{E_3}, \qquad \frac{\nu_{31}}{E_3} = \frac{\nu_{13}}{E_1}, \tag{A.30}$$

three constraints, so that a linear elastic orthotropic material is characterised by $12-3=9$ independent constants.

B

Solutions to the Exercises

This appendix collects worked solutions to all the exercises proposed in the chapters, in the same order. Each exercise ends with a link (▷) to its solution, and each solution starts with a link (◁) back to the statement. The solutions use the notation and the results of the chapters; references to the slides ("slide n") are to the course slides [10], and the textbook used to check them is mainly Holzapfel [3].

Chapter 1 Mathematical Preliminaries

Solution to Exercise 1.1

◁ statement on p. 9

For a right-handed orthonormal basis the products of the basis vectors are $\boldsymbol{e}_1 \times \boldsymbol{e}_2 = \boldsymbol{e}_3$, $\boldsymbol{e}_2 \times \boldsymbol{e}_3 = \boldsymbol{e}_1$, $\boldsymbol{e}_3 \times \boldsymbol{e}_1 = \boldsymbol{e}_2$, the same with a minus sign when the order is reversed, and $\boldsymbol{e}_i \times \boldsymbol{e}_i = \boldsymbol{0}$. Compare with $\varepsilon_{ijk}\boldsymbol{e}_k$:

- for $i = j$, $\varepsilon_{iik} = 0$ for every k, so $\varepsilon_{iik}\boldsymbol{e}_k = \boldsymbol{0}$;
- for $(i, j) = (1, 2)$ only $k = 3$ survives: $\varepsilon_{12k}\boldsymbol{e}_k = \varepsilon_{123}\boldsymbol{e}_3 = \boldsymbol{e}_3$; for $(2, 1)$, $\varepsilon_{213}\boldsymbol{e}_3 = -\boldsymbol{e}_3$. The other pairs are analogous (cyclic permutations give $+1$, anticyclic -1).

Hence $\boldsymbol{e}_i \times \boldsymbol{e}_j = \varepsilon_{ijk}\boldsymbol{e}_k$; taking the inner product with $\boldsymbol{e}_k$ gives $\varepsilon_{ijk} = (\boldsymbol{e}_i \times \boldsymbol{e}_j) \cdot \boldsymbol{e}_k$. By bilinearity of the vector product,

$$\boldsymbol{u} \times \boldsymbol{v} = (u_i \boldsymbol{e}_i) \times (v_j \boldsymbol{e}_j) = u_i v_j \, \boldsymbol{e}_i \times \boldsymbol{e}_j = \varepsilon_{ijk} u_i v_j \, \boldsymbol{e}_k .$$

In a left-handed basis $\boldsymbol{e}_1 \times \boldsymbol{e}_2 = -\boldsymbol{e}_3$ (and cyclically), so $\boldsymbol{e}_i \times \boldsymbol{e}_j = -\varepsilon_{ijk}\boldsymbol{e}_k$ and every sign in the formulas above is reversed.

■

Solution to Exercise 1.2

◁ statement on p. 14

With $(\boldsymbol{A}^{\mathrm{T}})_{ij} = A_{ji}$:

$$\operatorname{tr}(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{B}) = (\boldsymbol{A}^{\mathrm{T}})_{ij} B_{ji} = A_{ji} B_{ji} = A_{ij} B_{ij}, \qquad \operatorname{tr}(\boldsymbol{A}\boldsymbol{B}^{\mathrm{T}}) = A_{ij} (\boldsymbol{B}^{\mathrm{T}})_{ji} = A_{ij} B_{ij},$$
$$\operatorname{tr}(\boldsymbol{B}\boldsymbol{A}^{\mathrm{T}}) = B_{ij} A_{ij}, \qquad \operatorname{tr}(\boldsymbol{B}^{\mathrm{T}}\boldsymbol{A}) = B_{ji} A_{ji} = B_{ij} A_{ij},$$

(renaming dummy indices), all equal to $A_{ij}B_{ij} = \boldsymbol{A} : \boldsymbol{B} = \boldsymbol{B} : \boldsymbol{A}$. Furthermore $\boldsymbol{u} \cdot \boldsymbol{A}\boldsymbol{v} = u_i A_{ij} v_j = A_{ij}(u_i v_j) = A_{ij}(\boldsymbol{u} \otimes \boldsymbol{v})_{ij} = \boldsymbol{A} : (\boldsymbol{u} \otimes \boldsymbol{v})$, and $(\boldsymbol{u} \otimes \boldsymbol{v}) : (\boldsymbol{w} \otimes \boldsymbol{x}) = u_i v_j w_i x_j = (u_i w_i)(v_j x_j) = (\boldsymbol{u} \cdot \boldsymbol{w})(\boldsymbol{v} \cdot \boldsymbol{x})$.

■

Solution to Exercise 1.3

◁ statement on p. 22

From Eq. (1.55), $\operatorname{dev}\boldsymbol{A} = \boldsymbol{A} - \frac{1}{3}(\operatorname{tr}\boldsymbol{A})\boldsymbol{I}$. Since $\mathbb{I} : \boldsymbol{A} = \boldsymbol{A}$ and $(\boldsymbol{I}\otimes\boldsymbol{I}) : \boldsymbol{A} = \boldsymbol{I}\,(\boldsymbol{I} : \boldsymbol{A}) = (\operatorname{tr}\boldsymbol{A})\boldsymbol{I}$,

$$\operatorname{dev}\boldsymbol{A} = \mathbb{I} : \boldsymbol{A} - \tfrac{1}{3}(\boldsymbol{I}\otimes\boldsymbol{I}) : \boldsymbol{A} = (\mathbb{I} - \tfrac{1}{3}\boldsymbol{I}\otimes\boldsymbol{I}) : \boldsymbol{A} = \mathbb{P} : \boldsymbol{A}, \qquad \mathbb{P}_{ijkl} = \delta_{ik}\delta_{jl} - \tfrac{1}{3}\delta_{ij}\delta_{kl}.$$

Projector property (contracting the last two indices of the first factor with the first two of the second):

$$\mathbb{P}_{ijkl}\mathbb{P}_{klmn} = \delta_{im}\delta_{jn} - \tfrac{1}{3}\delta_{ij}\delta_{mn} - \tfrac{1}{3}\delta_{ij}\delta_{mn} + \tfrac{1}{9}\delta_{ij}\delta_{kk}\delta_{mn} = \delta_{im}\delta_{jn} - \tfrac{1}{3}\delta_{ij}\delta_{mn} = \mathbb{P}_{ijmn},$$

using $\delta_{kk} = 3$. Finally $\mathbb{P}_{ijkl}\delta_{kl} = \delta_{ij} - \frac{1}{3}\delta_{ij}\,3 = 0$, i.e., $\mathbb{P} : \boldsymbol{I} = \boldsymbol{O}$: the deviator of a spherical tensor is zero.

■

Solution to Exercise 1.4

◁ statement on p. 34

Both sides are zero if $i = j$ or $p = q$ (on the right, $\delta_{ip}\delta_{iq} - \delta_{iq}\delta_{ip} = 0$). Let $i \neq j$: then in the sum over k only the third index $k \notin \{i, j\}$ contributes, and the left side is non-zero only if $\{p, q\} = \{i, j\}$:

- $p = i$, $q = j$: left side $\varepsilon_{ijk}^2 = 1$; right side $1 \cdot 1 - 0 = 1$;
- $p = j$, $q = i$: left side $\varepsilon_{ijk}\varepsilon_{jik} = -1$; right side $0 - 1 = -1$;
- otherwise both sides vanish.

Then, using $(\boldsymbol{a}\times\boldsymbol{b})_i = \varepsilon_{ijk}a_jb_k$ and $(\boldsymbol{v}\times\boldsymbol{w})_k = \varepsilon_{pqk}v_pw_q$,

$$[\boldsymbol{u}\times(\boldsymbol{v}\times\boldsymbol{w})]_i = \varepsilon_{ijk}u_j\varepsilon_{pqk}v_pw_q = (\delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp})u_jv_pw_q = (u_jw_j)v_i - (u_jv_j)w_i,$$

i.e., $\boldsymbol{u}\times(\boldsymbol{v}\times\boldsymbol{w}) = (\boldsymbol{u}\cdot\boldsymbol{w})\boldsymbol{v} - (\boldsymbol{u}\cdot\boldsymbol{v})\boldsymbol{w}$.

■

Solution to Exercise 1.5

◁ statement on p. 35

$(\boldsymbol{u}\times\boldsymbol{v})\cdot\boldsymbol{w} = (\varepsilon_{ijk}u_iv_j\boldsymbol{e}_k)\cdot(w_l\boldsymbol{e}_l) = \varepsilon_{ijk}u_iv_jw_l\delta_{kl} = \varepsilon_{ijk}u_iv_jw_k$. For the matrix $[M]$ with rows $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$, $M_{1i} = u_i$, $M_{2j} = v_j$, $M_{3k} = w_k$, and the expansion of the determinant along the rows (equivalent to Eq. (1.39) applied to $[M]^{\mathrm{T}}$, since $\det[M] = \det[M]^{\mathrm{T}}$) is $\det[M] = \varepsilon_{ijk}M_{1i}M_{2j}M_{3k} = \varepsilon_{ijk}u_iv_jw_k$.

■

Solution to Exercise 1.6

◁ statement on p. 35

$\boldsymbol{A} : \boldsymbol{I} = A_{ij}\delta_{ij} = A_{ii} = \operatorname{tr}\boldsymbol{A}$; $\operatorname{tr}(\boldsymbol{u}\otimes\boldsymbol{v}) = (\boldsymbol{u}\otimes\boldsymbol{v})_{ii} = u_iv_i = \boldsymbol{u}\cdot\boldsymbol{v}$; $\operatorname{tr}(\boldsymbol{AB}) = (\boldsymbol{AB})_{ii} = A_{ij}B_{ji} = B_{ji}A_{ij} = (\boldsymbol{BA})_{jj} = \operatorname{tr}(\boldsymbol{BA})$.

■

Solution to Exercise 1.7

◁ statement on p. 35

$\varepsilon_{ijk} = +1$ for $(123), (231), (312)$, -1 for $(213), (132), (321)$, 0 otherwise. In the layer $k = 1$ the non-zero entries are $(i, j) = (2, 3)$ with $\varepsilon_{231} = 1$ and $(3, 2)$ with $\varepsilon_{321} = -1$; in the layer $k = 2$, $(3, 1)$: $+1$ and $(1, 3)$: -1; in the layer $k = 3$, $(1, 2)$: $+1$ and $(2, 1)$: -1. This is Eq. (1.8).

Expanding $\varepsilon_{ijk} A_{i1} A_{j2} A_{k3}$, only the six permutations survive:

$$A_{11}A_{22}A_{33} + A_{21}A_{32}A_{13} + A_{31}A_{12}A_{23} - A_{21}A_{12}A_{33} - A_{11}A_{32}A_{23} - A_{31}A_{22}A_{13},$$

which are the six terms of the rule of Sarrus Eq. (1.38). For the second form, note that $\varepsilon_{lrs} A_{il} A_{jr} A_{ks}$ is antisymmetric in (i, j, k) (exchanging i and j and renaming $l \leftrightarrow r$ changes the sign of ε) and equals $\det \boldsymbol{A}$ for $(i, j, k) = (1, 2, 3)$ (expansion along the rows); hence $\varepsilon_{lrs} A_{il} A_{jr} A_{ks} = \det \boldsymbol{A}\, \varepsilon_{ijk}$. Multiplying by ε_{ijk} and using $\varepsilon_{ijk}\varepsilon_{ijk} = 6$ gives $\det \boldsymbol{A} = \frac{1}{6} \varepsilon_{ijk} \varepsilon_{lrs} A_{il} A_{jr} A_{ks}$.

■

Solution to Exercise 1.8

◁ statement on p. 35

With $W_{ji} = -W_{ij}$, renaming the dummy indices, $\boldsymbol{v} \cdot \boldsymbol{W}\boldsymbol{v} = v_i W_{ij} v_j = v_j W_{ji} v_i = -v_i W_{ij} v_j$, hence it is zero. Similarly, with $S_{ji} = S_{ij}$, $\boldsymbol{S} : \boldsymbol{W} = S_{ij} W_{ij} = S_{ji} W_{ji} = -S_{ij} W_{ij} = 0$.

■

Solution to Exercise 1.9

◁ statement on p. 35

Let $\tilde{\boldsymbol{e}}_i = \boldsymbol{Q}\boldsymbol{e}_i = Q_{ji}\boldsymbol{e}_j$, so that $\boldsymbol{e}_j \cdot \tilde{\boldsymbol{e}}_i = Q_{ji}$. From $\boldsymbol{u} = u_j \boldsymbol{e}_j = \tilde{u}_j \tilde{\boldsymbol{e}}_j$,

$$\tilde{u}_i = \boldsymbol{u} \cdot \tilde{\boldsymbol{e}}_i = u_j\, \boldsymbol{e}_j \cdot \tilde{\boldsymbol{e}}_i = Q_{ji} u_j, \qquad u_i = \boldsymbol{u} \cdot \boldsymbol{e}_i = \tilde{u}_j\, \tilde{\boldsymbol{e}}_j \cdot \boldsymbol{e}_i = Q_{ij} \tilde{u}_j,$$

i.e., $[\tilde{\boldsymbol{u}}] = [\boldsymbol{Q}]^{\mathrm{T}} [\boldsymbol{u}]$ and $[\boldsymbol{u}] = [\boldsymbol{Q}] [\tilde{\boldsymbol{u}}]$. For a rotation by θ about $\boldsymbol{e}_3$, $\tilde{\boldsymbol{e}}_1 = \cos\theta\, \boldsymbol{e}_1 + \sin\theta\, \boldsymbol{e}_2$, $\tilde{\boldsymbol{e}}_2 = -\sin\theta\, \boldsymbol{e}_1 + \cos\theta\, \boldsymbol{e}_2$, $\tilde{\boldsymbol{e}}_3 = \boldsymbol{e}_3$, so

$$[\boldsymbol{Q}] = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \begin{Bmatrix} \tilde{u}_1 \\ \tilde{u}_2 \\ \tilde{u}_3 \end{Bmatrix} = \begin{Bmatrix} u_1 \cos\theta + u_2 \sin\theta \\ -u_1 \sin\theta + u_2 \cos\theta \\ u_3 \end{Bmatrix}.$$

■

Solution to Exercise 1.10

◁ statement on p. 35

Let $\boldsymbol{A}\boldsymbol{n} = \lambda \boldsymbol{n}$ with $\boldsymbol{n} \neq \boldsymbol{0}$, possibly complex. Since $\boldsymbol{A}$ is real, conjugation gives $\boldsymbol{A}\bar{\boldsymbol{n}} = \bar{\lambda}\bar{\boldsymbol{n}}$. Then, using the symmetry of $\boldsymbol{A}$,

$$\bar{\boldsymbol{n}} \cdot \boldsymbol{A}\boldsymbol{n} = \lambda\, \bar{\boldsymbol{n}} \cdot \boldsymbol{n}, \qquad \bar{\boldsymbol{n}} \cdot \boldsymbol{A}\boldsymbol{n} = \boldsymbol{A}\bar{\boldsymbol{n}} \cdot \boldsymbol{n} = \bar{\lambda}\, \bar{\boldsymbol{n}} \cdot \boldsymbol{n}.$$

Since $\bar{\boldsymbol{n}} \cdot \boldsymbol{n} = \sum_i |n_i|^2 > 0$, $\lambda = \bar{\lambda}$ is real. For two eigenpairs with $\lambda_1 \neq \lambda_2$, $\boldsymbol{n}_1 \cdot \boldsymbol{A}\boldsymbol{n}_2 = \lambda_2\, \boldsymbol{n}_1 \cdot \boldsymbol{n}_2$ and $\boldsymbol{n}_1 \cdot \boldsymbol{A}\boldsymbol{n}_2 = \boldsymbol{A}\boldsymbol{n}_1 \cdot \boldsymbol{n}_2 = \lambda_1\, \boldsymbol{n}_1 \cdot \boldsymbol{n}_2$; subtracting, $(\lambda_2 - \lambda_1)\, \boldsymbol{n}_1 \cdot \boldsymbol{n}_2 = 0$, so $\boldsymbol{n}_1 \cdot \boldsymbol{n}_2 = 0$.

■

Chapter 2 Dual Bases

Solution to Exercise 2.1

◁ statement on p. 44

(a) $\boldsymbol{g}_1\times\boldsymbol{g}_2=(\boldsymbol{e}_1+\boldsymbol{e}_2)\times(\boldsymbol{e}_2+\boldsymbol{e}_3)=\boldsymbol{e}_3-\boldsymbol{e}_2+\boldsymbol{e}_1$, so $g=(\boldsymbol{g}_1\times\boldsymbol{g}_2)\cdot\boldsymbol{g}_3=(\boldsymbol{e}_1-\boldsymbol{e}_2+\boldsymbol{e}_3)\cdot(\boldsymbol{e}_1+\boldsymbol{e}_3)=2\neq 0$: the vectors form a (right-handed) basis.

(b) $g_{11}=g_{22}=g_{33}=2$ and $g_{12}=g_{23}=g_{13}=1$:

$$[g_{ij}]=\begin{bmatrix}2&1&1\\1&2&1\\1&1&2\end{bmatrix},\qquad \left[g^{ij}\right]=[g_{ij}]^{-1}=\frac{1}{4}\begin{bmatrix}3&-1&-1\\-1&3&-1\\-1&-1&3\end{bmatrix},$$

and indeed $\det[g_{ij}]=4=g^2$.

(c) With Eq. (2.11):

$$\boldsymbol{g}^1=\frac{\boldsymbol{g}_2\times\boldsymbol{g}_3}{2}=\tfrac{1}{2}(\boldsymbol{e}_1+\boldsymbol{e}_2-\boldsymbol{e}_3),\qquad \boldsymbol{g}^2=\frac{\boldsymbol{g}_3\times\boldsymbol{g}_1}{2}=\tfrac{1}{2}(-\boldsymbol{e}_1+\boldsymbol{e}_2+\boldsymbol{e}_3),$$
$$\boldsymbol{g}^3=\frac{\boldsymbol{g}_1\times\boldsymbol{g}_2}{2}=\tfrac{1}{2}(\boldsymbol{e}_1-\boldsymbol{e}_2+\boldsymbol{e}_3).$$

Check with Eq. (2.14): $\boldsymbol{g}^1=g^{1j}\boldsymbol{g}_j=\frac{1}{4}(3\boldsymbol{g}_1-\boldsymbol{g}_2-\boldsymbol{g}_3)=\frac{1}{4}(2\boldsymbol{e}_1+2\boldsymbol{e}_2-2\boldsymbol{e}_3)$. ✓

(d) For $\boldsymbol{u}=\boldsymbol{e}_1+2\boldsymbol{e}_2+3\boldsymbol{e}_3$: $u_1=\boldsymbol{u}\cdot\boldsymbol{g}_1=3$, $u_2=\boldsymbol{u}\cdot\boldsymbol{g}_2=5$, $u_3=\boldsymbol{u}\cdot\boldsymbol{g}_3=4$; $u^1=\boldsymbol{u}\cdot\boldsymbol{g}^1=0$, $u^2=\boldsymbol{u}\cdot\boldsymbol{g}^2=2$, $u^3=\boldsymbol{u}\cdot\boldsymbol{g}^3=1$. Indeed $2\boldsymbol{g}_2+\boldsymbol{g}_3=\boldsymbol{e}_1+2\boldsymbol{e}_2+3\boldsymbol{e}_3$, and $u^iu_i=0+10+4=14=|\boldsymbol{u}|^2$.

■

Solution to Exercise 2.2

◁ statement on p. 44

Collect the Cartesian components of $\boldsymbol{g}_1,\boldsymbol{g}_2,\boldsymbol{g}_3$ as the columns of a matrix $[G]$; then $g=\det[G]$ (box product). The condition $\boldsymbol{g}^i\cdot\boldsymbol{g}_j=\delta^i_j$ says that the rows of $[G]^{-1}$ are the components of the $\boldsymbol{g}^i$. Hence $(\boldsymbol{g}^1\times\boldsymbol{g}^2)\cdot\boldsymbol{g}^3=\det[G]^{-1}=1/\det[G]=1/g$. Also $g_{ij}=\boldsymbol{g}_i\cdot\boldsymbol{g}_j=([G]^{\mathrm{T}}[G])_{ij}$, so $\det[g_{ij}]=(\det[G])^2=g^2$.

■

Solution to Exercise 2.3

◁ statement on p. 44

Under $\boldsymbol{g}'_i=\alpha_i{}^k\boldsymbol{g}_k$ the dual basis transforms as $\boldsymbol{g}'^i=(\alpha^{-1})^i_k\boldsymbol{g}^k$: indeed $\boldsymbol{g}'^i\cdot\boldsymbol{g}'_j=(\alpha^{-1})^i_k\alpha_j{}^m\delta^k_m=\alpha_j{}^k(\alpha^{-1})^i_k=\delta^i_j$. Then

$$A'^i_{\cdot j}=\boldsymbol{g}'^i\cdot\boldsymbol{A}\boldsymbol{g}'_j=(\alpha^{-1})^i_k\alpha_j{}^mA^k_{\cdot m},\qquad A'^i_{\cdot i}=\alpha_i{}^m(\alpha^{-1})^i_kA^k_{\cdot m}=\delta^m_kA^k_{\cdot m}=A^k_{\cdot k}.$$

For the covariant components, $A'_{ij}=\alpha_i{}^k\alpha_j{}^mA_{km}$ and $\sum_iA'_{ii}=\sum_i\alpha_i{}^k\alpha_i{}^mA_{km}$, which equals A_{kk} only if $\sum_i\alpha_i{}^k\alpha_i{}^m=\delta_{km}$ (orthogonal change of basis). Counterexample: $\boldsymbol{A}=\boldsymbol{I}$ in the plane basis of the example in Section 2.5: $A_{ij}=g_{ij}$, so $A_{11}+A_{22}=4+2=6$, while $\operatorname{tr}\boldsymbol{I}=A^i_{\cdot i}=\delta^i_i=2$.

■

Solution to Exercise 2.4

◁ statement on p. 44

The eigenvalues of the triangular matrix are $\lambda_1 = 2$, $\lambda_2 = 3$. Right eigenvectors: $(\boldsymbol{A}-2\boldsymbol{I})\boldsymbol{u} = \boldsymbol{0}$ gives $\boldsymbol{u}_1 = \boldsymbol{e}_1$; $(\boldsymbol{A}-3\boldsymbol{I})\boldsymbol{u} = \boldsymbol{0}$ gives $\boldsymbol{u}_2 = \boldsymbol{e}_1+\boldsymbol{e}_2$. The dual basis ($\boldsymbol{u}^\alpha\cdot\boldsymbol{u}_\beta = \delta^\alpha_\beta$) is $\boldsymbol{u}^1 = \boldsymbol{e}_1-\boldsymbol{e}_2$, $\boldsymbol{u}^2 = \boldsymbol{e}_2$. Then

$$2\,\boldsymbol{u}_1\otimes\boldsymbol{u}^1 + 3\,\boldsymbol{u}_2\otimes\boldsymbol{u}^2 = 2\begin{bmatrix}1 & -1\\ 0 & 0\end{bmatrix} + 3\begin{bmatrix}0 & 1\\ 0 & 1\end{bmatrix} = \begin{bmatrix}2 & 1\\ 0 & 3\end{bmatrix} = [\boldsymbol{A}]\,. \checkmark$$

With $\left[\boldsymbol{A}^{\mathrm{T}}\right] = \begin{bmatrix}2 & 0\\ 1 & 3\end{bmatrix}$: $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{u}^1 = 2\boldsymbol{e}_1 - 2\boldsymbol{e}_2 = 2\boldsymbol{u}^1$ and $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{u}^2 = 3\boldsymbol{e}_2 = 3\boldsymbol{u}^2$: the dual vectors are the left eigenvectors, with the same eigenvalues.

■

Chapter 3 Kinematics

Solution to Exercise 3.1

◁ statement on p. 74

Preservation of the area of the annulus between the cavity and the material circle of radius R: $\pi(R^2 - a_0^2) = \pi(r^2 - a^2)$, so

$$r = \left(R^2 + a^2 - a_0^2\right)^{1/2}, \qquad \boldsymbol{x}(\boldsymbol{X},t) = \frac{r}{R}\boldsymbol{X} = \left(1 + \frac{a^2(t) - a_0^2}{R^2}\right)^{1/2}\boldsymbol{X}.$$

Differentiating at fixed $\boldsymbol{X}$:

$$\boldsymbol{V} = \tfrac{1}{2}\left(1 + \frac{a^2 - a_0^2}{R^2}\right)^{-1/2}\frac{2a\dot{a}}{R^2}\boldsymbol{X} = \frac{R}{r}\,\frac{a\dot{a}}{R^2}\boldsymbol{X} = \frac{a\dot{a}}{rR}\boldsymbol{X},$$

and with $\boldsymbol{X} = (R/r)\boldsymbol{x}$, $\boldsymbol{v} = a\dot{a}\,\boldsymbol{x}/r^2 = a\dot{a}\,\boldsymbol{x}/\left|\boldsymbol{x}\right|^2$. The flux $2\pi r\, v_r = 2\pi a\dot{a}$ is the same through every circle.

For $\boldsymbol{F}$, write $\boldsymbol{x} = f(R)\boldsymbol{X}$ with $f = r/R$; then $\boldsymbol{F} = f\boldsymbol{I} + f'(R)\,\boldsymbol{X}\otimes\boldsymbol{X}/R$. Since $\mathrm{d}r/\,\mathrm{d}R = R/r$, $f' = (R^2/r - r)/R^2$, and

$$\boldsymbol{F} = \frac{r}{R}\,\boldsymbol{I} + \frac{R^2 - r^2}{rR^3}\,\boldsymbol{X}\otimes\boldsymbol{X} = \frac{R}{r}\,\boldsymbol{e}_R\otimes\boldsymbol{e}_R + \frac{r}{R}\,\boldsymbol{e}_\Theta\otimes\boldsymbol{e}_\Theta,$$

with $\boldsymbol{e}_R = \boldsymbol{X}/R$ and $\boldsymbol{e}_\Theta$ the hoop direction (in the plane). The radial stretch is R/r, the hoop stretch r/R, and $J = (R/r)(r/R) = 1$.

■

Solution to Exercise 3.2

◁ statement on p. 74

With $[\boldsymbol{F}] = \begin{bmatrix} -2 & -1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$ and $J = \frac{5}{2}$:

$$[\boldsymbol{C}] = [\boldsymbol{F}]^{\mathrm{T}}\,[\boldsymbol{F}] = \frac{1}{4}\begin{bmatrix} 25 & 5 \\ 5 & 5 \end{bmatrix}, \qquad [\boldsymbol{E}] = \tfrac{1}{2}([\boldsymbol{C}] - [\boldsymbol{I}]) = \frac{1}{8}\begin{bmatrix} 21 & 5 \\ 5 & 1 \end{bmatrix},$$
$$[\boldsymbol{b}] = [\boldsymbol{F}]\,[\boldsymbol{F}]^{\mathrm{T}} = \frac{1}{2}\begin{bmatrix} 10 & -5 \\ -5 & 5 \end{bmatrix}, \qquad [\boldsymbol{e}] = \tfrac{1}{2}([\boldsymbol{I}] - [\boldsymbol{b}]^{-1}) = \frac{1}{10}\begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix},$$

where $[\boldsymbol{b}]^{-1} = \frac{1}{5}\begin{bmatrix} 2 & 2 \\ 2 & 4 \end{bmatrix}$; check $\det \boldsymbol{C} = \det \boldsymbol{b} = \frac{25}{4} = J^2$. For $\boldsymbol{A} = \frac{1}{\sqrt{2}}(\boldsymbol{E}_1 + \boldsymbol{E}_2)$, $\lambda^2 = \boldsymbol{A}\cdot\boldsymbol{C}\boldsymbol{A} = \frac{1}{2}(C_{11} + 2C_{12} + C_{22}) = \frac{1}{2}\cdot\frac{40}{4} = 5$, as found with $|\boldsymbol{F}\boldsymbol{A}| = \sqrt{5}$. The eigenvalues of $\boldsymbol{C}$ solve $\mu^2 - \frac{15}{2}\mu + \frac{25}{4} = 0$:

$$\lambda_{1,2}^2 = \frac{15 \pm 5\sqrt{5}}{4}, \qquad \lambda_1 \approx 2.558, \quad \lambda_2 \approx 0.977, \qquad \lambda_1\lambda_2 = \tfrac{5}{2} = J.$$

■

Solution to Exercise 3.3

◁ statement on p. 74

Let $k(t) = a^2\dot{a}$, so that $\boldsymbol{v} = k\,\boldsymbol{x}/r^3$ with $r = |\boldsymbol{x}|$. Then $\partial\boldsymbol{v}/\partial t = \dot{k}\,\boldsymbol{x}/r^3$ with $\dot{k} = 2a\dot{a}^2 + a^2\ddot{a}$, and

$$\operatorname{grad}\boldsymbol{v} = k\Big(\frac{\boldsymbol{I}}{r^3} - 3\frac{\boldsymbol{x}\otimes\boldsymbol{x}}{r^5}\Big), \qquad (\operatorname{grad}\boldsymbol{v})\boldsymbol{v} = \frac{k^2}{r^3}\Big(\frac{\boldsymbol{x}}{r^3} - 3\frac{\boldsymbol{x}\,r^2}{r^5}\Big) = -\frac{2k^2}{r^6}\boldsymbol{x},$$

so that

$$\boldsymbol{a} = \Big(\frac{\dot{k}}{r^3} - \frac{2k^2}{r^6}\Big)\boldsymbol{x} = \Big(\frac{2a\dot{a}^2 + a^2\ddot{a}}{r^3} - \frac{2a^4\dot{a}^2}{r^6}\Big)\boldsymbol{x}.$$

Material check: the particle moves radially with $r^3 = R^3 + a^3 - a_0^3$, so $\dot{r} = k/r^2$ and $\ddot{r} = \dot{k}/r^2 - 2k\dot{r}/r^3 = \dot{k}/r^2 - 2k^2/r^5$; the material acceleration $\boldsymbol{A} = \ddot{r}\,\boldsymbol{x}/r$, expressed in spatial coordinates, coincides with $\boldsymbol{a}$.

■

Solution to Exercise 3.4

◁ statement on p. 74

Algebraically, $\boldsymbol{F}^{\mathrm{T}}\boldsymbol{e}\boldsymbol{F} = \frac{1}{2}\boldsymbol{F}^{\mathrm{T}}(\boldsymbol{I} - \boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{-1})\boldsymbol{F} = \frac{1}{2}(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} - \boldsymbol{I}) = \boldsymbol{E}$, and $\boldsymbol{e} = \boldsymbol{F}^{-\mathrm{T}}\boldsymbol{E}\boldsymbol{F}^{-1}$ follows by multiplying by $\boldsymbol{F}^{-\mathrm{T}}$ and $\boldsymbol{F}^{-1}$. Equivalently, $\mathrm{d}\boldsymbol{X}\cdot\boldsymbol{E}\,\mathrm{d}\boldsymbol{X} = \frac{1}{2}(|\,\mathrm{d}\boldsymbol{x}|^2 - |\,\mathrm{d}\boldsymbol{X}|^2) = \mathrm{d}\boldsymbol{x}\cdot\boldsymbol{e}\,\mathrm{d}\boldsymbol{x} = \mathrm{d}\boldsymbol{X}\cdot\boldsymbol{F}^{\mathrm{T}}\boldsymbol{e}\boldsymbol{F}\,\mathrm{d}\boldsymbol{X}$ for all $\mathrm{d}\boldsymbol{X}$, and both tensors are symmetric.

Let $\boldsymbol{H} = \operatorname{Grad}\boldsymbol{U}$, so $\boldsymbol{F} = \boldsymbol{I} + \boldsymbol{H}$. Then $\boldsymbol{E} = \frac{1}{2}(\boldsymbol{H} + \boldsymbol{H}^{\mathrm{T}} + \boldsymbol{H}^{\mathrm{T}}\boldsymbol{H}) = \operatorname{sym}\boldsymbol{H} + o(|\boldsymbol{H}|)$. Also $\boldsymbol{F}^{-1} = \boldsymbol{I} - \boldsymbol{H} + o(|\boldsymbol{H}|)$, so $\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{-1} = \boldsymbol{I} - \boldsymbol{H} - \boldsymbol{H}^{\mathrm{T}} + o(|\boldsymbol{H}|)$ and $\boldsymbol{e} = \operatorname{sym}\boldsymbol{H} + o(|\boldsymbol{H}|)$. Both reduce to $\boldsymbol{\varepsilon} = \operatorname{sym}\operatorname{Grad}\boldsymbol{U}$ (and $\operatorname{Grad}\boldsymbol{U}$, $\operatorname{grad}\boldsymbol{u}$ differ only at second order).

■

Solution to Exercise 3.5

◁ statement on p. 74

$$[\boldsymbol{F}] = \begin{bmatrix} 1 & \gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \; J = 1, \quad [\boldsymbol{C}] = \begin{bmatrix} 1 & \gamma & 0 \\ \gamma & 1+\gamma^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad [\boldsymbol{b}] = \begin{bmatrix} 1+\gamma^2 & \gamma & 0 \\ \gamma & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The fibre $\boldsymbol{E}_1$ is not stretched ($C_{11} = 1$); the fibre $\boldsymbol{E}_2$ has $\lambda = \sqrt{C_{22}} = \sqrt{1+\gamma^2}$. The angle between them, initially $\pi/2$, becomes θ with $\cos\theta = C_{12}/\sqrt{C_{11}C_{22}} = \gamma/\sqrt{1+\gamma^2}$; the decrease $\psi = \pi/2 - \theta$ satisfies $\sin\psi = \gamma/\sqrt{1+\gamma^2}$, i.e., $\tan\psi = \gamma$: γ is the tangent of the shear angle, which for small γ is the engineering shear strain.

■

Solution to Exercise 3.6

◁ statement on p. 75

From $\boldsymbol{RU} = \boldsymbol{vR}$: $\boldsymbol{v} = \boldsymbol{RUR}^{-1} = \boldsymbol{RUR}^{\mathrm{T}}$. Then $\boldsymbol{b} = \boldsymbol{v}^2 = \boldsymbol{RUR}^{\mathrm{T}}\boldsymbol{RUR}^{\mathrm{T}} = \boldsymbol{RU}^2\boldsymbol{R}^{\mathrm{T}} = \boldsymbol{RCR}^{\mathrm{T}}$ and $\boldsymbol{v}^{-1} = (\boldsymbol{RUR}^{\mathrm{T}})^{-1} = \boldsymbol{RU}^{-1}\boldsymbol{R}^{\mathrm{T}}$. Finally

$$\boldsymbol{R}^{\mathrm{T}}\,\tfrac{1}{2}(\boldsymbol{b}-\boldsymbol{I})\,\boldsymbol{R} = \tfrac{1}{2}(\boldsymbol{R}^{\mathrm{T}}\boldsymbol{RCR}^{\mathrm{T}}\boldsymbol{R} - \boldsymbol{R}^{\mathrm{T}}\boldsymbol{R}) = \tfrac{1}{2}(\boldsymbol{C}-\boldsymbol{I}) = \boldsymbol{E}.$$

■

Solution to Exercise 3.7

◁ statement on p. 75

Write $[\boldsymbol{R}] = \begin{bmatrix} c & -s \\ s & c \end{bmatrix}$, $c = \cos\beta$, $s = \sin\beta$. Then $\left[\boldsymbol{R}^{\mathrm{T}}\boldsymbol{F}\right] = \begin{bmatrix} c & c\gamma + s \\ -s & c - s\gamma \end{bmatrix}$, and its symmetry requires $c\gamma + s = -s$, i.e., $\tan\beta = -\gamma/2$. With $S = \sqrt{4+\gamma^2}$, $c = 2/S$, $s = -\gamma/S$, which gives $\boldsymbol{R}$, $\boldsymbol{U}$ and $\boldsymbol{v}$ of Example 3.11. $\operatorname{tr}\boldsymbol{U} = S$ and $\det\boldsymbol{U} = 1$, so $\lambda_{1,2} = \frac{1}{2}(S \pm \gamma)$, with $\lambda_1 - \lambda_2 = \gamma$ and $\lambda_1\lambda_2 = 1$.

Principal directions. From the first row of $(\boldsymbol{U} - \lambda_1\boldsymbol{I})\hat{\boldsymbol{N}} = \boldsymbol{0}$, $\hat{N}_2/\hat{N}_1 = (\lambda_1 S - 2)/\gamma = \lambda_1$; the referential axis is inclined by φ with $\tan\varphi = \lambda_1$. From the first row of $(\boldsymbol{v} - \lambda_1\boldsymbol{I})\hat{\boldsymbol{n}} = \boldsymbol{0}$, $\hat{n}_2/\hat{n}_1 = (\lambda_1 S - 2 - \gamma^2)/\gamma = \lambda_2 = 1/\lambda_1$; the spatial axis is inclined by $\vartheta = \pi/2 - \varphi$. Since $\lambda_1^2 - 1 = \gamma\lambda_1$, $\tan 2\varphi = 2\lambda_1/(1-\lambda_1^2) = -2/\gamma$, and

$$\tan(\vartheta - \varphi) = \tan\left(\tfrac{\pi}{2} - 2\varphi\right) = \cot 2\varphi = -\frac{\gamma}{2} = \tan\beta :$$

the spatial axes are the referential ones rotated by β, $\hat{\boldsymbol{n}}_\alpha = \boldsymbol{R}\hat{\boldsymbol{N}}_\alpha$. For $\gamma = 1$: $\lambda_1 = 1.618$, $\lambda_2 = 0.618$, $\varphi = 58.28°$, $\vartheta = 31.72°$, $\beta = -26.57°$.

■

Solution to Exercise 3.8

◁ statement on p. 75

Since $\boldsymbol{R} = \boldsymbol{I}$, $\boldsymbol{v} = \boldsymbol{U} = \boldsymbol{F} = \operatorname{diag}(\lambda, 1, 1)$ and $\boldsymbol{b} = \boldsymbol{C}$: the Eulerian measures coincide with the Lagrangian ones, $e_{11}^{(0)} = \ln\lambda$, $e_{11}^{(1)} = \lambda - 1$, $\frac{1}{2}(b_{11} - 1) = \frac{1}{2}(\lambda^2 - 1)$, $e_{11} = \frac{1}{2}(1 - \lambda^{-2})$, and the plot is Fig. 3.4. They differ only when the deformation involves a rotation (the two families are related by $\boldsymbol{e}^{(m)} = \boldsymbol{RE}^{(m)}\boldsymbol{R}^{\mathrm{T}}$).

■

Solution to Exercise 3.9 ◁ statement on p. 75

With $x_1 = X_1(1+\alpha t^3)$ and $v_1 = 3\alpha t^2 x_1/(1+\alpha t^3)$:

$$[\boldsymbol{l}] = \operatorname{diag}\Big(\frac{3\alpha t^2}{1+\alpha t^3}, 0, 0\Big) = [\boldsymbol{d}]\,, \qquad \boldsymbol{w} = \boldsymbol{O};$$

$[\boldsymbol{F}] = \operatorname{diag}(1+\alpha t^3, 1, 1)$, $\left[\dot{\boldsymbol{F}}\right] = \operatorname{diag}(3\alpha t^2, 0, 0) = [\boldsymbol{l}]\,[\boldsymbol{F}]$. ✓ Also $J = 1+\alpha t^3$, $\dot{J} = 3\alpha t^2$ and $J \operatorname{div} \boldsymbol{v} = (1+\alpha t^3)\, 3\alpha t^2/(1+\alpha t^3) = \dot{J}$. ✓

■

Solution to Exercise 3.10 ◁ statement on p. 75

Differentiating $\boldsymbol{F}\boldsymbol{F}^{-1} = \boldsymbol{I}$: $\dot{\boldsymbol{F}}\boldsymbol{F}^{-1} + \boldsymbol{F}\,\dot{\overline{\boldsymbol{F}^{-1}}} = \boldsymbol{O}$, so $\dot{\overline{\boldsymbol{F}^{-1}}} = -\boldsymbol{F}^{-1}\dot{\boldsymbol{F}}\boldsymbol{F}^{-1} = -\boldsymbol{F}^{-1}\boldsymbol{l}$; transposing, $\dot{\overline{\boldsymbol{F}^{-\mathrm{T}}}} = -\boldsymbol{l}^{\mathrm{T}}\boldsymbol{F}^{-\mathrm{T}}$. Then, since $\mathrm{d}\boldsymbol{S}$ does not depend on time,

$$\dot{\overline{\mathrm{d}\boldsymbol{s}}} = \dot{J}\boldsymbol{F}^{-\mathrm{T}}\,\mathrm{d}\boldsymbol{S} + J\,\dot{\overline{\boldsymbol{F}^{-\mathrm{T}}}}\,\mathrm{d}\boldsymbol{S} = \operatorname{div}\boldsymbol{v}\, J\boldsymbol{F}^{-\mathrm{T}}\,\mathrm{d}\boldsymbol{S} - \boldsymbol{l}^{\mathrm{T}} J\boldsymbol{F}^{-\mathrm{T}}\,\mathrm{d}\boldsymbol{S} = \big(\operatorname{div}\boldsymbol{v}\,\boldsymbol{I} - \boldsymbol{l}^{\mathrm{T}}\big)\,\mathrm{d}\boldsymbol{s}.$$

■

Chapter 4 The Concept of Stress

Solution to Exercise 4.1 ◁ statement on p. 82

Start from the definitions $\boldsymbol{\tau} = J\boldsymbol{\sigma}$, $\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}$, $\boldsymbol{S} = \boldsymbol{F}^{-1}\boldsymbol{\tau}\boldsymbol{F}^{-\mathrm{T}}$.

- Row $\boldsymbol{\tau}$: $\boldsymbol{\tau} = J\boldsymbol{\sigma}$; $\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}} = J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{\tau}$; $\boldsymbol{F}\boldsymbol{S}\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{F}\boldsymbol{F}^{-1}\boldsymbol{\tau}\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{\tau}$.
- Row $\boldsymbol{\sigma}$: divide the row $\boldsymbol{\tau}$ by J.
- Row $\boldsymbol{P}$: $\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}} = \boldsymbol{\tau}\boldsymbol{F}^{-\mathrm{T}}$, and $\boldsymbol{F}\boldsymbol{S} = \boldsymbol{F}\boldsymbol{F}^{-1}\boldsymbol{\tau}\boldsymbol{F}^{-\mathrm{T}} = \boldsymbol{\tau}\boldsymbol{F}^{-\mathrm{T}} = \boldsymbol{P}$.
- Row $\boldsymbol{S}$: $\boldsymbol{S} = \boldsymbol{F}^{-1}\boldsymbol{\tau}\boldsymbol{F}^{-\mathrm{T}} = J\boldsymbol{F}^{-1}\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}$ and, from $\boldsymbol{P} = \boldsymbol{F}\boldsymbol{S}$, $\boldsymbol{S} = \boldsymbol{F}^{-1}\boldsymbol{P}$.

■

Solution to Exercise 4.2 ◁ statement on p. 82

$\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}} = J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{\mathrm{T}} = J\boldsymbol{\sigma}$ and $\boldsymbol{F}\boldsymbol{P}^{\mathrm{T}} = \boldsymbol{F}\,J\boldsymbol{F}^{-1}\boldsymbol{\sigma}^{\mathrm{T}} = J\boldsymbol{\sigma}^{\mathrm{T}}$. Hence $\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{F}\boldsymbol{P}^{\mathrm{T}}$ if and only if $\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\mathrm{T}}$. Similarly $\boldsymbol{S}^{\mathrm{T}} = J\boldsymbol{F}^{-1}\boldsymbol{\sigma}^{\mathrm{T}}\boldsymbol{F}^{-\mathrm{T}}$, which equals $\boldsymbol{S} = J\boldsymbol{F}^{-1}\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}$ if and only if $\boldsymbol{\sigma}^{\mathrm{T}} = \boldsymbol{\sigma}$, because $\boldsymbol{F}$ is invertible.

■

Solution to Exercise 4.3 ◁ statement on p. 82

Here $J = 1$, $\boldsymbol{\tau} = \boldsymbol{\sigma}$, $[\boldsymbol{F}^{-1}] = \begin{bmatrix} 1 & -\gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $\boldsymbol{P} = \boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}$, $\boldsymbol{S} = \boldsymbol{F}^{-1}\boldsymbol{P}$.

(a) For $\boldsymbol{\sigma} = s(\boldsymbol{e}_1 \otimes \boldsymbol{e}_2 + \boldsymbol{e}_2 \otimes \boldsymbol{e}_1)$ (plane components):

$$[\boldsymbol{P}] = s \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\gamma & 1 \end{bmatrix} = s \begin{bmatrix} -\gamma & 1 \\ 1 & 0 \end{bmatrix}, \qquad [\boldsymbol{S}] = s \begin{bmatrix} -2\gamma & 1 \\ 1 & 0 \end{bmatrix}.$$

In this particular case $\boldsymbol{P}$ happens to be symmetric.

(b) For $\boldsymbol{\sigma} = \sigma_0 \boldsymbol{e}_2 \otimes \boldsymbol{e}_2$: $\boldsymbol{P} = \sigma_0\, \boldsymbol{e}_2 \otimes \boldsymbol{F}^{-1}\boldsymbol{e}_2 = \sigma_0\, \boldsymbol{e}_2 \otimes (-\gamma \boldsymbol{E}_1 + \boldsymbol{E}_2)$ and $\boldsymbol{S} = \sigma_0\, (\boldsymbol{F}^{-1}\boldsymbol{e}_2) \otimes (\boldsymbol{F}^{-1}\boldsymbol{e}_2)$:

$$[\boldsymbol{P}] = \sigma_0 \begin{bmatrix} 0 & 0 \\ -\gamma & 1 \end{bmatrix}, \qquad [\boldsymbol{S}] = \sigma_0 \begin{bmatrix} \gamma^2 & -\gamma \\ -\gamma & 1 \end{bmatrix}.$$

Now $\boldsymbol{P}$ is not symmetric, while $\boldsymbol{S}$ is symmetric in both cases. The symmetry of $\boldsymbol{P}$ is not a property of the stress state: it depends on both $\boldsymbol{\sigma}$ and $\boldsymbol{F}$.

■

Solution to Exercise 4.4

◁ statement on p. 82

In the principal frame of $\boldsymbol{\sigma}$, with $\sum_i n_i^2 = 1$,

$$|\boldsymbol{t}|^2 = \sum_i \sigma_i^2 n_i^2, \qquad \sigma = \sum_i \sigma_i n_i^2, \qquad \tau^2 = \sum_i \sigma_i^2 n_i^2 - \Big(\sum_i \sigma_i n_i^2\Big)^2.$$

Stationarity of $\tau^2 - \mu(\sum_i n_i^2 - 1)$ gives $n_i(\sigma_i^2 - 2\sigma\sigma_i - \mu) = 0$ for each i. If only one n_i is non-zero, $\boldsymbol{n}$ is principal and $\tau = 0$ (minimum). If $n_i, n_j \neq 0$, subtracting the two equations gives $(\sigma_i - \sigma_j)(\sigma_i + \sigma_j - 2\sigma) = 0$, i.e., $\sigma = \tfrac{1}{2}(\sigma_i + \sigma_j)$; with $\sigma = \sigma_i n_i^2 + \sigma_j n_j^2$ and $n_i^2 + n_j^2 = 1$ this yields $n_i^2 = n_j^2 = \tfrac{1}{2}$, and

$$\tau^2 = \tfrac{1}{2}(\sigma_i^2 + \sigma_j^2) - \tfrac{1}{4}(\sigma_i + \sigma_j)^2 = \tfrac{1}{4}(\sigma_i - \sigma_j)^2.$$

(Three non-zero components would require equal principal stresses.) The largest value is obtained for the pair $(1, 3)$: $\tau_{\max} = \tfrac{1}{2}(\sigma_1 - \sigma_3)$, on the planes with normal $\boldsymbol{n} = (\hat{\boldsymbol{n}}_1 \pm \hat{\boldsymbol{n}}_3)/\sqrt{2}$, which bisect the principal planes of the largest and smallest principal stresses; there the normal stress is $\tfrac{1}{2}(\sigma_1 + \sigma_3)$ (the centre of the largest Mohr circle).

■

Chapter 5 Balance Principles

Solution to Exercise 5.1

◁ statement on p. 98

For every part Ω of the body, $m = \int_\Omega \rho \,\mathrm{d}v$ is constant. By Reynolds' theorem Eq. (5.7) with $\Phi = \rho$,

$$0 = \frac{\mathrm{D}}{\mathrm{D}t} \int_\Omega \rho \,\mathrm{d}v = \int_\Omega (\dot{\rho} + \rho \operatorname{div} \boldsymbol{v}) \,\mathrm{d}v.$$

Since the part is arbitrary and the integrand continuous, it must vanish: $\dot{\rho} + \rho \operatorname{div} \boldsymbol{v} = 0$.

■

Solution to Exercise 5.2

◁ statement on p. 98

With $\boldsymbol{v} = k\boldsymbol{x}/r^3$ and $\operatorname{grad}(r^{-3}) = -3\boldsymbol{x}/r^5$,

$$\operatorname{div}\boldsymbol{v} = k\Big(\frac{\operatorname{div}\boldsymbol{x}}{r^3} + \boldsymbol{x}\cdot\operatorname{grad} r^{-3}\Big) = k\Big(\frac{3}{r^3} - \frac{3r^2}{r^5}\Big) = 0,$$

so $\dot J = J\operatorname{div}\boldsymbol{v} = 0$ and $\dot\rho = 0$: the motion is isochoric, as assumed in its construction.

■

Solution to Exercise 5.3

◁ statement on p. 98

The cofactor of $\boldsymbol{F}$ is $J\boldsymbol{F}^{-\mathrm{T}}$, with components $JF^{-1}_{Aa} = \frac{\partial J}{\partial F_{aA}} = \frac{1}{2}\varepsilon_{abc}\varepsilon_{ABC}F_{bB}F_{cC}$ (differentiate $J = \frac{1}{6}\varepsilon_{abc}\varepsilon_{ABC}F_{aA}F_{bB}F_{cC}$, which is symmetric in the three factors). Then

$$\frac{\partial(JF^{-1}_{Aa})}{\partial X_A} = \tfrac{1}{2}\varepsilon_{abc}\varepsilon_{ABC}\Big(\frac{\partial^2 x_b}{\partial X_A\partial X_B}F_{cC} + F_{bB}\frac{\partial^2 x_c}{\partial X_A\partial X_C}\Big) = 0,$$

because each second derivative is symmetric in the pair (A,B) or (A,C), while ε_{ABC} is antisymmetric in the same pair.

■

Solution to Exercise 5.4

◁ statement on p. 98

Write the balance of angular momentum in components with $\boldsymbol{r} = \boldsymbol{x} - \boldsymbol{x}_0$, $\boldsymbol{t} = \boldsymbol{\sigma}\boldsymbol{n}$:

$$\int_\Omega \varepsilon_{ijk} r_j \rho\dot v_k \,\mathrm{d}v = \int_{\partial\Omega} \varepsilon_{ijk} r_j \sigma_{kl} n_l \,\mathrm{d}s + \int_\Omega \varepsilon_{ijk} r_j b_k \,\mathrm{d}v.$$

By the divergence theorem and $r_{j,l} = \delta_{jl}$, $\int_{\partial\Omega}\varepsilon_{ijk}r_j\sigma_{kl}n_l\,\mathrm{d}s = \int_\Omega \varepsilon_{ijk}(\sigma_{kj} + r_j\sigma_{kl,l})\,\mathrm{d}v$. Hence

$$\int_\Omega \varepsilon_{ijk} r_j (\sigma_{kl,l} + b_k - \rho\dot v_k)\,\mathrm{d}v + \int_\Omega \varepsilon_{ijk}\sigma_{kj}\,\mathrm{d}v = 0.$$

The first integral vanishes by the balance of linear momentum; the second, for every part, gives $\varepsilon_{ijk}\sigma_{kj} = 0$, i.e., the axial vector of skew $\boldsymbol{\sigma}$ is zero: $\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\mathrm{T}}$. In the material description, $\boldsymbol{\sigma} = J^{-1}\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}}$ and the condition becomes $\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{F}\boldsymbol{P}^{\mathrm{T}}$ (Exercise 4.2).

■

Solution to Exercise 5.5

◁ statement on p. 99

In the reference configuration the domain is fixed. Using $\boldsymbol{T} = \boldsymbol{P}\boldsymbol{N}$ and the divergence theorem with $(P_{aA}V_a)_{,A} = P_{aA,A}V_a + P_{aA}V_{a,A}$ and $\operatorname{Grad}\boldsymbol{V} = \dot{\boldsymbol{F}}$:

$$\int_{\partial\Omega_0}\boldsymbol{T}\cdot\boldsymbol{V}\,\mathrm{d}S = \int_{\partial\Omega_0}\boldsymbol{P}\boldsymbol{N}\cdot\boldsymbol{V}\,\mathrm{d}S = \int_{\Omega_0}\operatorname{Div}(\boldsymbol{P}^{\mathrm{T}}\boldsymbol{V})\,\mathrm{d}V = \int_{\Omega_0}(\operatorname{Div}\boldsymbol{P}\cdot\boldsymbol{V} + \boldsymbol{P}:\dot{\boldsymbol{F}})\,\mathrm{d}V.$$

The kinetic energy has a time-independent domain and mass density, so $\frac{\mathrm{D}}{\mathrm{D}t}\int_{\Omega_0}\frac{1}{2}\rho_0\boldsymbol{V}\cdot\boldsymbol{V}\,\mathrm{d}V =$

$\int_{\Omega_0} \rho_0 \dot{\boldsymbol{V}} \cdot \boldsymbol{V}\,\mathrm{d}V$. Subtracting,

$$\int_{\partial\Omega_0} \boldsymbol{T}\cdot\boldsymbol{V}\,\mathrm{d}S + \int_{\Omega_0} \boldsymbol{B}\cdot\boldsymbol{V}\,\mathrm{d}V - \frac{\mathrm{D}}{\mathrm{D}t}\int_{\Omega_0} \tfrac{1}{2}\rho_0 \boldsymbol{V}\cdot\boldsymbol{V}\,\mathrm{d}V = \int_{\Omega_0} (\operatorname{Div}\boldsymbol{P} + \boldsymbol{B} - \rho_0\dot{\boldsymbol{V}})\cdot\boldsymbol{V}\,\mathrm{d}V + \int_{\Omega_0} \boldsymbol{P}:\dot{\boldsymbol{F}}\,\mathrm{d}V,$$

and the first integral on the right vanishes by Eq. (5.19): this is Eq. (5.24). Finally, with $\boldsymbol{P} = \boldsymbol{F}\boldsymbol{S}$ and $\boldsymbol{A}:(\boldsymbol{B}\boldsymbol{C}) = (\boldsymbol{B}^{\mathrm{T}}\boldsymbol{A}):\boldsymbol{C}$,

$$\boldsymbol{P}:\dot{\boldsymbol{F}} = \boldsymbol{F}\boldsymbol{S}:\dot{\boldsymbol{F}} = \boldsymbol{S}:\boldsymbol{F}^{\mathrm{T}}\dot{\boldsymbol{F}} = \boldsymbol{S}:\operatorname{sym}(\boldsymbol{F}^{\mathrm{T}}\dot{\boldsymbol{F}}) = \boldsymbol{S}:\dot{\boldsymbol{E}},$$

since $\boldsymbol{S}$ is symmetric and $\dot{\boldsymbol{E}} = \frac{1}{2}(\dot{\boldsymbol{F}}^{\mathrm{T}}\boldsymbol{F} + \boldsymbol{F}^{\mathrm{T}}\dot{\boldsymbol{F}}) = \operatorname{sym}(\boldsymbol{F}^{\mathrm{T}}\dot{\boldsymbol{F}})$.

■

Solution to Exercise 5.6

◁ statement on p. 99

Material form. With $\boldsymbol{T} = \boldsymbol{P}\boldsymbol{N}$ and $Q_N = -\boldsymbol{Q}\cdot\boldsymbol{N}$, the divergence theorem gives $\int_{\partial\Omega_0} \boldsymbol{T}\cdot\boldsymbol{V}\,\mathrm{d}S = \int_{\Omega_0}(\operatorname{Div}\boldsymbol{P}\cdot\boldsymbol{V} + \boldsymbol{P}:\dot{\boldsymbol{F}})\,\mathrm{d}V$ (Exercise 5.5) and $\int_{\partial\Omega_0} Q_N\,\mathrm{d}S = -\int_{\Omega_0}\operatorname{Div}\boldsymbol{Q}\,\mathrm{d}V$. The domain Ω_0 is fixed, so the global first law becomes

$$\int_{\Omega_0}\Big[(\operatorname{Div}\boldsymbol{P} + \boldsymbol{B} - \rho_0\dot{\boldsymbol{V}})\cdot\boldsymbol{V} + \boldsymbol{P}:\dot{\boldsymbol{F}} - \operatorname{Div}\boldsymbol{Q} + R - \dot{e}\Big]\,\mathrm{d}V = 0.$$

The first term vanishes by the balance of linear momentum; the part is arbitrary, hence $\dot{e} = \boldsymbol{P}:\dot{\boldsymbol{F}} - \operatorname{Div}\boldsymbol{Q} + R$.

Spatial form. By Reynolds' theorem $\frac{\mathrm{D}}{\mathrm{D}t}\int_\Omega(\frac{1}{2}\rho\boldsymbol{v}\cdot\boldsymbol{v} + e_c)\,\mathrm{d}v = \int_\Omega\big[\rho\dot{\boldsymbol{v}}\cdot\boldsymbol{v} + \dot{e}_c + e_c\operatorname{div}\boldsymbol{v}\big]\,\mathrm{d}v$ (mass conservation for the kinetic part), and $\int_{\partial\Omega}(\boldsymbol{t}\cdot\boldsymbol{v} + q_n)\,\mathrm{d}s = \int_\Omega(\operatorname{div}\boldsymbol{\sigma}\cdot\boldsymbol{v} + \boldsymbol{\sigma}:\boldsymbol{d} - \operatorname{div}\boldsymbol{q})\,\mathrm{d}v$. Using Cauchy's equation, $\dot{e}_c + e_c\operatorname{div}\boldsymbol{v} = \boldsymbol{\sigma}:\boldsymbol{d} - \operatorname{div}\boldsymbol{q} + r$.

Equivalence. With $e = Je_c$ and $\dot{J} = J\operatorname{div}\boldsymbol{v}$, $\dot{e} = J(\dot{e}_c + e_c\operatorname{div}\boldsymbol{v})$; moreover $\boldsymbol{P}:\dot{\boldsymbol{F}} = J\boldsymbol{\sigma}:\boldsymbol{d}$, $R = Jr$ and, by the Piola identity, $\operatorname{Div}\boldsymbol{Q} = \operatorname{Div}(J\boldsymbol{F}^{-1}\boldsymbol{q}) = J\operatorname{div}\boldsymbol{q}$. The material form is J times the spatial one.

■

Solution to Exercise 5.7

◁ statement on p. 99

By Reynolds' theorem and the divergence theorem, the spatial form of the second law gives, for every part, $\gamma_c = \dot{\eta}_c + \eta_c\operatorname{div}\boldsymbol{v} + \operatorname{div}(\boldsymbol{q}/\Theta) - r/\Theta \geq 0$. With $\operatorname{div}(\boldsymbol{q}/\Theta) = \Theta^{-1}\operatorname{div}\boldsymbol{q} - \Theta^{-2}\boldsymbol{q}\cdot\operatorname{grad}\Theta$ and $\Theta > 0$,

$$\Theta\gamma_c = \Theta(\dot{\eta}_c + \eta_c\operatorname{div}\boldsymbol{v}) + \operatorname{div}\boldsymbol{q} - r - \frac{1}{\Theta}\boldsymbol{q}\cdot\operatorname{grad}\Theta \geq 0.$$

The spatial energy balance (Exercise 5.6) gives $\operatorname{div}\boldsymbol{q} - r = \boldsymbol{\sigma}:\boldsymbol{d} - \dot{e}_c - e_c\operatorname{div}\boldsymbol{v}$, which yields the stated inequality. To compare with Eq. (5.55), use $\eta = J\eta_c$, so $\dot{\eta} = J(\dot{\eta}_c + \eta_c\operatorname{div}\boldsymbol{v})$; $\boldsymbol{P}:\dot{\boldsymbol{F}} - \dot{e} = J(\boldsymbol{\sigma}:\boldsymbol{d} - \dot{e}_c - e_c\operatorname{div}\boldsymbol{v})$; and, with $\operatorname{Grad}\Theta = \boldsymbol{F}^{\mathrm{T}}\operatorname{grad}\Theta$, $\boldsymbol{Q}\cdot\operatorname{Grad}\Theta = J\boldsymbol{F}^{-1}\boldsymbol{q}\cdot\boldsymbol{F}^{\mathrm{T}}\operatorname{grad}\Theta = J\,\boldsymbol{q}\cdot\operatorname{grad}\Theta$. Hence $\Theta\gamma = J\,\Theta\gamma_c$, i.e., $\gamma = J\gamma_c$, as expected for densities per unit reference and current volume.

■

Solution to Exercise 5.8

◁ statement on p. 99

From $\boldsymbol{E} = \frac{1}{2}(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} - \boldsymbol{I})$ and $\delta\boldsymbol{F} = \operatorname{Grad}\delta\boldsymbol{u}$, $\delta\boldsymbol{E} = \frac{1}{2}(\delta\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} + \boldsymbol{F}^{\mathrm{T}}\delta\boldsymbol{F}) = \operatorname{sym}(\boldsymbol{F}^{\mathrm{T}}\operatorname{Grad}\delta\boldsymbol{u})$. Since $\boldsymbol{S}$ is symmetric, $\boldsymbol{S} : \delta\boldsymbol{E} = \boldsymbol{S} : \boldsymbol{F}^{\mathrm{T}}\delta\boldsymbol{F} = (\boldsymbol{F}\boldsymbol{S}) : \delta\boldsymbol{F} = \boldsymbol{P} : \delta\boldsymbol{F}$. Then, with $\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}$ and $\operatorname{Grad}\delta\boldsymbol{u}\,\boldsymbol{F}^{-1} = \operatorname{grad}\delta\boldsymbol{u}$,

$$\boldsymbol{P} : \delta\boldsymbol{F} = J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}} : \operatorname{Grad}\delta\boldsymbol{u} = J\boldsymbol{\sigma} : \operatorname{Grad}\delta\boldsymbol{u}\,\boldsymbol{F}^{-1} = J\boldsymbol{\sigma} : \operatorname{grad}\delta\boldsymbol{u} = J\boldsymbol{\sigma} : \delta\boldsymbol{e},$$

the last step by the symmetry of $\boldsymbol{\sigma}$. Integrating over Ω_0 with $J\,\mathrm{d}V = \mathrm{d}v$: $\int_{\Omega_0} \boldsymbol{S} : \delta\boldsymbol{E}\,\mathrm{d}V = \int_{\Omega_0} \boldsymbol{P} : \delta\boldsymbol{F}\,\mathrm{d}V = \int_{\Omega} \boldsymbol{\sigma} : \delta\boldsymbol{e}\,\mathrm{d}v$.

■

Solution to Exercise 5.9

◁ statement on p. 99

The virtual work equation for the bar is $\int_0^L EA\,u'\eta'\,\mathrm{d}x = \int_0^L bA\,\eta\,\mathrm{d}x + F\eta(L)$ for all η with $\eta(0) = 0$. The exact solution follows from $(EAu')' + bA = 0$, $u(0) = 0$, $EAu'(L) = F$:

$$u(x) = \frac{1}{EA}\Big[Fx + bA\Big(Lx - \frac{x^2}{2}\Big)\Big], \qquad N(x) = EAu' = F + bA(L - x).$$

One term, $u = a_1 x$, $\eta = x$: $EAa_1 L = bAL^2/2 + FL$, so $a_1 = (F + bAL/2)/(EA)$. The approximate axial force is constant, $N = F + bAL/2$, equal to the exact one at mid-span; the tip displacement $u(L) = (FL + bAL^2/2)/(EA)$ is exact. *Two terms*, $u = a_1 x + a_2 x^2$, $\eta = x$ and $\eta = x^2$:

$$EA(a_1 L + a_2 L^2) = \frac{bAL^2}{2} + FL, \qquad EA\Big(a_1 L^2 + \frac{4}{3}a_2 L^3\Big) = \frac{bAL^3}{3} + FL^2,$$

whose solution is $a_1 = (F + bAL)/(EA)$, $a_2 = -b/(2E)$: the exact solution, which belongs to the space of trial functions.

■

Solution to Exercise 5.10

◁ statement on p. 99

For a rigid body at rest take $\boldsymbol{F} = \boldsymbol{I}$, so $\boldsymbol{Q} = \boldsymbol{q}$, Grad = grad, and the stress power vanishes. With $e = \hat{e}(\Theta)$ and $\eta = \hat{\eta}(\Theta)$, the Clausius–Duhem inequality reads

$$(\Theta\hat{\eta}'(\Theta) - \hat{e}'(\Theta))\dot{\Theta} - \frac{1}{\Theta}\boldsymbol{q}\cdot\operatorname{grad}\Theta \geq 0.$$

In a homogeneous process ($\operatorname{grad}\Theta = \boldsymbol{0}$) $\dot{\Theta}$ can have either sign, hence $\hat{e}' = \Theta\hat{\eta}'$ (i.e., $\eta = c\ln\Theta + \text{const}$ for $e = c\Theta$). What remains is $-\boldsymbol{q}\cdot\operatorname{grad}\Theta/\Theta = k\,|\operatorname{grad}\Theta|^2/\Theta \geq 0$ for every gradient, i.e., $k \geq 0$. Then $\boldsymbol{q}\cdot\operatorname{grad}\Theta \leq 0$: the heat flux has a non-positive component along the temperature gradient, heat flows from hot to cold. The energy balance $\dot{e} = -\operatorname{div}\boldsymbol{q} + r$ gives the *heat equation*

$$c\,\dot{\Theta} = k\,\Delta\Theta + r, \qquad \Delta\Theta = \operatorname{div}\operatorname{grad}\Theta.$$

■

Chapter 6 Objectivity

Solution to Exercise 6.1

◁ statement on p. 109

With $\boldsymbol{F}^+ = \boldsymbol{Q}\boldsymbol{F}$ and $\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q} = \boldsymbol{Q}\boldsymbol{Q}^{\mathrm{T}} = \boldsymbol{I}$:

$$\begin{aligned}
\boldsymbol{C}^+ &= \boldsymbol{F}^{+\mathrm{T}}\boldsymbol{F}^+ = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{F} = \boldsymbol{C}, \qquad \boldsymbol{E}^+ = \tfrac{1}{2}(\boldsymbol{C}^+ - \boldsymbol{I}) = \boldsymbol{E},\\
\boldsymbol{b}^+ &= \boldsymbol{F}^+\boldsymbol{F}^{+\mathrm{T}} = \boldsymbol{Q}\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\boldsymbol{Q}^{\mathrm{T}} = \boldsymbol{Q}\boldsymbol{b}\boldsymbol{Q}^{\mathrm{T}},\\
\boldsymbol{e}^+ &= \tfrac{1}{2}(\boldsymbol{I} - (\boldsymbol{Q}\boldsymbol{b}\boldsymbol{Q}^{\mathrm{T}})^{-1}) = \tfrac{1}{2}(\boldsymbol{Q}\boldsymbol{Q}^{\mathrm{T}} - \boldsymbol{Q}\boldsymbol{b}^{-1}\boldsymbol{Q}^{\mathrm{T}}) = \boldsymbol{Q}\boldsymbol{e}\boldsymbol{Q}^{\mathrm{T}}.
\end{aligned}$$

The eigenvalues of $\boldsymbol{C}$ (the squared principal stretches λ_i^2) are unchanged, and so are the Lagrangian principal directions $\boldsymbol{N}_i$. If $\boldsymbol{b}\boldsymbol{n}_i = \lambda_i^2\boldsymbol{n}_i$, then $\boldsymbol{b}^+(\boldsymbol{Q}\boldsymbol{n}_i) = \boldsymbol{Q}\boldsymbol{b}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{n}_i = \lambda_i^2\boldsymbol{Q}\boldsymbol{n}_i$: same eigenvalues, Eulerian directions rotated, $\boldsymbol{n}_i^+ = \boldsymbol{Q}\boldsymbol{n}_i$.

■

Solution to Exercise 6.2

◁ statement on p. 109

From $\boldsymbol{x}^+ = \boldsymbol{c} + \boldsymbol{Q}\boldsymbol{x}$: $\boldsymbol{v}^+ = \dot{\boldsymbol{c}} + \dot{\boldsymbol{Q}}\boldsymbol{x} + \boldsymbol{Q}\boldsymbol{v}$, and with $\dot{\boldsymbol{Q}} = \boldsymbol{\Omega}\boldsymbol{Q}$, $\boldsymbol{Q}\boldsymbol{x} = \boldsymbol{x}^+ - \boldsymbol{c}$, we get Eq. (6.6). Differentiating Eq. (6.6),

$$\boldsymbol{a}^+ = \dot{\boldsymbol{Q}}\boldsymbol{v} + \boldsymbol{Q}\boldsymbol{a} + \ddot{\boldsymbol{c}} + \dot{\boldsymbol{\Omega}}(\boldsymbol{x}^+ - \boldsymbol{c}) + \boldsymbol{\Omega}(\boldsymbol{v}^+ - \dot{\boldsymbol{c}}),$$

and, by Eq. (6.6) again, $\dot{\boldsymbol{Q}}\boldsymbol{v} = \boldsymbol{\Omega}\boldsymbol{Q}\boldsymbol{v} = \boldsymbol{\Omega}(\boldsymbol{v}^+ - \dot{\boldsymbol{c}}) - \boldsymbol{\Omega}^2(\boldsymbol{x}^+ - \boldsymbol{c})$: this gives Eq. (6.7). Now set $\boldsymbol{r} = \boldsymbol{x}^+ - \boldsymbol{c}$ and use the relative velocity $\boldsymbol{Q}\boldsymbol{v} = \boldsymbol{v}^+ - \dot{\boldsymbol{c}} - \boldsymbol{\Omega}\boldsymbol{r}$: $-\boldsymbol{\Omega}^2\boldsymbol{r} + 2\boldsymbol{\Omega}(\boldsymbol{v}^+ - \dot{\boldsymbol{c}}) = \boldsymbol{\Omega}^2\boldsymbol{r} + 2\boldsymbol{\Omega}\boldsymbol{Q}\boldsymbol{v}$, so that, with $\boldsymbol{\Omega}\boldsymbol{u} = \boldsymbol{\omega} \times \boldsymbol{u}$,

$$\boldsymbol{a}^+ = \boldsymbol{Q}\boldsymbol{a} + \ddot{\boldsymbol{c}} + \dot{\boldsymbol{\omega}} \times \boldsymbol{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r}) + 2\boldsymbol{\omega} \times \boldsymbol{Q}\boldsymbol{v}.$$

This is the classical composition of accelerations: the absolute acceleration equals the relative one ($\boldsymbol{Q}\boldsymbol{a}$) plus the transport acceleration of the moving frame (translational $\ddot{\boldsymbol{c}}$, Euler $\dot{\boldsymbol{\omega}} \times \boldsymbol{r}$, centripetal $\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r})$) plus the Coriolis acceleration $2\boldsymbol{\omega} \times \boldsymbol{Q}\boldsymbol{v}$. Seen from the rotating frame, these terms produce the fictitious forces (the centrifugal one, $-m\,\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r})$, points away from the axis). For a Galilean transformation $\ddot{\boldsymbol{c}} = \boldsymbol{0}$ and $\boldsymbol{\Omega} = \boldsymbol{O}$, so $\boldsymbol{a}^+ = \boldsymbol{Q}\boldsymbol{a}$.

■

Solution to Exercise 6.3

◁ statement on p. 110

$\boldsymbol{\Omega}^{\mathrm{T}} = \boldsymbol{Q}\dot{\boldsymbol{Q}}^{\mathrm{T}} = -\dot{\boldsymbol{Q}}\boldsymbol{Q}^{\mathrm{T}} = -\boldsymbol{\Omega}$ from the time derivative of $\boldsymbol{Q}\boldsymbol{Q}^{\mathrm{T}} = \boldsymbol{I}$. Then $\dot{\boldsymbol{F}}^+ = \dot{\boldsymbol{Q}}\boldsymbol{F} + \boldsymbol{Q}\dot{\boldsymbol{F}}$, $\boldsymbol{F}^{+-1} = \boldsymbol{F}^{-1}\boldsymbol{Q}^{\mathrm{T}}$ and

$$\boldsymbol{l}^+ = \dot{\boldsymbol{Q}}\boldsymbol{F}\boldsymbol{F}^{-1}\boldsymbol{Q}^{\mathrm{T}} + \boldsymbol{Q}\dot{\boldsymbol{F}}\boldsymbol{F}^{-1}\boldsymbol{Q}^{\mathrm{T}} = \boldsymbol{\Omega} + \boldsymbol{Q}\boldsymbol{l}\boldsymbol{Q}^{\mathrm{T}}.$$

The symmetric part of $\boldsymbol{\Omega}$ is zero and $\operatorname{sym}(\boldsymbol{Q}\boldsymbol{l}\boldsymbol{Q}^{\mathrm{T}}) = \boldsymbol{Q}\boldsymbol{d}\boldsymbol{Q}^{\mathrm{T}}$, $\operatorname{skew}(\boldsymbol{Q}\boldsymbol{l}\boldsymbol{Q}^{\mathrm{T}}) = \boldsymbol{Q}\boldsymbol{w}\boldsymbol{Q}^{\mathrm{T}}$: this gives Eq. (6.13). For $\boldsymbol{\chi} = \boldsymbol{Q}_0(t)\boldsymbol{X}$: $\boldsymbol{F} = \boldsymbol{Q}_0$, $\boldsymbol{l} = \dot{\boldsymbol{Q}}_0\boldsymbol{Q}_0^{\mathrm{T}}$, which is skew, so $\boldsymbol{d} = \boldsymbol{O}$ and $\boldsymbol{w} = \dot{\boldsymbol{Q}}_0\boldsymbol{Q}_0^{\mathrm{T}}$. A rigid rotation produces no rate of deformation but a spin, in agreement with Eq. (6.13) applied to the identity motion ($\boldsymbol{l} = \boldsymbol{O}$, $\boldsymbol{l}^+ = \boldsymbol{\Omega}$).

■

Solution to Exercise 6.4 ◁ statement on p. 110

$\boldsymbol{\tau}^+ = J^+\boldsymbol{\sigma}^+ = J\boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{Q}^{\mathrm{T}} = \boldsymbol{Q}\boldsymbol{\tau}\boldsymbol{Q}^{\mathrm{T}}$. For $\boldsymbol{P}$ and $\boldsymbol{S}$, see Section 6.6: $\boldsymbol{P}^+ = J\boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{Q}^{\mathrm{T}}(\boldsymbol{Q}\boldsymbol{F})^{-\mathrm{T}} = \boldsymbol{Q}\boldsymbol{P}$ and $\boldsymbol{S}^+ = (\boldsymbol{Q}\boldsymbol{F})^{-1}\boldsymbol{Q}\boldsymbol{P} = \boldsymbol{F}^{-1}\boldsymbol{P} = \boldsymbol{S}$. For the stress power, $\dot{\boldsymbol{F}}^+ = \dot{\boldsymbol{Q}}\boldsymbol{F} + \boldsymbol{Q}\dot{\boldsymbol{F}} = \boldsymbol{\Omega}\boldsymbol{Q}\boldsymbol{F} + \boldsymbol{Q}\dot{\boldsymbol{F}}$, hence

$$\boldsymbol{P}^+ : \dot{\boldsymbol{F}}^+ = \boldsymbol{Q}\boldsymbol{P} : \boldsymbol{\Omega}\boldsymbol{Q}\boldsymbol{F} + \boldsymbol{Q}\boldsymbol{P} : \boldsymbol{Q}\dot{\boldsymbol{F}} = (\boldsymbol{Q}\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}}\boldsymbol{Q}^{\mathrm{T}}) : \boldsymbol{\Omega} + \boldsymbol{P} : \dot{\boldsymbol{F}}.$$

$\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}} = J\boldsymbol{\sigma}$ is symmetric, so $\boldsymbol{Q}\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}}\boldsymbol{Q}^{\mathrm{T}}$ is symmetric and its contraction with the skew $\boldsymbol{\Omega}$ vanishes: $\boldsymbol{P}^+ : \dot{\boldsymbol{F}}^+ = \boldsymbol{P} : \dot{\boldsymbol{F}}$. More quickly, $\boldsymbol{P} : \dot{\boldsymbol{F}} = \boldsymbol{S} : \dot{\boldsymbol{E}}$, and both $\boldsymbol{S}$ and $\boldsymbol{E}$ are unaffected. ■

Solution to Exercise 6.5 ◁ statement on p. 110

From $\boldsymbol{\sigma}^+ = \boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{Q}^{\mathrm{T}}$ and $\dot{\boldsymbol{Q}} = \boldsymbol{\Omega}\boldsymbol{Q}$, $\dot{\boldsymbol{Q}}^{\mathrm{T}} = -\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{\Omega}$: $\dot{\boldsymbol{\sigma}}^+ = \boldsymbol{Q}\dot{\boldsymbol{\sigma}}\boldsymbol{Q}^{\mathrm{T}} + \boldsymbol{\Omega}\boldsymbol{\sigma}^+ - \boldsymbol{\sigma}^+\boldsymbol{\Omega} \neq \boldsymbol{Q}\dot{\boldsymbol{\sigma}}\boldsymbol{Q}^{\mathrm{T}}$. With $\boldsymbol{w}^+ = \boldsymbol{\Omega} + \boldsymbol{Q}\boldsymbol{w}\boldsymbol{Q}^{\mathrm{T}}$:

$$\dot{\boldsymbol{\sigma}}^+ - \boldsymbol{w}^+\boldsymbol{\sigma}^+ + \boldsymbol{\sigma}^+\boldsymbol{w}^+ = \boldsymbol{Q}\dot{\boldsymbol{\sigma}}\boldsymbol{Q}^{\mathrm{T}} + \boldsymbol{\Omega}\boldsymbol{\sigma}^+ - \boldsymbol{\sigma}^+\boldsymbol{\Omega} - \boldsymbol{\Omega}\boldsymbol{\sigma}^+ - \boldsymbol{Q}\boldsymbol{w}\boldsymbol{\sigma}\boldsymbol{Q}^{\mathrm{T}} + \boldsymbol{\sigma}^+\boldsymbol{\Omega} + \boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{w}\boldsymbol{Q}^{\mathrm{T}} = \boldsymbol{Q}\overset{\triangledown}{\boldsymbol{\sigma}}\boldsymbol{Q}^{\mathrm{T}}.$$

With $\boldsymbol{l}^+ = \boldsymbol{\Omega} + \boldsymbol{Q}\boldsymbol{l}\boldsymbol{Q}^{\mathrm{T}}$ and $\boldsymbol{l}^{+\mathrm{T}} = -\boldsymbol{\Omega} + \boldsymbol{Q}\boldsymbol{l}^{\mathrm{T}}\boldsymbol{Q}^{\mathrm{T}}$: $\boldsymbol{l}^+\boldsymbol{\sigma}^+ = \boldsymbol{\Omega}\boldsymbol{\sigma}^+ + \boldsymbol{Q}\boldsymbol{l}\boldsymbol{\sigma}\boldsymbol{Q}^{\mathrm{T}}$ and $\boldsymbol{\sigma}^+\boldsymbol{l}^{+\mathrm{T}} = -\boldsymbol{\sigma}^+\boldsymbol{\Omega} + \boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{l}^{\mathrm{T}}\boldsymbol{Q}^{\mathrm{T}}$, so

$$\dot{\boldsymbol{\sigma}}^+ - \boldsymbol{l}^+\boldsymbol{\sigma}^+ - \boldsymbol{\sigma}^+\boldsymbol{l}^{+\mathrm{T}} = \boldsymbol{Q}(\dot{\boldsymbol{\sigma}} - \boldsymbol{l}\boldsymbol{\sigma} - \boldsymbol{\sigma}\boldsymbol{l}^{\mathrm{T}})\boldsymbol{Q}^{\mathrm{T}}.$$

In both cases the terms in $\boldsymbol{\Omega}$ cancel.

■

Solution to Exercise 6.6 ◁ statement on p. 110

$\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}} = \det\boldsymbol{F}\,\boldsymbol{g}(\boldsymbol{F})\boldsymbol{F}^{-\mathrm{T}} \coloneqq \boldsymbol{G}(\boldsymbol{F})$. Objectivity: $\boldsymbol{P}^+ = \boldsymbol{Q}\boldsymbol{P}$ and $\boldsymbol{P}^+ = \boldsymbol{G}(\boldsymbol{Q}\boldsymbol{F})$, so $\boldsymbol{G}(\boldsymbol{Q}\boldsymbol{F}) = \boldsymbol{Q}\boldsymbol{G}(\boldsymbol{F})$; with $\boldsymbol{Q} = \boldsymbol{R}^{\mathrm{T}}$, $\boldsymbol{G}(\boldsymbol{U}) = \boldsymbol{R}^{\mathrm{T}}\boldsymbol{G}(\boldsymbol{F})$, i.e., $\boldsymbol{G}(\boldsymbol{F}) = \boldsymbol{R}\boldsymbol{G}(\boldsymbol{U})$. Directly from Eq. (6.17): $\boldsymbol{P} = \det\boldsymbol{U}\,\boldsymbol{R}\boldsymbol{g}(\boldsymbol{U})\boldsymbol{R}^{\mathrm{T}}\boldsymbol{R}\boldsymbol{U}^{-1} = \boldsymbol{R}\,[\det\boldsymbol{U}\,\boldsymbol{g}(\boldsymbol{U})\boldsymbol{U}^{-1}]$, so $\boldsymbol{G}(\boldsymbol{U}) = \det\boldsymbol{U}\,\boldsymbol{g}(\boldsymbol{U})\boldsymbol{U}^{-1}$. Then $\boldsymbol{S} = \boldsymbol{F}^{-1}\boldsymbol{P} = \boldsymbol{U}^{-1}\boldsymbol{R}^{\mathrm{T}}\boldsymbol{R}\boldsymbol{G}(\boldsymbol{U}) = \det\boldsymbol{U}\,\boldsymbol{U}^{-1}\boldsymbol{g}(\boldsymbol{U})\boldsymbol{U}^{-1}$, a function of $\boldsymbol{U} = \boldsymbol{C}^{1/2}$ only: $\boldsymbol{S} = \boldsymbol{H}(\boldsymbol{C})$. Conversely, for $\boldsymbol{g}(\boldsymbol{F}) = J^{-1}\boldsymbol{F}\boldsymbol{H}(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F})\boldsymbol{F}^{\mathrm{T}}$,

$$\boldsymbol{g}(\boldsymbol{Q}\boldsymbol{F}) = J^{-1}\boldsymbol{Q}\boldsymbol{F}\,\boldsymbol{H}(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{F})\,\boldsymbol{F}^{\mathrm{T}}\boldsymbol{Q}^{\mathrm{T}} = \boldsymbol{Q}\,\boldsymbol{g}(\boldsymbol{F})\,\boldsymbol{Q}^{\mathrm{T}},$$

for every $\boldsymbol{H}$: material laws are automatically objective.

■

Solution to Exercise 6.7 ◁ statement on p. 110

(a) Objective: $\boldsymbol{S}$ and $\boldsymbol{E}$ are both unaffected. (b) Not objective. Take $\boldsymbol{F} = \boldsymbol{I}$ (so $\boldsymbol{\sigma} = \boldsymbol{O}$) and superimpose a rotation: $\boldsymbol{F}^+ = \boldsymbol{Q}$, $\tilde{\boldsymbol{\varepsilon}}^+ = \operatorname{sym}\boldsymbol{Q} - \boldsymbol{I} \neq \boldsymbol{O}$ and $\boldsymbol{\sigma}^+ \neq \boldsymbol{O} = \boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{Q}^{\mathrm{T}}$. For a rotation by θ about $\boldsymbol{e}_3$, $\operatorname{sym}\boldsymbol{Q} - \boldsymbol{I} = (\cos\theta - 1)(\boldsymbol{e}_1 \otimes \boldsymbol{e}_1 + \boldsymbol{e}_2 \otimes \boldsymbol{e}_2)$: a rigid rotation would produce a compressive stress. (c) Objective: $\boldsymbol{\sigma}^+ = \mu(\boldsymbol{Q}\boldsymbol{b}\boldsymbol{Q}^{\mathrm{T}} - \boldsymbol{I}) = \boldsymbol{Q}\mu(\boldsymbol{b} - \boldsymbol{I})\boldsymbol{Q}^{\mathrm{T}}$. (d) Not objective: $\boldsymbol{\sigma}^+ = \mu(\boldsymbol{C} - \boldsymbol{I}) = \boldsymbol{\sigma} \neq \boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{Q}^{\mathrm{T}}$; a spatial tensor cannot be a function of a material tensor alone. (e) $\boldsymbol{\sigma} = 2\eta\boldsymbol{d}$ is objective because $\boldsymbol{d}^+ = \boldsymbol{Q}\boldsymbol{d}\boldsymbol{Q}^{\mathrm{T}}$; $\boldsymbol{\sigma} = 2\eta\boldsymbol{l}$ is not, since

$\boldsymbol{\sigma}^+ = \boldsymbol{Q}\boldsymbol{\sigma}\boldsymbol{Q}^{\mathrm{T}} + 2\eta\boldsymbol{\Omega}$ (and it would also give a non-symmetric stress): a rigid spin would produce stress.

■

Solution to Exercise 6.8

◁ statement on p. 110

The invariants of $\boldsymbol{Q}\boldsymbol{b}\boldsymbol{Q}^{\mathrm{T}}$ and $\boldsymbol{b}$ coincide, so the coefficients β_i are unchanged, and $(\boldsymbol{Q}\boldsymbol{b}\boldsymbol{Q}^{\mathrm{T}})^{-1} = \boldsymbol{Q}\boldsymbol{b}^{-1}\boldsymbol{Q}^{\mathrm{T}}$: $\boldsymbol{h}(\boldsymbol{Q}\boldsymbol{b}\boldsymbol{Q}^{\mathrm{T}}) = \beta_0\boldsymbol{Q}\boldsymbol{Q}^{\mathrm{T}} + \beta_1\boldsymbol{Q}\boldsymbol{b}\boldsymbol{Q}^{\mathrm{T}} + \beta_{-1}\boldsymbol{Q}\boldsymbol{b}^{-1}\boldsymbol{Q}^{\mathrm{T}} = \boldsymbol{Q}\boldsymbol{h}(\boldsymbol{b})\boldsymbol{Q}^{\mathrm{T}}$. Cayley–Hamilton, $\boldsymbol{b}^3 - I_1\boldsymbol{b}^2 + I_2\boldsymbol{b} - I_3\boldsymbol{I} = \boldsymbol{O}$, multiplied by $\boldsymbol{b}^{-1}$ gives $\boldsymbol{b}^2 = I_1\boldsymbol{b} - I_2\boldsymbol{I} + I_3\boldsymbol{b}^{-1}$; substituting into $\alpha_0\boldsymbol{I} + \alpha_1\boldsymbol{b} + \alpha_2\boldsymbol{b}^2$ gives $\beta_0 = \alpha_0 - I_2\alpha_2$, $\beta_1 = \alpha_1 + I_1\alpha_2$, $\beta_{-1} = I_3\alpha_2$. With $\boldsymbol{F}^{-1}\boldsymbol{F}^{-\mathrm{T}} = \boldsymbol{C}^{-1}$, $\boldsymbol{F}^{-1}\boldsymbol{b}\boldsymbol{F}^{-\mathrm{T}} = \boldsymbol{I}$, $\boldsymbol{F}^{-1}\boldsymbol{b}^{-1}\boldsymbol{F}^{-\mathrm{T}} = \boldsymbol{C}^{-2}$: $\boldsymbol{S} = J\boldsymbol{F}^{-1}\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}} = J(\beta_0\boldsymbol{C}^{-1} + \beta_1\boldsymbol{I} + \beta_{-1}\boldsymbol{C}^{-2})$.

Uniaxial tension: $\boldsymbol{b} = \mathrm{diag}(\lambda^2, \lambda^{-1}, \lambda^{-1})$, so $\sigma_{22} = \sigma_{33} = -p + \mu/\lambda = 0$ gives $p = \mu/\lambda$, and

$$\sigma_{11} = \mu\Big(\lambda^2 - \frac{1}{\lambda}\Big), \qquad P_{11} = \frac{\sigma_{11}}{\lambda} = \mu\Big(\lambda - \frac{1}{\lambda^2}\Big).$$

For $\lambda = 1 + \varepsilon$, $\sigma_{11} \approx 3\mu\varepsilon$: Young's modulus of an incompressible material, $E = 3\mu$.

■

Chapter 7 Constitutive Theory

Solution to Exercise 7.1

◁ statement on p. 122

Let $\eta^*(\boldsymbol{F}, \Theta, \boldsymbol{\xi})$ be the entropy at which the infimum is attained, $\partial e/\partial\eta(\boldsymbol{F}, \eta^*, \boldsymbol{\xi}) = \Theta$. Then $A = e(\boldsymbol{F}, \eta^*, \boldsymbol{\xi}) - \Theta\eta^*$ and, by the chain rule,

$$\frac{\partial A}{\partial \boldsymbol{F}} = \frac{\partial e}{\partial \boldsymbol{F}} + \Big(\frac{\partial e}{\partial \eta} - \Theta\Big)\frac{\partial \eta^*}{\partial \boldsymbol{F}} = \frac{\partial e}{\partial \boldsymbol{F}} = \boldsymbol{P}^e, \quad \frac{\partial A}{\partial \Theta} = \Big(\frac{\partial e}{\partial \eta} - \Theta\Big)\frac{\partial \eta^*}{\partial \Theta} - \eta^* = -\eta, \quad \frac{\partial A}{\partial \boldsymbol{\xi}} = \frac{\partial e}{\partial \boldsymbol{\xi}} = -\boldsymbol{\Xi}.$$

From $e = A + \Theta\eta$: $\dot e = \dot A + \dot\Theta\eta + \Theta\dot\eta$, hence $\mathcal{D}_{\mathrm{int}} = \boldsymbol{P} : \dot{\boldsymbol{F}} - \dot e + \Theta\dot\eta = \boldsymbol{P} : \dot{\boldsymbol{F}} - \dot A - \eta\dot\Theta$. Now take $A = A(\boldsymbol{F}, \Theta, \boldsymbol{\xi})$ and expand:

$$\mathcal{D}_{\mathrm{int}} = \Big(\boldsymbol{P} - \frac{\partial A}{\partial \boldsymbol{F}}\Big) : \dot{\boldsymbol{F}} - \Big(\eta + \frac{\partial A}{\partial \Theta}\Big)\dot\Theta - \frac{\partial A}{\partial \boldsymbol{\xi}} \cdot \dot{\boldsymbol{\xi}} \geq 0.$$

$\dot\Theta$ can be assigned arbitrarily and appears linearly (if η and $\boldsymbol{P}$ do not depend on it), so $\eta = -\partial A/\partial\Theta$. Defining $\boldsymbol{P}^e = \partial A/\partial \boldsymbol{F}$, $\boldsymbol{P}^v = \boldsymbol{P} - \boldsymbol{P}^e$ and $\boldsymbol{\Xi} = -\partial A/\partial\boldsymbol{\xi}$, $\mathcal{D}_{\mathrm{int}} = \boldsymbol{P}^v : \dot{\boldsymbol{F}} + \boldsymbol{\Xi}\cdot\dot{\boldsymbol{\xi}} \geq 0$. For an elastic material ($\boldsymbol{P} = \boldsymbol{P}(\boldsymbol{F}, \Theta)$, no $\boldsymbol{\xi}$) the arbitrariness of $\dot{\boldsymbol{F}}$ also gives $\boldsymbol{P} = \partial A/\partial \boldsymbol{F}$ and $\mathcal{D}_{\mathrm{int}} = 0$: the basis of hyperelasticity.

■

Solution to Exercise 7.2

◁ statement on p. 122

Enthalpy. Let $\boldsymbol{F}^*$ attain the infimum, $\partial e/\partial \boldsymbol{F}(\boldsymbol{F}^*, \eta, \boldsymbol{\xi}) = \boldsymbol{P}^e$, and $H = e(\boldsymbol{F}^*, \eta, \boldsymbol{\xi}) - \boldsymbol{P}^e : \boldsymbol{F}^*$. Differentiating (the terms in $\partial\boldsymbol{F}^*$ cancel because $\partial e/\partial\boldsymbol{F} = \boldsymbol{P}^e$ at $\boldsymbol{F}^*$):

$$\frac{\partial H}{\partial \boldsymbol{P}^e} = -\boldsymbol{F}^*, \qquad \frac{\partial H}{\partial \eta} = \frac{\partial e}{\partial \eta} = \Theta, \qquad \frac{\partial H}{\partial \boldsymbol{\xi}} = \frac{\partial e}{\partial \boldsymbol{\xi}} = -\boldsymbol{\Xi}.$$

Gibbs energy. Let $\boldsymbol{F}^*$ attain the supremum, $\boldsymbol{P}^e = \partial A/\partial \boldsymbol{F}(\boldsymbol{F}^*)$, and $G = \boldsymbol{P}^e : \boldsymbol{F}^* - A(\boldsymbol{F}^*, \Theta, \boldsymbol{\xi})$. Then

$$\frac{\partial G}{\partial \boldsymbol{P}^e} = \boldsymbol{F}^*, \qquad \frac{\partial G}{\partial \Theta} = -\frac{\partial A}{\partial \Theta} = \eta, \qquad \frac{\partial G}{\partial \boldsymbol{\xi}} = -\frac{\partial A}{\partial \boldsymbol{\xi}} = \boldsymbol{\Xi}.$$

The thermodynamic force is $+\partial G/\partial \boldsymbol{\xi}$, not $-\partial G/\partial \boldsymbol{\xi}$ as on slide 232. With the convention $\tilde{G} = \inf_{\boldsymbol{F}}[A - \boldsymbol{P}^e : \boldsymbol{F}] = -G$ all signs flip: $\boldsymbol{F} = -\partial \tilde{G}/\partial \boldsymbol{P}^e$, $\eta = -\partial \tilde{G}/\partial \Theta$, $\boldsymbol{\Xi} = -\partial \tilde{G}/\partial \boldsymbol{\xi}$. The slide mixes the two conventions.

■

Solution to Exercise 7.3

◁ statement on p. 122

Set $s = \varepsilon/\varepsilon_0$ and $\sigma_\infty = E_0\varepsilon_0$ (the asymptotic stress). Then

$$\omega(\varepsilon) = \int_0^\varepsilon \frac{E_0 \varepsilon}{1 + \varepsilon/\varepsilon_0}\, \mathrm{d}\varepsilon = E_0 \varepsilon_0^2 [s - \ln(1+s)].$$

Inverting the law, $\varepsilon = \varepsilon_0\, t/(1-t)$ with $t = \sigma/\sigma_\infty < 1$, and

$$\gamma(\sigma) = \int_0^\sigma \varepsilon\, \mathrm{d}\sigma = E_0\varepsilon_0^2[-t - \ln(1-t)], \qquad \frac{\mathrm{d}\gamma}{\mathrm{d}\sigma} = \frac{E_0\varepsilon_0^2}{\sigma_\infty}\Big(-1 + \frac{1}{1-t}\Big) = \varepsilon_0 \frac{t}{1-t} = \varepsilon.$$

At corresponding points $t = s/(1+s)$, $1 - t = 1/(1+s)$, so $\gamma = E_0\varepsilon_0^2[\ln(1+s) - s/(1+s)]$ and $\omega + \gamma = E_0\varepsilon_0^2\, s^2/(1+s) = \sigma\varepsilon$. For small strains, $\omega \approx \frac{1}{2}E_0\varepsilon^2$ and $\gamma \approx \sigma^2/(2E_0)$, the linear values. For finite strains they differ: at $s = 1$, $\omega = 0.307 E_0\varepsilon_0^2$ and $\gamma = 0.193 E_0\varepsilon_0^2$, while $\sigma\varepsilon/2 = 0.25 E_0\varepsilon_0^2$. For this softening (concave) curve, $\omega > \gamma$.

■

Solution to Exercise 7.4

◁ statement on p. 122

Write $\boldsymbol{d} = \frac{1}{3}(\operatorname{tr}\boldsymbol{d})\boldsymbol{I} + \operatorname{dev}\boldsymbol{d}$; then $\boldsymbol{d} : \boldsymbol{d} = \frac{1}{3}(\operatorname{tr}\boldsymbol{d})^2 + \operatorname{dev}\boldsymbol{d} : \operatorname{dev}\boldsymbol{d}$ and

$$\lambda(\operatorname{tr}\boldsymbol{d})^2 + 2\mu\, \boldsymbol{d} : \boldsymbol{d} = \Big(\lambda + \frac{2}{3}\mu\Big)(\operatorname{tr}\boldsymbol{d})^2 + 2\mu\, |\operatorname{dev}\boldsymbol{d}|^2 .$$

Necessity: a deviatoric $\boldsymbol{d} \neq \boldsymbol{O}$ gives $\mu \geq 0$; $\boldsymbol{d} = \boldsymbol{I}$ gives $9(\lambda + \frac{2}{3}\mu) \geq 0$, i.e., $3\lambda + 2\mu \geq 0$. *Sufficiency*: both terms are then non-negative. Fourier's law gives $-\boldsymbol{q} \cdot \operatorname{grad}\Theta/\Theta = k\,|\operatorname{grad}\Theta|^2/\Theta \geq 0$ for every gradient, i.e., $k \geq 0$. Finally, with $\boldsymbol{P}^v = J\boldsymbol{\sigma}^v\boldsymbol{F}^{-\mathrm{T}}$, $\boldsymbol{P}^v : \dot{\boldsymbol{F}} = J\boldsymbol{\sigma}^v : \dot{\boldsymbol{F}}\boldsymbol{F}^{-1} = J\boldsymbol{\sigma}^v : \boldsymbol{l} = J\boldsymbol{\sigma}^v : \boldsymbol{d}$, by the symmetry of $\boldsymbol{\sigma}^v$; similarly $\boldsymbol{Q} \cdot \operatorname{Grad}\Theta = J\boldsymbol{q} \cdot \operatorname{grad}\Theta$ (Exercise 5.7).

■

Solution to Exercise 7.5

◁ statement on p. 122

With $\boldsymbol{x}_\Theta = -\operatorname{grad}\Theta/\Theta$, $\psi = \frac{\lambda}{2}(\operatorname{tr}\boldsymbol{d})^2 + \mu\, \boldsymbol{d} : \boldsymbol{d} + \frac{k\Theta}{2}\,|\boldsymbol{x}_\Theta|^2$, and $\partial(\operatorname{tr}\boldsymbol{d})/\partial\boldsymbol{d} = \boldsymbol{I}$,

$$\frac{\partial \psi}{\partial \boldsymbol{d}} = \lambda(\operatorname{tr}\boldsymbol{d})\boldsymbol{I} + 2\mu\boldsymbol{d} = \boldsymbol{\sigma}^v, \qquad \frac{\partial \psi}{\partial \boldsymbol{x}_\Theta} = k\Theta\boldsymbol{x}_\Theta = -k\operatorname{grad}\Theta = \boldsymbol{q}.$$

ψ is a quadratic form; it is convex if and only if it is non-negative, i.e., (Exercise 7.4) $\mu \geq 0$, $3\lambda + 2\mu \geq 0$ and $k \geq 0$. Being homogeneous of degree two, $\boldsymbol{X} \cdot \partial\psi/\partial\boldsymbol{X} = 2\psi \geq 0$ (Euler's

theorem): the dissipation is twice the kinetic potential.

■

Solution to Exercise 7.6

◁ statement on p. 122

(a) $\sigma^e = \sigma(\dot\varepsilon = 0) = E\varepsilon$, $\sigma^v = \eta_v\dot\varepsilon$; $A = \frac{1}{2}E\varepsilon^2$ gives $\sigma^e = \frac{\mathrm{d}A}{\mathrm{d}\varepsilon}$. The dissipation is $\mathcal{D}_{\text{int}} = \sigma\dot\varepsilon - \dot A = \eta_v\dot\varepsilon^2 \geq 0$ for every $\dot\varepsilon$ if and only if $\eta_v \geq 0$.

(b) $\sigma = \partial A/\partial\varepsilon = E(\varepsilon - \xi)$ (no viscous stress at the level of the total strain) and $\Xi = -\partial A/\partial\xi = E(\varepsilon - \xi) = \sigma$: the thermodynamic force conjugate to the viscous strain is the stress in the dashpot. Then $\mathcal{D}_{\text{int}} = \sigma\dot\varepsilon - \dot A = \sigma\dot\varepsilon - E(\varepsilon - \xi)(\dot\varepsilon - \dot\xi) = \Xi\dot\xi$. The law $\dot\xi = \Xi/\eta_v$ derives from $\psi(\Xi) = \Xi^2/(2\eta_v)$, convex for $\eta_v > 0$, and $\mathcal{D}_{\text{int}} = \Xi^2/\eta_v \geq 0$. Eliminating ξ, $\dot\varepsilon = \dot\sigma/E + \sigma/\eta_v$: the Maxwell model.

■

Solution to Exercise 7.7

◁ statement on p. 123

For a thermoelastic material, Exercise 7.1 gives $\boldsymbol{P} = \partial A/\partial\boldsymbol{F}$, $\eta = -\partial A/\partial\Theta$, $\mathcal{D}_{\text{int}} = 0$. From $e = A + \Theta\eta$,

$$\dot e = \frac{\partial A}{\partial \boldsymbol{F}} : \dot{\boldsymbol{F}} + \frac{\partial A}{\partial\Theta}\dot\Theta + \dot\Theta\eta + \Theta\dot\eta = \boldsymbol{P} : \dot{\boldsymbol{F}} + \Theta\dot\eta,$$

and the energy balance $\dot e = \boldsymbol{P} : \dot{\boldsymbol{F}} - \operatorname{Div}\boldsymbol{Q} + R$ reduces to $\Theta\dot\eta = R - \operatorname{Div}\boldsymbol{Q}$. Differentiating $\eta = -\partial A/\partial\Theta$, $\Theta\dot\eta = -\Theta\frac{\partial^2 A}{\partial\Theta^2}\dot\Theta - \Theta\frac{\partial^2 A}{\partial\Theta\,\partial\boldsymbol{F}} : \dot{\boldsymbol{F}} = c\dot\Theta - \Theta\frac{\partial^2 A}{\partial\Theta\,\partial\boldsymbol{F}} : \dot{\boldsymbol{F}}$, which gives the coupled heat equation. The last term, $\Theta\,\partial\boldsymbol{P}/\partial\Theta : \dot{\boldsymbol{F}}$, is the *thermoelastic coupling*: under adiabatic conditions ($R = 0$, $\boldsymbol{Q} = \boldsymbol{0}$) a deformation changes the temperature. In linear thermoelasticity, $\partial\boldsymbol{\sigma}/\partial\Theta = -3K\alpha\boldsymbol{I}$ (with K the bulk modulus and α the thermal expansion coefficient) and $c\dot\Theta = r - \operatorname{div}\boldsymbol{q} - 3K\alpha\Theta\operatorname{tr}\dot{\boldsymbol{\varepsilon}}$: a metal bar cools when stretched adiabatically (Kelvin effect). Rubber, whose stiffness is entropic, warms up instead (Gough–Joule effect).

■

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